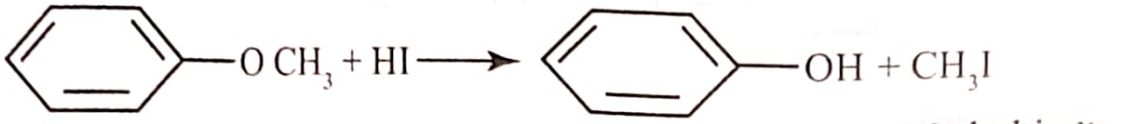
Chemical Properties of Ethers

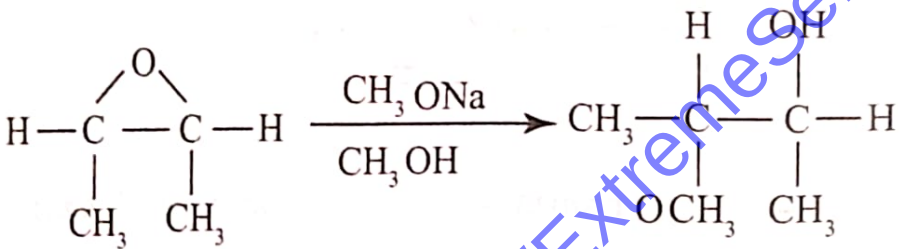
➥ Ethers are much less reactive than alcohols. For example, they are unaffected by treatment with oxidizing agents, reducing agents, bases and most acids. However, they undergo the following reactions.

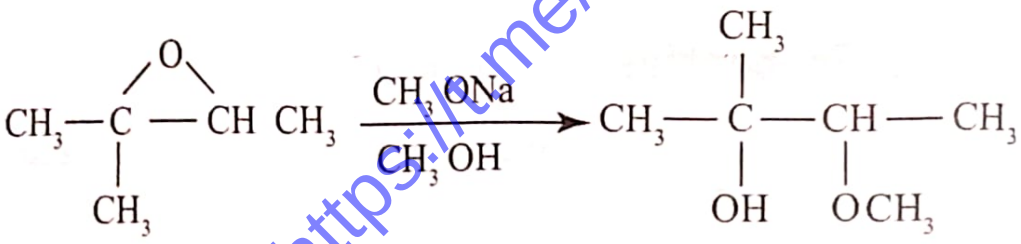
a. Reaction With Hydroiodic Acid (III)



anisole — Phenol — Methyl iodide

b. Acid - catalyzed Ring Opening Reactions

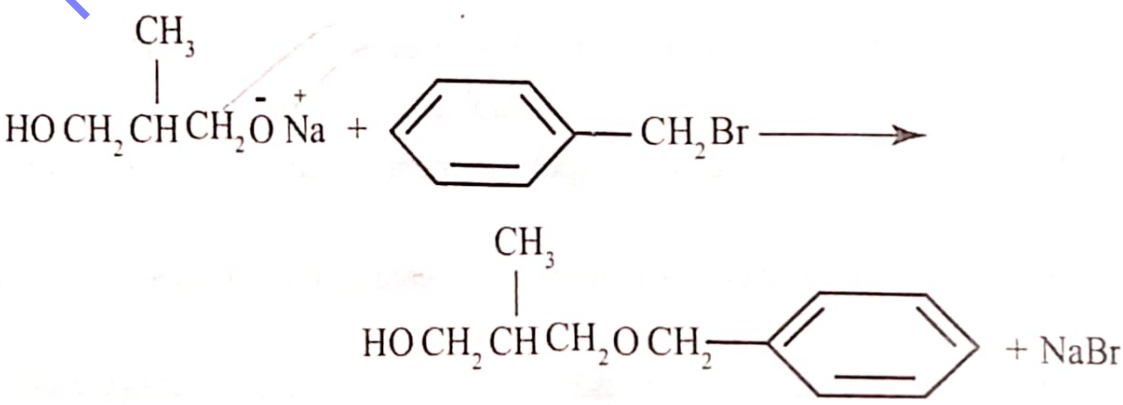




Preparation of Ethers

➥ The important method used to make ether is known as the **Williamson synthesis.**

➥ It is the reaction between alkoxide and alkyl halide.



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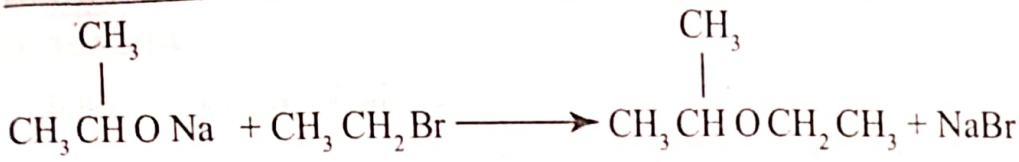

Uses of Ethers

Ethers are used as organic solvents because they are inert toward many reagents. Diethyl ether is used as aesthetic in hospitals.

Illustrative Examples

36. Which of the following ethers has the lowest boiling point?

A. CH_3OCH_3

B. $CH_3CH_2OCH_2CH_3$

C. $CH_3OCH_2CH_3$

D. $CH_3CH_2CH_2OCH_2CH_2CH_3$

Solution

Boiling point increasing with increasing carbon number.

Answer: A

37. The most soluble ether among the following in water is

A. CH_3OCH_3

B. $CH_3CH(CH_3)OCH_3$

C. $CH_3CH_2OCH_3$

D. $CH_3CH_2OCH_2CH_2CH_3$

Answer: A

38. Given the reaction: $C_6H_5OCH_3 + HI \longrightarrow X + CH_3I$

The name of the product X formed is:

A. Benzene

B. Phenyl

C. Phenol

D. Benzyl alcohol

Solution

$$C_6H_5OCH_3 + HI \longrightarrow \underset{\text{Phenol}}{C_6H_5OH} + CH_3I$$

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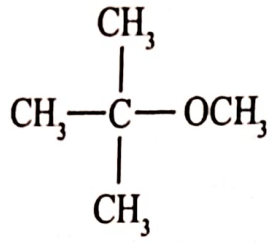
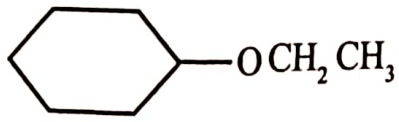
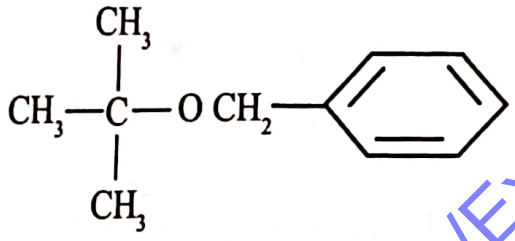
Answer: C

39. Synthesize each of the following ethers by the Williamson method using suitable alkoxide and alkyl halide.

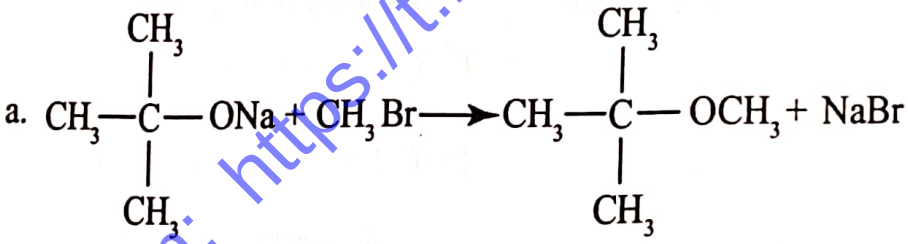
a. 

b. $CH_3CH_2OCH_2CH=CH_2$

c. 

d. 

Solution

a. 

b. $CH_2=CH-CH_2-ONa + CH_3CH_2Br \longrightarrow$

$CH_3CH_2OCH_2CH=CH_2 + NaBr$

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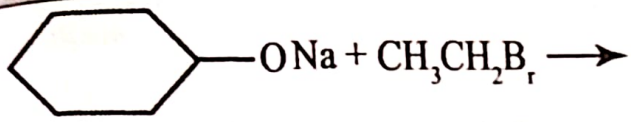
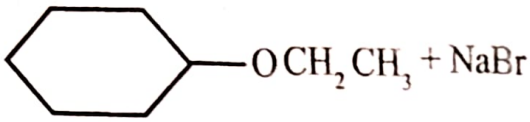
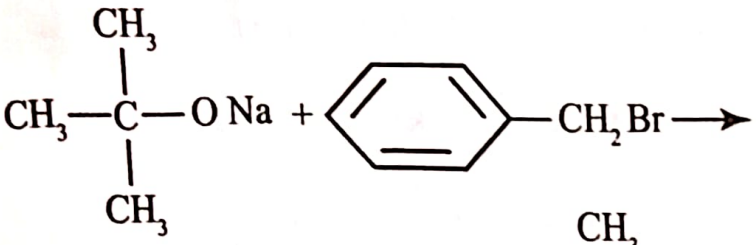
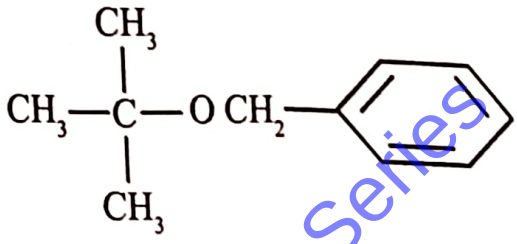
c. 



d. 



40. How dehydration of alcohols yields ether? What is its limitation?

Solution

Dehydration of alcohols yields alkene and alcohol at different temperature

$$CH_3CH_2OH \xrightarrow[170^oC]{H_2SO_4} CH_2{=}CH_2 + H_2O$$

$$2CH_3CH_2OH \xrightarrow[140^oC]{H_2SO_4} CH_3CH_2OCH_2CH_3 + H_2O$$

➥ This method is used only primary alcohol to synthesize ethers.

41. Explain how alkoxide ion is prepared for Williamson synthesis of ether.

Solution

➥ By reacting alcohols with active metal (sodium)

$$2ROH + Na \longrightarrow 2R\overset{-}{O} + \overset{+}{Na} + H_2$$

Alkoxide ion

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Examples

$$2CH_3OH + Na \longrightarrow 2CH_3O^{-} + Na^{+} + H_2$$

Methoxide ion

➥ By reacting alcohols with sodium hydride (NaH)

$$ROH + NaH \longrightarrow RO^{-} + Na^{+} + H_2$$

$$CH_3OH + NaH \longrightarrow 2CH_3O^{-} + Na^{+} + H_2$$

42. Ethers are excellent solvent to carry out many organic reactions. Why?

Solution

Because ethers are not affected by many reagents such as oxidizing agents, reducing agents, bases.

6.3 Aldehydes and Ketones

➥ Aldehydes and ketones have the carbonyl group ($-\overset{\overset{O}{\|}}{C}-$)

Table 6.3 General Formula and Function Group of Aldehydes and Ketones

Compound	General formula	Functional group
Aldehyde	RCHO or $R-\overset{\overset{O}{\|}}{C}-H$	$-CHO$ Or $-\overset{\overset{O}{\|}}{C}-H$
Ketone	RCOR' or $R-\overset{\overset{O}{\|}}{C}-R'$	$-CO$ Or $-\overset{\overset{O}{\|}}{C}-$

Nomenclature of Aldehdes and Ketones

➥ Simple aldehydes are named systematically by changing the last letter '-e' of hydrocarbons with '- al'

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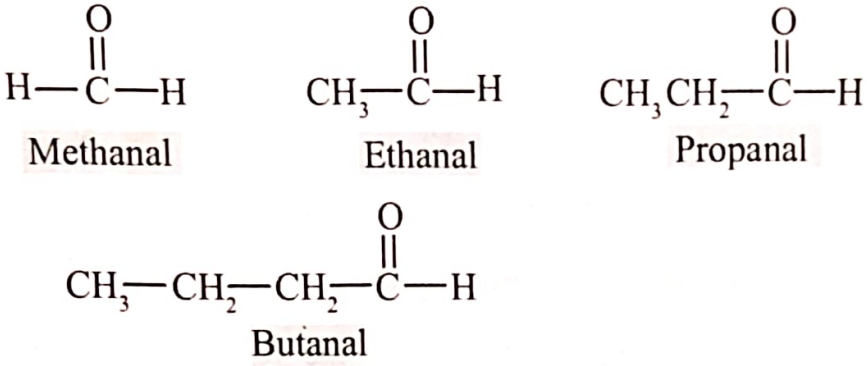
Example

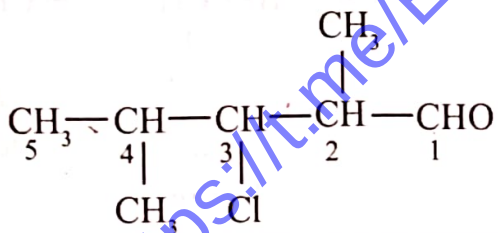


Note: The simplest aldehyde is methanal and commonly knowns as **formaldehyde.**

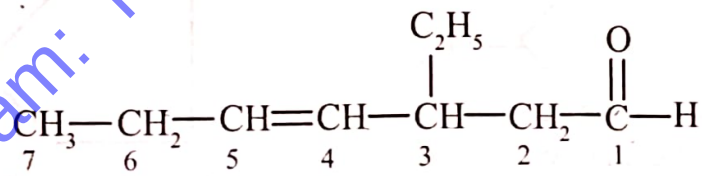
The second aldehyde, is ethanal and commonly called **acetaldehyde.**

➥ For long chain aldehydes, start from the carbonyl carbon and state the position of side chains and multiple bonds.

Example

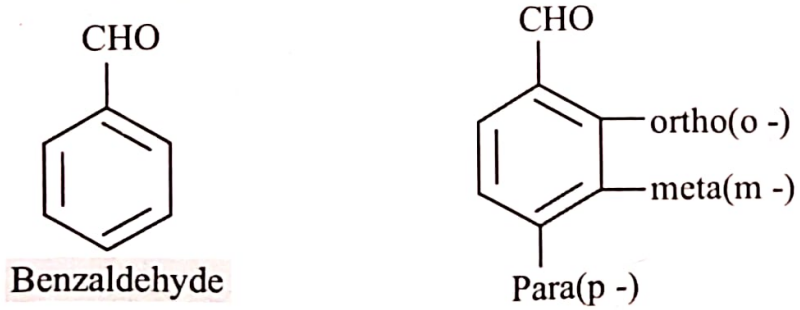
a) 

3 - chloro - 2, 4 - dimethyl pentanal

b) 

3 - ethyl - 4 - heptenal

➥ Benzene on which an aldehyde group attached is usually called **benzaldehyde.**



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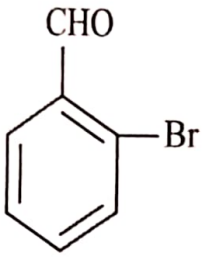
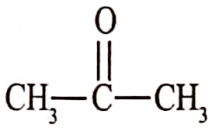
Example



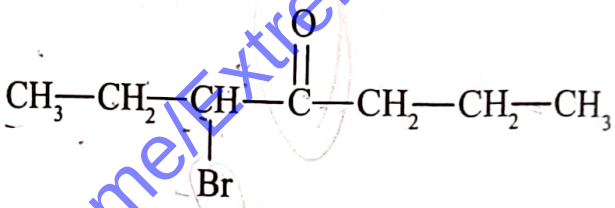
o - bromo benzaldehyde

- **Ketones** are named according to the IUPAC rules by substituting **' - one'** for the final ' -e' of the name of the hydrocarbon having the same carbon atoms.
- Count from the end nearer to the carbonyl carbon and indicate the position of carbonyl carbon and other side chains.

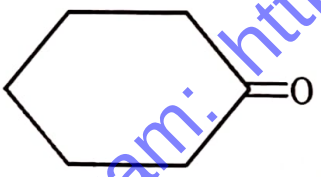
Example



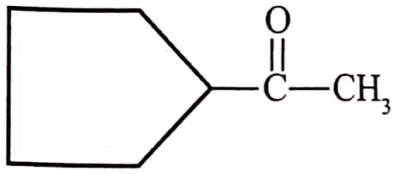
Propanone



3 - bromo - 4 - heptanone or
(3 - bromo heptan - 4 - one)



Cyclo hexanone



Cyclo pentyl methyl ketone

- Ketones having a phenyl group adjacent to the carbonyl group have common names ending in **"- one-"**

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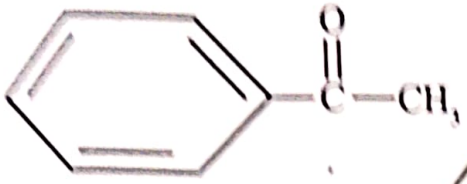
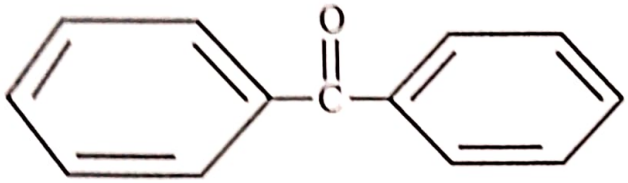
Example



Aceto phenone

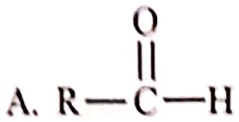


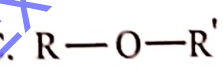
Benzo phenone

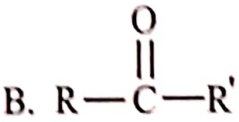
Note: The simplest ketone is propanone and commonly known as acetone.

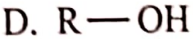
Illustrative Examples

43. The general formula of aldehyde is

A. 

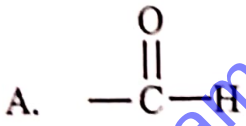
C. 

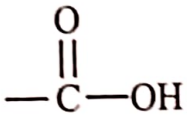
B. 

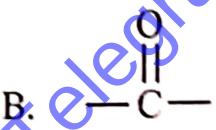
D. 

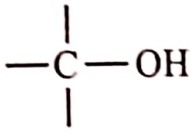
Answer: A

44. The functional group of ketones is

A. 

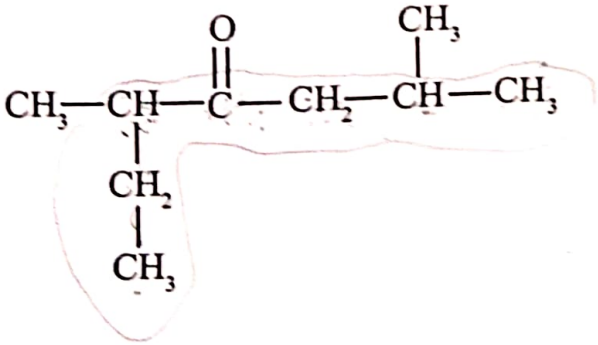
C. 

B. 

D. 

Answer: B

45. The IUPAC name of the following organic compound is



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A. 3, 6 - dimethyl - 4 - heptanone

B. 2,5 - dimethyl - 4 - heptanone

C. 2 - ethyl - 5 - methyl - 3 - hexanone

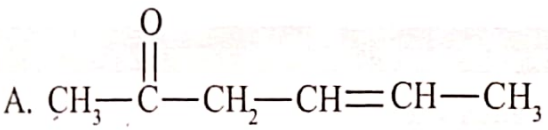
D. 5 - ethyl - 2 - methyl - 3 - hexanone.

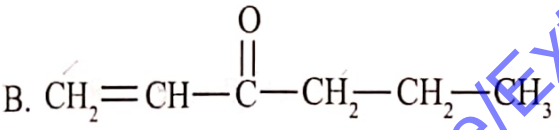
Solution

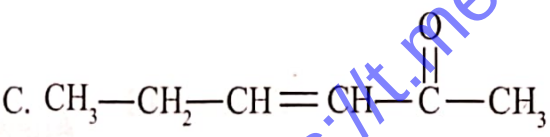
The longest chain that includes the carbonyl carbon has seven carbon atoms and start from the end nearer to the functional group.

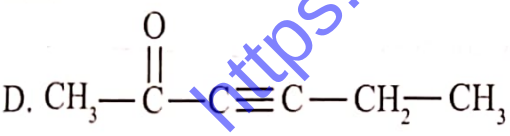
Answer: B

46. Which of the following is the structural formula of 3 - hexen - 2 - one?

A. 

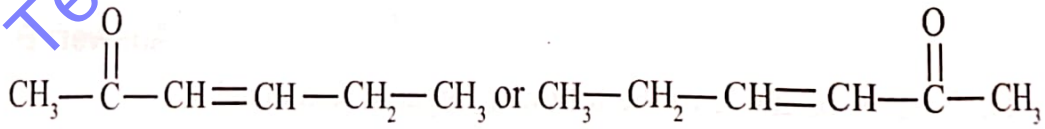
B. 

C. 

D. 

Solution

The given IUPAC name has two functional groups alkene and ketone. Thus its structure can be



Answer: C

47. Give the IUPAC name of following aldehydes

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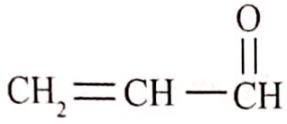
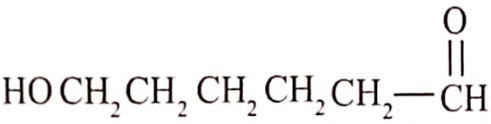
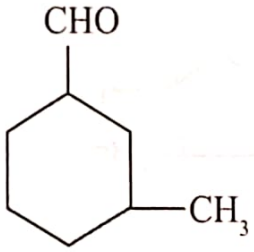
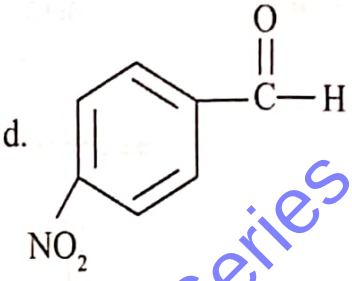
a. 

b. 

c. 

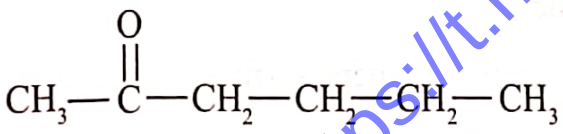
d. 

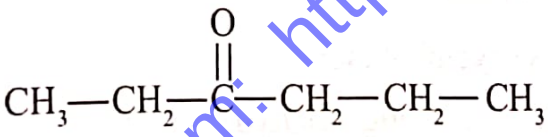
Solution

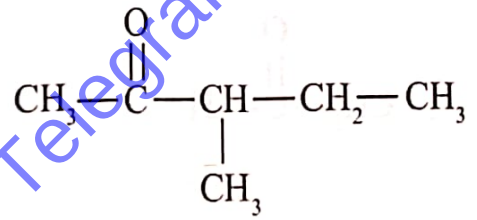
a. 2 - propenal
b. 6 - hydroxylhexanal
c. 3 - methyl cyclohexanal
d. 4 - nitrobenzaldehyde

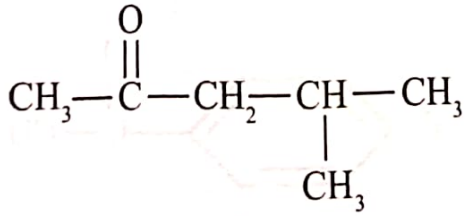
48. Write the IUPAC names all ketones that contain 6 carbon atoms.

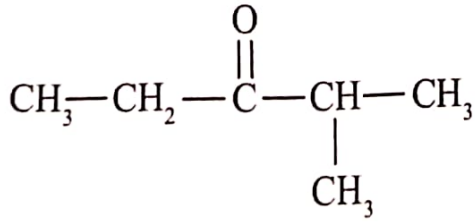
Solution

 2 - hexanone

 3 - hexanone

 3 - methyl - 2 - pentanone

 4 - methyl - 2 - pentanone

 2 - methyl - 3 - pentanone

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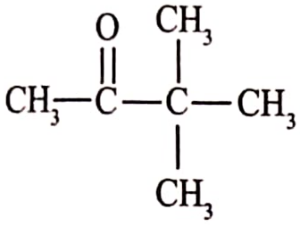
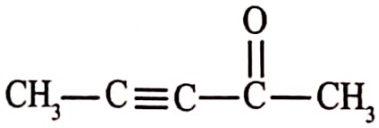
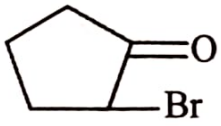
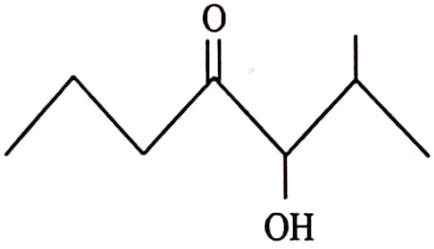


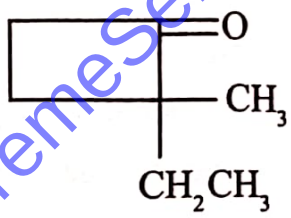
3, 3 - dimethyl - 2 - butanone

49. Give the IUPAC name for the following ketones.

a. 

c. 

b. 

d. 

Solution

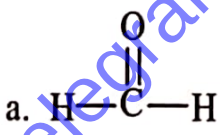
a. 3 - pentyn - 2 - one

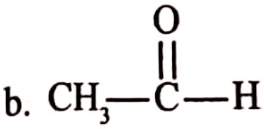
b. 3 - hydroxy - 2 - methyl - 4 - heptanone

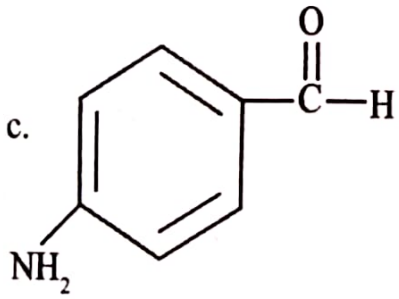
c. 2 - bromo cyclopentanone

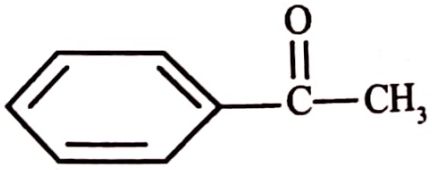
d. 2 - ethyl - 2 - methyl cyclobutanone.

50. Write the common name of the following aldehydes and ketones.

a. 

b. 

c. 

d. 

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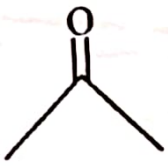
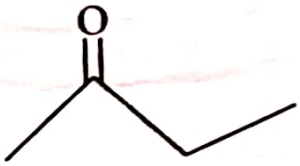
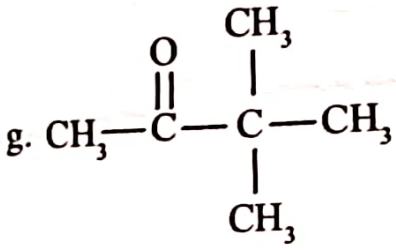
e. 

f. 

g. 

Solution

a. Formaldehyde
b. Acetaldehyde
c. P - aminobenzaldehyde
d. Methyl phenyl ketone (acetophenone)
e. Acetone (Dimethyl ketone)
f. Ethyl methyl ketone
g. t - butyl methyl ketone

51. Draw the structural formulas of the following compounds

a. 3 - methyl cyclohexanone
b. Phenyl cylopentyl ketone
c. 1 - hydroxy - 3 - pentanone
d. 1, 4 - cyclohexanedione
e. Cyclopentanal
f. 3 - hexynal
g. hexanedial
h. P - nitro benzaldehyde.

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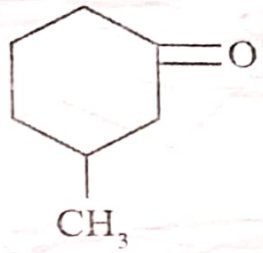
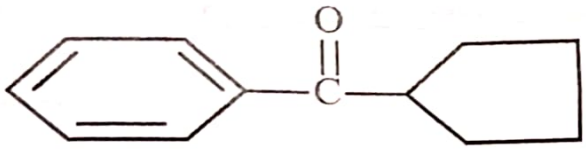
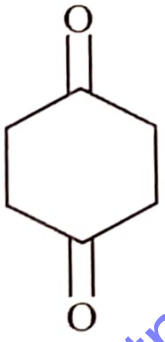
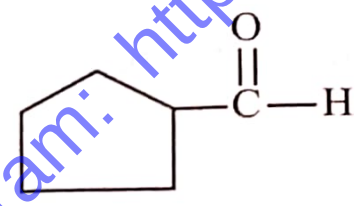
Solution

a. 

b. 

c. $HO\,CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_2-CH_3$

d. 

e. 

f. $H-\overset{\overset{O}{\|}}{C}-CH_2-C\equiv C-CH_2-CH_3$

g. $H-\overset{\overset{O}{\|}}{C}-CH_2-CH_2-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-H$

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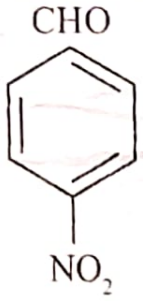
h. 

52. Write the general name of aldehydes and ketones as derivative of alkanes.

Solution

Aldehydes: Alkanals

Ketones: Alkanones

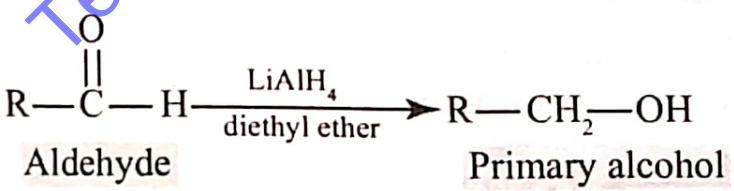
Physical Properties of Aldehydes and Ketones

- Aldehydes and ketones have higher boiling points than hydrocarbons due to the polarity of carbonyl carbon but lower boiling points than alcohols due to lack of hydrogen bond.
- As carbon number increases boiling point of aldehydes and ketones also increases.
- Aldehydes and ketones with lower mass are more soluble in water than those with higher mass.

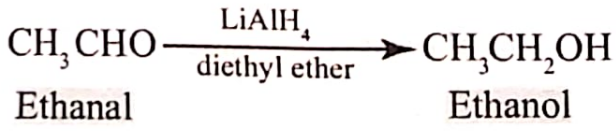
Chemical Properties of Aldehydes and Ketones

a. Reduction

- Reduction of aldehydes with lithium aluminum hydride ($LiAlH_4$) yields **primary alcohol**.



Example



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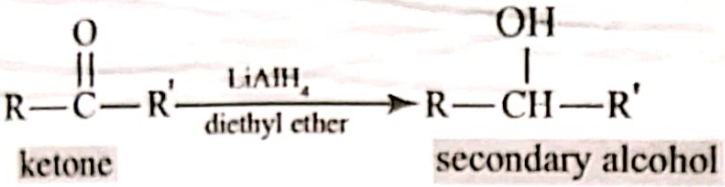
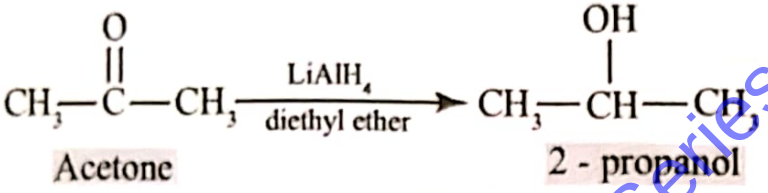
➥ Reduction of ketone with lithium aluminum hydride gives **secondary alcohol**



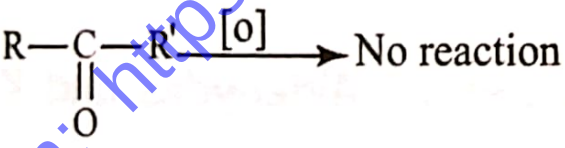
Example



b. Oxidation

➥ Oxidation of aldehydes results carboxylic acid

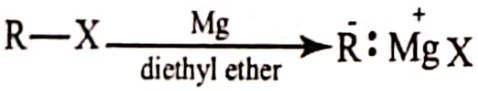
➥ Ketones do not undergo oxidation reaction under normal conditions.



c. Addition Reaction

i. Addition of Grignard Reagents to Aldehydes and Ketones

➥ The organometallic reagents with magnesium are named Grignard reagents, after victor Grignard, a french chemist who developed them.



Grignard reagent

(X = I, Br, Cl) R = alkyl or aryl

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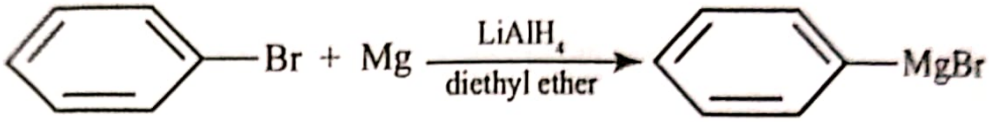
Example

$$CH_3{-}I + Mg \xrightarrow[\text{diethyl ether}]{LiAlH_4} CH_3MgI$$

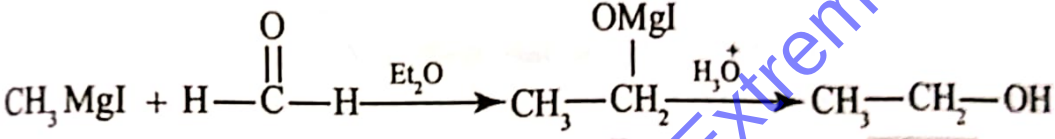
Methyliodide — Methyl magnesium iodide (a methyl Grignard reagent)



Phenyl magnesium bromide (a phenyl Grignard reagent)

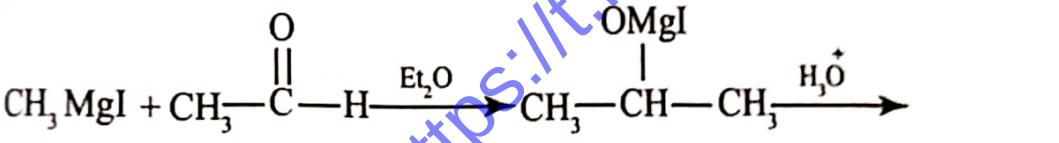
Consider CH_3MgI

➥ **Addition of formaldehyde yields primary alcohol.**



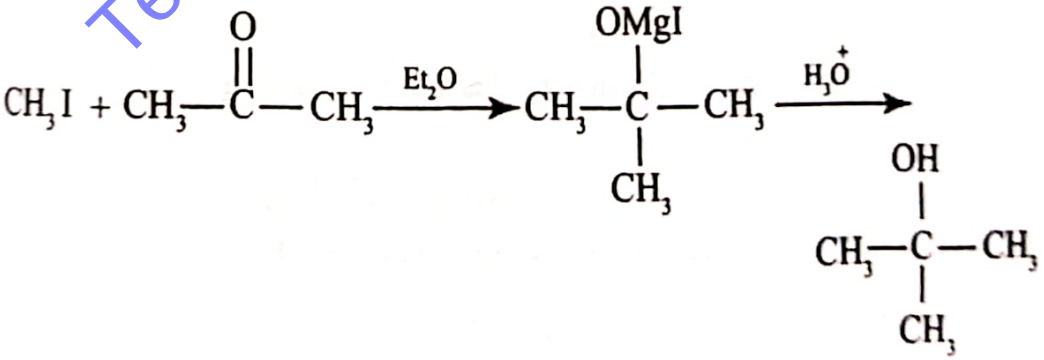
Ethanol

➥ **Addition of other aldehydes gives secondary alcohol.**



2 - propanol

➥ **Addition of ketones yields teritiary alcohol**



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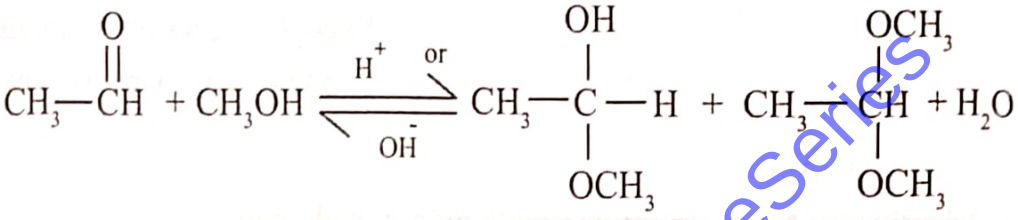
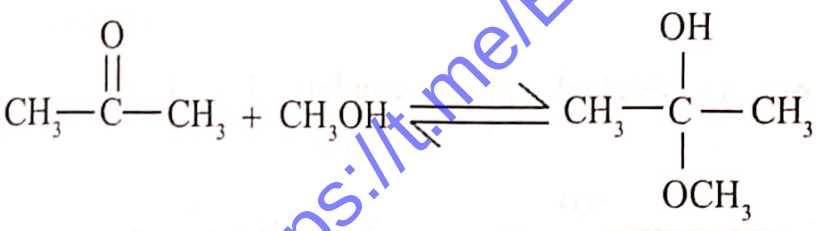
ii. Addition of Alcohol: Formation of Acetals and Ketals

- Alcohol react with aldehydes in equilibrium process, to yield compounds known as **hemiacetals.** In a hemiacetal, a hydroxyl group and an alkoxyl group are bonded to the same carbon atom.
- Hemiacetals, after losing a molecule of water, react with yet another molecule of alcohol to give **acetals**. In an acetal, two alkoxyl groups are bonded to the same carbon atom.

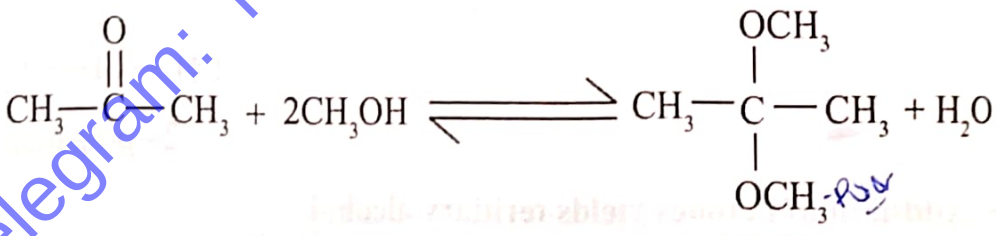


a hemiacetal an acetal

- **Ketones** give **hemiketals** and **ketals** with alcohols. Hemiketals and ketals do not form as spontaneously as hemiacetals and acetals do

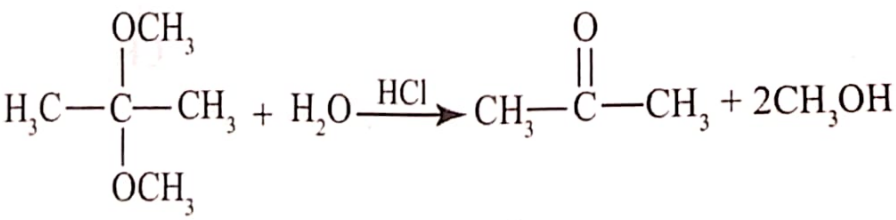


a hemiketal



a ketal

- An acetal or a ketal structurally resembles a diether and as such is stable to bases and nucleophilic reagents.
- Acetals and ketals, however, are sensitive to dilute acid are easily converted to the orginal carbonyl compound and alcohol.



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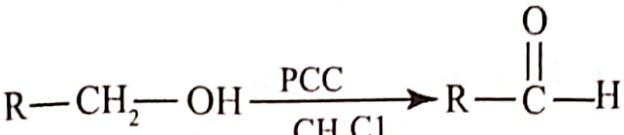
Preparation of Aldehydes and Ketones

➥ Aldehydes and ketones are after prepared by oxidation of alcohols.

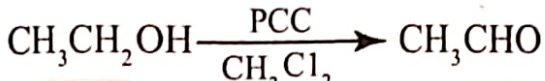


Primary alcohol — Aldehyde

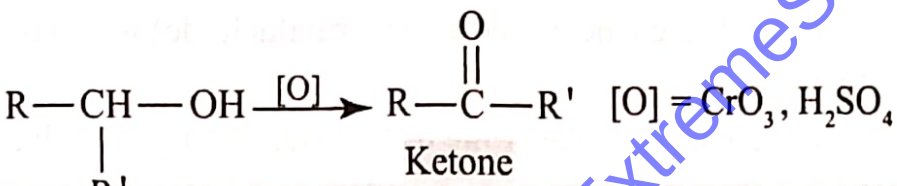
$[O] = C_6H_5NHCrO_3Cl$

PCC = Pyridinium chloro chromate

Example



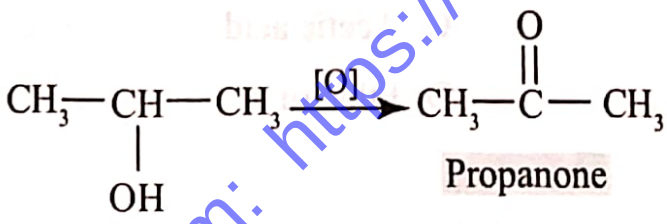
Ethanol — Ethanal



Ketone

Secondary alcohol

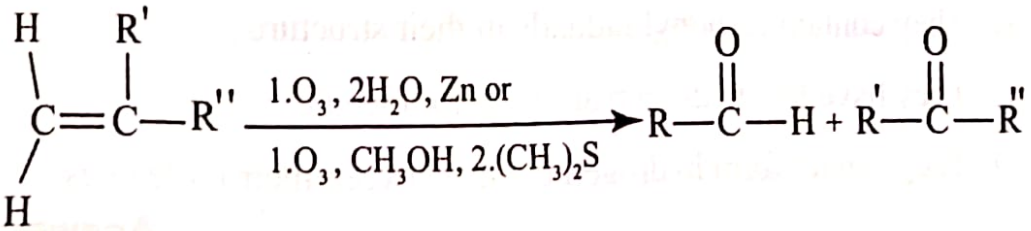
Example



Propanone

2 - propanol

➥ **Ozonolysis:** is also used to prepare aldehyes and ketones. Ozonolysis is most useful as a method of preparing an aldehyde or a ketone when the second fragment from the ozonolysis is a low - molecular weight compound that is easily separated from the desired product of the reaction.



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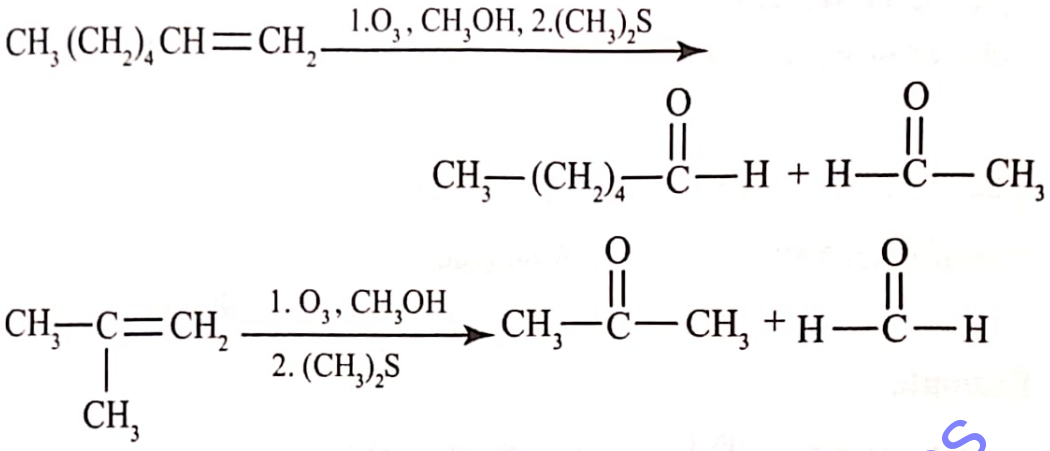
Example



Uses of Aldehydes and Ketones

- The most important aldehyde is methanal (formaldehyde) which is used to make **formalin**
- The most important ketone is **acetone** that is used as organic solvent.

Illustrative Examples

53. Which one of the following organic compound is used to preserve dead organisms?

A. Formal dehyde
B. Acetone
C. Acetic acid
D. Ethanol

Solution

A 40% formaldehyde is called formalin that preserves dead organisms in biological laboratories.

Answer: A

54. Aldehydes usually have lower boiling points than the corresponding alcohol because

A. They are liquids at room temperature
B. They contain carbonyl radicals in their structure
C. They have less hydrogen atoms per molecule
D. They cannot form hydrogen bonds between their molecules.

Answer: D

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55. Which of the following yields a primary alcohol upon reduction?

A. Alkene
B. Ether
C. Aldehyde
D. Ketone

Solution

$$RCHO \xrightarrow{LiAlH_4} RCH_2OH$$

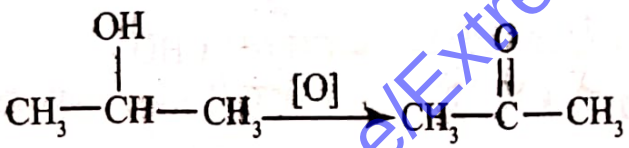
Aldehyde 1^0 alcohol

Answer: C

56. Which one of the following molecules yields propanone up on oxidation?

A. Propanal B. 1 - propanol C. 2 - propanol D. Propane

Solution



2 - propanol propanone

Answer: C

57. Addition of formaldehyde to a Grignard reagent gives.

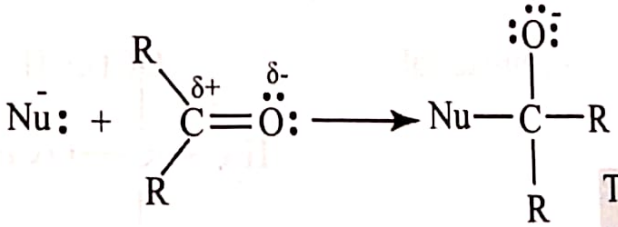
A. Primary alcohol
B. Secondary alcohol
C. Tertiary alcohol
D. Ketone

Answer: A

58. How could you prepare tetraheadral cardonyl addition intermediate?

Solution

The partially positive charge on the carbonyl carbon is the cause of the most common reaction theme of the carbonyl group, the addition of a nucleophile (Nu:) to form tetrahedral carbonyl addition intermediate.



Tetrahedral carbonyl addition intermediate

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59. Complete and balance the following

a. $CH_3CH_2CH_2OH \xrightarrow[CH_2Cl_2]{PCC}$

b. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3 \xrightarrow[H_2SO_4]{CrO_3}$

c. $CH_3-CH_2-CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2 \xrightarrow[2.\ H_2O,\ Zn]{1.\ O_3}$

d. $H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-O\,CH_2\,CH_3 + HO\,CH_2\,CH_3 \overset{H^+}{\rightleftharpoons}$

Solution

a. $CH_3CH_2CH_2OH \xrightarrow[CH_2Cl_2]{PCC} CH_3CH_2CHO$

1 - propanol 1 - propanal

b. $CH_3-\underset{\underset{OH}{|}}{CH}-CH_2-CH_3 \xrightarrow[H_2SO_4]{CrO_3} CH_3-\overset{\overset{O}{\|}}{C}-CH_2CH_3$

c. $CH_3-CH_2-CH_2-\underset{\underset{CH_3}{|}}{C}=CH_2 \xrightarrow[2.\ H_2O,\ Zn]{1.\ O_3}$

$CH_3-CH_2-CH_2-\overset{\overset{O}{\|}}{C}-CH_3 + H-\overset{\overset{O}{\|}}{C}-H$

d. $H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{OH}{|}}{C}}-O\,CH_2\,CH_3 + HO\,CH_2\,CH_3 \overset{H^+}{\rightleftharpoons}$

A hemiacetal

$H_3C-\underset{\underset{CH_3}{|}}{\overset{\overset{OCH_2CH_3}{|}}{C}}-O\,CH_2\,CH_3 + H_2O$

A diethyl acetal

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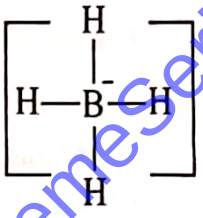
60. Write the common oxidizing agents that converts secondary alcohols to ketones.

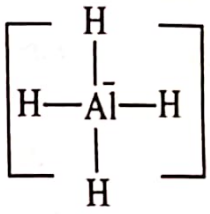
Solution

Chromic acid (H_2CrO_3). It is formed when Chromium trioxide (CrO_3) or sodium dichromate ($Na_2Cr_2O_7$) is dissolved in equeous acid

61. What are the two common metal hydrides that reduce aldhyde or ketone to a hydroxyl group?

Solution

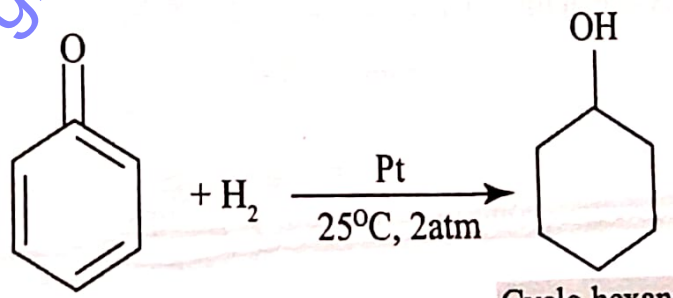
➥ Sodium boron hydride ($NaBH_4$) $\Longrightarrow Na^+$ 

➥ Lithium aluminium hydride ($LiAlH_4$) $\Longrightarrow Li^+$ 

62. Explain catalytic reduction of aldehyde or ketone.

Solution

It is reduction of aldehyde or ketone to a hydroxyl group with hydrogen in the presence of a transition metal catalyst such as palladium, platinum, nickel or rhodium



Cyclo hexanone Cyclo hexanol

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6.4 Carboxylic Acids

- They are organic acids containing carboxyll group, (- COOH)
- Their general formula is RCOOH or $R-\overset{\overset{O}{\|}}{C}-OH$
- Their functional group is - COOH or $-\overset{\overset{O}{\|}}{C}-OH$

Table 6.4 Some Fruits and Carboxylic Acids Found in them.

Fruit	Carboxylic acids present
Lemon	Citric, malic, tartatic and oxalic acids
Grape	Citric, tartaric, malic and oxalic actids
Orange	Citric malic and oxalic acids
Apple	Fumaric, lactic and succnic acids malic, quinic, aketoglutatric, oxalacetic, citric, pyruvic
Pineapple	Citric and melic acids
Banana	malic, citric, taratatic and traces of facetic and formic acids
Salad	Citric and malic acid
Avocado	Tartaric acid

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Classification of Carboxylic Acids

a. Based on Number of Carboxyl groups

1. **Monocarboxylic acids** - have one carboxyl group

   Example

   CH_3COOH

2. **Dicarboxylic acids** - contain two carboxyl groups

   Example

   $HOOC-CH_2-COOH$

3. **Tricarbooxylic acids** - possess three carboxyl groups

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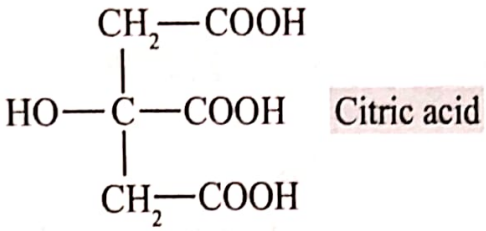
Example

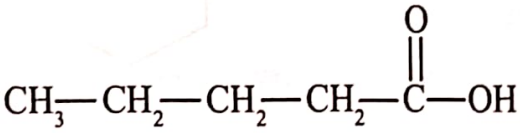


Citric acid

b. Based on carbon - carbon bond

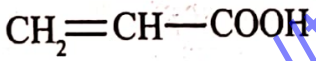
1. **Saturated carboxylic acids** are acids with all carbon - carbon single bonds.

Example



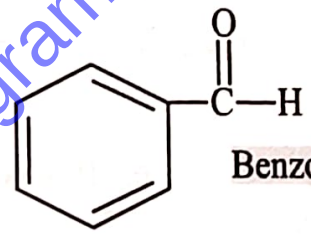
2. **Unsaturated carboxylic acids** are organic acids containing carbon - carbon double or triple bond.

Example



Note: Carboxylic acids that contain benzene ring are called **aromatic carboxylic acids**

Example



Benzoic acid

Illustrative Examples

63. The organic compound CH_3COOH is

A. Aldehyde
C. Carboxylic acid
B. Ketone
D. Ester

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Solution

It is carboxylic acid because it has carboxyl group.

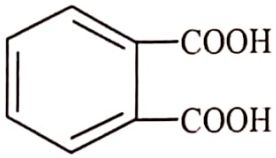
Answer: C

64. The general formula of carboxylic acid is

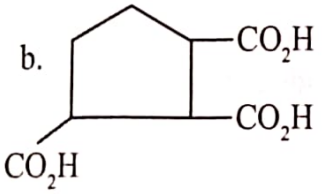
A. RCOOR' B. RCOOH C. RCOR' D. RCHO

Answer: B

65. Classify the following as mono,di or tricarboxylic acids.

a. 

c.

b. 

Solution

a. Dicarboxlic acid
b. Tricaboxylic acid
c. Monocarboxylic acid

66. Write the organic acid found in

a. Ants, bees
b. Vinegar
c. Sour milk
d. Goat
e. Rancid butter

Solution

a. Formic acid
b. Acetic acid
c. Lactic acid
d. Caproic acid
e. Butanoic acid

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Nomenclature of Carboxylic Acids

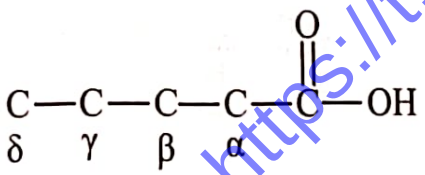
A. Common Name

i. Common Name of Straght Chain Carboxiylic Acids

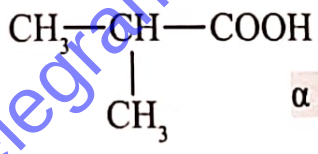
Formula	Common Name
HCOOH	Formic acid
CH_3COOH	Acetic acid
CH_3CH_2COOH	Propionic acid
$CH_2(CH_2)COOH$	Butyric acid
$CH_3(CH_2)_3COOH$	Valeric acid
$CH_3(CH_2)_4COOH$	Caproic acid

ii. Branched Chain Carboxylic Acids

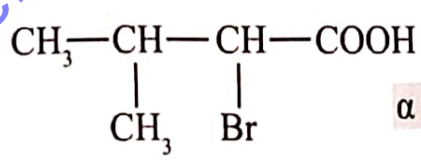
- Use the Greek letters **ά , β, γ, δ,** ------- on the 2nd, 3d, 4th, 5th, ----- carbon to state position of side chains or substitutes.
- If the same side chain appears twice three times, four times use the prefixes, di, tri, tetra- respectively.
- Arrange the side chains in alphabetical or size order.



Example



α - methyl propionic acid



α - bromo - β - methyl butyric acid

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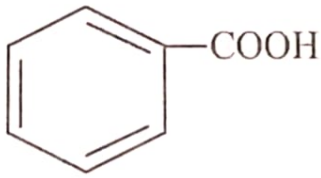
iii. Dicarboxylic Acids

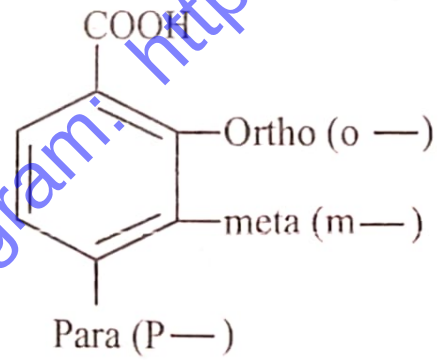
Formula	Common Name
HOOC—COOH	Oxalic acid
HOOC—CH_2—COOH	Malonic acid
HOOC—CH_2—CH_2—COOH	Succinic acid
HOOC—CH_2—CH_2—CH_2—COOH	Glutaric acid
HOOC—CH_2—CH_2—CH_2—CH_2—COOH	Adipic acid
HOOC$\leftarrow CH_2 \rightarrow_5$ COOH	Pimeric acid

iv. Aromatic Carboxylic Acids

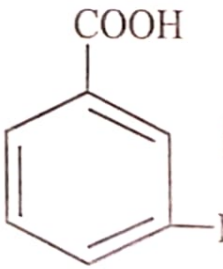
➥ The simplest aromatic carboxylic acid is commonly known as **benzoic acid**



➥ If side chains appear, we use the prefix ortho (o -), mata (m -), and para (p -) , on the second, third and fourth carbon atoms respectively.



Example



m - bromo benzoic acid

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Illustrative Examples

67. The common name of the second monocarboxylic acid

A. Formic acid C. Oxlic acid

B. Acetic acid D. Malonic acid

Solution

The second monocarboxylic has a formula of CH_3COOH. Its common name is acetic acid.

Answer: B

68. The structural formula of β - ethylvaleric acid is:

A. $CH_3-CH_2-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-COOH$

B. $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-CH_2-COOH$

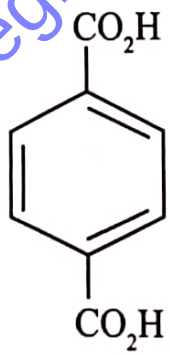
C. $Br\,CH_2-CH_2-CH_2-COOH$

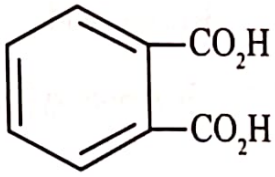
D. $CH_3-\underset{\displaystyle Br}{\underset{|}{CH}}-CH_2-COOH$

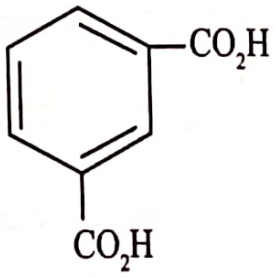
Answer: A

69. Write the common name for the following carboxylic acids.

a. Cl_3CCOOH

b. 

c. 

d. 

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Solution

a. Trichloroacetic acid
b. Terephthalic acid
c. Phthalic acid
d. Isophthalic acid

70. Draw the structural formula for the following organic acids.

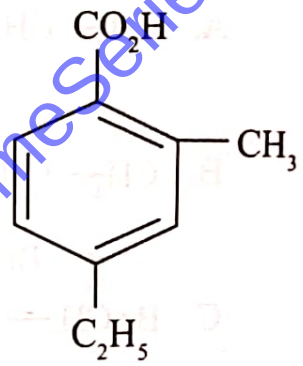
a. ά - bromo - β - chlorobutyric acid
b. Succinic acid
c. p - ethyl - o - methyl benzoic acid
d. Caproic acid

Solution

a. $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{Br}{|}}{CH}-CO_2H$

b. $HO_2C-CH_2-CH_2-CO_2H$

c.



d. $CH_3(CH_2)_4CO_2H$

71. The simplest aromatic carboxylic acid is _____

Solution

Benzoic acid

2. IUPAC Name of Carboxylic Acids

Rules

1. The last letter of hydrocarbons " - e" is replaced by " - **oic acid**"
2. Identify the longest chain and start with the carbonyl carbon

   $(-\overset{\overset{O}{\|}}{C}-)$ for open chain carboxylic acids.
3. Arrange the side chains in alphabetical or size order.
4. If the same side chain appears twice, three, times, four times use the prefixes di - tri - and tetra respectively.

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5. Number and number is separated by comma (,) and number and letter is separated by hyphen (-).

6. For dicarboxylic acids use the simple method below.

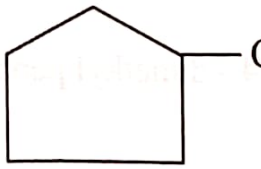
Name of alkane + dioic acid

Example

$HO_2C—CH_2—CH_2—CO_2H$ Butanedioic acid

7. For cylic carboxylic acids, name of the cyclic alkanes followed by carboxylic acid.

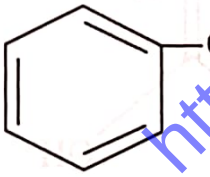
Example



Cyclo pentane carboxylic acid

8. For aromatic hydrocarbon, benzene followed by the word carbxylic acid.

Example



Benzene carboxylic acid

9. For unsaturated carboxylic acid, state the position of double or triple bond.

Example

$CH_3—CH=CH—CO_2H$

2 - butenoic acid

Illustrative Example

72. What is the IUPAC name for the compound

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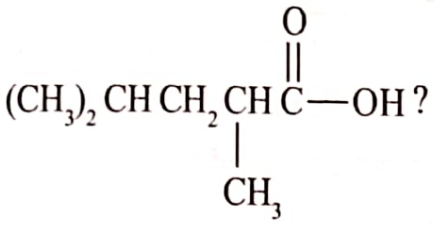


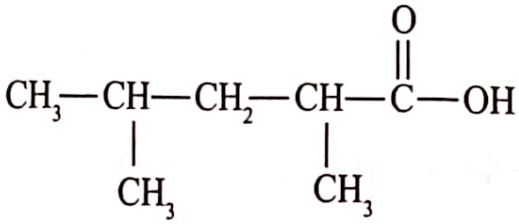
A. 2,4 - dimethyl pentanoic acid

B. 1, 1, 3 - trimethyl butanoic acid

C. 1 - hydroxyl - 2,4 - dimethyl pentanone

D. 2 - carboxyisohexane

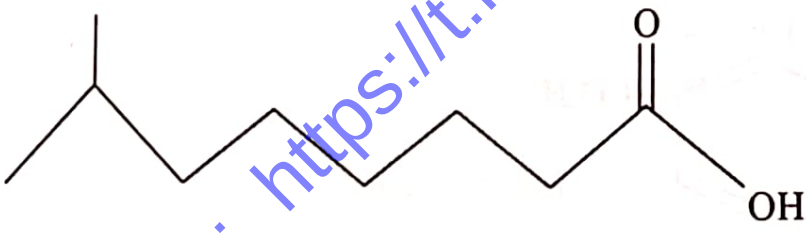
Solution



2, 4 - dimethyl pentanoic acid

Answer: A

73. Which of the following is an acceptable IUPAC name for the organic compounds below?



A. Octanoic acid

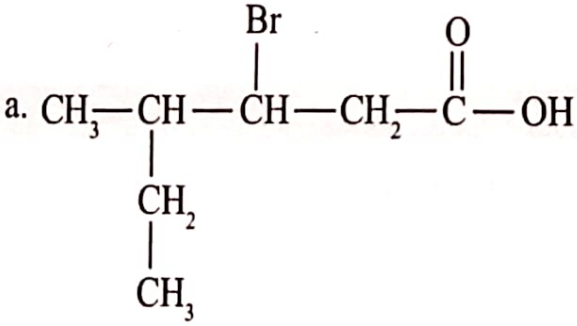
B. 7 - methyl pentanoic acid

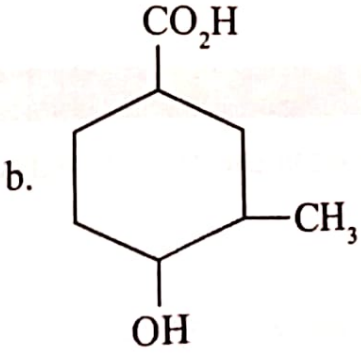
C. 8 - methyl heptanoic acid

D. 7 - methyl octanoic acid

Answer: D

74. Give IUPAC name for the carboxylic acids given below.

a. 

b. 

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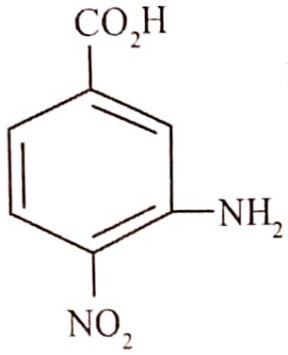
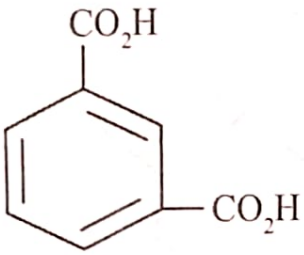
c. 

d. $CH_3—(CH_2)_{14}CO_2H$

e. 

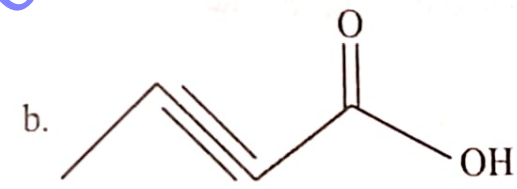
f. $(CH_3)_3CCO_2H$

Solution

a. 3 - bromo - 4 - methyl hexanoic acid
b. 4 - hydroxyl - 3 - methyl cyclohexane carboxylic acid
c. 3 - amino - 4 - nitro benzene carboxylic acid.
d. Hexadecanoic acid
e. 1,3 - benzene dicarboxylic acid
f. 2,2 - dimethyl propanoic acid.

75. Write the IUPAC name of the following unsaturated and dicarboxylic acids.

a.

b. 

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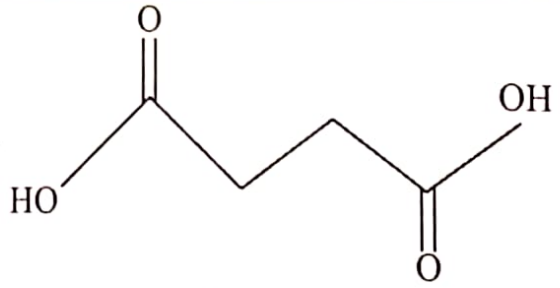
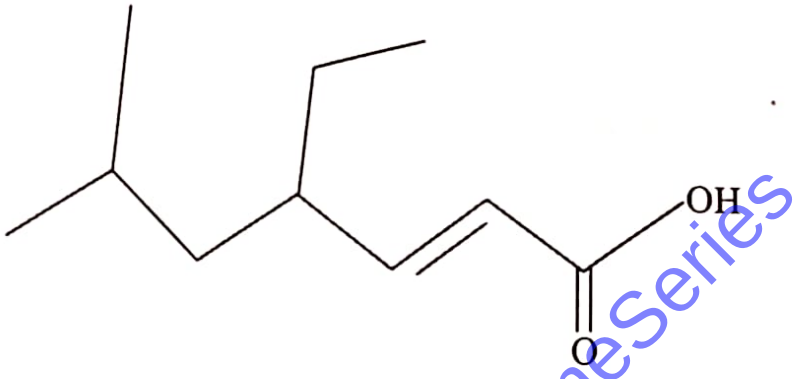
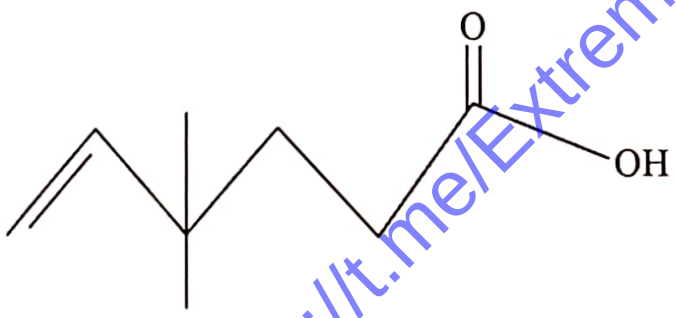
c.



d.



e.



Solution

a. 2 - hexenoic
b. 2 - butynoic acid
c. Butane dioic acid
d. 4 - ethyl - 6 - methyl - 2 - heptenoic acid
e. 4,4 - dimethyl - 5 - hexenoic acid

76. Draw the structures of the carboxylic acids below.

a. Dichloroethanoic acid
b. 1,2 - benzene dicarboxylic acid
c. 2 - hydroxyl propanoic acid
d. 5 - bromo - 2 - chloro - 2 - methylheptanoic acid
e. Pentane dioic acid
f. 3 - heptenoic acid.

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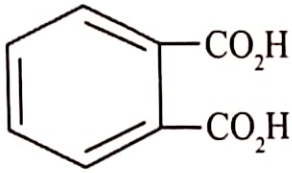
g. Cyclopentane carboxylic acid

Solution

a. $CH\ Cl_2COOH$

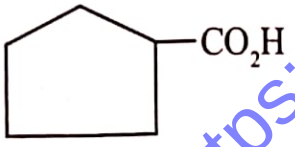
b. 

c. $CH_3-CH(OH)-CO_2H$

d. $CH_3-CH_2-CH(Br)-CH_2-CH_2-C(CH_3)(Cl)-COOH$

e. $HOOC-CH_2-CH_2-CH_2-COOH$

f. $CH_3-CH_2-CH_2-CH=CH-CH_2-COOH$

g. 

Physical Properties of Carboxylic Acids

a. State

* C_1 — C_9 aliphatic carboxylic acids are liquid at room temperature.
* **C_{10} and above** aliphatic carboxylic acids as well as aromatic carboxylic acids are colorless waxy solids.

b. Odour

- Formic acid and acetic acid have sharp irritating odour.
- Propanoic acid, butanoic acid, pentanoic acid, and hexanoic acid have unpleasant odour
- Higher member carboxylic acids are odourless because as carbon number rises, volatility decreases.

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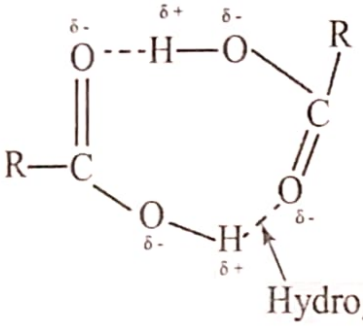
c. Melting and Boiling Points

- Carboxylic acids have higher melting and boiling point than hydrocarbons and oxygen containing hydrocarbons of similar size. The reason for this is that, they form double hydrogen bond with their molecules.



- As carbon number increases, melting and boiling points also increase.
- Boiling point of dicarboxylic acid is higher than mono carboxylic acid

Example

$$CH_3COOH < HOOC-COOH$$
$$(118.2^oC) \quad (187^oC)$$

d. Solubility

- The first four carboxylic acids are completely miscible with water.
- As carbon number increases, solubility of carboxylic acid in water decreases because the non - polar tail dominates the polar head.

Table 6.5 Physical Properties Some Carboxylic Acids

Carboxylic acid	Melting point (^{0}C)	Boiling point (^{0}C)	Solubility per 100g of ware
HCOOH	8.4	100.5	Miscible
CH_3COOH	16.7	118.2	Miscible
CH_3CH_2COOH		141	Miscible
$CH_3CH_2CH_2COOH$	- 7.9	163	Miscible
$CH_3CH_2CH_2CH_2COOH$	- 34.5	186.4	3.7g
HOOC—COOH	187		9.0g

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C_6H_5— COOH	122		0.29g

Illustrative Example

77. When the following substance are arranged in the order of increasing melting point, the correct order is

I. $CH_3 CH_2 CH_3$

II. $CH_3 CH_2 OH$

III. $CH_3 CO CH_3$

IV. $CH_3 CH_2 COOH$

A. I < III < II < IV

B. II < IV < III < I

C. IV < III < II < I

D. I < III < II < IV

Solution

Boiling point: carbonxylic acid > alcohol > ketone > alkane

Answer: A

78. Which of the following is the most soluble in water?

A. HCOOH

B. CH_3 COOH

C. $CH_3 CH_2$ COOH

D. $CH_3 CH_2 CH_2$ COOH

Solution

Lower mass carboxylic acids are more soluble than higher mass carboxylic acids in water

Answer: A

79. Which of the following is the most polar molecule?

A. CH_4 B. CH_3 OH C. HCOOH D. HCHO

Solution

Carboxylic acids are more polar than alcohols and aldehydes.

Answer: C

80. Arrange the following in decreasing order of volatility.

I. HCOOH

II. CH_3 COOH

III. $CH_3 CH_2$ COOH

IV. $CH_3 (CH_2)_2$ COOH

Solution

I > II > III > IV

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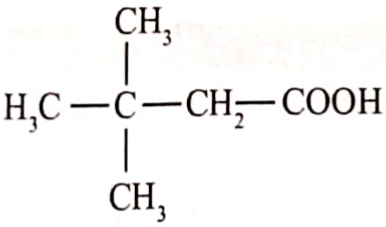
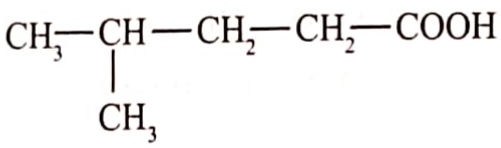
81. Arrange the following in increasing order of boiling point.

a. 

b. $H_3C-CH_2-CH_2-CH_2-CH_2-COOH$

c. 

Solution

a < c < b

82. Predict which compound in each of the following series will have the highest solubility in water.

a. $CH_3CH_2CH_2CH_2COOH$, $CH_3CH_2COCH_2CH_3$, $CH_3CH_2CH_2CH_2COO^-Na^+$

b. $CH_3CH_2CH_2COOH$, $CH_3CH_2CH_2CH_2OH$, $CH_3COOCH_2CH_3$

Solution

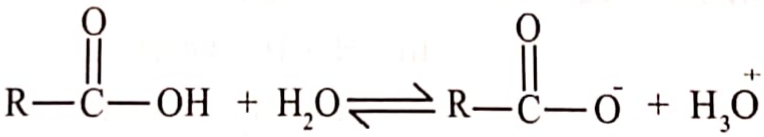
a. Carboxylic acids treated with NaOH or KOH are more soluble in water because they become more ionic.

$\Rightarrow CH_3CH_2CH_2CH_2COO^-Na^+$

b. $CH_3CH_2CH_2COOH$

Chemical Properties of Carboxylic Acids

Carboxylic acids are generally weak acids



Note: The strength of carboxylic acids can be increased by replacing hydrogen with halogens.

➥ The more the electronegative element replaced, the stronger is the acid.

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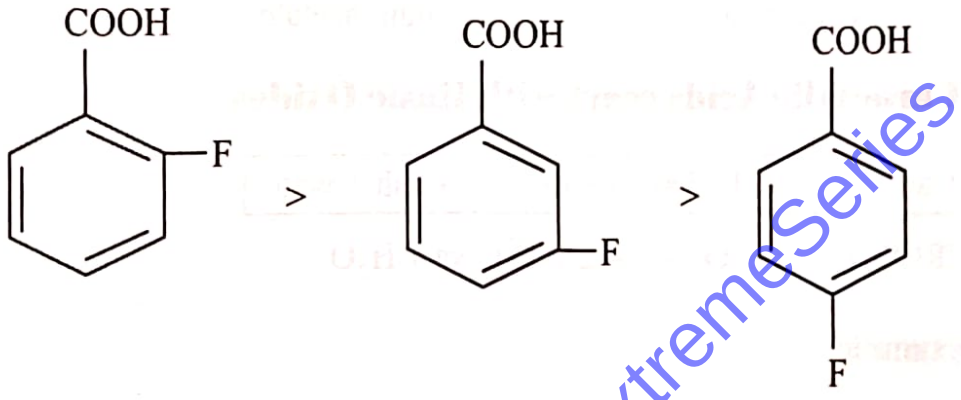
Example

$CH_2F\,COOH > CH_2ClCOOH > CH_2Br\,COOH > CH_3\,COOH$

$CF_3\,COOH > CCl_3COOH > CBr_3\,COOH > CH_3\,COOH$

➥ For benzoic acid, ortho substituted is stronger than meta, and metal substituted is stronger than that of para.

Example



'Tella' or 'Tej' turns sour after some times if left open due to oxidation of ethanol to ethanoic acid.

$$CH_3\,CH_2\,OH \xrightarrow{[O]} CH_3\,COOH$$

Ethanol — Ethenoic acid

The Common Reactions of Carboxylic are:

1. Reaction as Acid

i. Carboxylic Acids React with Active Metals

Carboxylic acid + active metals ⟶ Salt + hydrogen

$$2\,RCOOH + Mg \longrightarrow (RCOO)_2\,Mg + H_2$$

carboxylic acid — Magnesium caboxylate

Example

$$2\,CH_3COOH + Mg \longrightarrow (CH_3COO)_2\,Mg + H_2$$

Acetic acid — Magnesium acetate

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ii. Carboxylic Acids React with Bases.

Carboxylic acid + Base ⟶ Salt + water

$RCOOH + NaOH \longrightarrow RCOONa + H_2$

Example

$CH_3COOH + NaOH \longrightarrow CH_3COONa + H_2O$

Acetic acid — Sodium acetate

iii. Carboxylic Acids react with Basic Oxides

Carboxylic acid + basic oxide ⟶ salt + water

$2RCOOH + Na_2O \longrightarrow 2RCOONa + H_2O$

Example

$2CH_3COOH + Na_2O \longrightarrow 2CH_3COONa + H_2O$

iv. Caboxylic Acids react with Carbonates and Hydrogen Carbonates.

Carboxylic acid + Carbonate/ hydrogen carbonate ⟶ salt + carbondioxide + water

$2RCOOH + CaCO_3 \longrightarrow (RCOO)_2Ca + CO_2 + H_2O$

Example

$2CH_3COOH + CaCO_3 \longrightarrow (CH_3COO)_2Ca + CO_2 + H_2O$

$CH_3COOH + NaHCO_3 \longrightarrow CH_3COONa + CO_2 + H_2O$

2. Formation of Ester

➥ The reaction between carboxylic acid and alcohol to yield ester is known as **esterification.**

$$R-\overset{\overset{O}{\|}}{C}-OH + R'-OH \underset{}{\overset{H^+}{\rightleftharpoons}} R-\overset{\overset{O}{\|}}{C}-OR' + H_2O$$

Carboxylic acid — Alcohol — Ester

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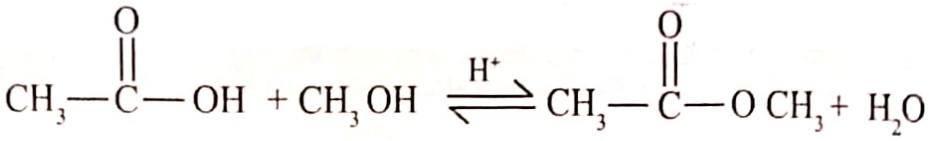
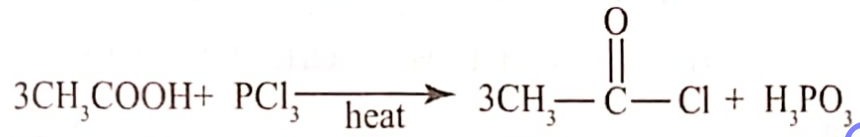
Example

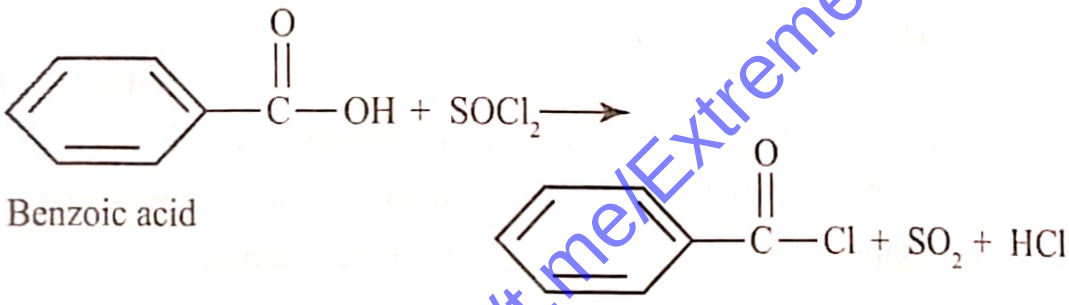


3. Converting Carboxylic Acids to Acid Chlordes and Acid Anhydrides

➥ Phoshorus halides and thionyl chloride change carboxylic acids in to acid halides.

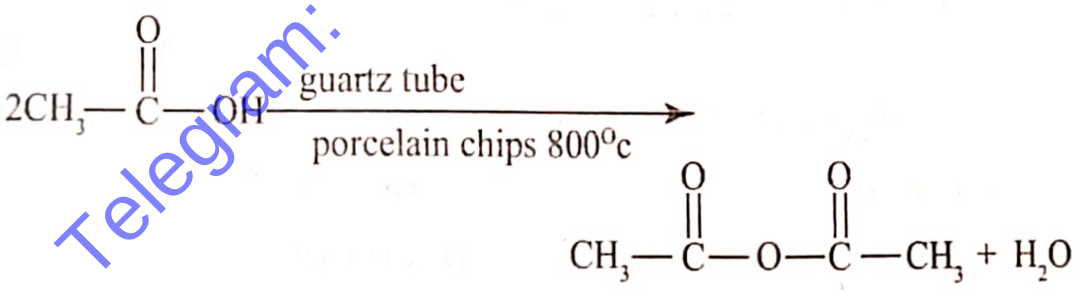


Acetic acid — Acetyl chloride



Benzoic acid — Benzoyl chloride

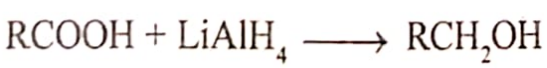
➥ **Acetic anhydride,** a compound of great commercial importance is prepared industrially by heating acetic acid to 800°c.



acetic anhydride

4. Reduction of Carboxylic Acids

➥ Carboxylic acids undergo reduction reaction with lithium aluminum hydride ($LiAlH_4$) or sodium boron hydride ($NaBH_4$)



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Example

$$CH_3COOH + LiAlH_4 \longrightarrow CH_3\,CH_3\,OH$$

Illustrative Example

83. Which of the following gives the correct order of decreasing acidity of cardoxylic acids?

A. $Cl_3CCOOH, Cl_2CHOOH, FCH_2COOH, CH_3COOH$

B. $F\,CH_2COOH, CH_3\,COOH, Cl_2\,CH\,COOH, Cl_3\,CCOOH$

C. $CH_3\,COOH, FCH_2\,COOH, Cl_2\,CH\,COOH, Cl_3\,CCOOH$

D. $Cl_2\,CH\,COOH, CH_3\,COOH, FCH_2\,COOH, Cl_3\,CCOOH$

Solution

The more substituted acid with electronegative element,the stronger acid.

Answer: A

84. Which of the description of the reaction between C_3H_7OH and $CH_3\,COOH$ is correct?

A. Addition

B. Hydrolysis

C. Esterification

D. Elimination

Solution

The reaction between alcohol (C_3H_7OH)and carboxylic acid (CH_3COOH) is esterification.

Answer: C

85. Which of these organic compounds bubbles when calcium is added to it?

A. CH_3CH_3

B. CH_3COOH

C. CH_3COCH_3

D. CH_3CHO

Solution

Acids react with active metals and bubble hydrogen gas.

Answer: B

86. Which type of organic substance would be produced by heating a mixture of a carboxylic acid and an alcohol in the presence of H_2SO_4?

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A. Ethers B. Esters C. Aldehydes D. Ketones

Solution

Carboxyli acid + Alcohol $\longrightarrow$ Ester + water

Answer: B

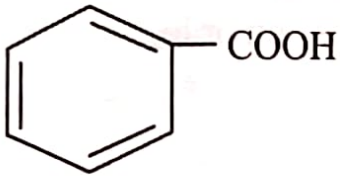
87. Complete and balance the following

a. $CH_3\ CH_2\ COOH \xrightarrow{LiAlH_4}$

b. $CH_3\ CH_2\ COOH + Mg \longrightarrow$

c. $CH_3\ CH_2\ COOH + CaCO_3 \longrightarrow$

d. $CH_3\ CH_2\ COOH + NH_3 \longrightarrow$

e.  $+\ CH_3\ CH_2\ OH \overset{H^+}{\rightleftharpoons}$

f. $CH_3\ CH_2\ COOH + Ca(OH)_2 \longrightarrow$

g. $CH_3\ CH_2\ COOH + Ba(HCO_3)_2 \longrightarrow$

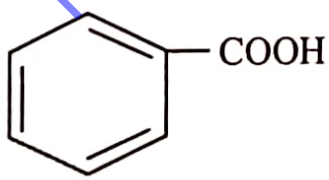
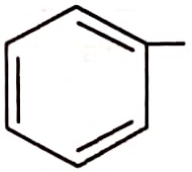
Solution

a. $CH_3\ CH_2\ COOH \xrightarrow{LiAlH_4} CH_3\ CH_2\ CH_2\ OH$

b. $2CH_3\ CH_2\ COOH + Mg \longrightarrow (CH_3\ CH_2\ COO)_2\ Mg + H_2O$

c. $2CH_3\ CH_2\ COOH + CaCO_3 \longrightarrow (CH_3\ CH_2\ COO)_2\ Ca + CO_2 + H_2O$

d. $CH_3\ CH_2\ COOH + NH_3 \longrightarrow CH_3\ CH_2\ COONH_4$

e.  $+\ CH_3\ CH_2\ OH \overset{H^+}{\rightleftharpoons}$  $COOCH_2CH_3$ $+\ H_2O$

f. $2CH_3\ CH_2\ COOH + Ca(OH)_2 \longrightarrow$

$(CH_3\ CH_2\ COO)_2\ Ca + H_2O$

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g. $2CH_3\,CH_2\,COOH + Ba(HCO_3)_2 \longrightarrow (CH_3\,CH_2\,COO)_2\,Ba + CO_2 + H_2O$

88. What are the two important reducing agents that reduce carboxylic acids into primary alcohols?

Solution

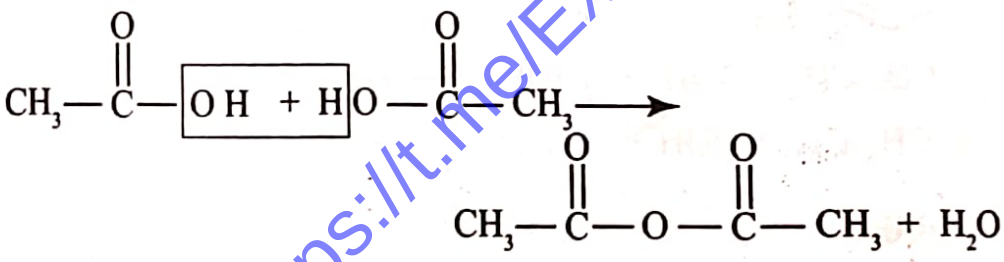
- Lithium aluminum hydride ($LiAlH_4$)
- Sodium boron hydride ($NaBH_4$)

89. Explain the formation of acid anhydride

Solution

Two carboxylic acids combine together by losing water molecule to form acid anhydride. In this word, an means anti or loss.

Example

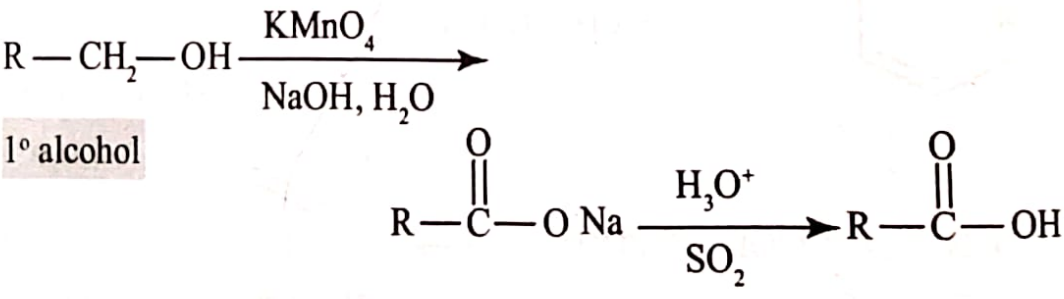


Preparation of Carboxylic Acids

There are two methods used to prepare carboxylic acids.

a. Carboxylic Acids as Product of Oxidation Reactions

- Primary alcohols and aldehydes are oxidized to carboxylic acids containing the same number of carbon atoms.



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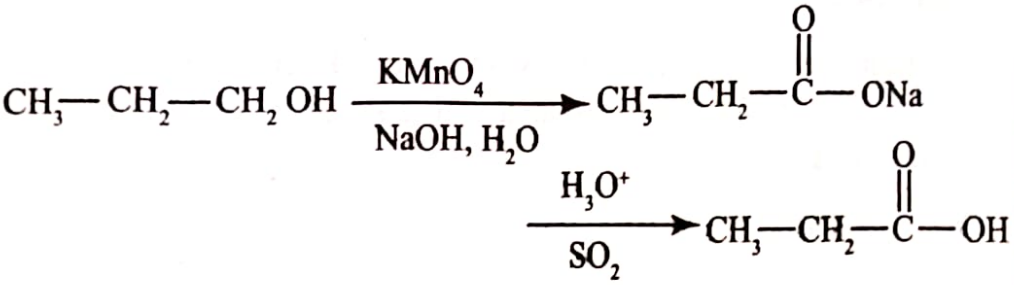
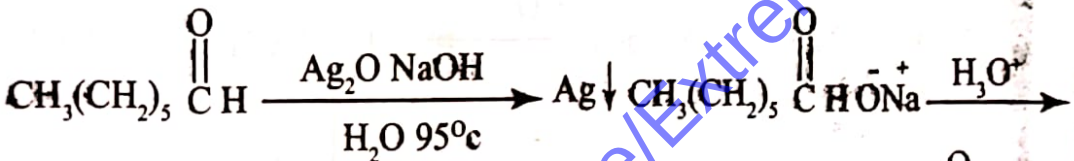
Example

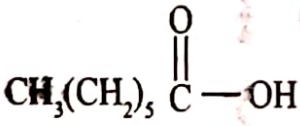


Aldehyde

Example

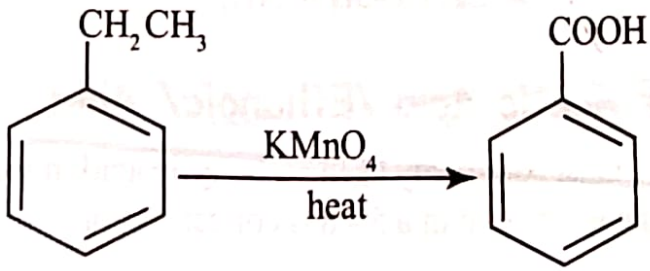


Heptanal



Heptanoic acid

- In this reaction, silver (I) is reduced to metallic silver. If the reaction is carried out in a clean test tube or flask, a sliver mirror deposits on the glass The reaction is therefore, used as a test to distinguish aldehydes, The reagent is known as **Tollens reagent** and the test as the **Tollens test** or the **silver mirror test**.
- **Oxidation of alkyl benzene yields benzoic acid.**
- Any alkyl benzene oxidized with $KMnO_4$ or $Na_2Cr_2O_7$ to give benzoic acid.



Ethyl benzene Benzoic acid

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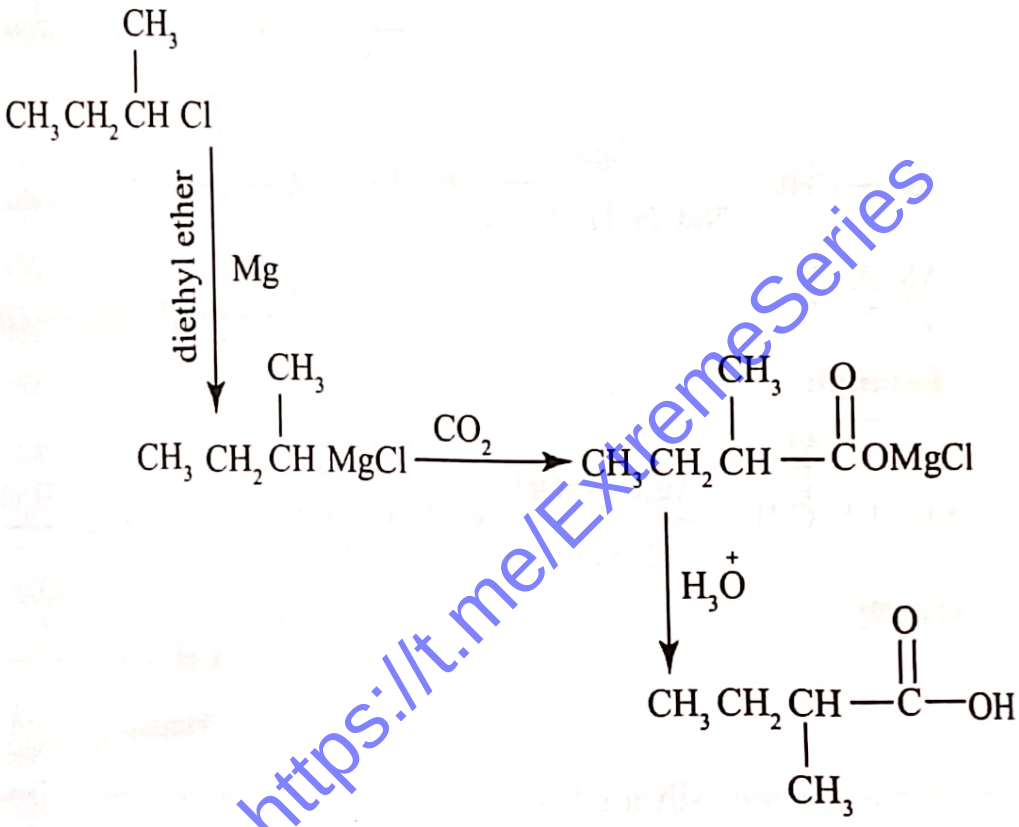
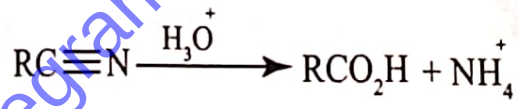
b. Reaction of Organometallic Reagents with Carbon dioxide.

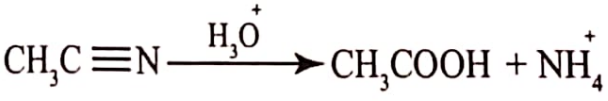
Organometallic regents react with carbondioxide to given salts of carboxylic acids. The salt is treated with a strong mineral acid to recover the carboxylic acid.



c. Hydrolysis of Cyano Compounds



Example



Preparation of Acetic Acid /Ethanoic/ Acid

➥ **Acetic** acid is the most commonly used carboxylic acid. It is commonly known as **vinegar** when used in a 5 - 8% concentration.

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1. By Oxidation of Ethyl alcohol

$$C_2H_5OH \xrightarrow[K_2Cr_2O_7]{[O]} CH_3CHO \xrightarrow[K_2Cr_2O]{[O]} CH_3COOH$$

Ethanol Ethanal Ethanoic acid

2. Hydrolysis of eEthyl Acetate

$$CH_3COOC_2H_5 + H_2O \xrightarrow{conc.H_2SO_4} CH_3COOH + C_2H_5OH$$

Ethyl acetate Ethanoic acid Ethanol

3. Quick Vinegar Fermentation Process

➥ This method is used to manufacture large quantities of acetic acid

Illustrative Example

90. All of the following methods are used to prepare carboxylic acids. **EXCEPT**

A. Oxidation of primary alcohols

B. Oxidation of aldehydes

C. Oxidation of alkyl benzene

D. Oxidation of alkenes

Answer: D

91. Large quantities of acetic acid is obtained in industry from

A. Oxidation of ethanol

B. Oxidation of ethanol

C. Hydrolysis of ethanol

D. Fermentation of ethanol

Answer: D

a. $CH_3CH_2CH_2OH \xrightarrow{KMno_4}$

b. $CH_3CH_2CHO \xrightarrow{KMnO_4}$

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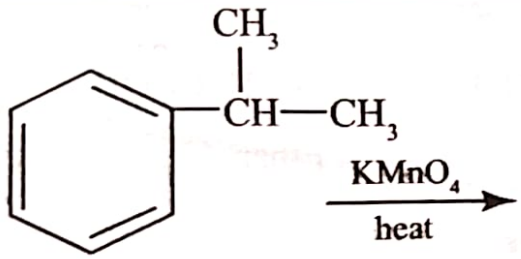
c. 

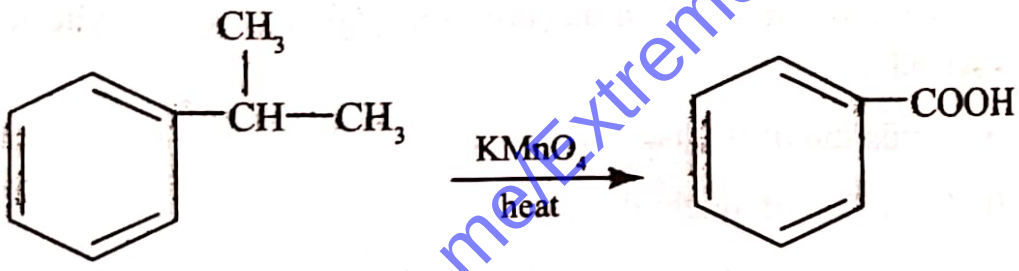
d. $CH_3CH_2Br \xrightarrow{Mg} x \xrightarrow[2)H_2O]{1)\,CO_2} y$

e. $CH_3CH_2C \equiv N \xrightarrow{H_3O^+}$

Solution

a. $CH_3CH_2OH \xrightarrow{KMnO_4} CH_3CH_2COOH$

b. $CH_3CH_2CHO \xrightarrow{KMnO_4} CH_3CH_2COOH$

c. 

d. $CH_3 - CH_2Br \xrightarrow{Mg} CH_3CH_2MgBr \xrightarrow[2)H_2O]{1)\,CO_2} CH_3CH_2COOH$

e. $CH_3CH_2C \equiv N \xrightarrow{H_3O^+} CH_3CH_2COOH + NH_4^+$

Fatty Acids

- They are long chain carboxylic acids
- Fatty acids that contain carbon - carbon double bond are called **unsaturated fatty acids.**

Example

$CH_3(CH_2)_5CH=CH(CH_2)_7COOH$ Palmitoleic acid

$CH_3-(CH_2)_7CH=CH-(CH_2)_7COOH$ Oleic acid

$CH_3CH_2CH=CHCH_2CH=CHCH_2CH=CH(CH_2)_7COOH$

Linolenic acid

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Note: If there is only one carbon - carbon double bond, it is usually formed between the 9th and 10th carbon atoms.

- Fatty acids containing all carbon - carbon single bonds are known as **saturated fatty acids.**

Table 6.6 Common Examples of Fatty Acids

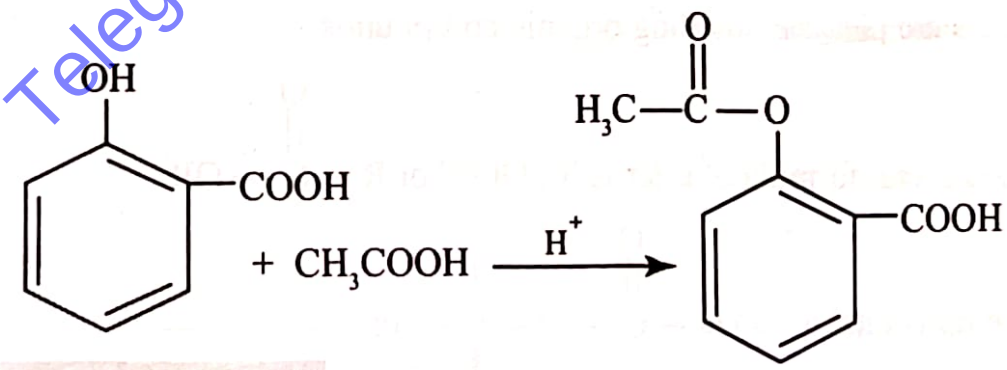
Formula	Common name	IUPAC name
$CH_3(CH_2)_{10}COOH$	Lauric acid	Dodecanoic acid
$CH_3(CH_2)_{12}COOH$	Myristic acid	Tetradecanoic acid
$CH_3(CH_2)_{14}COOH$	Palmitic acid	Hexadecanoic acid
$CH_3(CH_2)_{16}COOH$	Stearic acid	Octadecanoic acid
$CH_3(CH_2)_{16}COOH$	Arachidic acid	Eicosanoic acid

Note: The melting point of saturated fatty acids increases with increases in size.

- The melting point of unsaturated fatty acids decreases with increasing double bonds.

Uses of Carboxylic Acids

- Fatty acids are used to make soaps, detergents and shampoos.
- Carboxylic acids are used to make greases, crayons, plastics and synthetic odors and flavors
- The most widely used carboxylic acid acetic acid. It is used to make acetates, acetic anhydride, vinyl acetate polymer and the common pain killer acetyl salicylic acid (Aspirin)



2 - hydroxyl benzoic acid
(Sali cylic acid)

Acetyl salicylic acid
(Aspirin)

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Illustrative Examples

92. What is the chemical name for Aspirin?

A. Acetyl salicylic acid
B. Salicyli acid
C. Methyl salicylate
D. Sodium salicylic acid

Solution

The chemical name of a spirin is acetyl salicylic acid.

Answer: A

93. Which of the following acid is used as chemical defense for ants?

A. HCOOH
B. $CH_3(CH_2)_2COOH$
C. CH_3COOH
D. CH_3CH_2COOH

Solution

Formic acid is derived from Latin, formica meaning ants or bees which is used as chemical defense for them.

Answer: A

94. Write the name of the carboxylic acid found in

a. Cheese b. Beef fat c. Palm oil

Solution

a. Propionic acid b. Stearic acid c. Palmitic acid

6.5 Esters

Esters are pleasant smelling organic compounds.

The general formula of ester is RCOOR' or $R-\overset{\overset{O}{\|}}{C}-OR'$

The functional group is $-\overset{\overset{O}{\|}}{C}-O-\overset{|}{C}-$ or $-COO-$

Esters are responsible for the aroma of many fruits.

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Table 6.7 Fruits and Esters Found in them

Ester	Fruit
Methyl butanoate	Pineapple
Pentyl ethanoate	Banana
Octyl butanoate	Orange
Isoamyl methanoate	Apple
Methyl methanoate	Rum essence
Methyl anthranilate	Grape

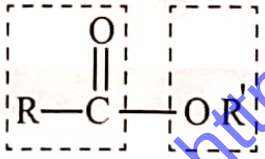
Nomenclature of Esters

I. Common Name

- The common name of the acid ' - ic acid' of carboxylic acid is replaced with ' -ate'
- The alcohol portion of the ester is converted in to alkyl or aryl group.

Methanol ⟶ methyl

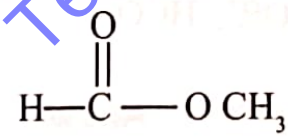
Ethanol ⟶ Ethyl

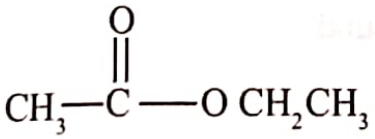


Acid portion Alcohol portion

Example

Give common name for the ester below

a.  Methyl formate

b.  - Ethyl acetate

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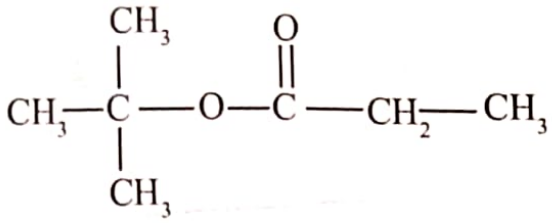
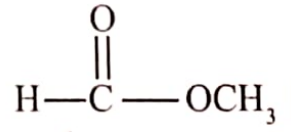
c.  - t - butyl propionate

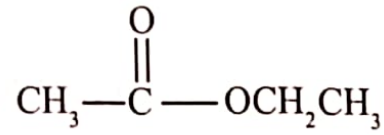
2. IUPAC Name of Esters

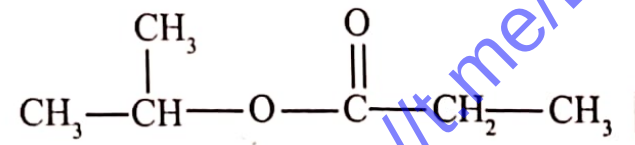
➥ The IUPAC name of carboxylic acid '-ic acid' is replaced with '-ate'

Example

Given the IUPAC name of the esters below

a.  Methyl methanoate

b.  Ethyl ethanoate

c.  Isopropyl propanoate

Illustrative Examples

95. Which compound is an ester?

A. CH_3COOH C. C_2H_5CHO

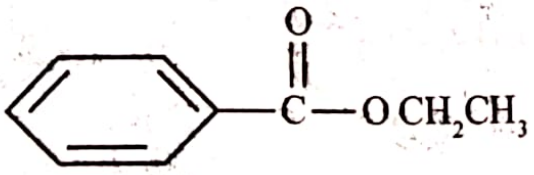
B. $CH_3OC_2H_5$ D. $HCOOCH_3$

Solution

Based on the general forumula of ester RCOOR', $HCOOCH_3$ is an esters.

Answer: D

96. What is the name of the following compound?



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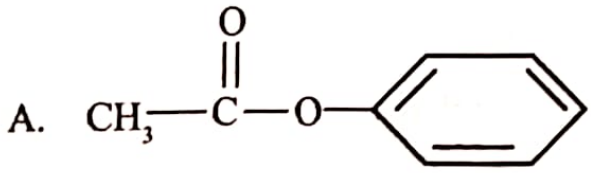
A. Benzoate ester
C. Phenyl butyrate
B. Ethyl benzoate
D. Ethyl benzyl ketone

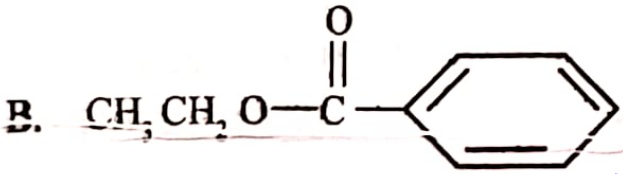
Solution

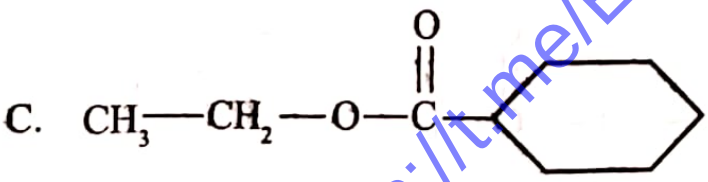
The acid portion is benzoic acid and the alcohol portion is ethanol. Thus, the name of the ester becomes ethylbenzoate.

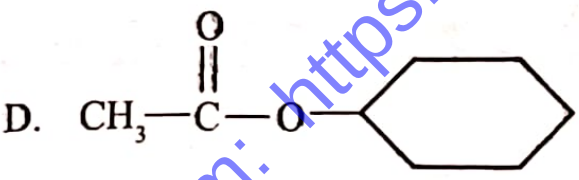
Answer: B

97. Which of the following is the structure of phenyl acetate?

A. 

B. 

C. 

D. 

Solution

A. Phenyl acetate
C. Ethyl cyclohexane carboxylate
B. Ethyl benzoate
D. Cyclohexyl acetate

Answer: A

98. Which of the following is responsible for the pleasant odour of many fruits?

A. Carboxylic acid
C. Ester
B. Ether
D. Aldehyde

Answer: C

99. Write the common name of the following esters.

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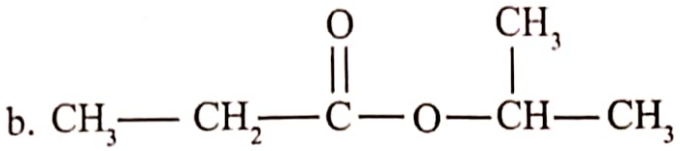
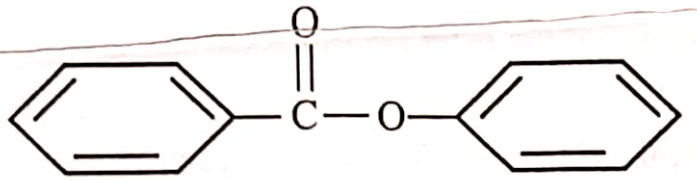
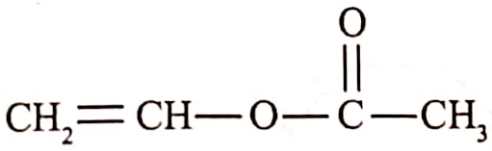
a. $CH_3\,CO_2\,CH_2\,CH_3$

b. 

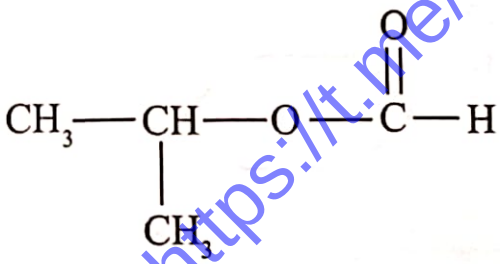
c. 

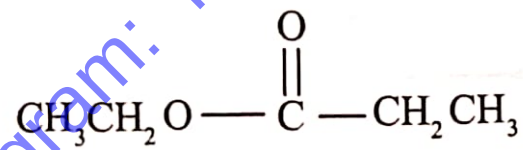
d. 

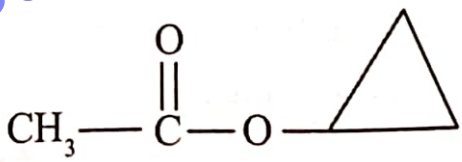
Solution

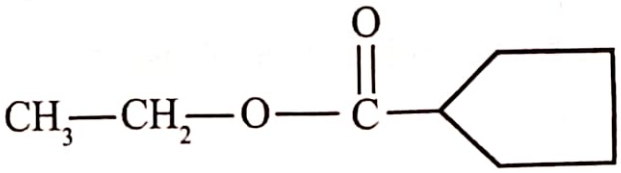
a. Ethyl acetate
b. Isopropyl propionate
c. Pheyl benzoate
d. Vinyl acetate

100. Give the IUPAC name for the following esters

a. 

b. 

c. 

d. 

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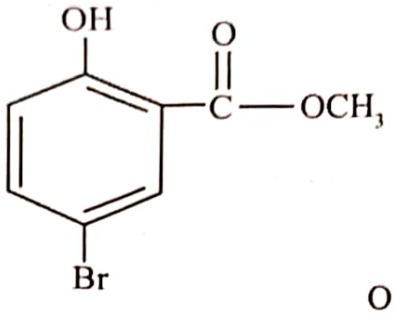
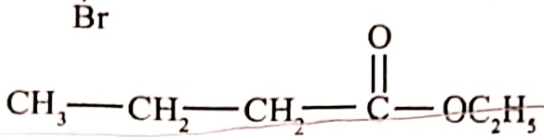
e. 

f. 

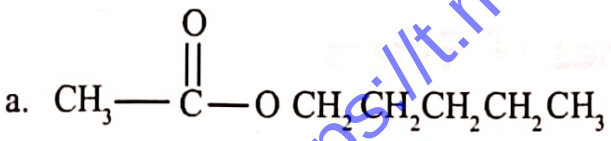
Solution

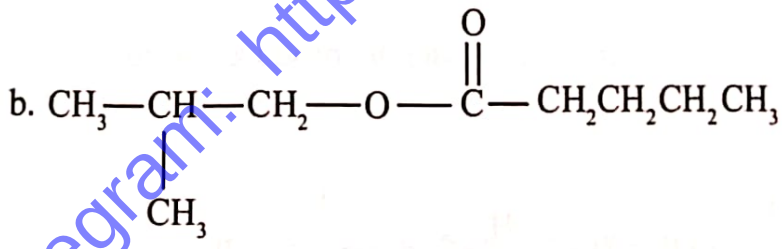
a. Isopropyl methanoate
b. Ethyl propanoate
c. Cycloproyl ethanoate
d. Ethyl cyclopentanoate
e. Methyl - 3 - bromo benzene carboxylate
f. Ethyl butanoate

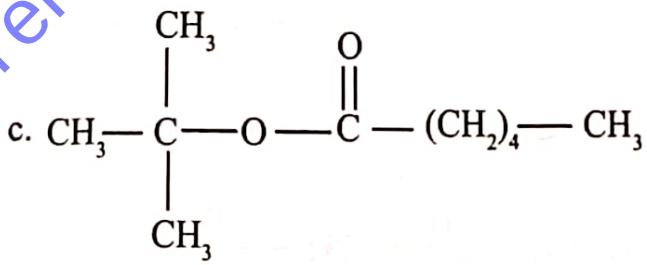
101. Draw the structures of the following esters.

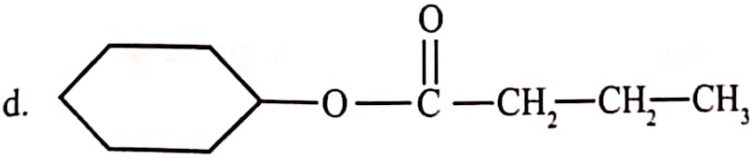
a. n - pentyl ethanoate
b. Iso - butyl valerate
c. t - butyl caproate
d. Cyclohexyl butyrate

Solution

a. 

b. 

c. 

d. 

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Physical Properties of Esters

a. Boiling Point

Esters have lower boiling point than alcohol and carboxylic acids of similar mass.

Example

Boiling point: $HCOOCH_3 < CH_3CH_2OH < CH_3COOH$

- As carbon number increases, boiling point of ester also increases.
- As branching increases, boiling point decreases.

b. Solubility

- Esters with lower mass are fairly soluble in water.
- As carbon number increases, solubility of esters in water decreases.
- All esters are soluble in non - polar or organic solvents.

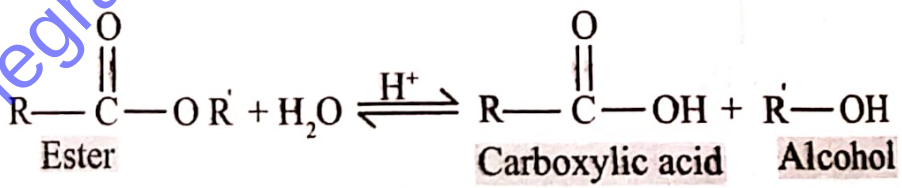
c. Odour

Esters are generally pleasant smelling organic compounds. The odours of most fruits and flowers is due to esters in them.

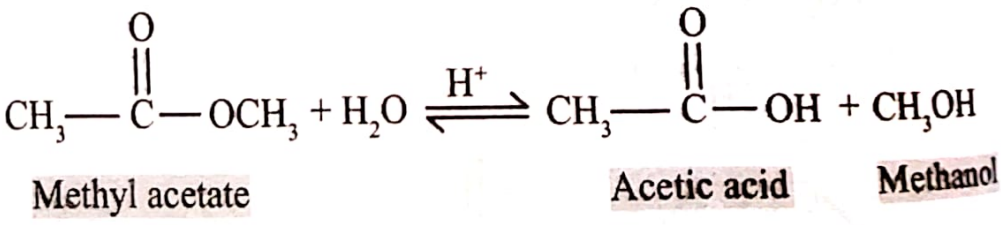
Chemical Properties of Esters

I. Hydrolysis

It is the reaction of esters with water to produce carboxylic acid and alcohol.



Example



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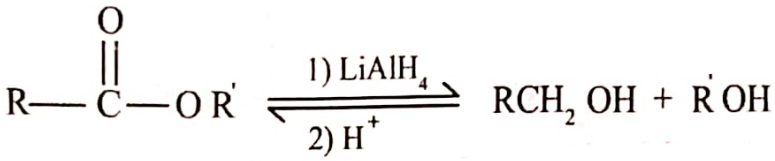
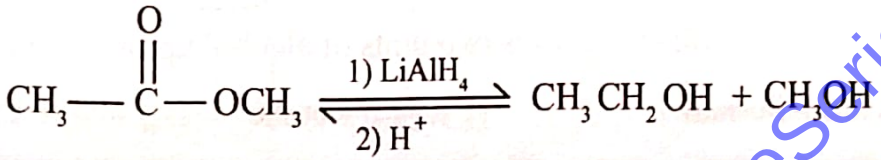
Note: Base hydrolysis of esters is known as saponification because a soap is obtained by this process.

Reduction of ester yield two units of alcohol



Example



Illustrative Example

102. What is the name of a base - promoted ester hydrolysis reaction?

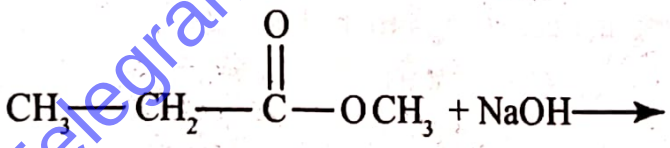
A. Acylation
C. Condensation
B. Esterification
D. Saponification

Solution

Base hydrolysis of ester is known as saponifiaction

Answer: D

103. Consider the reaction



What are the products of this reaction?

A. Sodium acetate and ethanol
B. Sodium acetate and methanol
C. Sodium propionate and methanol
D. Propionic acid and methanol

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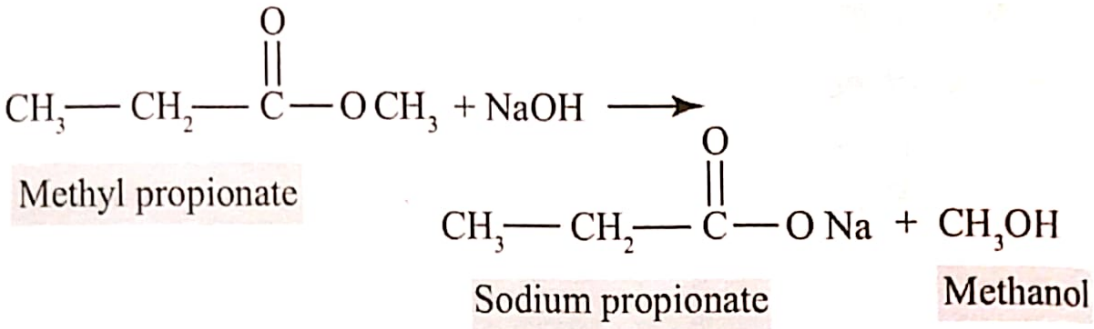
Solution



Answer: C

104. Which of the following gives two units of alcohol up on reduction?

A. Carboxlic acid
C. Ketone
B. Ester
D. Aldehyde

Solution

Esters $\xrightarrow{\text{Reduction}}$ two units of alcohols

Answer: B

105. Arrange the following esters in increasing order of boiling point

a. $HCOOCH_3$
c. $CH_3COOCH_2CH_3$
b. $HCOOCH_2CH_3$
d. $CH_3COOCH_2CH_2CH_3$

Solution

As carbon number increases, boiling point also increases. a < b < c < d.

106. Arrange the following in decreasing order of solubility in water.

a. CH_3COOCH_3
c. CH_3COOH
e. $CH_3CH_2COOCH_3$
b. $HCOOCH_3$
d. CH_3CH_2OH

Solution

Solubility of: carboxylic alcohol > acid > ester

$\Rightarrow$ c > d > b > a > e

107. Complete and balance the following.

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a. $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OCH_2CH_3 + H_2O \underset{H^+}{\rightleftharpoons}$

b. $CH_3CH_2CH_2OOCCH_2CH_2CH_3 + KOH \longrightarrow$

c. $C_6H_5\overset{O}{\overset{\|}{C}}-O-CH_2-CH_2-CH_3 \underset{H^+}{\overset{LiAlH_4}{\rightleftharpoons}}$

Solution

a. $CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OCH_2CH_3 + H_2O \overset{H^+}{\rightleftharpoons} CH_3-CH_2-\overset{O}{\overset{\|}{C}}-OH + CH_3CH_2OH$

b. $CH_3CH_2CH_2OOCCH_2CH_2CH_3 + KOH \longrightarrow CH_3CH_2CH_2OH + CH_3CH_2CH_2\overset{O}{\overset{\|}{C}}-OK$

c. $C_6H_5\overset{O}{\overset{\|}{C}}-O-CH_2-CH_2-CH_3 \underset{H^+}{\overset{LiAlH_4}{\rightleftharpoons}} C_6H_5CH_2OH + CH_3CH_2CH_2OH$

Preparation of Esters

1. Esterification

➥ It is the reaction between carboxylic acids and alcohol to form esters.

$$\underset{\text{Acyl chloride}}{R-\overset{O}{\overset{\|}{C}}-Cl} + \underset{\text{Alcohol}}{R'-OH} \overset{H^+}{\rightleftharpoons} \underset{\text{Ester}}{R-\overset{O}{\overset{\|}{C}}-OR'} + HCl$$

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Example

$$CH_3CH_2OH + CH_3COOH \underset{}{\overset{H^+}{\rightleftharpoons}} CH_3CH_2OOCCH_3 + H_2O$$

Ethanol Acetic acid Ethyl acetate

2. The Reaction Between Acyl Chloride and Alcohol

$$R-\overset{\overset{O}{\|}}{C}-Cl + R'-OH \overset{H^+}{\rightleftharpoons} R-\overset{\overset{O}{\|}}{C}-OR' + HCl$$

Acyl chloride Alcohol Ester

Example

$$CH_3-\overset{\overset{O}{\|}}{C}-Cl + CH_3OH \longrightarrow CH_3-\overset{\overset{O}{\|}}{C}-OCH_3 + HCl$$

Acetyl chloride Methanol Methyl acetate

3. The Reaction Between Acid Anhyidrde and Alcohol.

$$R-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-R + R'OH \longrightarrow R-\overset{\overset{O}{\|}}{C}-OR' + R-\overset{\overset{O}{\|}}{C}-OH$$

Acid anhydride

Example

$$CH_3-\overset{\overset{O}{\|}}{C}-O-\overset{\overset{O}{\|}}{C}-CH_3 + CH_3CH_2OH \longrightarrow$$

$$CH_3-\overset{\overset{O}{\|}}{C}-O\,CH_2CH_3 + CH_3-\overset{\overset{O}{\|}}{C}-OH$$

Uses of Esters

- Ester is pleasant smelling organic compounds. Some of them are used as food flavourings and others are used as fragrances or perfumers.
- Esters are used to make polymers (polyesters) which can be used to make cans or plastic bottles.

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➥ Esters are used as organic solvents.

Illustrative Examples

108. The IUPAC name of the alcohol and carboxylic used to prepare isopropyl acetate are

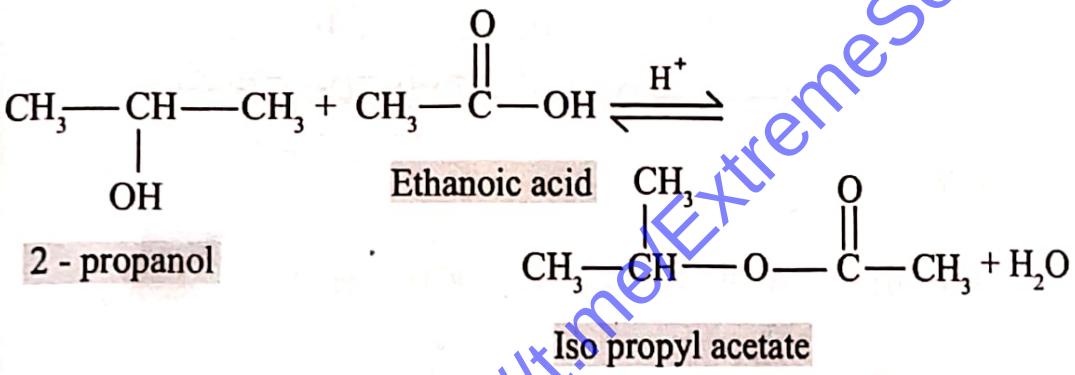
A. 1 - Propanol and ethanoic acid

B. 2 - propanol and ethanoic acid

C. Ethanol and propanoic acid

D. Ethyl alcohol and propanoic acid

Solution



Answer: B

109. The reaction between acyl chloride and alcohol yields

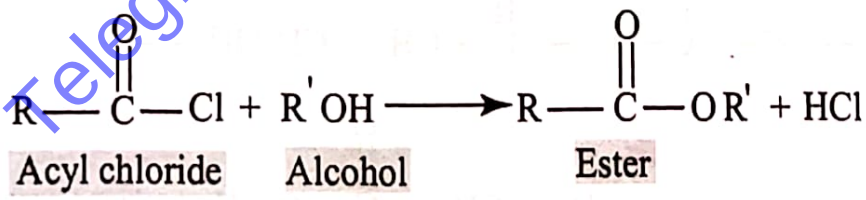
A. Acid - anhydride

B. Ester

C. Carboxylic acid

D. Acetyl chloride

Solution



Answer: B

110. Write the name of the alcohol and acid used to prepare the following esters.

a. Benzyl benzoate

b. Cyclohexyl cyclohexane carboxylate

c. Isobutyl pentanoate

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Solution

a. Benzyl acohol and benzoic acid.

b. Cyclohexyl alcohol and cyclohexane carboxylic acid

c. Isobutyl alcohol and pentanoic acid

iii. Complete and balance the following

a. $HCOOH + CH_3OH \overset{H^+}{\rightleftharpoons}$

b. $CH_3—CH_2—\overset{\overset{O}{\|}}{C}—Cl + CH_3CH_2OH \longrightarrow$

c. $CH_3—CH_2—\overset{\overset{O}{\|}}{C}—O—\overset{\overset{O}{\|}}{C}—CH_3 + CH_3OH \longrightarrow$

Solution

a. $HCOOH + CH_3OH \overset{H^+}{\rightleftharpoons} HCOOCH_3 + H_2O$

b. $CH_3—CH_2—\overset{\overset{O}{\|}}{C}—Cl + CH_3CH_2OH \longrightarrow$

$$CH_3—CH_2—\overset{\overset{O}{\|}}{C}—O\,CH_2CH_3 + HCl$$

c. $CH_3—CH_2—\overset{\overset{O}{\|}}{C}—O—\overset{\overset{O}{\|}}{C}—CH_3 + CH_3OH \longrightarrow$

$$CH_3—CH_2—\overset{\overset{O}{\|}}{C}—O\,CH_3 + CH_3CH_2OH$$

Fats and Oils

Lipids are substances that are found in living organisms and are insoluble in water but soluble in organic solvents. A lipid contains a long hydrocarbon chain as part as part of its structure but can have a variety of functional groups.

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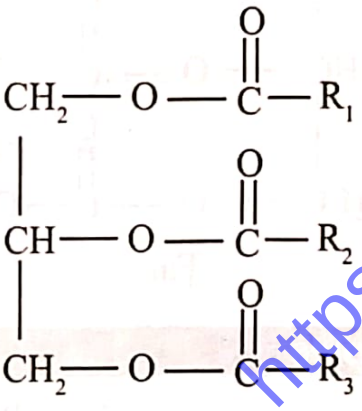
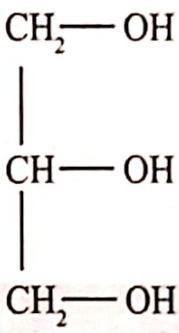
- **Fats and oils** are esters of grycerol(1, 2, 3 - propanetriol) and long - chain carboxylic acids, usually having twelve or more carbon atoms.
- Fats and oils are widely found in living things.
- **Fats** are a class of energy - rich organic compounds occurring in organic tissue especially in the adipose tissue of animals and in the seeds, nuts and fruits of plants.
- Fats are solids or semi - solids while oils are liquids at room temperature
- Animal fats include butter, lard and tallow, oils obtained from animals are **liver oils,blubber oils and fish oils.**
- Oils obtained from plants are peanut oil, soya bean oil and olive oil.

Structure of Fats and Oils

- Fats and oils are triesters of glycerol, triacyl glycerols or triglycerides.

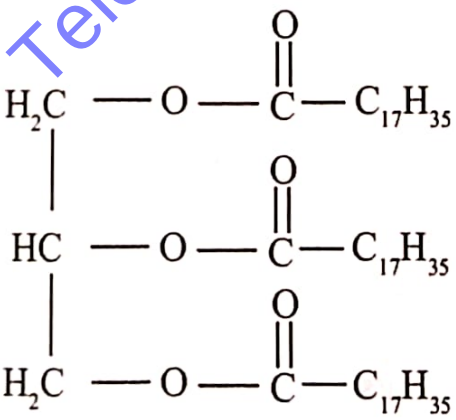


A triglyceride

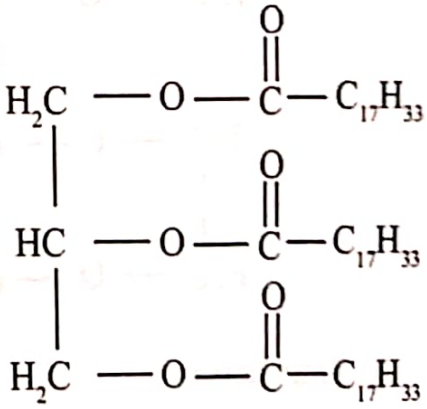


Glycerol

- **Fats** are ester of glycerol of saturated fatty acids but **oils** are ester of glycerol of unsaturated fatty acids.



Giyceryl tri streate (stearin) fat



Glyceryl trioleate (olein) oil

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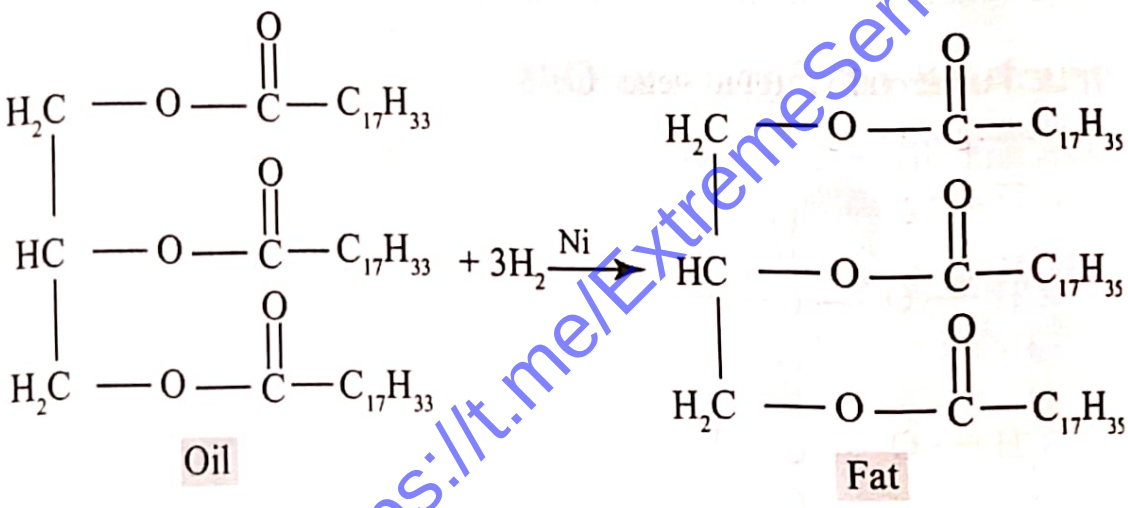
Physical Properties of Fats and Oils

- Fat is solid while oil is liquid at room temperature.
- Fats and oils are insoluble in water but soluble in non - polar solvents.
- They are greasy to touch and are non - volatile
- They are colorless, ordourless and neutral in pure form.

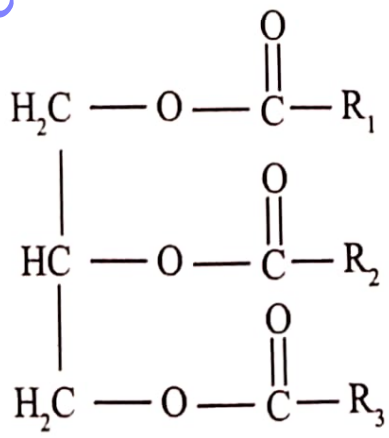
Hardening of Oils

Oils are converted in to fats by addition of hydrogen (hydrogenation) in the presence of metal catalyst.



Illustrative Examples

112. Which type of substance has the following general formula? (R_1 , R_2 and R_3 hydrocarbon groups.) (ESSLCE)



A. Fatty acids
B. Alcohols
C. Fats
D. Carboxylic acids

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Solution

The structure represents a triglyceride (fat or oil).

Answer: C

113. Which of the following statements concerning fats and oils is incorrect?

A. They are also called triacyl glycerols

B. They are also called triglycerides

C. They are fatty acid salts

D. They are glycerol triesters.

Answer: C

114. Commercially, liquid vegetable oils are converted to solid fats such as margarine by

A. Hydrogenation

B. Hydration

C. Saponification

D. Oxidation

Solution

$$\text{Oil} \xrightarrow{\text{Hydrogenetionation}} \text{Fat}$$

Answer: A

115. Which catalyst is used in hydrogenation of vegetable oils?

A. Iron

B. Nickel

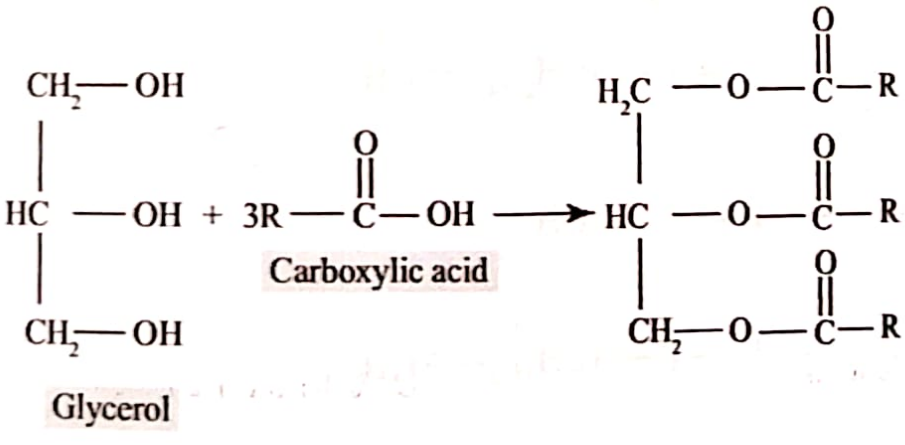
C. Platimum

D. Molybdenum

Answer: B

116. What is the general formula used to make fats and oils.

Solution



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117. What is the main cause for rancidity of fats and oils.

Solution

It is caused mainly because of the hydrolysis of ester bond or linkage and oxidation across the double bonds. Other causes may be raw materials used to make fats and oils and storage under unfavourable conditions.

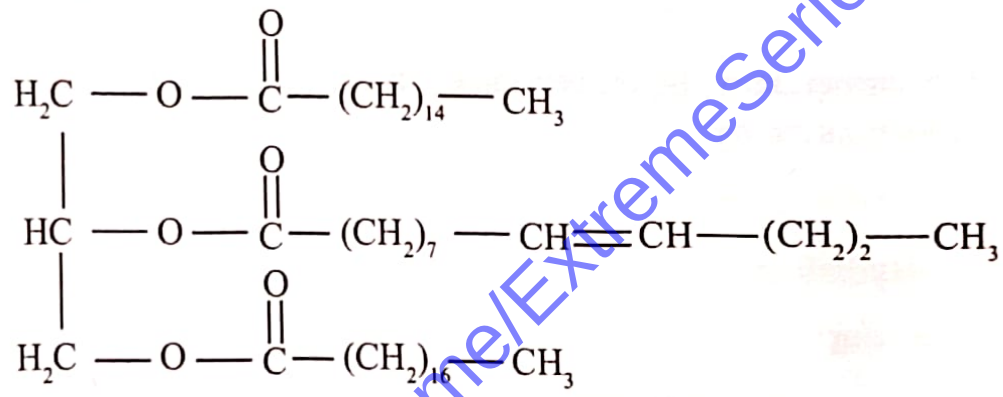
118. Write the structural formula for the following fats and oils.

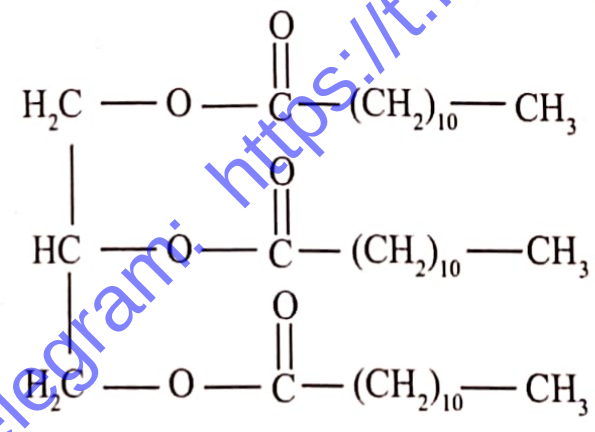
a. Glyceryl palmitoo lestreate

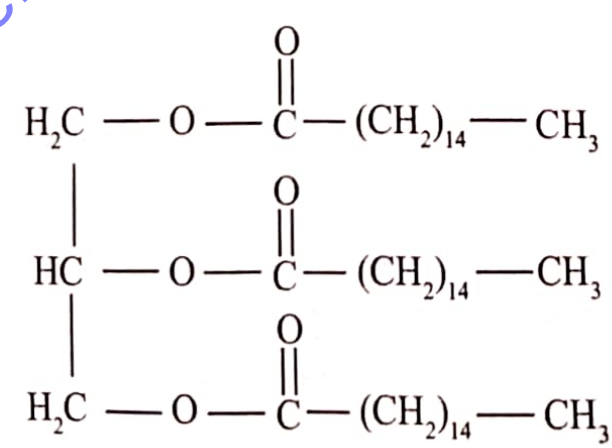
b. Glyceryl triluarate

c. Glyceyl triplamitate.

Solution

a. 

b. 

c. 

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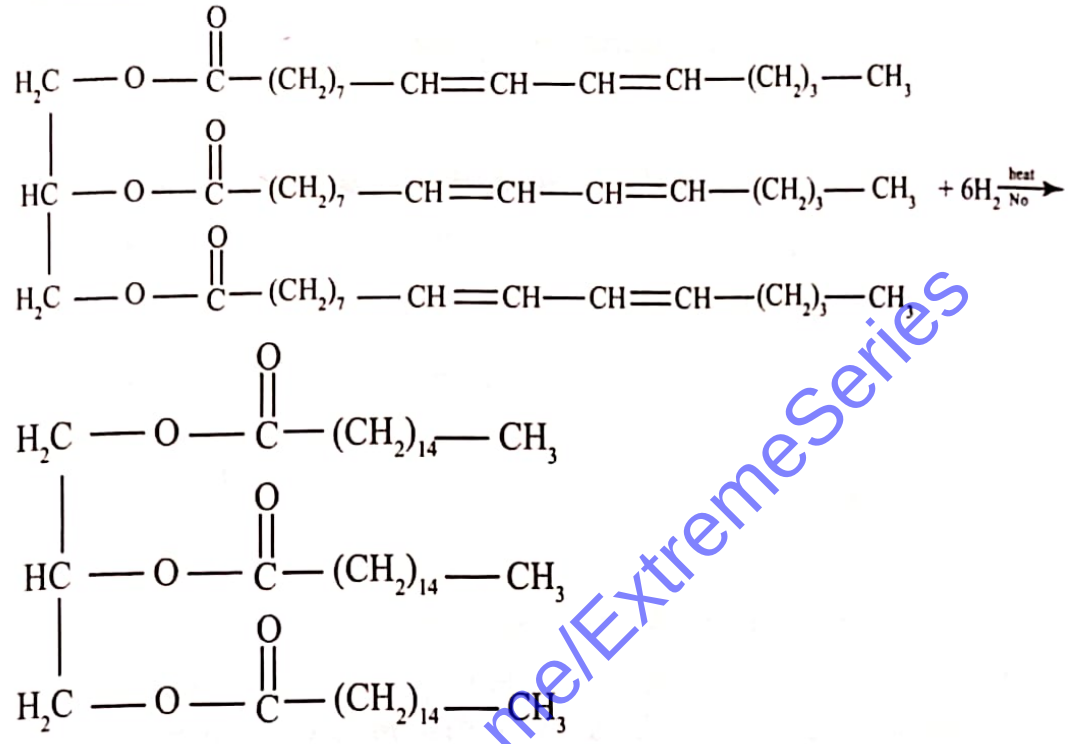
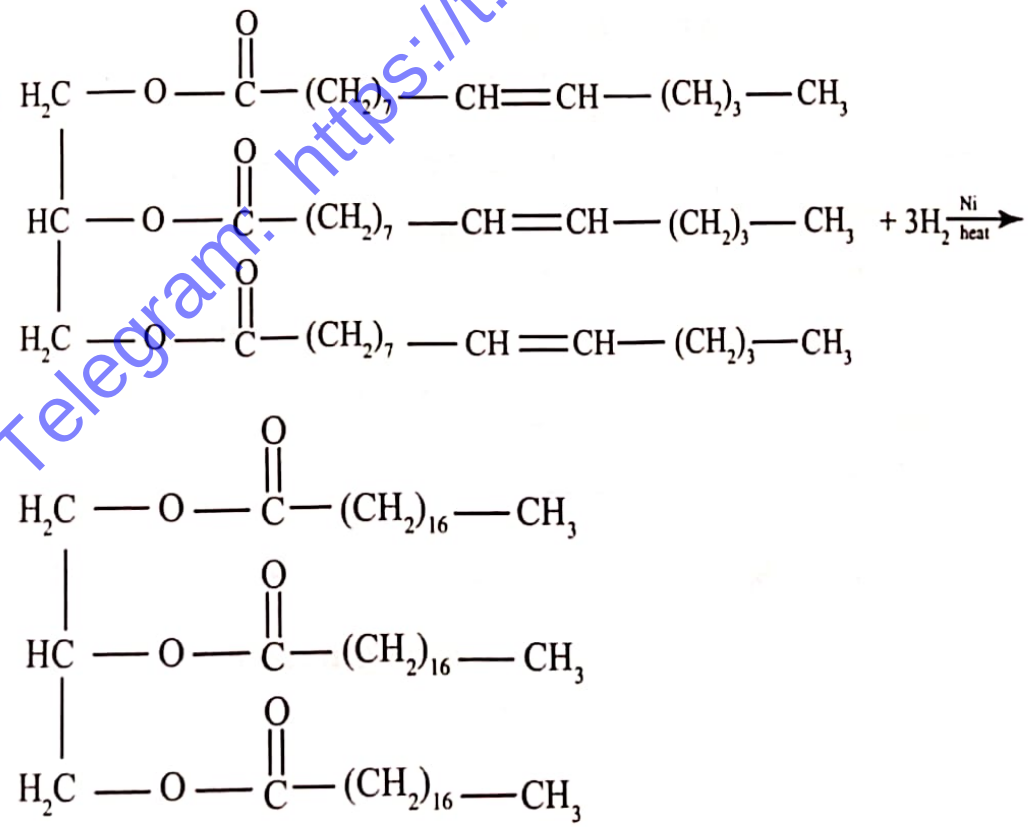
119. Write chemical equation for hydrogenation of:

a. Glyceryl trilinoleate

b. Glycery trioleate

Solution

a. 

b. 

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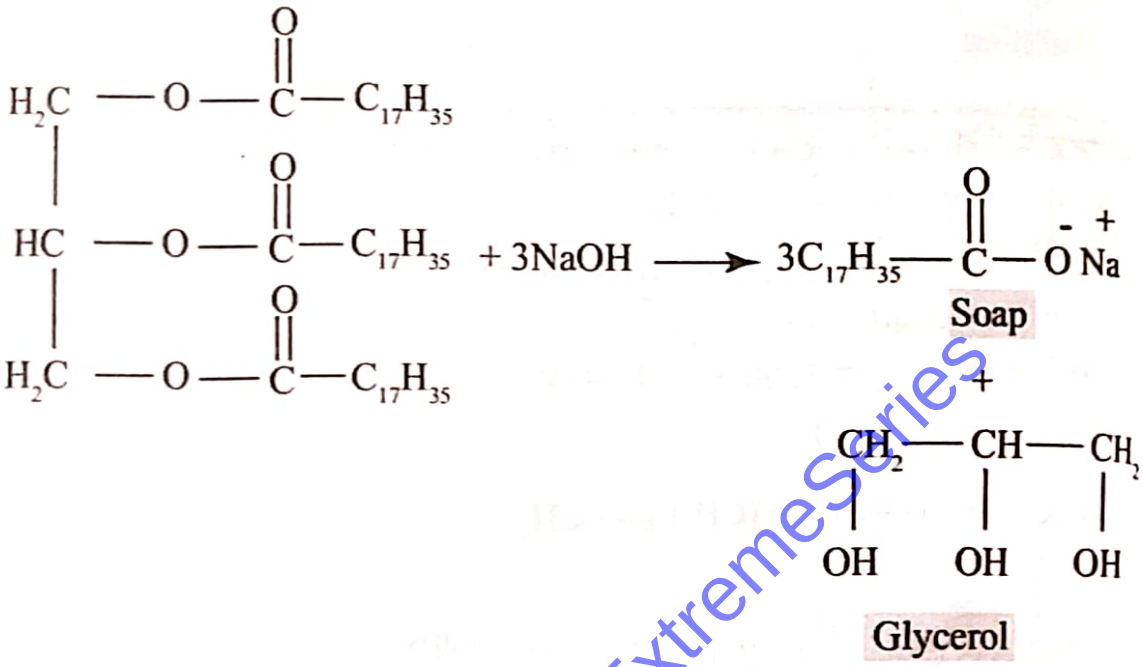
6.6 Soaps and Detergents

Soaps are sodium or potassium salts of fatty acids.



- Sodium soaps are **hard soap** while potassium soaps are **soft soaps.**
- The polar end of a soap molecule is said to be **hydrophilic or water loving** and the non - polar tail is **hydro phobic or water loving.**
- Polar head dissolves polar dirt and non - polar tail dissolves non - polar dirt

A sodium soap is a solid and potassium soaps are liquids at room temperature.

- Soaps and detergents affect the properties of surface of the dispered material. Due to this, they are often called **surfactants.**
- **Detergents** are synthetic substance that contain either sulphate ion or sulphonate groups.

$$CH_3(CH_2)_{12}CH_2OH + H_2SO_4 \longrightarrow$$

Myristyl alcohol

$$CH_3(CH_2)_{12}CH_2 — O — SO_2 — OH + H_2O$$

Myristyl hydrogen sulphate

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$CH_3(CH_2)_{12}CH_2 — O — SO_2 — OH + NaOH \longrightarrow$

$CH_3(CH_2)_{12}CH_2\,O\,SO_3Na$

Sodium myristil sulphate

Illustrative Example

120. Which of the following types of compounds are expected products from the saponification of a fat?

A. Glycerol and fatty acid salts

B. Glycerol and fatty acids

C. Fatty acid salts and fatty acids

D. Glycerol, fatty acids and fatty acids salts

Answer: A

121. Which of the following is false about soaps?

A. They are alkaline

B. They are not used in hard

C. They have no draw backs

D. They are neutral

Solution

Soaps are alkaline because they are formed from weak acid and strong base. They have no draw backs and used in soft water.

Answer: D

122. Write the cleaning action of soap

Solution

Soaps have hydrophilic (water loving) head and hydrophobic (water hating) tail. The hydrophilic head removes polar dirt and the hydrophobic tail removes non - polar dirt.

Hydrophobic part (Non - polar tail)

Hydrophilic part (polar head)

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123. The concentration of molecules at the surface of the water has the effect of lowering the surface tension of water. Compounds that behave in this water are said to be _____ .

Solution

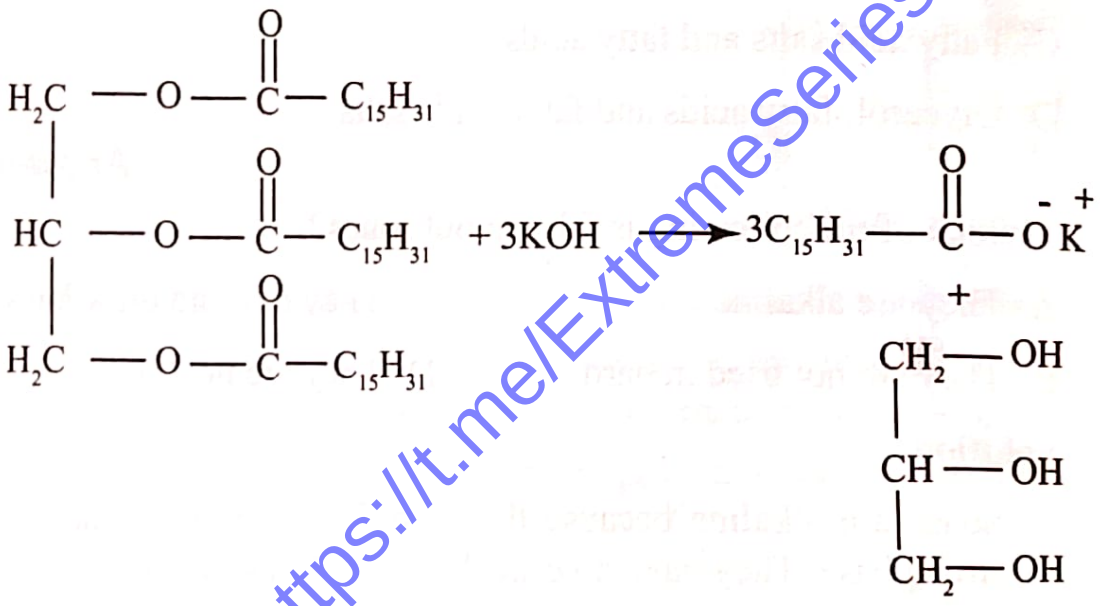
Surface - active compounds or surfactants.

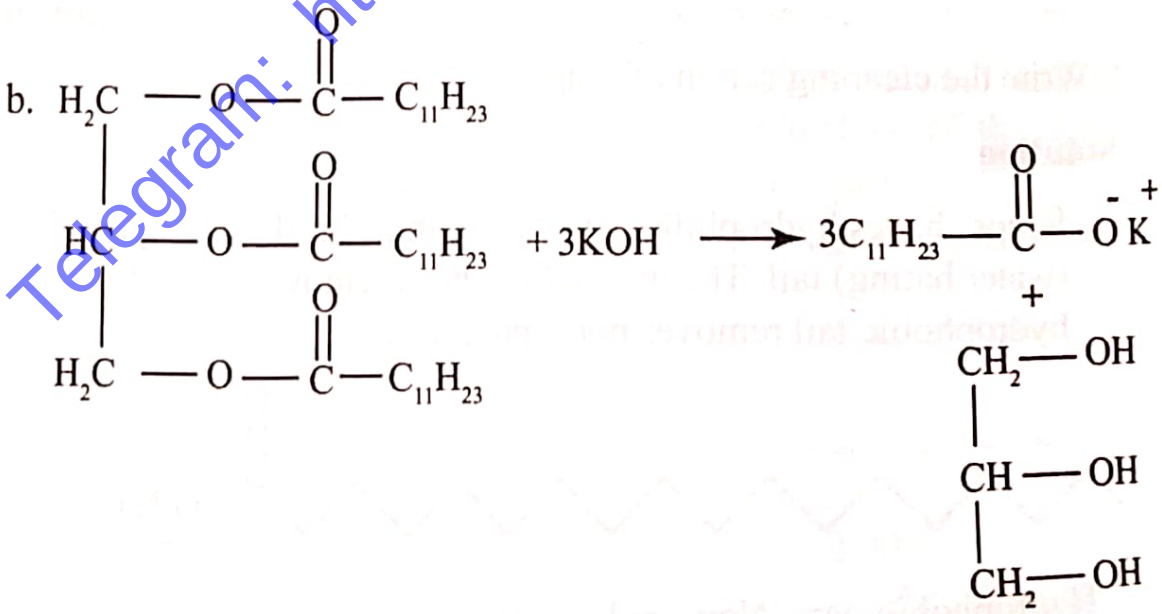
124. What are saponification products obtained when the following fats are boiled with KOH?

a. Glyceryltrilaurate

b. Glycetyl triplamitate

Solution

a. 

b. 

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Solved Examples

125. The simplest alcohol, a compound that is highly toxic and can cause blindness and death. It can be generated by heating wood and is thus also called wood alcohol. This alcohol is _____ .

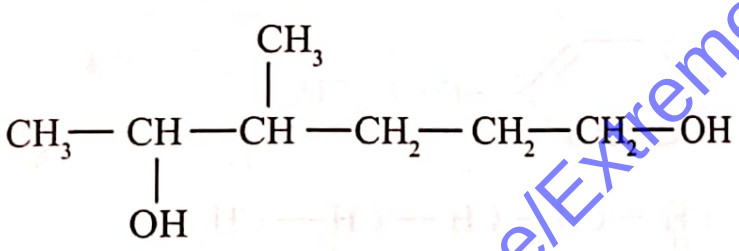
Solution

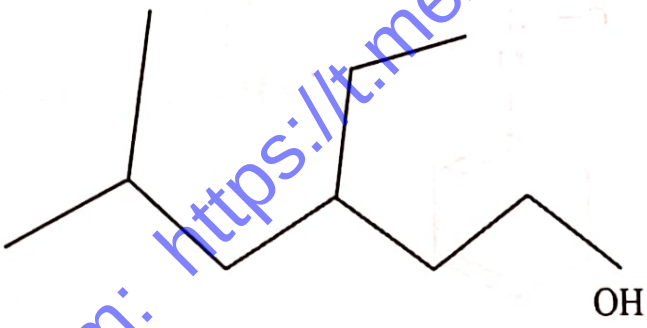
Methanol or methyl alcohol, CH_3OH

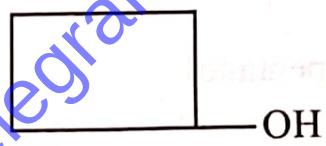
126. Write the IUPAC name of the following alcohols.

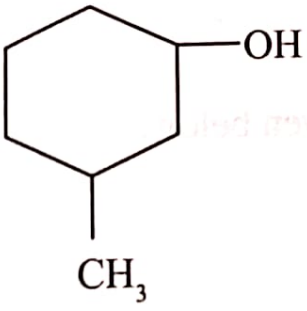
a. $HOCH_2C \equiv C\ CH_2CH_2\ OH$

b. $CH_2=CH-CH_2-CH_2OH$

c. 

d. 

e. 

f. 

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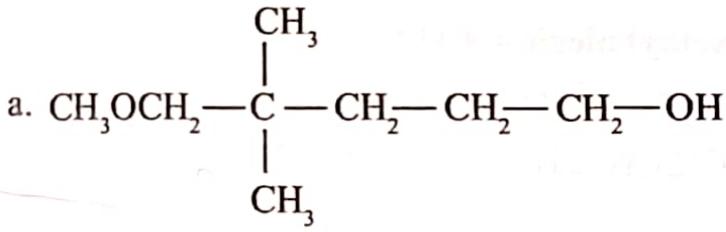
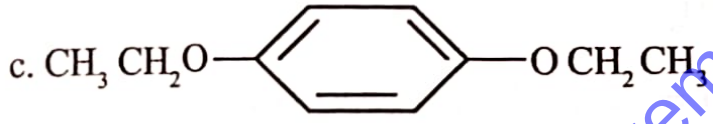
Solution

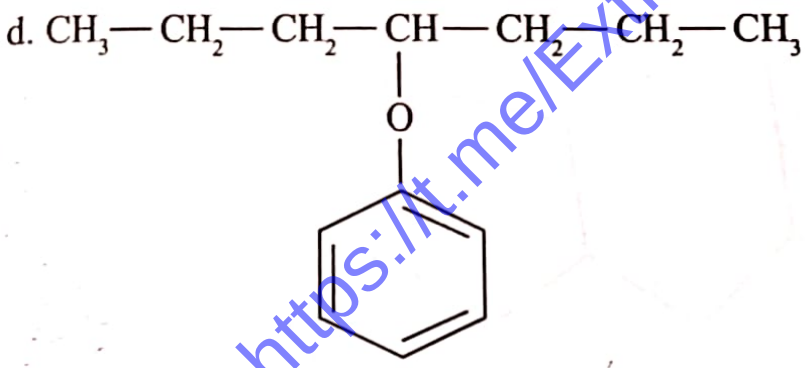
a. 2 - pentyne - 1,5 - diol
b. 3 - buten - 1 - 01
c. 4 - methyl hexan - 1,5 - diol
d. 3 - ethyl - 5 - methyl - 1 - hexanol
e. cyclobutanol
f. 3 - methyl cyclohexanol

127. Given the name of the following ethers

a. 

b. $CH_3O\ CH_2CH_2O\ CH_2\ CH_2\ CH_3$

c. 

d. 

Solution

a. 5 - methoxy - 4,4 - dimethyl - 1 - pentanol
b. 1 - methoxy - 2 - propoxyl ethane
c. 1,4 - diethoxyl benzene
d. 4 - phenoxyheptane

128. Draw the structure of alcohols and ethers given below.

a. 4,5 - dichloro 1 - hexanol
b. 2 - methylcylopentanol
c. Diphenyl ether
d. 1 - chloro - 3 - ethoxy propane
e. 1, 3 - dimethoxyl propane

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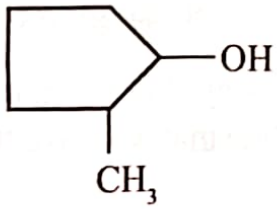
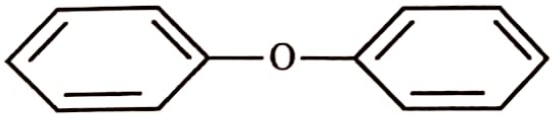
f. Hexane - 1,3,5 - triol

Solution

a. $CH_3-\underset{\underset{Cl}{|}}{CH}-\underset{\underset{Cl}{|}}{CH}-CH_2-CH_2-CH_2-OH$

b. 

c. 

d. $CH_3\,CH_2\,O\,CH_2\,CH_2\,CH_2Cl$

e. $CH_3\,O\,CH_2\,CH_2\,CH_2O\,CH_3$

f. $HO\,CH_2CH_2\,\underset{\underset{OH}{|}}{CH}\,CH_2\,\underset{\underset{OH}{|}}{CH}\,CH_3$

129. What would be the structure of the Markovinkov addition product for reaction of 2 - methyl - 1 - butene with dilute acid?

Solution

a. $CH_2=\underset{\underset{CH_3}{|}}{CH}-C=CH_2 \xrightarrow[60\%\ H_2SO_4]{H_2O} CH_3-CH_2-\underset{\underset{OH}{|}}{\overset{\overset{CH_3}{|}}{C}}-CH_3$

130. How would you prepare potassium t - butoxide?

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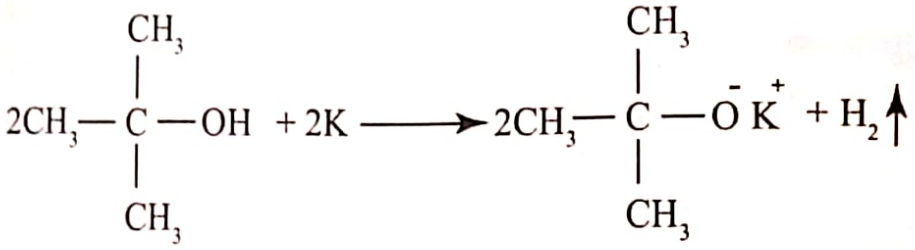
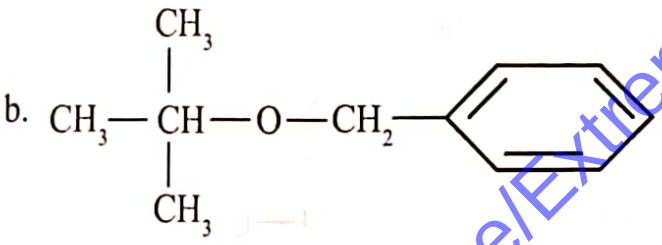
Solution



t - butyl alcohol — Potassium t - butoxide

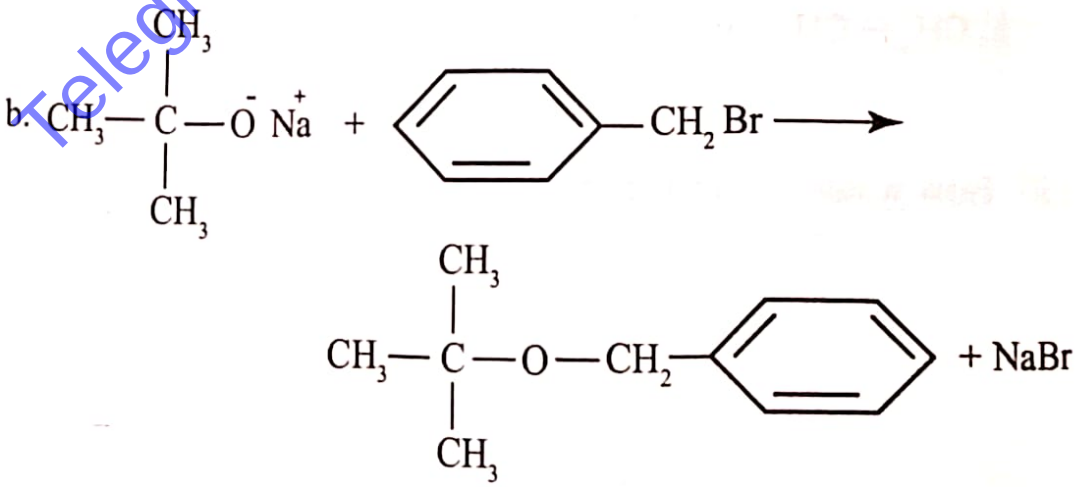
131. Synthesize each of the following ethers by the Williamson method, choosing the alkoxide anion and akyl halide that will give the best yield

a. $CH_3-CH(CH_3)-O\ CH_2CH_2CH_3$



Solution

a. $CH_3-CH(CH_3)-O^- Na^+ + CH_3CH_2CH_2Br \longrightarrow CH_3-CH(CH_3)-O-CH_2-CH_2-CH_3 + NaBr$

b. 

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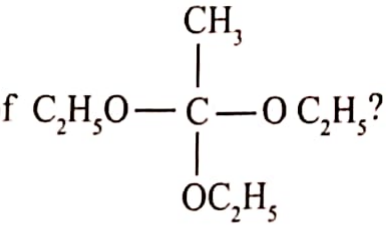
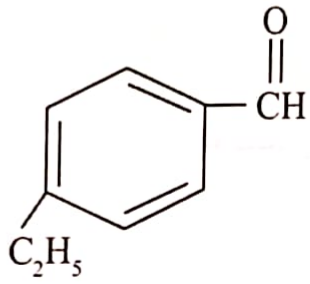
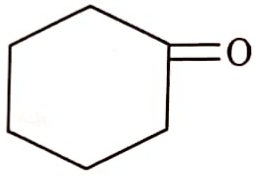
132. What is the name of  ?

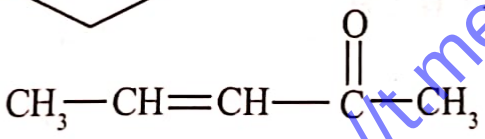
Solution

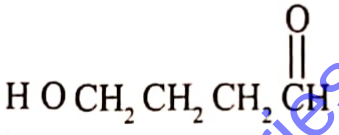
1,1,1,- triethoxy ethane

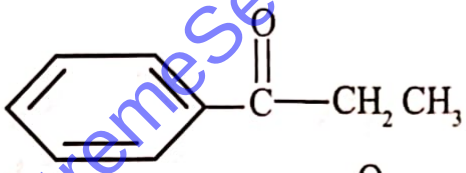
133. Write the IUPAC name of the following aldehydes and ketones

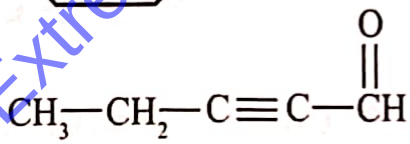
a. 

b. 

c. 

d. 

e. 

f. 

Solution

a. 4 - ethylbenzaldehyde
b. Cyclohexanone
c. 3 - penten - 2 - one
d. 4 - hydroxy butanal
e. 1 - phenyl propanone
f. 2 - pentynal

134. Draw structural formulas for the following compounds.

a. 3 - methyl cycclobutanone
b. Phenyl cyclopentyl ketone
c. Hexanedial
d. p - nitro benzaldehyde
e. 4 - heptenal
f. 1.4 - cyclohexanedione

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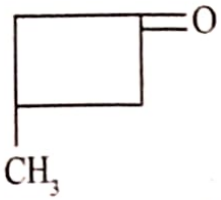
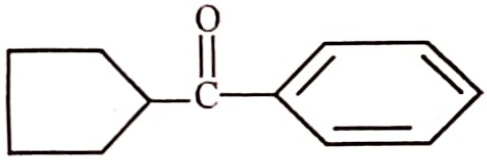
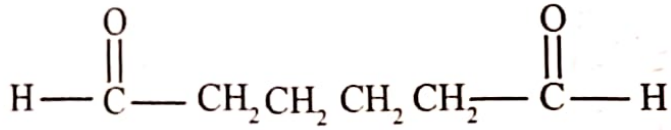
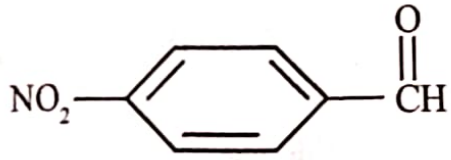
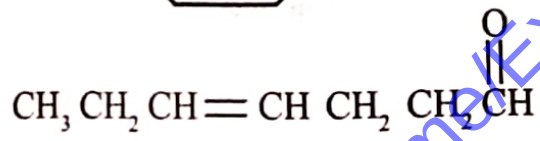
Solution

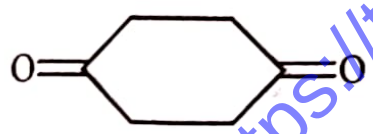
a. 

b. 

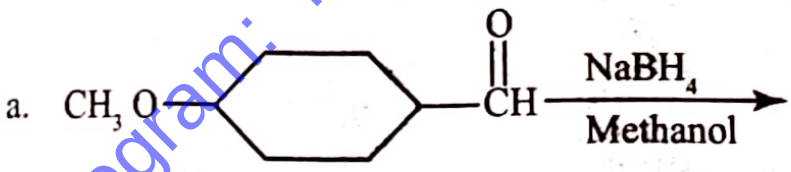
c. 

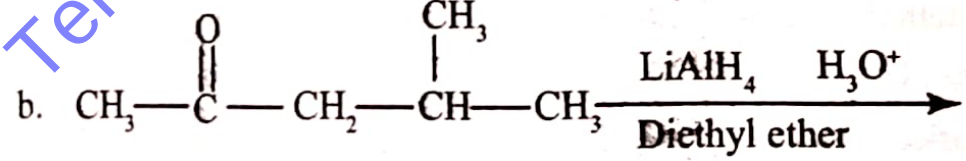
d. 

e. 

f. 

135. Complete the following equations

a. 

b. 

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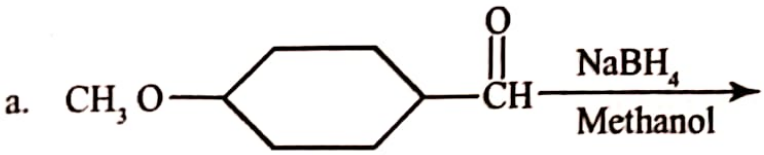
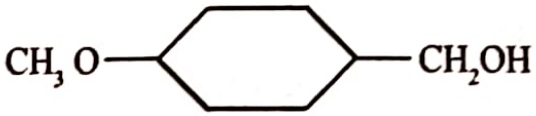
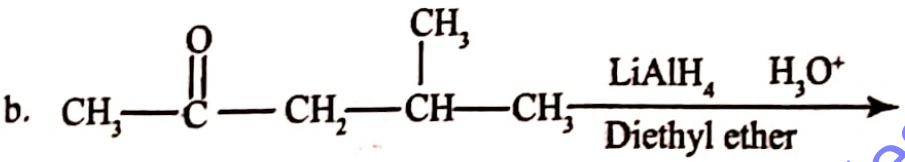
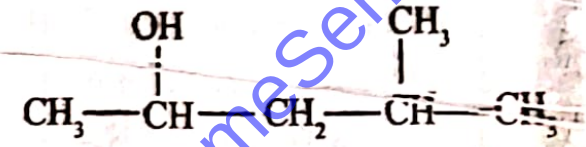
Solution

a. 



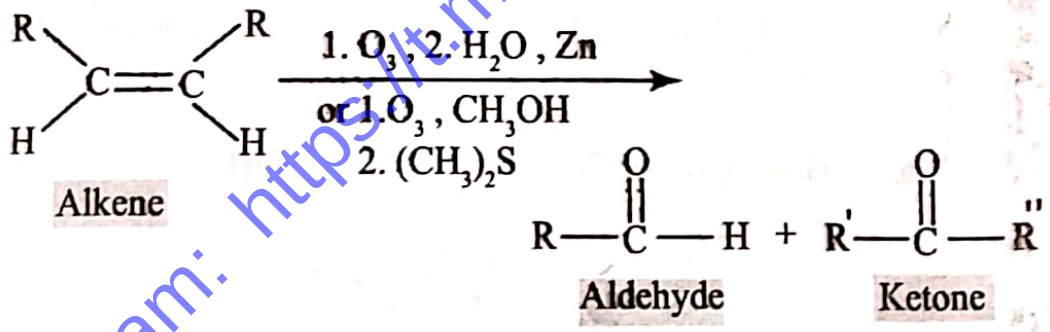
b. 



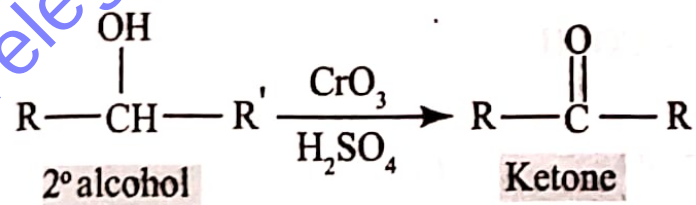
136. Explain the methods used to prepare aldehydes and ketones

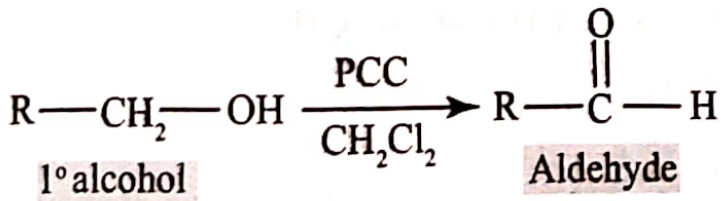
Solution

a. Ozonolysis of Alkenes



b. Oxidation of Alcohols





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37. Write the results of the following of the combinations addition of Grignard reagent to

a. Formaldehyde

b. Aldehyde except formaldehyde

c. Ketone

Solution

a. Primary alcohol

b. Secondary alcohol

c. Tertairy alcohol

38. Which member has higher boiling point?

a. CH_3OH or CH_3COOH

b. CH_3CH_2OH or CH_3COOH

c. CH_3CH_2OH or CH_3COOH

d. $HCOOCH_3$ or $CH_3CH_2CH_2OH$

Solution

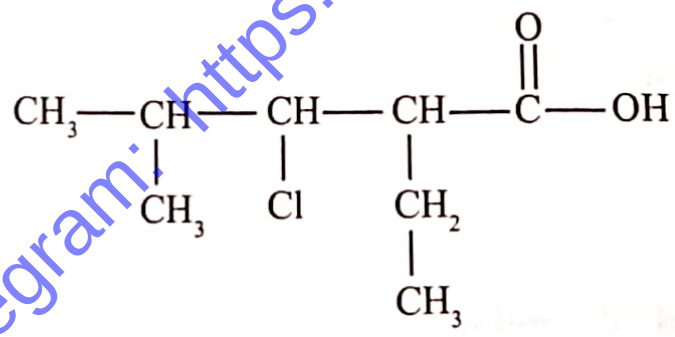
a. CH_3COOH

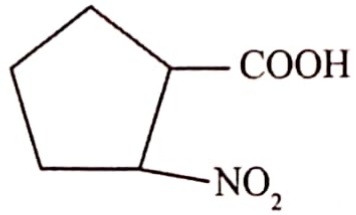
b. CH_3COOH

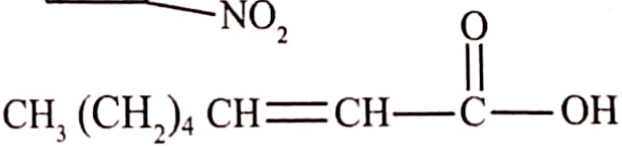
c. CH_3COOH

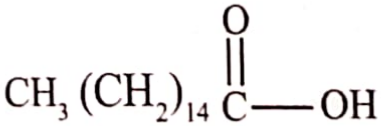
d. $CH_3CH_2CH_2OH$

39. Give the IUPAC name of the following carboxylic acids.

a. 

b. 

c. 

d. 

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Solution

a. 3 - chloro - 2 - ethyl - 4 - methyl pentanoic acid

b. 2 - nitro cyclopertane carboxylic acid

c. 2 - octenoic acid

d. Hexadecanoic acid

140. Draw the structure of the following carboxylic acids

a. 2,3, - dibromo propanoic acid

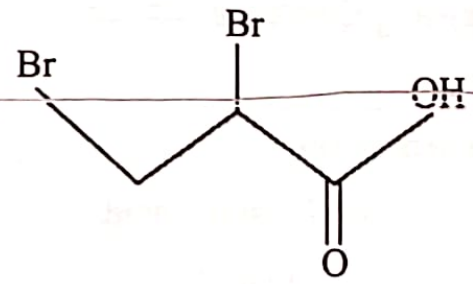
b. 2 - heptynoic acid

c. Pentanedioic acid

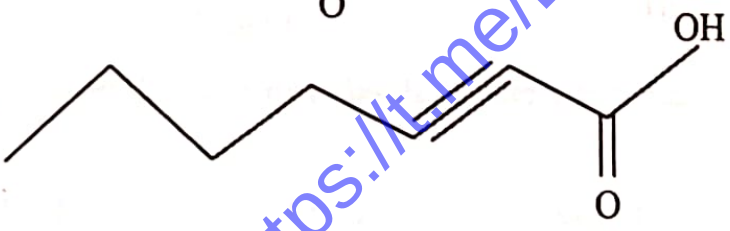
d. 0 - amino benzoic acid

Solution

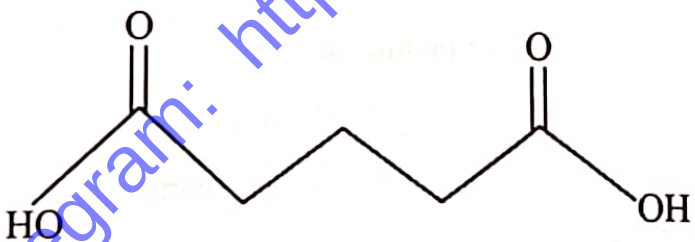
a.



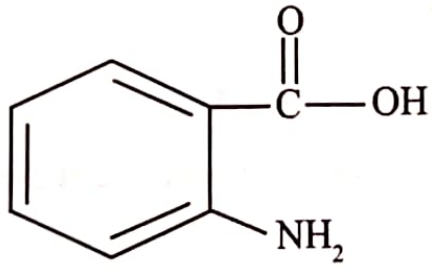
b.



c.



d.



141. Write the common name of the following aromatic carboxylic acids

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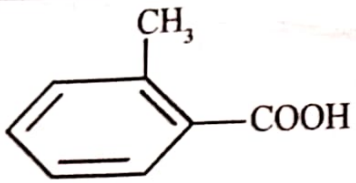
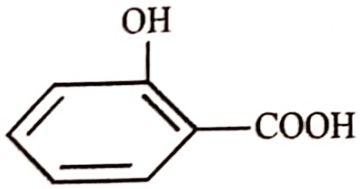
a. 

b. 

Solution

a. O - toluic acid

b. Salicylic acid

Supplementary Exercise

I. Multiple Choice items.

1. Acid catalyzed hydration of ethylene gives.

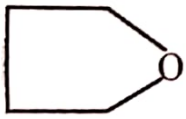
   A. Ethyne
   C. Ethanoic acid
   B. Ehanal
   D. Ethanol

2. When a piece of calcium metal is added to an alcohol, the gas that given off is

   A. CH_4 B. CO_2 C. O_2 D. H_2

3. The alcohol formed during soap production is

   A. Ethyleneglcol
   C. Ethanol
   B. Glycerol
   D. 1 - propanol

4. The structure  is

   A. Ether B. Alcohol C. Aldehyde D. Ketone

5. Which of the following compounds reacts with active metal to give salt?

   A. CH_3CH_3
   C. CH_3COOH
   B. CH_3CHO
   D. CH_3COOCH_3

6. A compound has the molecular formula C_3H_6O. All of the following

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can be the structural formula for the compound. **EXCEPT.**

A. $CH_3\,CO\,CH_3$ C. $CH_3\,CH_2\,CHO$

B. $CH_3\,CH_2\,CH_2\,OH$ D. $CH_2{=}CH\,O\,CH_3$

7. Which one represents secondary alcohol?

A. 1 - Propanol C. 2 - methyl - 2 - propanol

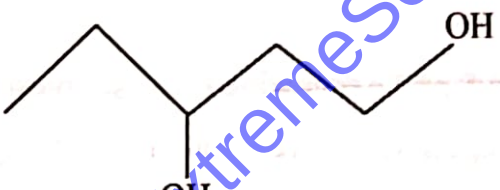
B. 2 - Propanol D. 2 - methyl - 1 - propanol

8. The general formula of saturated carboxylic acid is:

A. RCOOOH C. HOOCRCOOH

B. RCOOR' D. R2COOR

9. The alcohol with the formula



A . Primary alcohol

B) Secondary alcohol

C) Trihydric alcohol

D) Dihydric alcohol

10. Neopentyl alcohol is

A. Primary alcohol C. Tertairy alcohol

B. Secondary alcohol D. Dihydric alcohol

11. All of the following are distilled alcoholic beverages. EXCEPT

A. Tella B. Katikalla C. Whisky D. Gin

12. Which of the following alcohols is the most difficult to dehydrate

A. $CH_3CH_2CH_2CH_2OH$

C. $CH_3-\underset{\displaystyle CH_3}{\overset{\displaystyle CH_3}{\overset{|}{\underset{|}{C}}}}-OH$

B. $CH_3\,\underset{\displaystyle OH}{\underset{|}{CH}}\,CH_2\,CH_3$

D. $CH_3CH_2\,\underset{\displaystyle OH}{\underset{|}{CH}}\,CH_2CH_3$

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13. Identify the unsymmetrical ether among the following.

A. CH_3OCH_3 C. $CH_3OCH_2CH_3$

B. $C_6H_5OC_6H_5$ D. $C_2H_5OC_2H_5$

14. Which of the following statements is true?

A. The carbonyl carbon in all aldehydes is bonded to a hydrogen

B. Ketones are more soluble in water than alcohols of comparable size

C. Any reaction oxidizes aldehydes can oxidize ketones to carboxylic acid.

D. Aldehydes are ethers of similar carbon number are isomers to each other.

15. Which of the following is the simplest monocarboxylic acid?

A. HCOOH C. CH_3COOH

B. $HCOOCH_3$ D. CH_3COOCH

16. Which of the following aliphatic carboxylic acid has the highest boiling point?

A. Hexanoic acid C. Acetic acid

B. Valeric acid D. Heptanoic acid

17. Which of the following compounds would react with product of its own oxidation to form sweet smelling liquid?

A. Propanal C. Propanone

B. 1- propanol D. 2 - propanol

18. Acid - catalyzed hydrolysis of phenyl acetate yields

A. C_6H_5OH and CH_3COOH

B. C_6H_5COOH and CH_3CH_2OH

C. $C_6H_5CH_2OH$ and CH_3COOH

D. $C_6H_5CH_2COOH$ and CH_3CH_2OH

19. Slight oxidation of primary alcohol gives

A. Ketone B. Aldehyde C. Organic acid D. Ether

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20. An ester can be prepared by the reaction of

A. Two alcohols

B. An alcohol and aldehyde

C. An alcohol and an organic acid

D. An organic acid and an aldehyde

21. An isomer of ethanol is

A. Methanol C. Ethanol

B. Ethoxy ethane D. Methoxy methane

22. The alcohol that forms fats with fatty acids is

A. Ethanol C. Glycerol

B. Methanol D. Isopropanol

23. Ketones can be obtaned by:

A. Dehydrogenation of secondary alcohols

B. Oxidation of secondary alcohols

C. Dehydrogenation of primary alcohols

D. Oxidation of primary alcohols

24. Grignard reagent on reaction with acetone forms

A. Trtiairy alcohol C. Secondary alcohol

B. Acetic acid D. Acetaldehyde

25. Which of the following reagents will convert butanoic acid to 1 - butanol?

A. $KMnO_4$ B. Cr_2O_3 C. MnO_2 D. $Li Al H_4$

26. Which of the following compounds is oxidized to prepare ethyl ketone?

A. 2 - propanol B. 2 - butanol C. 2 - pentanol D. 2 - hexanol

27. How many isomeric compounds are possible for $C_4 H_{10}O$?

A. 4 B. 6 C. 5 D. 7

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28. Which organic compounds has no oxygen?

A. Methane
C. Methanoic acid
B. Methanol
D. Methanal

29. The structure: $C_{17}H_{35}\overset{\overset{O}{\|}}{C}—O^{-}K^{+}$ kepresents

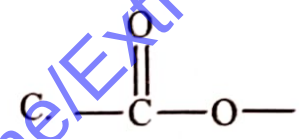
A. Detergent B. Polymer C. Soap D. Fats and oils

30. The conversion of ester in to alcohol and carboxylic acid is:

A. Esterification
C. Dehydration
B. Saponification
D. Hydrolysis

31. The carbonyl group is represented by the structure

A. $—\overset{\overset{O}{\|}}{C}—$



B. $—\overset{\overset{O}{\|}}{C}—OH$

D. $—\overset{|}{\underset{|}{C}}—OH$

32. Which of the following is not true about fats and oils?

A. Hydrolytic rancidity occurs when water spilits fatty acid chains away from the glycerol back bone in glycerides.

B. Oxidative ranidify occurs when the double bonds of unsaturated fatty acid react chemically with oxygen.

C. Microbial rancidity happens when micro - organisms such as bacteria use their enzyme to breakdown chemical structure in the fat.

D. Rancidity is the chemical decomposition of fats and oils to result desirable odours and flavors.

33. The IUPAC name of $(CH_3)_3C\ CH_2\ COOH$

A. 3,3 - dimethyl butanoic acid

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B. 3 - methyl pentanoic acid

C. Hexanoic acid

D. 2,3 - dimethyl butanoic acid

34. Which of the following is not natural ester?

A. Bees wax C. Vegetable oil

B. Animal fat D. Glycerol

35. Which of the following elements is necessarily found in all organic compounds?

A. Hydrogen B. Carbon C. Oxygen D. Nitrogen

II. Short answer items.

36. Give common name for

a. $CH_3\,CH(OH)\,CH_2\,CH_3$

b. $CH_3CH\,O$

c. $CH_3CO\,CH_3$

d. $CH_3\,O\,CH_3$

e. $HCOOCH_3$

f. CH_3CH_2COOH

37. Draw the structure of:

a. ά - methyl valeric acid

b. m - nitro benzoic acid

38. Write the general formula for

a. Alcohol c. Aldehyde e. Carboxylic acid

b. Ether d. Ketone f. Ester

39. Write the balanced chemical equation for the reaction between methanoic acid and

a. Mg b. NaOH c. K_2CO_3 d. NH_3

40. The oxidizing agent that converts methyl benzene (tolune) to benzoic acid is _____

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Answer to supplementary exercise

Part I

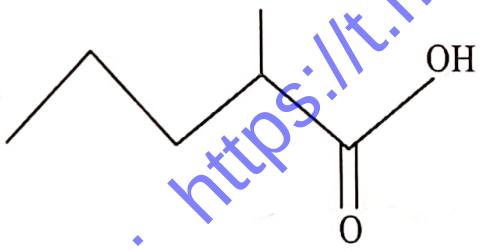
1. D	8. A	15. A	22. C	29. C
2. D	9. D	16. D	23. B	30. D
3. B	10. C	17. B	24. A	31. A
4. A	11. A	18. A	25. D	32. D
5. C	12. A	19. B	26. B	33. A
6. B	13. C	20. C	27. D	34. D
7. B	14. A	21. D	28. A	35. B

Part II

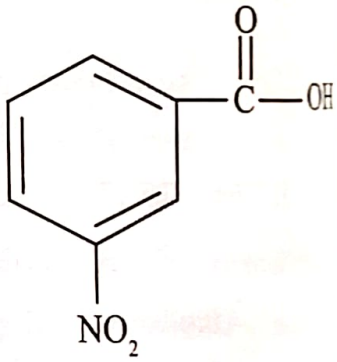
36.

a. Sec - butyl alcohol
b. Acetaldehyde
c. Acetone
d. Dimethyl ether
e. Methyl formate
f. Propionic acid

37. a.



b.



38.

a. ROH
b. ROR
c. RCHO
d. RCOR
e. RCOOH
f. RCOOR

39.

a. $2HCOOH + Mg \longrightarrow (HCOO)_2 Mg + H_2$
b. $HCOOH + NaOH \longrightarrow HCOONa + H_2O$
c. $2HCOOH + K_2CO_3 \longrightarrow 2HCOOK + CO_2 + H_2O$
d. $HCOOH + NH_3 \longrightarrow HCOONH_4$

40. $KMnO_4$

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UNIT 7

Acid - Base Equilibria

7.1 Introduction

Acids and bases are important chemicals used to make new substances such as salts, fertilizers, soaps, detergents, explosives, drugs.

They are used as raw materials and catalysts in many industrial processes.

They are found in living and non - living things

Example

- Citric acid - Lemonds
- Acetic acid - vinegar
- Lactic acid - sour milk
- Ascorbic acid - Vatimin C
- Hydrochloric acid - Human stomach
- Sulphuric acid - carbattery
- Formic acid - ants, bees
- Phospheric acid - carbonated beverages
- Sodium hydroxide - drain cleaner / soap/

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- Ammonia - glass cleaner
- Milk of magnesia - anti acid
- Sodium bicarbonate - baking powder
- Sodium carbonate - washing powder
- Strong acids and strong bases are strong electrolytes while weak acids and weak bases are weak electrolytes.
- Acids and bases change colors of indicators.

Table 7.1 Colors of Some indicators in Acidic and Basic Solutions.

Indicator	Color in acidic solution	Color in basic solution
Litmus paper	red	blue
Phenol phthalien	colorless	pink
Methyl orange	red	yellow
Universal indicator	yellow, orange, red	blue, purple

Illustrative example

1. All of the following are common house hold acids. **EXCEPT**

A. Acetic acid
C. Citric acid
B. Sulphuric acid
D. Phospheric acid

Solution

Acetic acid - vinegar
Citric acid - citrus fruits
Phospheric acid - carbonated beverages
Sulphuric acid - poisonous but important acid.

Answer : B

2. Which of the following does not change the color of indicators ?

A. Acetic acid solution
C. Sodium chloride solution
B. Ammonia solution
D. Lemon juice

Solution

Sodium chloride solution is a neutral solution that doest not change

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color of indicators.

Answer : C

3. Which of the following acids is the component of digestive juice ?

A. HCl B. H_2SO_4 C. HNO_3 D. HI

Answer : A

4. Write the formula and name of two house hold bases.

Solution

NaOH - Sodium hydroxide

NH_3 - Ammonia

7.2 Acid - Base Concepts

There are six different definitions of acids and bases. These are:

1. Arrhenius definition
2. Bronsted - Lowery definition
3. Lewis - definition
4. Solvent system definition
5. Aniontropic definition
6. Usanovich definition

In this unit, we will discuss the first three, the rest you will study in higher education levels.

I. Arrhenius Definition

Table 7.2 Arrhenius Definition of Acid and Base

Acid	Base
➥ Substance that yields hydrogen ion in an aqueous solution	➥ Substance that gives hydroxide ions in aqueous solution
➥ Increases concentration of hydrogen ion in water.	➥ Increases concentration of hydroxide ion in water.
$HCl(aq) + H_2O(l) \longrightarrow H_3O^+(aq) + Cl^-(aq)$	$NaOH(aq) \xrightarrow{H_2O(l)} Na^+(aq) + OH^-(aq)$
Some Examples	Some Examples
HBr, HCl, HI, HNO_3, H_2SO_4	NaOH, KOH, $Ca(OH)_2$, $Ba(OH)_2$, $Sr(OH)_2$

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Note : The general acid - base reaction according to Arrhenius is :

$$\text{Acid} + \text{base} \longrightarrow \text{salt} + \text{water}$$

Limitation of Arrhenius Definition

1. The definition can be applied only to the reaction that occurs in water. It does not explain reactions in non - aqueous and gaseous phases.

Example

$$CH_3COOH\ (l) + NH_3\ (g) \longrightarrow CH_3COONH_4\ (s)$$

$$HCl\ (g) + NH_3\ (g) \longrightarrow NH_4Cl\ (s)$$

2. The theory does not explain why some compounds containing hydrogen such as H Cl are acidic but others such as CH_4 are not acid.

Illustrative Examples

5. Which of the following can not serve as Arrhenius acid ?

A. HCl B. HNO_3 C. $AlCl_3$ D. H_2SO_4

Answer : C

6. All of the following are Arrhenius bases **EXCEPT**

A. NH_3 B. Na OH C. $Ba(OH)_2$ D. H_2O

Answer : D

7. Identify the Arrhenius acid and base in the following solutions

a. $HNO_3\ (l) + H_2O\ (l) \longrightarrow H_3O^+\ (aq) + NO_3^-\ (aq)$

b. $Ba(OH)_2\ (s) + H_2O\ (l) \longrightarrow Ba^{2+}\ (aq) + 2OH^-\ (aq)$

c. $K\ Cl\ (s) + H_2O\ (l) \longrightarrow K^+\ (aq) + Cl^-\ (aq)$

Solution

a. $HNO_3\ (l)$ is Arrhenius acid

b. $Ba(OH)_2\ (s)$ is Arrhenius base

c. KCl (s) neither Arrhenius acid nor Arrhenius base.

8. Write the complete reaction that shows the formation of hydronium ion (H_3O^+) using an acid, HX.

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Solution

$$HX + H_2O \rightleftharpoons H_3O^+ + X^-$$

2. Bronsted - Lowry Definition

Acid is a substance that donates a proton (H^+) to other substances.

- This definition agrees with the aqueous solution definition of an acid giving up hydrogen ions in solutions but goes beyoned to other cases as well.
- Based on Bronsted - Lowry definition, to be an acid it must contain ionizable hydrogen.

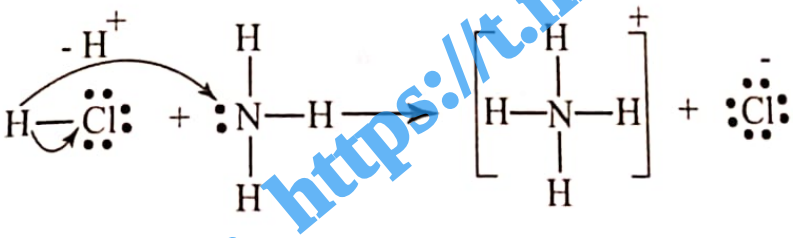
Base is a substance that accepts a proton from other substance.

- It should contain a pair of electrons to accept a proton.
- Metal hydroxides are not Bronsted - Lowry bases but they contain hydroxide ion which is a base

Example

$$HCl\,(g) + NH_3\,(g) \longrightarrow NH_4^+\,(aq) + Cl^-\,(aq)$$

Acid Base



- The general acid - base reaction based on Bronsted - Lowry definition is:

$$Acid_1 + Base_1 \longrightarrow Acid_2 + Base_2$$

- In an acid - base reaction, the original acid (acid1) gives up its proton to become **a conjugate base (CB)**

Conjugate base = Acid - H^+

- The original base accepts a proton, so it now is classified as **conjugate acid (CA)**

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Conjugate acid = Base + H^+

Example

a. Identify the CA and CB of HS^-.

Solution

CA = Base + H^+ = HS^- + H^+ = H_2S, add H and write as subscript to get charge, add the charges in the superscript.

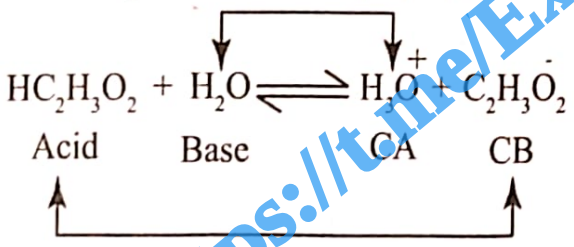
CB = Acid - H^+ = HS^- - H^+ = S^{2-}, net charge = (-1 - (+1) = 2-).

b. Identify the acid, base, CA and CB in the following reaction

$HC_2H_3O_2(aq) + H_2O(l) \rightleftharpoons H_3O^+(aq) + C_2H_3O_2^-(aq)$

Solution

- $HC_2H_3O_2$ is an acid because it is a proton donor to H_2O
- H_2O is a base because it is a proton acceptor
- H_3O^+ is CA and $C_2H_3O_2^-$ is CB



Note :

i. CA has one more hydrogen and positive charge than its base.

ii. CB has one fewer hydrogen and one more negative charge than its base

iii. In conjugate acid - base pairs only hydrogen and charge change but the rest of the atoms remain same

Illustrative Examples

9. For the reaction shown below, which of the following is a conjugate acid - base pair ? (ESSLCE)

$HC_2O_4^-$ and $C_2O_4^{2-}(l) \rightleftharpoons H_3O^+(aq) + C_2O_4^{2-}(aq)$

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A. H_2O and $C_2O^{2-}_4$

B. $HC_2O^-_4$ and H_2O

C. $HC_2O^-_4$ and $C_2O^{2-}_4$

D. H_3O^+ and $HC_2O^-_4$

Solution

HC_2O_4 - $C_2O^{2-}_4$ and H_2O - H_3O^+ are conjugate acid - base pairs.

Answer : C

10. For the acid - base equilibrium :

$HCO^-_3 + H_2O \rightleftharpoons H_2CO_3 + OH^-$, the Bronsted - Lowry acids are

A. H_2O and OH^-

B. HCO^-_3 and OH^-

C. H_2O and H_2CO_3

D. HCO^-_3 and H_2CO_3

Solution

In a reversible reaction, there are two Bronsted - Lowry acids and bases for the forward and backward reactions.

H_2O and H_2CO_3 are acids and HCO^-_3 and OH^- are bases

Answer : C

11. Which of the following is not a conjugate - acid base pair ?

A. HNO_3 / NO^-_3

B. H_2SO_4 / HSO^-_4

C. NH_3 / NH^-_2

D. H_3O^+ / OH^-

Solution

The difference between conjugate acid - base pair is a single hydrogen and charge. The conjugate base of H_3O^+ is H_2O.

Answer : D

12. A Bronsted - Lowry base refers to any substance that :

A. increases $[H_3O^+]$ ions in aqueous solution

B. increase $[OH^-]$ ions in aqueous solution

C. decreases $[OH^-]$ ions in aqueous solution

D. accepts a hydrogen ion.

Answer : D

13. Identify the Bronsted - Lowry acid, base, CA and CB in the following reactions

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a. $H_2O + (CH_3)_3N \rightleftharpoons (CH_3)_3NH^+ + OH^-$

b. $NH_3 + H_2O \rightleftharpoons NH^+_4 + OH^-$

c. $SO^{2-}_4 + HSO^-_3 \rightleftharpoons HSO^-_4 + SO^{2-}_3$

d. $CH_3NH_2 + HClO \rightleftharpoons CH_3NH^+_3 + ClO^-$

Solution

	Acid	CB	Base	CA
a.	H_2O	OH^-	$(CH_3)_3N$	$(CH_3)_3NH^+$
b.	H_2O	OH^-	NH_3	NH^+_4
c.	HSO^-_3	SO^{2-}_3	SO^{2-}_3	HSO^-_4
d.	$HClO$	ClO^-	CH_3NH_2	$CH_3NH^+_3$

14. Determine the CA and CB of $H_2PO^-_4$.

Solution

CA = base + $H^+ = H_2PO^-_4 + H^+ = H_3PO_4$

CB = acid - $H^+ = H_2PO^-_4 - H^+ = HPO^{2-}_4$

15. Find the CA of the following bases

a. NH^-_2 c. $C_2O^{2-}_4$ e. N_2H_4

b. HCO^-_3 d. CN^- f. NO^+_3 CH_2COO

Solution

a. NH_3 c. $HC_2O^-_4$ e. $N_2H^+_5$

b. H_2CO_3 d. HCN f. NH_2 CH_2COO

16. Write the formula of the CB of the following acids.

a. HNO_2 c. HTe^- e. H_3O^+

b. H_3PO_2 d. $[Al(H_2O)_6]^{3+}$ f. $C_6H_5NH^+_3$

Solution

a. NO^-_2 c. Te^{2-} e. H_2O

b. $H_2PO^-_2$ d. $[Al(H_2O)_5OH]^{2+}$ f. $C_6H_5NH_2$

17. Why H_2CO_3 and NH^+_4 cannot act as Bronsted - Lowry bases ?

Solution

Bronsted - Lowry base is proton acceptor, but H_2CO_3 and NH^+_4

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not accept proton because their central atoms have no lone pair of electrons and cannot expand their octet.

18. Aluminum chloride ($AlCl_3$) does not act as Bronsted - Lowry acid. Why ?

Solution

An acid is a proton donor according to Bronsted -Lowry concept, however $AlCl_3$ has no proton.

Strength of Conjugate acid - base Pairs

If the acid is strong, its conjugate base becomes weak and vice - versa.

If the base is strong, its conjugate acid is weak and vice - versa.

Example

1. Given the strength of acids : $HNO_3 > H_3O^+ > HSO_4^-$. Arrange the CB of these acids in increasing order of strength.

Solution

First, find the CB of each acid and then reverse your answer. That is, the stronger the acid, the weaker is its CB and vice - versa

$\Rightarrow NO_3^- < H_2O < SO_4^{2-}$

2. Given the strength of bases : $OH^- < S^{2-} < O^{2-}$

   a. The strongest CA is ____

   b. The weakest CA is ____

Solution

a. HOH or H_2O b. OH^-

Auto- ionization (self - ionization) of substances

It is interaction of two molecules of a substance to produce its CA and CB by transfer of proton.

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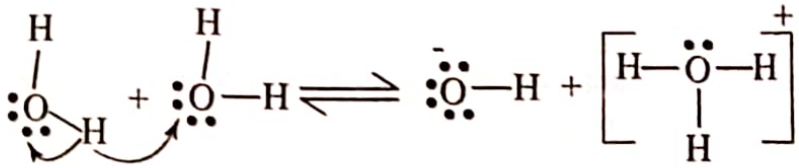
Example

Water and ammonia

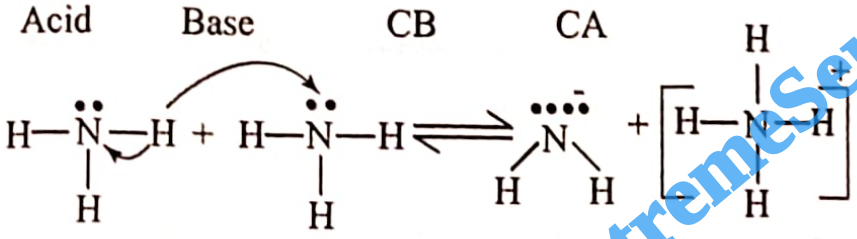
a. $H_2O + H_2O \rightleftharpoons H_3O^+ + OH^-$

Acid Base CA CB



b. $NH_3 + NH_3 \rightleftharpoons NH_2^- + NH_4^+$

Acid Base CB CA



Amphiprotic Species

- Substances that behave as either a Bronsted - Lowry acid or a Bronsted - Lowry base are known as **amphiprotic species.**
- They can donate or accept a proton.

Amphiprotic species include

i. Self - ionized substances - H_2O, NH_3

ii. Anions of polyprotic acids having ionizable hydrogen - HS^-, HCO_3^-, HSO_4^-, HSO_3^-, $HC_2O_4^-$, $H_2PO_4^-$, HPO_4^{2-} etc

iii. Amino acids - $NH_2CH\,COOH$

Limitation of Bronsted - Lowry concept

This definition restricted the acid - base concept only to proton transfer.

Illustrative Examples

19. Each of the following can act as both a Bronsted acid and a Bronsted base

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A. HCO^-_3 B. NH^+_4 C. $H_2PO^-_4$ D. H_2O

Solution

NH^+_4 can donate a proton but it does not accept.

Answer : B

20. Which species is amphoteric ?

A. H_2S B. HS^- C. Cl^- D. H Cl

Answer : B

21. Given the strength of acids : HCN < CH_3COOH < HF < HNO_2 the weakest CB is :

A. CN^- B. CH_3COO^- C. F^- D. NO^-_2

Solution

The strongest acid is HNO_2, Thus, the weakest CB is NO^-_2.

Answer : D

22. _____ is a reaction that involves the transfer of a proton from one molecule to another molecules of the same substance

Solution

Auto - ionization or self - ionization of substance.

23. Given the strength of bases : $ClO^- > ClO^-_2 > ClO^-_3 > ClO^-_4$ Arrange the conjugate acids of these bases in increasing order of strength.

Solution

$HClO < HClO_2 < HClO_3 < HClO_4$

24. Explain how the following substances act as a Bronsted - Lowry acid and base by using chemical reactions.

a. H_2O c. $H_2PO^-_4$

b. $HC_2O^-_4$ d. NH_2CH_2COOH

Solution

a. $H_2O + HCN \rightleftharpoons H_3O^+ + CN^-$, H_2O is base

$H_2O + NH_3 \rightleftharpoons NH^+_4 + OH^-$, H_2O is acid

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b. $HC_2PO^-_4 + H^+ \rightleftharpoons H_2C_2O_4$, $HC_2O^-_4$ acts as a base

$HC_2O^-_4 \rightleftharpoons C_2O^{2-}_4 + H^+$, It acts as an acid

c. $H_2PO^-_4 + H^+ \rightleftharpoons H_3PO_4$ base

$H_2PO^-_4 \rightleftharpoons HPO^{2-}_4 + H^+$, acid

d. $NH_2\ CH_2COOH + H^+ \rightleftharpoons N^+H_3\ CH_2COOH$, base

$NH_2\ CH_2COOH \rightleftharpoons NH_2\ CH_2COO^- + H^+$, acid

25. Write the limitations of the Bronsted - Lowry concept of acid and base

Solution

- The definition restricted the acid - base concept only to proton transfer
- It does not explain why some species such as CO_2, SO_2, Cu^{2+}, $AlCl_3$ are acidic without proton.

3. Lewis Concept of Acids and Bases

Acid is a substance that accepts a pair of electrons.

- It is electrophile (loves electrons)
- It should contain vacant or unfulfilled orbital to accept a pair of electrons.

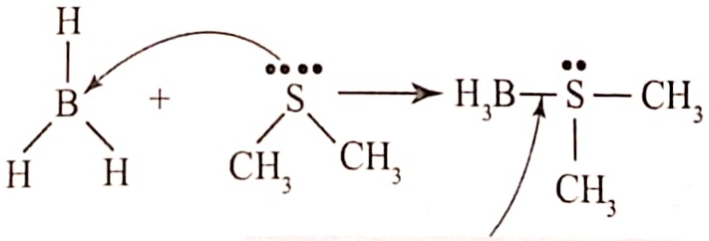
Base is a substance that donates a pair of electrons.

- It is nucleophile ((loves nucleus)
- It should have lone pair of electrons
- It is electron rich species.

Example

$$BH_3 + (CH_3)_2S \longrightarrow H_3BS(CH_3)_2$$

Lewis acid — **Lewis base** — **Lewis acid - base complex**



Coordinate covalent bond

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Note :

i. When a Lewis acid reacts with a Lewis base, coordinate covalent bond is formed

ii. Lewis acids include the following

a. Molecules Below Octet

Example

BCl_3, $AlCl_3$, $BeCl_2$, etc

$BeCl_2 + 2Cl^- \longrightarrow BeCl_4^{2-}$

Lewis acid Lewis base

b. Molecules that Expand their Octet

Example

$SiCl_4$, PCl_5, SF_6, $FeCl_3$, etc

$FeCl_3 + Cl^- \longrightarrow FeCl_4^-$

Lewis acid Lewis base

c. Many Simple Positive Ions (cations)

Example

H^+, Fe^{3+}, Cu^{2+}, Ag^+, etc

$Fe^{3+} + 6CN^- \longrightarrow [Fe(CN)_6]^{3-}$

L. acid L. base.

d. Some Metals

Example

Ni

$Ni + 4CO \longrightarrow Ni(CO)_4$

L. acid L. base

e. Some Molecules with Multiple Bonds

Example

CO_2, SO_2, SO_3, etc

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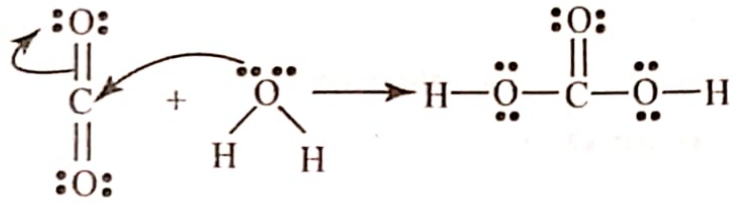
$CO_2 + H_2O \longrightarrow H_2CO_3$



Lewis acid Lewis base

Anions (negative ions) and other electron rich species are Lewis bases.

Illustrative Examples

26. Which of the following is a Lewis acid ?

A. NH_3 B. CCl_4 C. $AlCl_3$ D. Li Cl

Solution

Lewis acid should possess vacant or unfulfilled orbitals.

Answer : C

27. Which species cannot act a Lewis acid ?

A. NH_3 B. BF_3 C. Fe^{2+} D. $AlCl_3$

Answer : A

28. What type of bond is formed when a Lewis acid reacts with a Lewis base ?

A. Ionic bond C. Covalent bond

B. Hydrogen bond D. Coordinate covalent bond

Solution

When a Lewis acid reacts with a Lewis base coordinate covalent bond is formed.

Answer: D

29. In the reaction : $FeCl_3 + Cl^- \longrightarrow FeCl^-_4$, $FeCl_3$

A. a Lewis acid C. a Bronsted - Lowry acid

B. a Lewis base D. a Bronsted - Lowry base

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Answer : A

30. Identify the Lewis acid and base in the reactions below :

a. $HS^- + CH_3Cl \longrightarrow CH_3SH + Cl^-$

b. $(CH_3)_3N + BF_3 \longrightarrow (CH_3)_3N\ BF_3$

c. $Cu^{2+} + 4NH_3 \longrightarrow [Cu(NH_3)_4]^{2+}$

d. $SO_2 + H_2O \longrightarrow H_2SO_3$

e. $CO_2 + O^{2-} \longrightarrow CO_3^{2-}$

f. $PF_5 + F^- \longrightarrow PF_6^-$

Solution

Letter	Lewis base	Lewis acid
a.	HS^-	CH_3Cl
b.	$(CH_3)_3N$	BF_3
c.	NH_3	Cu^{2+}
d.	H_2O	SO_2
e.	O^{2-}	CO_2
f.	F^-	PF_5

7.3 Ionic Equilibria of Weak Acids and Weak Bases

1. Ionization of Water

Water self - ionizes, transferring a proton from one water molecule to another and producing a hydronium (H_3O^+) and hydroxide ion (OH^-)

$$2H_2O(l) \rightleftharpoons H_3O^+(aq) + OH^-(aq)$$

$K_c = \dfrac{[H_3O^+][OH^-]}{[H_2O]^2}$, but H_2O (1) is not included in K_c expression

$$\Rightarrow K_c\,[H_2O]^2 = [H_3O^+][OH^-]$$

The product of K_c and $[H_2O]^2$ is known as **ion - product of water (K_w)**

$$K_w = [H_3O^+][OH^-]$$

K_w is affected by temperature . As temperature increases, the value of

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also increases and vice - versa

$K_w = 1 \times 10^{-14} M^2$ at 25^0C (for most calculations)

$= 2.5 \times 10^{-14} M^2$ at 37^0C

$= 2.92 \times 10^{-14} M^2$ at 40^0C

The concentration of water is constant (55.56M), This can be determined from density of water.

$\rho = 1kg/L = 1000g/L \quad \Rightarrow m = 1000g, V = 1L$

$M_{wt} = 18g/mol$

$$M = \frac{m}{M_{wt} \times V} = \frac{1000g}{18g/mol \times /L} = 55.56M$$

For pure water, $[H_3O^+] = [OH^-]$

$\Rightarrow [H_3O^+][OH^-] = 1 \times 10^{-14} M^2$

$\Rightarrow [H_3O^+] = [OH^-] = \sqrt{1 \times 10^{-14} M^2} = 1 \times 10^{-7} M$

Note :

$[H_3O^+] = [OH^-] = 1 \times 10^{-7} M$ neutral solution

$$\left.\begin{array}{l} [H_3O^+] > [OH^-] \\ [H_3O^+] > 1 \times 10^{-7} M \\ [OH^-] < 1 \times 10^{-7} M \end{array}\right\} \text{acidic solution}$$

$$\left.\begin{array}{l} [H_3O^+] < [OH^-] \\ [H_3O^+] < 1 \times 10^{-7} M \\ [OH^-] > 1 \times 10^{-7} M \end{array}\right\} \text{basic solution}$$

Illustrative Examples

31. All of the following represent acidic solution. **EXCEPT**

A. $[H_3O^+] = 100\,[OH^-]$ B. $[OH^-] = 1 \times 10^{-8} M$

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C. $[OH^-] = 0.01M\ [H_3O^+]$ D. $[H_3O^+] = 1 \times 10^{-7}M$

Solution

For acidic solution $[H_3O^+] > [OH^-]$, $[H_3O^+] > 1 \times 10^{-7}M$ and $[OH^-] < 1 \times 10^{-7}M$

Answer : D

32. A solution that contains $[OH^-] = 6.5 \times 10^{-3}M$ is

A. acidic B. basic C. neutral D. none

Solution

$[OH^-] > 1 \times 10^{-7}M$ is basic solution.

Answer : B

33. Calculate the concentration of hydronium, H_3O^+ in a solution with :

a. $[OH^-] = 5 \times 10^{-5}M$
b. $[OH^-] = 100\ [H_3O^+]$
c. $0.05M\ Sr(OH)_2$
d. $0.01M\ H_2SO_4$

Solution

a. $[H_3O^+] = \frac{K_w}{[OH^-]} = \frac{1 \times 10^{-14}M^2}{5 \times 10^{-5}M} = 2 \times 10^{-10}M$

b. $[H_3O^+] = \frac{K_w}{[OH^-]} = \frac{1 \times 10^{-14}M^2}{100[H_3O^+]} = 1 \times 10^{-8}M$

c. $Sr(OH)_2 \longrightarrow Sr^{2+} + 2OH^-$, $Sr(OH_2)$, is strong base

1M 2M

0.05M ?

$\Rightarrow [OH^-] = \frac{0.05M \times 2M}{1M} = 0.10M = 1.0 \times 10^{-1}M$

$[H_3O^+] = \frac{K_w}{[OH^-]} = \frac{1 \times 10^{-14}M^2}{1.0 \times 10^{-1}M} = 1 \times 10^{-13}M$

d. $H_2SO_4 + 2H_2O \longrightarrow 2H_3O^+ + SO_4^{2-}$, it is strong acid

1M 2M

0.01M ?

$\Rightarrow [H_3O^+] = \frac{2M \times 0.01M}{1M} = 0.02M = 2 \times 10^{-2}M$

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34. Determine the concentration of hydroxide ion in the following solution.

a. $[H_3O^+] = 2.5 \times 10^{-4}$ b. $[H_3O^+] = 400\ [OH^-]$

Solution

a. $[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1 \times 10^{-14} M^2}{2.5 \times 10^{-4} M} = 4.0 \times 10^{-11} M$

b. $[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1 \times 10^{-14} M^2}{400[OH^-]} = 5 \times 10^{-9} M$

35. Classify the following solutions as acidic, basic or neutral

a. $[H_3O^+] = 5.4 \times 10^{-3} M$
b. $[OH^-] = 4.8 \times 10^{-5} M$
c. $[H_3O^+] = 0.2\ [OH^-]$
d. $[OH^-] = [H_3O^+]$
e. $[H_3O^+] = 6.2 \times 10^{-3} M$
f. $[OH^-] = 8.0 \times 10^{-10} M$

Solution

- a, e and f are acidic solutions
- b and c are basic solutions
- d is neutral solution.

36. The value of K_w at 37°C is $2.5 \times 10^{-4} M^2$. Find the concentrations of H_3O^+ and OH.

Solution

$$[H_3O^+] = [OH^-] = \sqrt{Kw} = \sqrt{2.5 \times 10^{-14} M^2} = 1.58 \times 10^{-7} M$$

pH and pH Scale

pH is defined as the negative of the base - 10 logarithm of the hydrogen ion or hydronium ion concentration

$pH = \log \frac{1}{[H^+]} = -\log [H^+]$	$[H^+] = 10^{-ph}$

Example

Calculate the pH of a solution that contains 0.0001M hydrogen ion concentration

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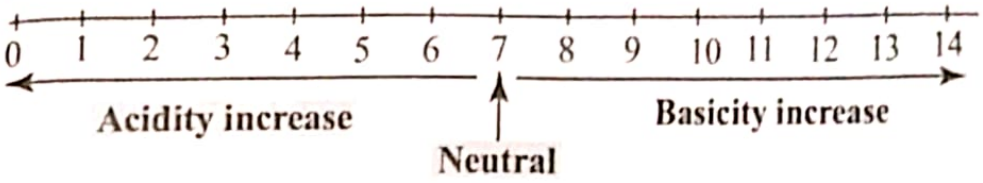
Solution

$pH = -\log [H^+] = -\log 0.0001 = 4$

Note : As hydrogen ion concentration increase, pH decreases and vice - Versa.

pH scale s the scale between 0 to 14.



The instrument used to measure pH of solution is called **pH meter.**

pOH is defined as the negative of the base - 10 logarithm of the hydroxide ion concentration.

$pOH = \log \frac{1}{[OH^-]} = \log [OH]$	$[OH-] = 10^{-pOH}$

Example

The pOH of a solution is 2. Determine the hydroxide ion concentration

Solution

$[OH^-] = 10^{-pOH} = 10^{-2}M = 0.01M$

For pure water, $[H^+] = [OH^-] = 1 \times 10^{-7}M$, Thus, $pH = pOH = 7$

Relationship Between pH and pOH

The relationship between pH and pOH can be derived from ion - product of water at 25^0c

$K_w = [H^+] [OH^-] = 10^{-14}$, Put logarithm on both sides

$\log K_w = \log [H^+] [OH^-] = 10^{-14}$

$\log K_w = \log [H^+] + \log [OH^-] = \log 0^{-14}$, multiply both sides with (-1)

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$- \log Kw = - \log [H^+] - \log [OH^-] = - \log 10^{-14}$

$$\boxed{pKw = pH + pOH = 14}$$

$pH = 14 - pOH = 14 - (-\log [OH^-]) = 14 + \log [OH^-]$

$pOH = 14 - pH = 14 - (-\log [H^+]) = 14 + \log [H^+]$

✓ As concentration of hydroxide ion increases, pH also increases and vice - versa.

✓ As concentration of hydrogen ion increases, pOH also increases and vice - versal.

$pH < 7$, $pOH > 7$ } Acidic solution

$pH > 7$, $pOH < 7$ } Basic solution

$pH = 7$, $pOH = 7$ } Neutral solution

pH and pOH of strong Acids and Strong Bases

Strong acids and strong bases ionize almost 100 % in water.

➥ For strong monoprotic acid : $[H^+] = [\text{acid}]$

Example

$[H^+] = [HCl]$

For - strong diprotic acid : $[H^+] = 2\ [\text{acid}]$

Example

$[H^+] = 2 \times [H_2SO_4]$

Note : The most common strong acids are :

HCl - hydrochloric acid
HBr - hydro bromic acid
HI - hydro iodic acid
HNO_3 - nitric acid
H_2SO_4 - sulphuric acid
$HClO_4$ - per chloric acid

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For strong mono basic base : $[HO^-] = [base]$

Example

$[OH^-] = [NaOH]$

For strong dibasic base : $[OH^-] = 2 \times [base]$

Example

$[OH^-] = 2 \times [Ca(OH)_2]$

Note : The most common strong bases are

LiOH - Lithium hydroxide

NaOH - sodium hydroxide

KOH - potassium hydroxide

$Ca(OH)_2$ - calcium hydroxide

$Sr(OH_2)$- strontium hydroxide

$Ba(OH)_2$ - barium hydroxide

Illustrative Example

37. The negative logarithm of the molar concentration of hydrogen ion to the base ten is known as ___. (ESSLCE)

A. pOH B. Ka C. K_w D. pH

Solution

$pOH = -\log [OH^-]$ $K_w = [H^+][OH^-]$

$K_a = \frac{[H^+][A^-]}{[HA]}$ $pH = -\log [H^+] = -\log [H_3O^+]$

Answer : D

38. A 1×10^{-4} solution has a pH of 10.00. The solute is a

A. Weak acid C. Strong base

B. Weak base D. Strong acid

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Solution

The pH is greater than 7. Thus, it is a base.

$pH = 14 + \log [OH^-] = 14 + \log 10^{-4} = 10.0$, this happens only for strong base.

Answer : C

39. An aqueous solution contains 0.10M NaOH at 25°C, the pH of the solution is

A. 0.100 B. 1.00 C. 7.00 D. 13.0

Solution

$[OH^-] = [Na\,OH] = 0.10M$

$pH = 14 + \log [OH^-] = 14 + \log 0.1 = 13$

40. A 0.1M HCl is dissolved in water, what species of ions are present at equilibrium and what will be their concentrations ?

A. $[H_3O^+] = 0.1M, [OH^-] = 1 \times 10^{-13} M, [Cl^-] = 0.1M$

B. $[H_3O^+] = 0.1M, [OH^-] = 0.1M, [Cl^-] = 0.1M$

C. $H_3O^+] = 0.1M, [OH^-] = 1 \times 10^{-3} M, [Cl^-] = 0.0M$

D. $[H_3O^+] = 1 \times 10^{-13} M, [OH^-] = 0.1M, [Cl^-] = 0.1M$

Solution

H Cl is strong acid

$HCl + H_2O \longrightarrow H_3O^+ + Cl^-$

$\Rightarrow [HCl] = [H_3O^+] = [Cl^-] = 0.1M$ becuse the coefficient ratio is 1:1:1

$$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1 \times 10^{-14} M^2}{0.1M} = 1.0 \times 10^{-13} M$$

Answer : A

41. A solution with a pH of 7.5 would be described as

A. Very basic C. Slightly acidic

B. Slightly basic D. Very acidic

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Solution

pH > 7 is basic and pH and basicity are directly related.

Answer : B

42. Which solutions below has the highest concentration of hydroxide ions?

A. pH = 3.21
C. pH = 7.93
B. pH = 12.59
D. pH = 9.82

Solution

The higher the pH value, the larger is the concentration of hydroxide ion.

Answer : B

43. A solution which as a pH of 5.0

A. is basic
B. has a hydrogen ion concentration of 10^{5}
C. has a hydroxide ion concentration of 10^{-9}
D. is neutral

Solution

pH < 7 is acid.

$\Rightarrow [H^+] = 10^{-pH} = 10^{-5}$ M

$[OH^-] = 10^{pH-14} = 10^{5-14} = 10^{-9}$ M

Answer : C

44. What is the hydrogen ion concentration for a solution with a pOH of 10 at 25°C ?

Solution

$pOH = 14 + \log [H^+]$

$\Rightarrow [H^+] = 10^{pOH-14} = 10^{10-4} = 10^{-4}$M

45. Solution A has a pH of 3 and solution Z has a pH of 6. How many times greater is the hyronium ion concentration in solution A than the hydronium ion concentration in solution Z ?

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Solution

$$\frac{[H_3O^+]_A}{[H_3O^+]_Z} = \frac{10^{-pHA}}{10^{-pHZ}} = \frac{10^{-3}}{10^{-6}} = 1000$$

$$\Rightarrow [H_3O^+]_A = 1000\,[H_3O^+]_Z$$

46. Calculate $[H^+]$, pH, pOH and $[OH^-]$ of 0.005M H_2SO_4 solution Assume complete ionization.

Solution

- $[H^+] = 2 \times [H_2SO_4] = 2 \times 0.005M = 0.010M$
- $pH = -\log [H^+] = -\log 0.01 = 2$
- $pOH = 14 - pH = 14 - 2 = 12$
- $[OH^-] = 10^{-pOH} = 10^{-12}\,M$

47. Find the $[OH^-]$, pOH, pH and $[H^+]$ of 0.001M Sr $(OH)_2$ solution.

Solution

- $[OH^-] = 2 \times [Sr(OH)_2] = 2 \times 0.001M = 0.002M$
- $pOH = -\log [OH^-] = -\log 0.002 = 2.7$
- $pH = 14 - pOH = 14 - 2.7 = 11.3$
- $[H^+] = 10^{-pH} = 10^{-11.3} = 5.0 \times 10^{-12}M$ or
- $[H^+] = \frac{K}{[OH^-]} = \frac{1 \times 10^{-14}M^2}{2 \times 10^{-3}M} = 5.0 \times 10^{-12}M$

48. The pH of the blood is usually close to 7.35.Determine the concentration of H_3O^+ and OH^- in the blood.

Solution

$[H_3O^+] = 10^{-pH} = 10^{-7.35} = 4.47 \times 10^{-8}\,M$

$[OH^-] = 10^{pH-14} = 10^{7.35-14} = 10^{-6.65} = 2.24 \times 10^{-7}\,M$

49. What is pH of lemon juice solution in which $[H_3O^+]$ is 5×10^{-3} M.

Solve by your own : pH = 2.3

50. The pH of a solution of nitric acid (HNO_3) is 2.What is the concentration of the acid ?

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Solution

$[H^+] = [HNO_3] = 10^{-pH} = 10^{-2}\ M = 0.01\ M$

51. What is the pH of a 500ml of aqueous solution of 0.02 mol of KOH ?

Solution

$$[OH^-] = [KOH] = \frac{n}{V} = \frac{0.02\,mol}{0.5\,L} = 0.04M$$

$$pH = 14 + \log [OH^-] = 14 + \log 0.04 = 12.6$$

Measures of the Strength of Acids and Bases in Aqueous Solution

Strength refers to the degree of ionization or dissociation of acids or bases in water.

Strong acids ionize almost completely when dissolved in water.

Example

$HClO_4$, H_2SO_4, HNO_3, HI, HBr, HCl

$$HClO_4 + H_2O \longrightarrow H_3O^+ + ClO_4^-$$

Weak acids ionize slightly when dissolved in water.

Example

HF, H_2CO_3, HNO_2, H_2SO_3, H_3PO_4, H_3PO_3, HCOOH, CH_3COOH, HCN etc

$$HF + H_2O \rightleftharpoons H_3O^+ + F^-$$

Factors Affecting the Strength of Acids

i. Number of Oxygen Atoms

As the number of oxygen atoms increase, strength of acids also increases.

Example

- $HClO_4 > HClO_3 > HClO_2 > HClO$
- $H_2SO_4 > H_2SO_3$

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➥ $HNO_3 > HNO_2$

Exception: $H_3PO_2 > H_3PO_3 > H_3PO_4$

ii. Electro Negativity of the Central Atom

For oxyacids and acids in the same period, the more electronegative the central atom, the stronger will be the acid.

Example

➥ $HClO_4 > H_2SO_4 > H_3PO_4$

➥ $HClO_3 > H\,BrO_3 > HIO_3$

➥ $HF > H_2O > NH_3$

iii. Size of the Central Atom for Binary Acids in the Same Group

As size increases, strength of binary acids also increases.

Example

$HI > H\,Br > H\,Cl > HF$

$H_2Te > H_2Se > H_2S > H_2O$

Strong bases dissociate almost completely in water.

Example

LiOH, NaOH, KOH, RbOH, CSOH, $Ca(OH)_2$, $Sr(OH)_2$, Ba(OH)

$LiOH\,(s) \xrightarrow{H_2O} Li^+\,(aq) + OH^-\,(aq)$

Weak bases ionize slightly in water.

Example

$Mg(OH)_2$, $Al(OH)_3$, $Fe(OH)_2$, $Cu(OH)_2$, NH_3 etc

$NH_3(g) + H_2O(1) \rightleftharpoons NH_4^+(aq) + OH^-(aq)$

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The Strength of Acids and Bases in Aqueous Solution can be Measured by:

1. Concentration of Hydrogen and Hydroxide ions

With same concentration :

Strong acids produce greater amount of hydrogen ion than weak acid.

For strong acid : [acid] ≈ [H^+]

For weak acid : [acid] > [H^+]

Strong bases give more hydroxide amount than weak bases.

For strong base : [Base] ≈ [OH^-]

For weak base : [Base] < [OH^-]

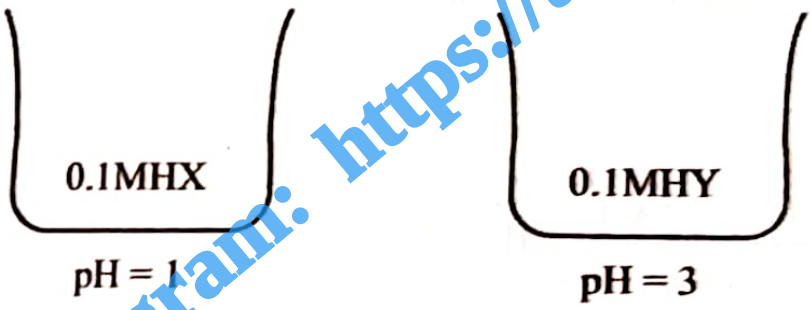
2. pH and pOH

With same concentration

Strong acid : lower pH

Weak acid : higher pH

Example



pH = 1 pH = 3

Which acid is stronger ? **Answer : HX**

Strong base : lower pOH

Weak base : higher pOH

3. Percent of Ionization or Dissociation

$$\% \text{ acid} = \frac{[H^+]}{[\text{acid}]_o} \times 100$$

$[\text{acid}]_o$ - initial concentration of acid

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$$\% \text{ base} = \frac{[OH^-]}{[\text{base}]_o} \times 100$$

$[\text{base}]_o$ - initial concentration of base

Strong acids and strong bases ionize almost 100%

Weak acids and weak bases ionize less than equal to 5%

Example

The pH of 0.5MHA solute is 3. Calculate the percent of ionization of HA.

Solution

$$[H^+] = 10^{-pH} = 10^{-3}M = 0.001M$$

$$\% \text{ HA} = \frac{[H^+]}{[HA]_o} = \frac{0.001M}{0.5M} \times 100 = 0.2\%$$

4. Dissociation or Ionization Constants

a. Acid - dissociation Constant (K_a)

Consider an acid HA

$$HA\ (aq) + H_2O\ (l) \rightleftharpoons H_3O^+\ (aq) + A^-\ (aq)$$

$$K = \frac{[H_3O^+]\,[A^-]}{[H_2O]\,[HA]}$$

$$K[H_2O] = \frac{[H_3O^+]\,[A^-]}{[HA]},\quad K[H_2O] = K_a$$

$$\boxed{K_a = \frac{[H_3O^+]\,[A^-]}{[HA]}}$$

The larger the value of K_a, the stronger is the acid and vice - versa.

Example

Acid	K_a
HF	7.2×10^{-4}
HNO_2	4.5×10^{-4}
HCN	6.2×10^{-10}

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Identify the strongest and weakest acid.

Solution

Strongest acid = HF because it has the largest K_a value.

Weakest acid = HCN because it has the smallest K_a value

$$pK_a = \log \frac{1}{K_a} = -\log K_a$$

Note : As the value of pKa increases, the strength of the acid decreases and vice - versa.

Example

Given three acids A, B and C with pK_a values 5.72, 3.68 and 8.92 respectively. Arrange these acids in increasing order of strength.

Solution

B > A > C

b. Base Dissociation Constant (K_b)

Consider the base B :

$$B + H_2O \rightleftharpoons BH^+ + OH^-$$

$$K = \frac{[BH^+]\,[OH^-]}{[B]\,[H_2O]}$$

$$K\,[H_2O] = \frac{[BH^+]\,[OH^-]}{[B]}, \quad K\,[H_2O] = K_b$$

$$K_b = \frac{[BH^+]\,[OH^-]}{[B]}$$

The larger the K_b value, the stronger is the base and vice - versa.

$$pK_b = \log \frac{1}{K_b} = -\log K_b$$

The pK_b value is inversely related to the strength of bases.

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Illustrative Example

52. Which of the following is true about 0.1M HA weak acid solution ?

A. $[H^+] = 0.1M$

B. $[A^-] = 0.1M$

C. pH = 1

D. $[HA] > [H^+]$

Solution

For weak acid : $[acid] > [H^+]$

Answer : D

53. Three acids, HA, HB and HC have the following K_a values. K_a (HA) = 1×10^{-6}, K_a (H B) = 1×10^{-5} and K_a (H C) = 1×10^{-7}. What is the correct order of increasing acid strength (weakest first) ?

A. HA, H B, H C

B. H C, H B, HA

C. H C, HA, H B

D. H B, HA, H C

Solution

Acid strength α K_a

Answer : C

54. Which of the following statements is true about the percent of ionization of weak acid ?

A. It decreases with increasing concentrations

B. It increases with decreasing concentration

C. It increases with increasing concentration

D. It decreases with decreasing concentration

Solution

$$\% \text{ Acid} = \frac{[H^+]}{[acid]_o} \times 100$$

$\Rightarrow$ As $[acid]_o$ increases, % acid decreases.

Answer : A

55. Which of the following is the weakest Bronsted - Lowry acid ?

A. H_2SO_4 B. HNO_3 C. HF D. HI

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Answer : C

56. Consider the following solids and their K_b values for the dissociation which one will give the most basic aqueous solution ?

A. $Fe(OH)_2$, $K_b = 7.9 \times 10^{-15}$

B. $Bi(OH)_3$, $K_b = 2.7 \times 10^{-40}$

C. $Pb(OH)_2$, $K_b = 2.8 \times 10^{-16}$

D. $Mg(OH)_2$, $K_b = 1.5 \times 10^{-11}$

Solution

$K_b \propto$ base strength

Answer : D

57. Given the acid dissociation constant (Ka)

Acid	K_a
HClO	3.5×10^{-8}
$HClO_2$	1.2×10^{-2}
H CN	6.2×10^{-8}
CH_3COOH	1.8×10^{-5}

a. The strongest acid is _____

b. The strongest conjugate base is _____

c. The weakest conjugate base is _____

d. The weakest acid is _____

Solution

a. $HClO_2$ b. ClO^- c. ClO^-_2 d. HClO

58. Consider the following acids

i. CH_3COOH, $K_a = 1.8 \times 10^{-5}$

ii. HCO_2H, $K_a = 1.8 \times 10^{-4}$

iii. HOBr, $K_a = 2.4 \times 10^{-9}$

iv. C_6H_5OH, $K_a = 1.0 \times 10^{-10}$

a. Arrange these acids in increasing order of strength

b. Arrange the conjugate bases of these acids in decreasing order of strength

c. 0.10M of which acid will have the highest pH value

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Solution

a. IV < III < I < II

b. $C_6H_5O^-$ > OBr > CH_3COO^- > HCO^-_2

c. The weakest acid has the highest pH value . That is, C_6H_5OH.

59. Arrange the following acids in decreasing order of acid strength.

a. $HBrO_4$, $HBrO_3$, HBrO, $HBrO_2$

b. HIO_4, $HClO_4$, $HBrO_4$

c. CH_3COOH, CF_3COOH, CCl_4COOH

Solution

a. $HBrO_4 > HBrO_3 > HBrO_2 > HBrO$

b. $HClO_4 > HBrO_4 > HIO_4$

c. $CF_3COOH > CCl_3COOH > CH_3COOH$

60. The pKb values of $C_2H_5NH_2$, NH_3 and C_5H_5N are 3.37, 4.74 and 8.82 respectively.

a. Arrange these bases in increasing order of strength.

b. Arrange the conjugate acids of these bases in decreasing order of strength.

Solution

a. $C_5H_5N < NH_3 < C_2H_5NH_2$

b. $C_5H_5NH^+ < NH_4^+ > C_2H_5N^+H_3$

61. Predict which acid in each of the following pairs is stronger ?

a. NH_3 or H_2O

b. H_2O or HF

c. PH_3 or NH_3

d. H Cl or H_2S

Solution

a. H_2O b. HF c. PH_3 d. H Cl

Determination of pH of Weak Acids

The pH of weak acid is not determined by stoichiometric relationship li strong acids because they ionize slightly when dissolved in water. Th we follow the following steps in order to calculate the pH of weak acids

Step 1 : Identify the measure species that can affect the pH of the

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solution. In most cases we may ignore the dissociation of water.

Step 2 : State the initial concentrations, the change in terms of **x** and equilibrium concentrations of the species at equilibrium

Step 3 : Write the acid dissociation constant, K_a in terms of equilibrium concentration

Table 7.3 Ka and pKa Values of Some Acids

Name of acid	Formula	K_a	pK_a
Hydrofluoric acid	HF	6.8×10^{-4}	3.18
Formic acid	HCOOH	1.7×10^{-4}	3.77
Nitrous acid	HNO_2	4.5×10^{-4}	3.35
Ascorbic acid	$C_6H_8O_6$	8.0×10^{-5}	4.52
Benzoic acid	C_6H_5COOH	6.5×10^{-5}	4.19
Acatic acid	CH_3COOH	1.8×10^{-5}	4.74
Hydrobromouse acid	HOBr	2.5×10^{-9}	8.60
Hydro bromous acid	HOCl	3.0×10^{-8}	7.52
Hydrocyanic acid	HCN	4.9×10^{-10}	9.31

Step 4: Calculate the value of x, then the concentration of all species

Step 5: check the validity of the percent of ionization of the acid find the pH of the solution

Example

What is the pH of 0.2M HA solution.($K_a = 2 \times 10^{-5}$)

Solution

	$HA\ (aq) + H_2O\ (l) \rightleftharpoons H_3O^+\ (aq) + A^-\ (aq)$			
Initial conc.(M)	0.2	-	0	0
Change conc.(M)	-x	-	+x	+x
Equilibrium con.(M)	0.2-x	-	x	x

$$K_a = \frac{[H_3O^+]\,[A^-]}{[HA]}$$

$$2 \times 10^{-5} = \frac{x \cdot x}{0.2-x} = \frac{x^2}{0.2-x}$$

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Use approximation method and check the validity of percent of ionization. If the percent of dissociation less than or equal to 5 %, it is valid. Assume : $0.2 - x \approx 0.2$

$$2 \times 10^{-5} = \frac{x^2}{0.2}$$

$$x^2 = 0.2 \times 2 \times 10^{-5} = 4 \times 10^{-6}$$

solving for x, $x = 2 \times 10^{-3}\ M = [H_3O^+] = [A^-]$

$$\% \text{ HA} = \frac{[H_3O^+]}{[HA]_o} \times 100 = \frac{2 \times 10^{-3}}{0.2} \times 1.0\%$$

$\therefore$ Theapproximation is valid

At equilibrium : $[H_3O^+] = [A^-] = 2 \times 10^{-3}\ M$

$[HA] = [HA]_o = x = 0.2\,M - 0.002M$

$[]HA] = 0.198\,M$

$pH = - \log [H^+] = - \log 2 \times 10^{-3} = 2.7$

If the approximation is invalid, use the quadratic formula :

$$x = -b \pm \sqrt{\frac{b^2 - 4ac}{2a}}$$

Determination of pH of Weak Bases

We use the same procedure as weak acids to determine the pH of weak bases.

Table 7.4 K_b and pK_b Values of Some Common Weak Bases

Name of base	Formula	K_b	pK_b
Ammonia	NH_3	1.8×10^{-5}	4.74
Aniline	$C_6H_5NH_2$	4.2×10^{-10}	9.38
Ethylamine	$C_2H_5NH_2$	4.7×10^{-4}	3.33
Hydrazine	N_2H_4	1.7×10^{-6}	5.77
Hydroxylamine	NH_2OH	1.1×10^{-8}	7.96
Methyl amine	CH_3NH_2	4.4×10^{-4}	3.36

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Pyridine	C_5H_5N	1.7×10^{-9}	8.77
Urea	$(NH_2)_2CO$	1.5×10^{-4}	3.82

Example

Calculate the pH and pOH of 0.2 M NH_3 solution

Solution

	NH_3 (g)	+ H_2O (l)	$\rightleftharpoons$ NH_4^+ (aq)	+ OH^- (aq)
Initial conc.(m)	0.2	-	0	0
Change con.(m)	-x	-	+x	+x
Equiliriu conc(m)	-0.2 - x	-	x	x

$$K_b = \frac{[NH_4^+][OH^-]}{[NH_3]}$$

$1.8 \times 10^{-5} = \frac{x^2}{0.2-x}$, Approximation : $0.2 - x \approx 0.2$

$1.8 \times 10^{-5} = \frac{x^2}{0.2}$, solving for x , $x = 1.9 \times 10^{-3}$

$\Rightarrow x = [NH_4^+] = [OH^-] = 1.9 \times 10^{-3}$ M

check the approximation

The approximation is valid

$pH = 14 + \log [OH^-] = 14 + \log 1.9 \times 10^{-3} = 11.3$

$pOH = 14 - pH = 14 - 11.3 = 2.7$

Illustrative Example

62. The pH of a 0.10M solution of formic acid (HCOOH) is 3. What is the K_a of the acid ? (ESSLCE)

A. 9.99×10^{-5} C. 1.01×10^{-7}

B. 9.99×10^{-4} D. 1.01×10^{-5}

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Solution

$$HCOOH \rightleftharpoons HCOO^- + , [H^+] = [HCOO^-] = 10^{-pH} = 10^{-3}$$

$$K_a = \frac{[HCOO^-]\,[H^+]}{HCOOH]}, \quad [CH_3COO^-] = [H^+]$$

$$[HCOOH] = [HCOOH]_o - [H^+]$$

$$K_a = \frac{[H^+]^2}{[HCOOH]_o - [H^+]} = \frac{(10^{-3})^2}{(0.1 - 0.001)} = 1.01 \times 10^{-5}$$

Answer : D

63. Nitrous acid, HNO_2 is 0.37 % in 3.0M solution. What is the value of K_a for this acid ?

A. 3.7×10^{-3} C. 6.8×10^{-6}

B. 6.4×10^{-4} D. 4.05×10^{-5}

Solution

$$[H^+] = 0.37\% \times [HNO_2]_o = \frac{0.37}{100} \times 3M = 0.0111M$$

$$K_a = \frac{[H^+]^2}{HNO_2]_o - [H^+]} = \frac{(0.011)^2}{3 - 00111)} = 4.05 \times 10^{-5}$$

64. The K_a of hypo chlorous acid (HClO) is 3×10^{-8} at 25°C. What is the % ionzation of H ClO in a 0.015M aqueous solution of HClO at 25°C ?

Solution

$$HClO \rightleftharpoons H^+ + ClO^-$$

$$K_a = \frac{[H^+]\,[OH^-]}{[HClO}, \quad [H^+] = [ClO^-]$$

$$K_a = \frac{[H^+]^2}{[HClO]}, [H^+] = \sqrt{Ka\,[HClO]} = \sqrt{3 \times 10^{-8} \times 0.015}$$

$$[H^+] = 2.12 \times 10^{-5}\,M$$

$$\%\ HClO = \frac{2.12 \times 10^{-5}\,M}{0.015\,M} \times 100 = 0.141\%$$

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65. What is ionization constant for a weak acid, HA that is 1.60 % ionized in 0.095M solution ?

Solution

$$[H^+] = 1.60\% \times 0.095\,M = 1.52 \times 10^{-3}\,M$$

$$K_a = \frac{[H^+]^2}{[HA]_o - [H^+]} = \frac{(1.52 \times 10^{-3})^2}{0.095 - 0.00152} = 2.47 \times 10^{-5}$$

66. The pH at room temperature of x M solution of formic acid (H COOH) was measured to be 4.50.Find the initial concentration x. ($K_a = 6.8 \times 10^{-4}$)

Solution

$$[H^+] = 10^{-pH} = 10^{-4.50} = 3.16 \times 10^{-5}\,M$$

$$K_a = \frac{[H^+]^2}{[HCOOH]_o - [H^+]}$$

$$[HCOOH]_o - [H^+] = \frac{[H^+]^2}{K_a}$$

$$[HCOOH]_o = \frac{[H^+]^2}{K_a} + [H^+] = \frac{(3.16 \times 10^{-5})^2}{6.8 \times 10^{-4}} + 3.16 \times 10^{-5}$$

$$= 3.31 \times 10^{-5}\,M$$

67. What molar concentration of ammonia provides a hydroxide concentration of 2×10^{-3} M ? ($K_b = 1.8 \times 10^{-5}$)

Solution

$$K_b = \frac{[OH^-]^2}{[NH_3]_o - [OH^-]}$$

$$1.8 \times 10^{-5} = \frac{(2 \times 10^{-3})^2}{[NH_3]_o - 2 \times 10^{-3}} \Rightarrow [NH_3]_o = 0.224M$$

68. What is the factor that affects the values of acid and base dissociation constants ?

Solution

Temperature

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7.4 The Common ion Effect and Buffer Solution

The common ion effect is the suppression of the ionization weak acid or weak base caused by adding an ion that is product of dissociation.

It shifts the equilibrium to the right.

It does not affect the equilibrium constant of weak acid or weak base.

The sources of common ion can be strong acid, strong base or soluble salt that has similar ion with the weak acid or weak base.

Consider the ionization of acetic acid and ammonia

Acetic Acid (CH_3COOH)

When strong acid is added to a solution of acetic acid, the common ion is H^+.

$$CH_3COOH\ (aq) \underset{\text{Added } H^+}{\rightleftharpoons} CH_3COO^-(aq) + H^+\ (aq)$$

When sodium acetate (CH_3COONa) is added to a solution of acetic acid, the common ion is **CH_3COO^-**

$$CH_3COOH\ (aq) \underset{\text{Added } CH_3COO^-}{\rightleftharpoons} CH_3COO^-\ (aq) + H^+\ (aq)$$

Table 7.5 The Effect of Added Acetate Ion on the Dissociation of Acetic Acid

Initial concentration of CH_3COOH	Concentration of acetate ion added	Percent of ionization	$[H^+]$	pH
0.10	0.00	1.3	1.3×10^{-3}	2.89
0.10	0.05	0.036	3.6×10^{-5}	4.44
0.10	0.10	0.018	1.8×10^{-5}	4.74
0.10	0.012	0.012	1.2×10^{-5}	4.92

Example

Calculate the pH of :

a. 0.1M CH_3COOH ($K_a = 1.8 \times 10^{-5}$)

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b. 0.1M CH_3COOH and 0.3 M CH_3COONa

Solution

a. $K_a = \frac{[CH_3COO^-][H^+]}{[CH_3COOH]}$, but at equilibrium, $[H^+] = [CH_3COO^-]$

$$K_a = \frac{[H^+]^2}{[CH_3COOH]}$$

$$[H^+] = \sqrt{Ka[CH_3COOH]} = \sqrt{1.8 \times 10^{-5} \times 0.1} = 1.34 \times 10^{-3}m$$

$$pH = -\log [H^+] = -\log 1.34 \times 10^{-3} = 2.87$$

b. $CH_3COONa \longrightarrow CH_3COO^- + Na^+$

$$\Rightarrow [CH_3COONa] = \left[CH_3COO^{-1}\right] = 0.3M$$

$$K_a = \frac{[CH_3COO^-]\,[H^+]}{[CH_3COOH]}$$

$$[H^+] = K_a \times \frac{[CH_3COOH]}{[CH_3COO^-]} = 1.8 \times 10^{-5} \times \frac{(0.1)}{(0.3)}$$

$$[H^+] = 5.4 \times 10^{-5}\,M$$

$$pH = -\log 5.4 \times 10^{-5}\,M = 4.23$$

$$[H^+] = \frac{K_a\,[acid]_o}{[anion]}$$

Ammonia (NH_3)

Adding strong base shifts the equilibrium to the left and the common ion is hydroxide ion (OH^-) Added OH^-

$$NH_3\,(g) + H_2O\,(1) \rightleftharpoons NH_4^+\,(aq) + OH^-\,(aq)$$

Adding ammonium chloride (NH_4Cl) shifts the equilibrium to the left and the common ion is ammonium ion (NH_4^+)

Added NH_4^+

$$NH_3\,(g) + H_2O\,(1) \rightleftharpoons NH_4^+\,(aq) + OH^-\,(aq)$$

The common ion effect is very important to adjust pH or concentration of hydronium ion

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Illustrative Example

69. The suppression of ionization of a weak acid or a weak base by adding one of its own ions is known as

A. Buffer effect
C. Common ion effect
B. Ionization effect
D. Association effect

Answer : C

70. In which direction will the following equilibrium shift, if a solution of CH_3COONa is added ?

$$CH_3COOH\ (aq) \rightleftharpoons CH_3COO^-\ (aq) + H^+\ (aq)$$

A. The equilibrium shifts to the right (more products
B. No change
C. The equilibrium shifts to the left (more reactants)
D. Cannot be predicted

Solution

CH_3COONa is used as a source of CH_3COO^-. As the concentration of acetate ion, CH_3COO^- increases, the equilibrium shifts to the left.

Answer : C

71. If $NaNO_2$ is added to a solution of HNO_2, Which of the following statements is true ?

A. The pH of the solution will increase
B. The pH of the solution will decrease
C. The pH will remain same
D. The equilibrium will not be affected

Solution

➥ If the common ion s cation, the pH of the solution will decrease, because cations are Lewis acids.

➥ If the common ion is anion the pH of the solution will increase, because anions, are Lewis bases.

Answer : A

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72. Consider the reaction : $HOCl + H_2O \rightleftharpoons H_3O^+ + OCl^-$. What will happen if NaOCl is added to the above reaction at equilibrium ?

Solution

- The equilibrium shifts to the left
- The concentration of HOCl increases
- The concentration of H_3O^+ decrease
- The pH of the solution increases
- The pOH of the solution decreases.

73. a) Determine the pH of 0.30M solution of pyridine, C_5H_5N

b) Predict the affect of the addition of pyridinium ion, $C_5H_5NH^+$ has been added

c) Calculate the pH of 0.30M pyridine solution to which 0.15M C_5H_5NHCl has been added.

($K_b = 1.52 \times 10^{-9}$)

Solution

a. $K_b = \dfrac{[C_5H_5NH^+]\,[OH^-]}{[C_5H_5N]}$, $[OH^-] = [C_5H_5N^+H]$

$[OH^-] = \sqrt{K_a \times [C_5H_5N]} = \sqrt{(0.3) \times (1.52 \times 10^{-9})} = 2.14 \times 10^{-5}M$

$pH = 14 + \log [OH^-] = 14 + \log 2.14 \times 10^{-5} = 9.33$

b. The equilibrium shifts to the left.

c. $[OH^-] = K_b \times \dfrac{[[C_5H_5N]}{C_5H_5NH^+]} = 1.52 \times 10^{-9} \times \dfrac{(0.30)}{(0.15)}$

$[OH^-] = 3.04 \times 10^{-9}M$

$pH = 14 + \log [OH^-] = 14 + \log 3.04 \times 10^{-9} = 5.48$

Buffer Solution

Buffer solution is a solution that resists pH change upon the addition of a small amounts of strong acid (hydronium ion) or strong base (hydroxide ion)

- It has two portions , **acidic portion** that neutralizes the added base

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and **basic portion** that neutralizes the added acid.

Example

HA / A^-

Acidic portion is HA and its basic form is A^-.

➥ When an acid is added to this buffer, the basic portion neutralizes it

$$A^- + H_3O^+ \longrightarrow HA + H_2O$$

➥ When a base is added to the above buffer, the acidic component of the buffer neutralizes it.

$$HA + OH^- \longrightarrow A^- + H_2O$$

Note : Buffer solution can be prepared from :

1. Weak Acid and its Conjugate base

Example

CH_3COOH / CH_3COO^- or CH_3COOH/ CH_3COONa because in CH_3COONa we have the conjugate base CH_3COO^-

2. Weak Base and its Conjugate Acid

Example

NH_3 / NH_4^+ or NH_3 / NH_4Cl because in NH_4Cl there is conjugate acid, NH_4^+.

3. Excess Weak acid and Limited Amount of Strong Base

At the end of the reaction, the weak acids is left with certain amount and the strong base is consumed. The conjugate base is formed by the reaction.

Example

CH_3COOH / Na OH in 2 : 1 mole ratio

$$CH_3COOH + Na\,OH \longrightarrow CH_3COO\,Na + H_2O$$

2mol 1mol

⇒ At the end of the reaction :

n CH_3COOH left = 2 - 1 = 1mol

n Na OH = 0 because it is limiting reactant.

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n CH_3COO Na formed = 1mol

CH_3COOH/ CH_3COO Na is a buffer

4. Excess Weak Base and Limited Amount of Strong Acid

Example

NH_3 / H Cl in 2 : 1 mole ratio

$NH_3 + H\,Cl \longrightarrow NH_4Cl$

2mol 1mol

At the end of the reaction :

n NH_3 left = 2 - 1 = 1mol

n HCl = 0 becuase it is limiting reactant

n NH_4Cl formed = NH_4Cl

$\therefore$ NH_3 / NH_4Cl is a buffer.

Buffer Capacity is the measure of the strength of the buffer

- It depends on component concentrations, both absolute and relative concentrations.
- The more concentrated the buffer components or the closer the component concentrations to each other the greater the capacity.
- Buffer range is the pH range in which the buffer is effective.

Applications of Buffer Solution

- It is used in human blood.
- Protein studies often must be performed in buffer media.
- Enzymes activity is usually working with pH 6 and pH 8 which is required buffered system.
- The control of pH is often important in industrial processes.
- Buffer solutions are important in solubility precipitation processes.

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Strong acids and strong bases are not used to prepare buffer solutions because they ionize almost completely when dissolved in water. Thefore there is no unionized acid or base to neutralize the base or acid.

The Henderson - Hasselbalch Equation

It is an equation used to detrmine the pH and pOH of butter solutions.

For and weakacid HA, the dissocciation equation and Ka expression are given below

$$HA + H_2O \rightleftharpoons H_3O^+ + A^-$$

$$K_a = \frac{[H_3O^+][A^-]}{[HA]}$$

$$[H_3O^+] = K_a \frac{[HA]}{[A^-]}$$

$$\log[H_3O^+] = -\log K_a \frac{[HA]}{[A^-]}$$

$$\log[H_3O^+] = -\log K_a + \log \frac{[HA]}{[A^-]}$$

For any conjugate acid - base pair

$$\boxed{pH = pKa + \log \frac{[CB]}{[acid]}}$$

To find the concentration ratio of acid to CB use the formula below

$$\boxed{\frac{[H_3O^+]}{K_a} = \frac{[CB]}{[acid]}}$$

For any weak base, B, the dissociation equation and the K_b expression are:

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$$B + H_2O \rightleftharpoons BH^+ + OH^-$$

$$K_b = \frac{[BH^+][OH^-]}{[B]}$$

$$[OH^-] = K_b \frac{[B]}{[BH^+]}$$

$$\log[OH^-] = \log K_b \frac{[B]}{[BH^+]}$$

$$\log[OH^-] = pK_b + \log \frac{[BH^+]}{[B]}$$

$$\boxed{pOH = pK_b + \log \frac{[CA]}{[Base]}}$$

To find the concentration ratio of base to CA use the formula below.

$$\boxed{\frac{[OH^-]}{K_b} = \frac{[Base]}{[CA]}}$$

Illustrative Examples

74. Which of the following combinations CANNOT produces a buffer solution?

A. $HClO_4$ and $NaClO_4$
C. HNO_2 and $NaNO_2$
B. HCN and NaCN
D. NH_3 and $(NH_4)_2 SO_4$

Solution

Strong acid and its conjugate base cannot form a buffer solution

Answer: A

75. The following bases and their conjugate acids (as chlorides) are available in the laboratory.

NH_3, $K_b = 1.8 \times 10^{-5}$
$CH_3 CH_2 NH_2$, $K_b = 6.4 \times 10^{-4}$
C_6H_5N, $K_b = 1.7 \times 10^{-9}$
$CH_3 NH_2$, $K_b = 4.4 \times 10^{-4}$

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Which of the these acid - base pairs are best to prepara a buffer solution having a pH of about 9?

A. NH_3 + Conjugate acid
C. $CH_3CH_2NH_2$ + conjugate acid
B. C_6H_5N + Conjugate acid
D. CH_3NH_2 + conjugate acid

Solution

- Buffer has the highest capacity when the component concentrations are equal.
- $pH = pK_a$ or $pOH = pK_b$

Answer: B

76. Which of following procedures will produce a buffered solution?

I. Equal volumes of 0.5M NaOH and 1MHCl solutions are mixed.

II. Equal volumes of 0.5M NaOH and 1M CH_3COOH solutions are mixed.

III. Equal volumes of 1M Na CH_3CO_2 and 1M CH_3COOH Solution

IV. Equal volumes of 1.0M NaOH and 1M HCl solutions are mixed.

A. I
B. III
C. I and II
D. II and III

Solution

Buffer solution = weak acid + conjugate base
= excess weak acid + strong base
= weak base + conjugate acid
= excess weak base + strong acid

Answer: D

77. To a 50 ml of a buffer solution that is 0.25M in HF and 0.10M in NaF, If 5ml of 0.050M NaOH is added, what is the final pH of this buffer solution? (pKa for HF = 3.17)

A. 2.77
B. 2.80
C. 2.74
D. 3.17

$$nHF = M \times V = 0.25\frac{mol}{\cancel{L}} \times 50\cancel{L} = 12.5mmol$$

$$nNaF = M \times V = 0.10\frac{mol}{L} \times 50ml = 5mmol$$

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$$nNaOH = M \times V = 0.05\frac{mol}{L} \times 5ml = 0.25mmol$$

$$HF + NaOH \longrightarrow NaF + H_2O$$

12.5mmol 0.25mmol

nHF left = 12.50 - 0.25 = 12.25mmol

nNaF = 0.25 mmol + 5mmol = 5.25 mmol

$$[HF] = \frac{12.25 mmol}{(5 + 50) mL} = 0.223M$$

$$[NaF] = \frac{5.25 mmol}{(5 + 50) mL} = 0.0955M$$

$$pH = pK_a + \log \frac{[NaF]}{[HF]}$$

$$pH = 3.17 + \log \frac{0.0955}{0.223} = 2.80$$

78. The pH of a buffer solution that contains 0.13M of B and unknown concentration of BH^+ is 8. Determine the concentration of BH^+. ($pK_b = 5$)

Solution

$$pOH = 14 - pH = 14 - 8 = 6$$

$$pOH = pK_b + \log \frac{[BH^+]}{[B]}$$

$$\frac{[BH^+]}{[B]} = 10^{pOH - pk_b} \times [B] = 10^{6-5} \times 0.13M = 1.3M$$

79. How many moles must be added to 1.5L of 0.2M solution of NH_3 to form of a buffer whose pH is 9.00?

Solution

$$pOH = 14 - pH = 14 - 9 = 5$$

$$[OH^-] = 10^{-pOH} = 10^{-5}M$$

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$$K_a = \frac{[NH_4Cl][OH^-]}{[NH_3]}$$

$$[NH_4Cl] = K_a \times \frac{[NH_3]}{[OH^-]} = 1.8 \times 10^{-5} \times \frac{0.2M}{1 \times 10^{-5}} = 0.26M$$

$$nNH_4Cl = M \times V = 0.36\frac{mol}{L} \times 1.5L = 0.54mol$$

80. The Pk_a of a weak monoptotic acid is 4.8. What should be the ratio of [Acid]/[salt] of a buffer, if pH = 5.8 is required?

Solution

$$pH = pKa + \log \frac{[salt]}{[Acid]}$$

$$\frac{[salt]}{[Acid]} = 10^{pH-pKa} \Rightarrow \frac{[acid]}{[salt]} = 10^{pKa-pH} = 10^{4.8-5.8} = \frac{1}{10}$$

81. From preparting a buffer pH 6 by mixing soduium acetate and acetic acid, what would be the ratio of the concentration of salt and acid? ($K_a = 10^{-5}$)

Solution

$$\frac{[H_3O^+]}{K_a} = \frac{[CH_3COOH]}{[CH_3COONa]}, [H_3O^+] = 10^{-pH} = 10^{-6}$$

$$\Rightarrow \frac{[CH_3COONa]}{[CH_3COOH]} = \frac{K_a}{[H_3O^+]} = \frac{10^{-5}}{10^{-6}} = 10:1$$

82. Two buffer solutions, X and Y have the following concentration of their components.

Buffer	[HA]	$[A^-]$
X	0.7M	0.87M
Y	0.1M	0.1M

Which buffer resists change in pH upon the addition of 0.01mol of KOH in 500ml of buffer?

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Solution

The higher the conceration or the buffer solution, the better is its buffer capacity. As aresult, buffer X resists pH change better than butter Y.

83. Calculate the pH of buffer solution prepared by mixing 60ml of CH_3 COOH and 40ml of 0.1M CH_3COONa? ($K_b = 1.8 \times 10^{-5}$)

Solution

From type of question use the formula below

$$[acid] = \frac{C_a V_a}{V_a + V_s}, [salt] = \frac{C_s V_s}{V_a + V_s}$$

a - acid, s-salt

$$[CH_3COOH] = \frac{C_a V_a}{V_a + V_s} = \frac{0.1M \times 60mL}{(60+40)mL} = 0.06M$$

$$[CH_3COOH] = \frac{C_s V_s}{V_a + V_s} = \frac{0.1M \times 40mL}{(60+40)mL} = 0.04M$$

$$pH = pKa + \log \frac{[CH_3COONa]}{[CH_3COOH]} = 4.7 + \log \frac{0.04}{0.06} = 4.54$$

84. Calculate the pH

a. of a buffer solution containing 0.1M NH_3 and 0.1M NH_4Cl

b. When 1ml of 0.10M HCl is added to 100mol of the buffer in (a).

c. When 1ml of 0.10M NaOH is added to 100ml of the buffer in (a).

d. of unbuffered solution containing 1.8×10^{-5}M NaOH solution.

e. i. When 1ml of 0.1M HCl is added to 100ml of solution (d).

ii. when 1ml of 0.1M NaOH is added to 100ml of solution (d)

Solution

a) $pOH = pK_b + \log \frac{[NH_4Cl]}{[NH_3]}$, $pK_b = -\log 1.8 \times 10^{-5} = 4.74$

$$= 4.74 + \log \frac{0.1}{0.1} = 4.74$$

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$pH = 14 - pOH = 14 - 4.74 = 9.26$

b) $nNH_3 = M \times V = 0.1 \frac{mol}{\cancel{L}} \times 100\cancel{mL} = 10mmol$

$nNH_4Cl = M \times V = 0.1 \frac{mol}{\cancel{L}} \times 100\cancel{mL} = 10mmol$

$nHCl = M \times V = 0.1 \frac{mol}{\cancel{L}} \times 1\cancel{mL} = 0.1mmol$

$NH_3 + HCl \rightarrow NH_4Cl$

10mmol 0.1mmol

$\Rightarrow$ HCl is limiting reactant.

$\Rightarrow$ At the end of the reaction

nNH_3 left $= 10 - 0.1 = 9.9$ mmol

$nNH_4Cl = nNH_4Cl$ (buffer) $+ nNH_4Cl$

$= 10mmol + 0.1mmol = 10.1mol$

$$[NH_3] = \frac{nNH_3}{V_{HCl} + V_{buffer}} = \frac{9.9mmol}{(1+100)\cancel{mL}} = 0.098M$$

$$[NH_4Cl] = \frac{nNH_4Cl}{V_{HCl} + V_{buffer}} = \frac{10.1mmol}{(1+100)\cancel{mL}} = 0.10M$$

$$pOH = pKb + \log \frac{[NH_4Cl]}{[NH_3]} = 4.74 + \log \frac{0.10}{0.0980} = 4.75$$

$pH = 14 - 4.75 = 9.25$

c. When a base is added to a buffer, the acid portion of the buffer neutralize it.

$nNaOH = M \times V = 0.1 \frac{mol}{\cancel{L}} \times 1\cancel{mL} = 0.1mmol$

$NH_4Cl + NaOH \longrightarrow NH_3 + NaCl + H_2O$

10mmol 0.1mmol

At the end of the reaction:

$nNH_4Cl = 10 - 0.1 = 9.9mmol$

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$nNH_3 = 0.1 + 10 = 101mmol$

$[NH_4Cl] = \frac{9.9mmol}{(1+100)ml} = 0.098M$

$[NH_3] = \frac{10.1mmol}{(1+100)ml} = 0.10M$

Note: NaCl and H_2O do not affect pH because they are natural substances.

$$pOH = pKb + \log\frac{[NH_4Cl]}{[NH_3]} = 4.74 + \log\frac{0.098}{0.10} = 4.73$$

$pH = 14 - 4.73 = 9.27$

d. $[NaOH] = [OH^-] = 1.8 \times 10^{-5}M$

$pH = 14 + \log[OH^-] = 14 + \log 1.8 \times 10^{-5} = 9.26$

e. i $nNaOH = M \times V = 1.8 \times 10^{-5}\frac{mol}{L} = 0.0018mmol$

$nNH_4Cl = M \times V = 0.1\frac{mol}{L} \times 1mL = 0.1mmol$

$NaOH + HCl \longrightarrow NaCl + H_2O$

0.0018mmol 0.1mmol

$\Rightarrow nHCl = 0.1mmol - 0.0018mmol = 0.0982mmol$

$[HCl] = [H^+] = \frac{0.0982mmol}{(1+100)mol} = 9.72 \times 10^{-4}M$

$pH = -\log[H^+] = -\log 9.72 \times 10^{-4} = 3.0$

ii. NaOH + NaOH

0.1mmol 0.0018mmol

$[NaOH] = [OH^-] = \frac{(0.1 + 0.0018)mmol}{(1+100)mL} = 0.001M$

$pH = 14 + \log[OH^-] = 14 + \log 0.001 = 11$

85. *Identify* wether the following combinations form a buffer or not.

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a. C_5H_5N/C_5H_5NHCl
b. HCOOH/ HCOOK
c. H_3PO_4/ NaH_2PO_4
d. $NaC_2H_3O_2/NaH_2PO$
e. RbOH/ RbBr
f. NH_3/NH_4NO_3
g. HNO_3/KNO_3
h. HCOOH/ KOH in 3:1 mole ration
i. NH_3/HNO_3 in 1:2 mole ratio

Solution

- a, b, c, d, f and h are buffer solutions.
- a, g and i are not buffer solutions.

7.5 Hydrolysis of Salts

Hydrolysis is the reaction between an ion and water in which an oxygen - hydrogen bond is broken.

- It is derived from a Greek word **hydro** that means water and **- lysis** meaning split.
- **Hydrolysis of salt** is the reaction of cation or anion or both ions of salt with water to produce hydronium (H_3O^+) or hydroxide (OH^-) ion.

1. Hydrolysis of salts of Strong Acid and Strong Base

- Salt obtained from strong acid and strong base are neutral salts (pH = 7)
- These salts do not undergo hydrolysis.
- These salts do not affect pH of water when they are added to it.

Strong acid + strong base ⟶ Neutral salt + water

Example:

$$NaOH + HCl \rightarrow NaCl + H_2O$$

$$NaCl \rightarrow Na^+ + Cl$$

$$Na^+ + H_2O \rightarrow \text{No reaction}$$

$$Cl^- + H_2O \rightarrow \text{No reaction}$$

What is the pH of 0.1M NaCl solution at 25°C

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Solution

NaCl is neutral salt an it does not affect the pH of water. Thus, the concentration of hydrogen ion is the one obtained from self - ionization of water.

$[H^+] = 1.0 \times 10^{-7}M \Rightarrow pH = -\log 10^{-7} = 7$

NaCl, KCl, $NaNO_3$, $CaSO_4$, $BaSO_4$, $Sr(ClO_4)_2$ are some examples of neutral salt.

2. Hydrolysis of Salts of Weak acids and Strong Bases

- This type of hydrolysis of salts is also called **anion hydrolysis** because only the anion (negative ion) of the salt undergoes hydrolysis.
- Salts obtained by this process are called basic salt (pH > 7)
- These salts increases the pH of water when they are added to it.

Weak acid + strong base ⟶ Basic salt + water

Example

$CH_3COOH + KOH \rightarrow CH_3COOK + H_2O$

$CH_3COOK \rightarrow CH_3COO^- + K^+$

$K^+ + H_2O \rightarrow$ No reaction

$CH_3COO^- + H_2O \rightleftharpoons CH_3COOH + OH^-$

$$K_b \text{ for } CH_3COO^- = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-]}$$

$$K_b = \frac{K_w}{K_a} = \frac{[CH_3COOH][OH^-]}{[CH_3COOH]}$$

K_a = acid dissociation constant for weak acid

K_w = Ion - product of water

Example

Calculate the pH of 0.1M CH_3COOK solution ($K_a = 1.8 \times 10^{-5}$)

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Solution

$$K_b = \frac{K_w}{K_a} = \frac{1 \times 10^{-14}}{1.8 \times 10^{-5}} = 5.6 \times 10^{-10}$$

$$[CH_3COOK] = [CH_3COO^-] = 0.1M$$

$$K_b = \frac{[CH_3COOH][OH^-]}{[CH_3COO^-]}$$

$$K_b = \frac{[OH^-]^2}{[CH_3COO^-]}, \text{ because } [OH^-] = [CH_3COOH]$$

$$[OH^-] = \sqrt{Kb \times [CH_2COO^-]} = \sqrt{5.6 \times 10^{-10} \times 0.1}$$

$$[OH^-] = 7.5 \times 10^{-6} M$$

$$pH = 14 + \log[OH^-] = 14 + \log^{7.5 \times 10^{-6}} = 8.88$$

$HCOONa$, K_2CO_3, Na_2S, BaF_2, $Sr_3(PO_4)_2$ are some examples of acidic salt.

3. Hydrolysis of Salts of Weak Bases and Strong Acids

- It is also known as **cation hydrolysis** because only the cation of the salt undergoes hydrolysis.
- The salts obtained from weak bases and strong acids are **acidic salts (pH < 7)**
- Dissolving these salts in water lowers the pH of water.

Example:

$$NH_3 + HNO_3 \rightarrow NH_4NO_3$$

$$NH_4NO_3 \rightarrow NH_4^+ + NO_3^-$$

$$NO_3^- + H_2O \rightarrow \text{No reaction}$$

$$NH_4^+ + H_2O \rightleftharpoons NH_3 + H_3O^+$$

$$\boxed{K_a = \frac{K_w}{K_b}}, K_b \text{ - ionization of constant of weak base.}$$

$$K_a = \frac{[NH_3][H_3O^+]}{[NH_4^+]}$$

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Calculate the pH of 0.1M NH_4NO_3 solution. ($K_b = 1.8 \times 10^{-5}$)

Solution

$$[NH_4NO_2] = [NH_4^+] = 0.1M$$

$$K_a = \frac{K_w}{K_b} = \frac{1 \times 10^{-14}}{1.8 \times 10^{-5}} = 5.6 \times 10^{-10}$$

$$K_a = \frac{[NH_3][H_3O^+]}{[NH_4^+]} = \frac{[H_3O^+]^2}{[NH_4^+]}, [NH_3] = [H_3O^+]$$

$$[H_3O^+] = \sqrt{K_a \times [NH_4^+]} = \sqrt{5.6 \times 10^{-10} \times 0.1} = 7.5 \times 10^{-6} M$$

$$pH = -\log[H_3O^+] = -\log 7.5 \times 10^{-6} = 5.12$$

NH_4Cl, $Al_2(SO_4)_3$, $AlCl_3$, $CuSO_4$, $Zn(NO_3)_2$ are some examples of acidic salts.

4. Hydrogen of Salts of Weak Acids and Weak Bases

In this type of hydrolysis both cation and anion undergo hydrolysis.

Salts obtained from weak acids and weak bases can be:

i. Neutral if K_a of Cation is Equal to K_b of Anion

Example

CH_3COONH_4

$$K_a(NH_4^+) = K_b(CH_3COO^-) = 5.6 \times 10^{-10}$$

ii. Acidic if K_a greater than K_b

Example:

NH_4F

$$\left.\begin{aligned} K_a(NH_4^+) &= 5.6 \times 10^{-10} \\ K_b(F^-) &= 1.47 \times 10^{-11} \end{aligned}\right\} K_a > K_b$$

iii. Basic if K_b is greater than K_a

Example:

NH_4CN

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Solution

Acid + Base → salt + water

The cation of salt is derived from base and its anion from acid .

Answer: D

88. The salt of which of the following weak acid will be the most hydrolyzed?

A. HA, $K_a = 4 \times 10^{-10}$

B. HB, $K_a = 2 \times 10^{-5}$

C. HC, $K_a = 3 \times 10^{-7}$

D. HD, $K_a = 6 \times 10^{-4}$

Solution

The weaker the acid, the more hydrolyzed is the salt because the conjugate base is stronger.

Answer: A

89. Which of the following gives pH > 7 when dissolved in water?

A. $CaCl_2$ B. NH_4Cl C. NaF D. $NaNO_3$

Solution

Strong acid + strong base → Neutral salt (pH = 7)

Strong acid + weak base → Acidic salt salt (pH < 7)

Strong acid + strong base → Basic salt (pH > 7)

Answer: C

90. Which of the following salts affects the pH of an aqueous solution?

A. KCl B. $CaCl_2$ C. NaCl D. NH_4Cl

Solution

Acidic and basic salts affect the pH of water.

Answer: D

91. CH_3COOH is a weak acid, $K_a = 1.8\ 10^{-5}$ NH_3 is a weak base, $K_b = 1.8 \times 10^{-5}$. A 0.10M solution of CH_3COONH_4 would be

A. Strongly acidic

B. Neutral

C. Weakly acidic

D. Weakly basic

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$$\left.\begin{array}{l} K_a(NH_4^+) = 5.6 \times 10^{-10} \\ K_b(CN^-) = 2.04 \times 10^{-5} \end{array}\right\} K_b > K_a$$

Illustrative Examples

86. Sodium carbonate is a basic salt because it is a salt of (ESSLCE)

A. A strong base and a strong acid and the anion of the strong acid can be hydrolyzed.

B. A strong base and a weak acid and the anion of the weak acid can be hydrolyzed

C. A strong acid and a weak base and the anion of the strong acid can be hydrolyzed.

D. A weak acid and weak base and the cation of the weak base can be hydrolyzed

Solution

Sodium hydroxide + carbonic acid → sodium carbonate + water

(strong base) (weak acid) (basic salt)

$Na_2CO_3 \rightarrow 2Na^+ + CO_3^{2-}$

$CO_3^{2-} + H_2O \rightarrow$ No reaction

$CO_3^{2-} + H_2O \rightleftharpoons HCO_3^- + OH^-$

Answer: B

87. What are salts? salts are (ESSLCE)

A. ionic compounds that result from the reaction between metal hydroxide and water

B. ionic compounds formed by complete replacement of ionization hydrogen anions of the acid

C. ionic compounds formed by partial replacement of ionization hydroxide by metal cations.

D. ionic compounds that derive their cations from a base and their anions from an acid.

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Solution

Acid + Base → salt + water

The cation of salt is derived from base and its anion from acid .

Answer: D

88. The salt of which of the following weak acid will be the most hydrolyzed?

A. HA, $K_a = 4 \times 10^{-10}$

B. HB, $K_a = 2 \times 10^{-5}$

C. HC, $K_a = 3 \times 10^{-7}$

D. HD, $K_a = 6 \times 10^{-4}$

Solution

The weaker the acid, the more hydrolyzed is the salt because the conjugate base is stronger.

Answer: A

89. Which of the following gives pH > 7 when dissolved in water?

A. $CaCl_2$ B. NH_4Cl C. NaF D. $NaNO_3$

Solution

Strong acid + strong base → Neutral salt (pH = 7)

Strong acid + weak base → Acidic salt salt (pH < 7)

Strong acid + strong base → Basic salt (pH > 7)

Answer: C

90. Which of the following salts affects the pH of an aqueous solution?

A. KCl B. $CaCl_2$ C. NaCl D. NH_4Cl

Solution

Acidic and basic salts affect the pH of water.

Answer: D

91. CH_3COOH is a weak acid, $K_a = 1.8\ 10^{-5}$ NH_3 is a weak base, $K_b = 1.8 \times 10^{-5}$. A 0.10M solution of CH_3COONH_4 would be

A. Strongly acidic

B. Neutral

C. Weakly acidic

D. Weakly basic

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Solution

$K_a = K_b$, thus the salt obtained is neutral.

Answer: B

92. Which of the following substances undergoes hydrolysis in aqueous solution?

A. NaCl B. NH_3 C. HCl D. NH_4NO_3

Solution

Substance that undergo hydrolysis are acidic salts, basic salts and neutral salts obtained from weak acid and weak base.

Answer: D

93. Which of the following compounds would be the most basic?

A. 0.10M acetic acid C. 0.10M hydrochloric acid

B. 0.10M sodium acetate D. 0.10M ammonium chloride

Answer: B

94. Classify the following salts as acidic, basic or neutral.

a. $Ba(NO_3)_2$ c. K_2CO_3 e. $Al_2(SO_4)_3$

b. LiCl d. $FeCl_3$ f. NaCN

Solution

➥ a and b are neutral

➥ c and f are basic salts

➥ d and e are acidic salts

95. What is the concentration of hydroxide ion in a 0.05M solution of sodium formate (K_b for formate ion is 5.6×10^{-11})

Solution

$[HCOONa] = [HCOO^-] = 0.05M$

$HCOO^- + H_2O \rightleftharpoons HCOOH + OH^-$

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$$K_b = \frac{[HCOOH][OH^-]}{[HCOO^-]}$$

$$K_b = \frac{[OH^-]^2}{[CH_3COO^-]}, [HCOOH][OH^-]$$

$$[OH^-] = \sqrt{Kb \times [CH_3COO^-]} = \sqrt{5.6 \times 10^{-11} \times 0.05} = 1.67 \times 10^{-6} M$$

96. What is the pH of 0.2M NaF solution? (K_a for HF is 6.8×10^{-4})

Solution

$$[NaF] = [F^-] = 0.2M$$

$$F^- + H_2O \rightleftharpoons HF + OH^-$$

$$K_b = \frac{[HF][OH^-]}{[F^-]}, K_b = \frac{K_w}{K_a} = \frac{1 \times 10^{-14}}{6.8 \times 10^{-4}} = 1.47 \times 10^{-11}$$

$$\Rightarrow [OH^-] = \sqrt{Kb \times [F^-]} = \sqrt{1.47 \times 10^{-11} \times 0.2} = 1.71 \times 10^{-6} M$$

$$pH = 14 + \log[OH^-] = 14 + \log 1.71 \times 10^{-6} = 8.23$$

7.6 Acid - Base Indicators

Indicators are weak organic acids or weak organic bases that show color change in acidic, basic or neutral solution.

- They do not affect the pH of a solution.
- They are represented by two forms, acidic form with a general form, HIn and base form with a formula of In-

$$HIn + H_2O \rightleftharpoons In^- + H_3O^+$$

Acid color Bases color

- In acidic medium, the equilibrium shifts to the left and the acid color is visible.
- In basic medium, the equilibrium shift to the right and base color is visible.

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Indicators are used in the laboratory to estimate the pH of a solution and to show the end point of a titration.

➥ The equilibrium constant (K_{In}) for this dissociation of HIn is:

$$K_{In} = \frac{[In^-][H_3O^+]}{[HIn]}$$

$$\frac{K_{In}}{[H_3O^+]} = \frac{[In^-]}{[HIn]}$$

Example

To illustrate how an indicator works, let us consider chlorophenol red. The acid form (HIn) of this inicator is yellow and the base form (In^-) is red. ($K_{In} = 1 \times 10^{-6}$). Identify the color of the indicator visible at pH

a. 3 b. 6 c. 8

Solution

If the acid form dominates the base form, the acid color will be visible and vice - versa

a) $[H_3O^+] = 10^{-PH} = 10^{-3}$

$$\frac{K_{In}}{[H_3O^+]} = \frac{[In^-]}{[HIn]}$$

$$\frac{10^{-6}}{10^{-3}} = \frac{[In^-]}{[HIn]}$$

$\Rightarrow [HIn] = 1000\,[In^-]$, yellow color is visible

b) $[H_3O^+] = 10^{-PH} = 10^{-6}$

$$\frac{[In^-]}{[HIn]} = \frac{10^{-6}}{10^{-6}} \Rightarrow [HIn] = [In^-]$$

$\therefore$ Neither yellow norred color is visible.

The mixture of the two color is observede

c) $[H_3O^+] = 10^{-PH} = 10^{-8}$

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$$\frac{[In^-]}{[H_3O^+]} = \frac{10^{-6}}{10^{-8}} \Rightarrow 100[HIn] = [In^-]$$

$\therefore$ Red color is visble

Table 7.6 Some common Indicators

Indicator	Acid color	Base color	pH range of color change
Methyl violet	Yellow	Violet	0.0 - 1.6
Methyl orange	Red	Yellow	3.2 - 4.4
Bromocresol green	Yellow	Blue	3.8 - 5.4
Methyl resd	Red	Yellow	4.8 - 6.0
Litmus	Red	Blue	5.0 - 8.0
Bromothyl bule	Yellow	Blue	6.0 - 7.6
Themol blue	Yellow	Blue	8.0 - 9.6
Phymolphthalien	Yellow	Pink	8.2 - 10.0
Thymolphthalien	Colorless	Blue	9.4 - 10.6
Alizarin yellow R	Colorless	Red	10.1 - 12.0

Illustrative Examples

97. The acid - base indicator bromocresol green is a weak acid. The yellow acid and blue base forms of the indicator are present in equal concentration in a solution when the pH is 4.68. What is the pK_a of bromoaresol green?

A. 4.48 B. 4.68 C. 5.68 D. 6.68

Solution

$$pH\ pKa + 10g \frac{[In^-]}{[HIn]}, [HIn] = [In^-]$$

$$\Rightarrow pH = pK_a = 4.68$$

Answer: B

98. Which of the statement is true?

A. A universal indicator is a mixture of indicators that will give a

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different color for different pH.

B. Phenolphalien is a universal indicator

C. A universal indicator can only be used in either strongly acidic or basic solution

D. The color of a universal indicator is red in a weak acid.

Solution

A universal indicator gives different color for different pH.

Answer: A

99. The indicator bromthymol blue (HBb) is a weak acid whose ionization can be represented as follows: $HBb\,(aq) \rightleftharpoons H^+(aq) + Bb^-(aq)$

Which way will the equilibrium shift when NaOH is added?

A. To the left B. To the right

C. Initially to the left and a while to the right

D. Initially to the right and while to the left

Solution

When a base is added, the equilibrium shift to the right and the base color is visible

Answer: B

100. The acid - base indicator methyl red has a Ka of 1×10^{-5} Its acidic form is red while its alkaline form is yellow. If methyl red is added to a solution with a pH = 7, the color will be:

A. Red B. Pink C. Orange D. Yellow

Solution

$$[H_3O^+] = 10^{-pH} = 10^{-7}$$

$$\frac{[In^-]}{[HIn]} = \frac{10^{-5}}{10^{-7}} \Rightarrow [In^-] = 100[HIn]$$

∴ yellow color is visible

Answer: D

101. Write the applications of acid - base indicators

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Solution

- To determine end point in acid - base titration
- To identify acids and bases.
- For quality control.

7.7 Equivalent of Acids and Bases

$$\text{Normality} = \frac{\text{Number of equivalents of solution}}{\text{Volume of solution in liter}}$$

$$\boxed{N = \frac{No.eq}{V}}$$

$$\boxed{No.eq = \frac{m}{E}}, m \text{ -given mass}$$

E-equivalenet mass

$$\boxed{E = \frac{M}{Z}}, M \text{ - molar mass}$$

Z- number of H^+ for acid

- Number of OH^- for base
- total positive change for salt
- Electrons lost for reducing agent
- Electrons gained for oxidization agent

$N = \frac{No.eq}{V}$, $\boxed{No.\ eq = N \times V}$

$N = \frac{m}{E \times V}$, $\boxed{No\ eq = \frac{m}{E}}$

$N = \frac{m \times Z}{M \times V}$, $\boxed{No\ eq = \frac{M \times Z}{M}}$

$N = \frac{n \times Z}{V}$, $n = \frac{m}{M}$, $\boxed{No\ eq = n \times Z}$

$\boxed{N = M \times Z}$, M - molarity

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Example

Calculate the number of equivalents of 2mol of H_3PO_4 if it ionizes to give

a. HPO^{2-}_4 b. $H_2PO^{-}_4$ c. PO^{3-}_4

Solution

a. $H_3PO_4 \longrightarrow HPO_4^{2-} + 2H^+$

$Z = 2eq/mol$

$$\text{No eq} = n \times Z = 2mol \times \frac{2eq}{mol} = 4eq$$

b. $H_3PO_4 \longrightarrow H_2PO_4^- + H^+$

$Z = 1eq/mol$

$$\text{No eq} = n \times Z = 2mol \times \frac{eq}{mol} = 2eq$$

c. $H_3PO_4 \longrightarrow PO_4^{3-} + 3H^+$

$Z = 3eq/mol$

$$\text{No eq} = n \times Z = 2mol \times \frac{3eq}{mol} = 6eq$$

Note: *Always one equivalent of acid reacts with one equivalent of base.*

$(No . eq)\,acid = (No, eq)\,base$

$$\boxed{N_a \times V_a = N_b \times V_b}$$

a - acid

b - base

Example

Calculate the volume of 0.5N H_2SO_4 that reacts completely with 20ml of 0.1N NaOH solution.

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Solution

$$N_a = 0.5N,\ V_a = ?,\ N_b = 0.1N,\ V_b = 20ml$$

$$N_a \times V_a = N_b \times V_b$$

$$V_a = \frac{N_b \times V_b}{N_a} = \frac{20ml \times 0.1N}{0.5N} = 4mL$$

Illustrative Examples

102. The normality of a solution obtained by dissolving 0.2mol H_2SO_4 in 500ml of solution is (Assume complete ionization)

A. 0.4N B. 0.2N C. 0.1N D. 0.8N

Solution

$$N = \frac{n \times Z}{V}$$

$$H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$$

$$Z = \frac{2eq}{mol}$$

$$V = 500mol = 0.5L$$

$$N = \frac{0.2mol \times 2\ eq/mol}{0.5L} = 0.8N$$

Answer: D

103. What volume of 0.4N HCl reacts completely with 100ml of 0.5 NKOH solution?

Solution

$$V_a = \frac{N_b \times V_b}{N_a} = \frac{0.5N \times 100ml}{0.4N} = 125ml$$

104. Calculate the equivalent mass of the following compounds.

a. H_2SO_4 b. $Ca(OH)_2$ c. $Al_2(SO_4)_3$

Solution

a. $H_2SO_4 \longrightarrow 2H^+ + SO_4^{2-}$, $Z = 2eq/mol$

$M = 98glmol$

$$E = \frac{M}{Z} = \frac{98g/mol}{2eq/mol} = 49/eq$$

b. $Ca(OH)_2 \longrightarrow Ca^{2+}(aq) + 2OH^-(aq)$

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$$E = \frac{M}{Z} = \frac{74\,g/mol}{2eq/mol} = 37g/eq$$

c. $Al_2(SO_4)_3 \longrightarrow 2Al^{3+} + 3SO_4^{2-}$, $Z = 2 \times (+3) = \frac{6eq}{mol}$

$M = 342\,g/mol$

$$E = \frac{M}{Z} = \frac{342\ g/mol}{6eq/mol} = 57\ g/eq$$

105. Determine the number of equivalents of 1.5N NaCl in 1.5 dm^3 solution

Solution

$V = 1.5dm^3 = 1.5L$

$$No.eq = N \times V = 1.5\frac{eq}{L} \times 1.5L = 2.25eq$$

7.8 Acid - Base Titrations

Titration is a method of determining the concentration of a solution by monitoring relative amounts during its reaction with a known concentration.

- The known concentration is known as **titrant** and is placed in **burette**.
- The unknown concentration is called **analyte** and is placed in erlenmeyer flask.

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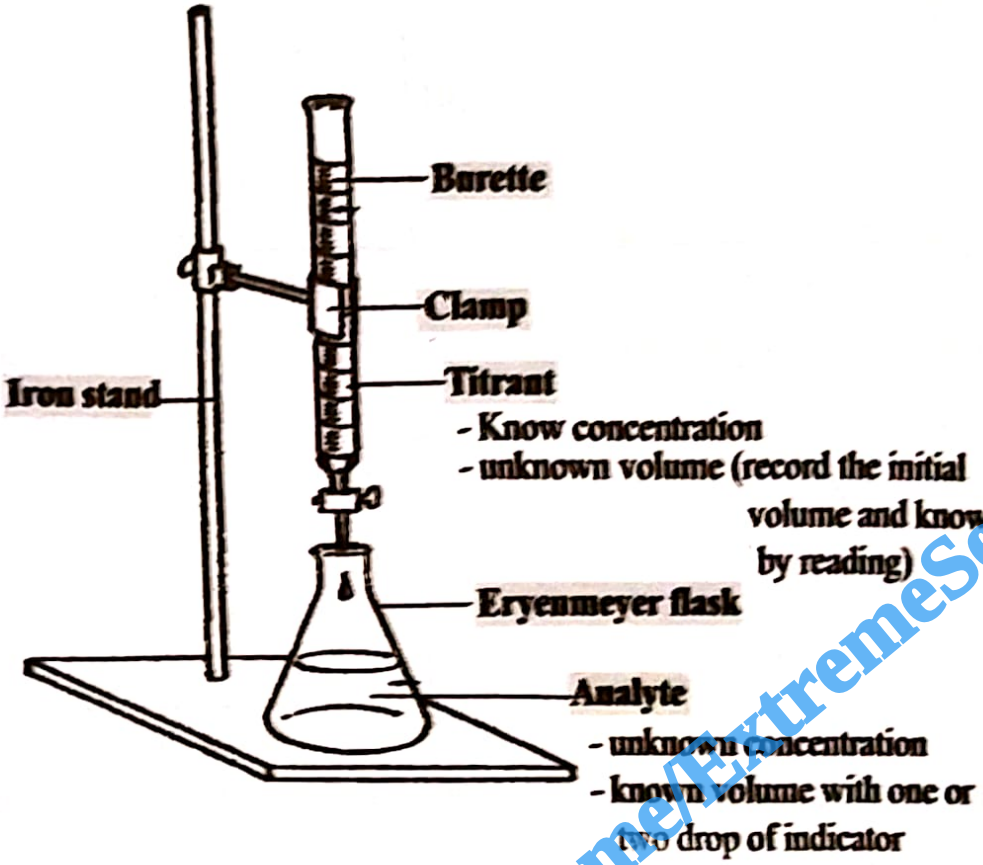

Fig 7.2 Titration set up.

Equivalence point is the point at which both the acid and base have been consumed.

- It is the point at which the number of moles of OH^- ions added to a solution is equal to the number moles of H^+ ion originally present.

End point is the point at which the indicator changes color permanently.

Titration curve is the graph of pH versus volume of titrant (the solution in the burette)

Table 7.7 Types of titrations and pH the equivalent point

Titration	pH at the equivalence point
Strong acid - strong base	pH = 7
Weak acid - strong base	pH > 7
Weak base - strong acid	pH < 7

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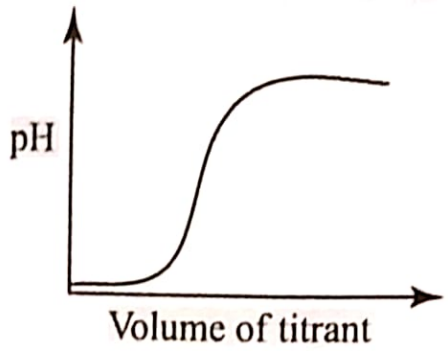
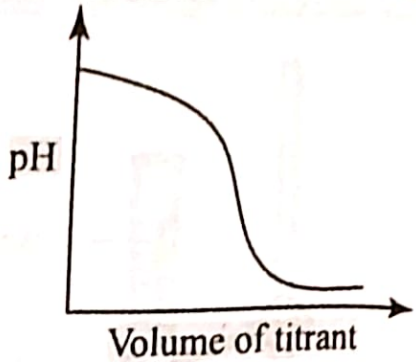
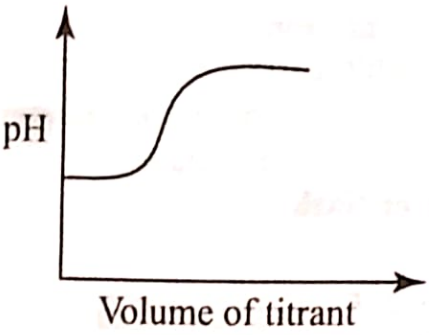


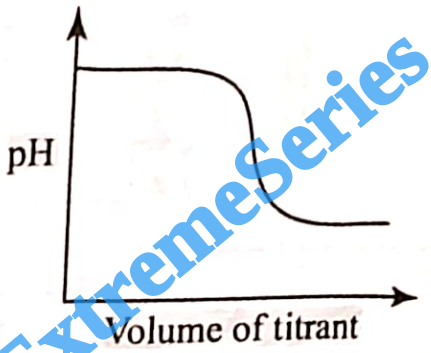
Strong acid - strong base titration



Strong base- strong acid titration



weak acid - strong base titration



weak base - strong base titration

Fig 7.3 Acid - base titration curves.

Illustrative Examples

106. The point in a titration at which the number of moles of OH^- ions added to a solution is equal to the number of moles of H^+ ions originally present is (ESSLE)

A. Equivalence point
C. End point
B. Triple point
D. Zero point

Answer: A

107. Which of the following statements is true for an acid - base titration? (ESSLCE)

A. Acid - base titration cannot be carried out in the absence of indicators.

B. Indicators caralyze an acid - base titration to show the inflection point of the titration curve

C. An acid - base titration is a method of qualitative analysis for determining the concentration of an acid or base.

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D. An acid - base titration is a method quantitative anylsis for determining the concentration of an acid or a base

Solution

➥ Acid - base titration is important to determine unknown concentration from known concentration.

Answer: D

108. Which of the following titration will have an equivalence point at a pH < 7?

A. Strong acid with weak base
C. Strong with strong base
B. Weak acid with weak base
D. Weak acid with strong base

Answer: A

109. The point at which the indicator changes color in acid - base titration is _____

Answer: End point

110. In an acid - base titration why is at the equivalent point as follows.

a. pH = 7 for strong acid - strong base titration
b. pH < 7 for weak base - strong acid titration
c. pH > 7 for weak acid - strong bases

Solution

a. Because of a neutral salt formed
b. Because an acidic salt is formed
c. Because a basic salt is formed.

Solved problems

111. Soluble bases are called _____ .

Answer: Alkalis

112. At all conditions for pure water, what is true?

Solution

$[H_3O^+] = [OH^-]$

113. What is the pH of 1.0×10^{-3} MNaOH solution?

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Solution

$[NaOH] = 1.0 \times 10^{-3}M = [OH^-]$

$pH = 14 + \log [OH^-] = 14 + 10g\ 10^{-3} = 11.0$

114. An aqueous solution of window cleaner has $[H_3O^+] = 1.8 \times 10^{-10}$ M at 25°c. Calculate $[OH^-]$.

Solution

$$[OH^-] = \frac{K_w}{[H_3O^+]} = \frac{1 \times 10^{-14}M^2}{1.8 \times 10^{-10}M} = 5.56 \times 10^{-5}\ M$$

115. Determine the conjugate acid and base of HCO^-_3.

Solution

$CA = base + H^+ = HCO^-_3 + H^+ = H_2CO_3$

$CB = acid - H^+ = HCO^-_3 - H^+ = CO^{2-}_3$

116. Write the two methods used to measure pH

Solution

pH meter and Acid - base indicators.

117. Calculate the pH of water at 40°c, given that K_w is 3.8×10^{-14}

Solution

$$[H^+] = \sqrt{Kw} = \sqrt{3.8 \times 10^{-14}} = 1.95 \times 10^{-7}M$$

$$pH = -\log[H^+] = -10g\ 1.95 \times 10^{-7} = 6.71$$

118. Arrange the following 0.1M solutions in order of increase pH.

a. NaCl
b. NH_4Cl
c. CH_3COONa
d. HCl
e. NaOH

Solution

d < b < a < c < e

119. What is the pH of 1×10^{-8}M HCl dilute solution

Solution

For very small amounts of acid or base, the self - ionization of wate

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must be considered. The acid or base becomes source of common ion

$[HCl] = [H^+] = 1 \times 10^{-8}\ M$

$$H_2O \rightleftharpoons H^+ + OH$$

$$K_w = x(x + 10^{-8}) = 10^{-14} \Rightarrow x^2 + 10^{-8}\,x - 10^{-14} = 0$$

Solving for x, $x = 9.5 \times 10^{-8}\ M$

$$[H^+] = x + 10^{-8} = 9.5 \times 10^{-8} + 10^{-8} = 1.05 \times 10^{-7}\ M$$

$$pH = -\log 1.05 \times 10^{-7} = 6.98$$

120. Arrange the following bases in increasing order of base strength

a. BrO^-_3, ClO^-_3, IO^-_3

b. OH^-, H^- Cl^-

c. HSO^-_4, $H_2PO^-_4$, ClO^-_4

d. NH^-_2, HS^-, PH^-_2

Solution

a. $ClO^-_3 < BrO^-_3 < IO^-_3$

b. $Cl^- < OH^- < H^-$

c. $ClO^-_4 < HSO^-_4 < H_2PO^-_4$

d. $HS^- < PH^-_2 < NH^-_2$

121. What is the difference between monoprotic acid and polyprotic acid?

Solution

Monoprotic acid have one ionizable hydrogen

Example: HCl, HBr, HI, HNO_3 etc

Polyprotic acids have more than one ionizable hydrogen

Example: H_2SO_4, H_2S, H_3PO_4 etc

122. Write the equations for the ionization of phospheric acid (H_3PO_4)

Solution

$$H_3PO_4 \rightleftharpoons H^+ + H_2PO^-_4,\ Ka_1 = 7.5 \times 10^{-3}$$

$$H_2PO^-_4 \rightleftharpoons H^+ + HPO^{2-}_4,\ Ka_2 = 6.2 \times 10^{-8}$$

$$HPO^{2-}_4 \rightleftharpoons H^+ + PO^{3-}_4,\ Ka_3 = 7.5 \times 10^{-13}$$

123. What is $[NH^+_4]$ in a solution that is 0.2M NH_3 and 0.1M NaOH? ($K_b = 1.8 \times 10^{-5}$)

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Solution

$$[NaOH] = [OH^-] = 0.1M$$

$$K_b = \frac{[NH_4^+]}{[NH_3]}$$

$$[NH_4^+] = K_b \times \frac{[NH_3]}{[OH^-]} = 1.8 \times 10^{-5} \left(\frac{0.2}{0.1}\right) = 3.6 \times 10^{-5} M$$

124. For which type of salt, changing concentration does not change the pH of the solutions?

Answer: Neutral salt

125. Calculate the normality of 100ml of H_2SO_4 required to neutralize 80ml of 0.5 N KOH Solution.

Solution

$$N_a \times V_a = N_b \times V_b$$

$$N_a = \frac{N_b \times V_b}{V_a} = \frac{0.5N \times 80ml}{100} = 0.40N$$

126. What is the difference between strong acid and concentrated acid?

Solution

- Strong acid ionizes almost completely in water .
- Concentrated acid contains relative more acid than water.

127. Calculate the pH of 0.02M CH_3NH_3Cl solution. ($K_b = 5.2 \times 10^{-4}$)

Solution

$$K_a = \frac{K_w}{K_b} = \frac{[CH_3NH_2][H_3O^+]}{[CH_3N^+H_3Cl]}, [H_3O^+] = 1.96 \times 10^{-6} M$$

$$pH = -\log [H_3O^+] = -\log 1.96 \times 10^{-6} = 5.71$$

128. What mass of NH_4Cl must be added to 500ml of 0.15M NH_3 to prepare a buffer with a pH of 9.2? ($K_b = 1.8 \times 10^{-5}$)

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Solution

$pOH = 14 - pH = 14 - 9.2 = 4.8$

$\Rightarrow [OH^-] = 10^{-POH} = 10^{-4.8} = 1.585 \times 10^{-5} M$

$[NH_4Cl] = [NH_4^+] = Kb \times \frac{[NH_3]}{[OH^-]} = 1.8 \times 10^{-5} \times \frac{0.15M}{1.585 \times 10^{-5}}$

$[NH_4Cl] = 0.17M$; $V = 500ml = 0.5L$

$nNH_4Cl = M \times V = 0.17 \frac{mol}{L} \times 0.5L = 0.085mol$

$m = n \times M = 0.085 mol \times 53.5 \frac{g}{mol} = 4.55g$

129. The amount of an indicator used in acid - base titration must be small, why?

Solution

If the amount of the indicator is large, it affects the volume of analyte.

130. A 0.10M solution of a weak acid, HX is 0.059% ionized. What is the dissociation constant for the acid?

Answer: $Ka = 3.5 \times 10^{-8}$

131. What is the percent of dissociation in 1M of HF containing $[H^+] = 2.7 \times 10^{-2} M$?

Answer: 2.7

132. What is the factor that affects K_a and K_b?

Answer: Temperature

133. Given, $HF + H_2O \xrightarrow{K_a} H_3O^+ + F^-$

$F^- + H_2O \xrightarrow{K_b} HF + OH^-$

Explain the relation ship between Ka, Kb and Kw

Answer: Kw = Ka × Kb

134. If $K_a = 10^{-5}$ for a weak acid then what is the pKd for its conjugate base?

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Solution

$$K_b = \frac{K_w}{K_a} = \frac{1 \times 10^{-14}}{1 \times 10^{-5}} = 1 \times 10^{-9}$$

$$pK_b = -\log K_b = -\log 1 \times 10^{-9} = 9$$

135. The pH of a solution is 4. What is the concentration of hydroxide ion?

Solution

$$[OH^-] = 10^{PH-14} = 10^{4-14} = 1 \times 10^{-10} M$$

136. A certain butter solution contains equal concentrations of weak acid and its conjugate base. The K_b of the conjugate base is 1×10^{-10}. What is the pH of the buffer?

Solution

$$K_a = \frac{K_w}{K_b} = \frac{1 \times 10^{-14}}{1 \times 10^{-10}} = 1 \times 10^{-4}$$

If [acid] = [conjugate base], then pH = pKa

$$\Rightarrow pH = -\log K_a = -\log 10^{-4} = 4$$

137. 100ml of 0.2M HCl solution is added to 100ml of distilled water. The pH of the resulting solution is _____

Solution

$$[HCl] = [H^+] = \frac{M \times VHCl}{V_{HCl} + VH_2O} = \frac{0.2M \times 100ml}{(100 + 100)ml}$$

$$[HCl] = [H^+] = 0.1M = 01M^{-1}$$

$$pH = -\log[H^+] = -\log 10^{-1} = 1.0$$

138. When 4 grams of NaOH is dissolved in 10L of solution, what is the pOH?

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Solution

$$[NaOH] = [OH^-] = \frac{m}{M \times V} = \frac{4g}{40\frac{g}{mol}} = 0.01M$$

$$pOH = -\log[OH^-] = -\log 10^{-2} = 2$$

139. An acid solution of pH 6 is diluted thousand times. The pH of solution becomes approximately _____

Solution

$$[H^+] = 10^{-pH} = 10^{-6}M$$

Diluted 1000 times means: $[H^+] = 10^{-6}\ M \times \frac{1}{1000}$

$$[H^+] = 1 \times 10^{-9}\ M$$

$$pH = -\log[H^+] = -\log 10^{-9} = 9$$

140. One liter of water contains 10^{-7} moles of H^+ ions percent of ionization of water is _____

Solution

$$[H^+] = \frac{1 \times 10^{-7}}{1L} = 1 \times 10^{-7}M$$

$$[H_2O] = \frac{\rho}{M} = \frac{1000g/L}{189/mol} = 55.6M$$

$$\%H_2O\ \frac{[H^+]}{[H_2O]} \times 100 = \frac{1 \times 10^{-7}M}{55.6M} \times 100$$

$$\%H_2O = 1.8 \times 10^{-7}\%$$

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Supplementary Exercise

I. Multiple choice item

1. Which of the following is not amphiprotic species?

   A. HSO^-_3 B. $H_2PO^-_4$ C. CH_3COO^- D. HCO^-_3

2. Which one is the strongest Bronsted - Lowry base out of the following?

   A. ClO^-_4 B. ClO^- C. ClO^-_3 D. ClO^-_2

3. The conjugate acid of HSO^-_3 is

   A. H_2SO_3 B. H_2SO_4 C. SO^{2-}_3 D. SO^{2-}_4

4. The conjugate base H_3PO_3 is

   A. $H_2PO^-_3$ B. $H_2PO^-_4$ C. H_3P D. H_2P^-

5. In the reaction, $AlCl_3 + Cl^- \rightarrow AlCl^-_4$, $AlCl_3$ acts as a

   A. Salt C. Lewis acid

   B. Lewis base D. Bronsted base

6. Which of the following compounds does not act as Bronsted - Lowry base?

   A. KOH B. OH^- C. NH_3 D. HS^-

7. In the reaction: $CN^-(aq) + H_2O(1) \rightleftharpoons HCN^+(aq) + OH^-(aq)$ The conjugate acid base pairs are

   A. CN^-/OH and H_2O/HCN C. CN^-/HCN and H_2O/OH^-

   B. HCN/H_2O and CN^-/OH^- D. CN^-/H_2O and HCN/OH

8. Consider the following equilibria

   I. $HS^- + H_2O \rightleftharpoons H_2S + OH^-$

   II. $NH^+_4 + H_2O \rightleftharpoons H_3O^+ + NH_3$

   III. $HCO^-_3 + H_2O \rightleftharpoons H_3O^+ + CO^{2-}_3$

   In which of the equilibrium (equilibria) does/ do water act as a Broned - Lowry base?

   A. I only B. II only C. II and III D. I,II,II

9. Which of the following solutions is basic?

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A. $[H^+] = 5 \times 10^{-3}M$ C. $[OH^-] = 2.0 \times 10^{-8}M$

B. $[OH^-] = 1 \times 10^{-7}\ M$ D. $[H^+] = 5 \times 10^{-8}M$

10. The pH of a solution is increased from 3 to 6, its hydrogen ion concentration will be

A. Reduced to half C. Doubled

B. Reduced by 1000 D. Increased by 1000

11. Given the reaction: $B(OH)_3 + H_2O \rightleftharpoons B(OH)_4^- + H^+$, the Lewis base

A. $B(OH)_3$ B. H_2O C. $B(OH)^-_4$ D. H^+

12. Which of the following salts does not undergo hydrolysis?

A. MgF_2 B. NH_4NO_3 C. $NaClO_4$ D. HCOOK

13. A solution with pOH > 7 is

A. Acidic B. Basic C. Neutral D. None

14. Which one of the following represents Bronsted - Lowry acid - base reaction?

A. $NH_3 + H^+ \rightarrow NH^+_4$

B. $H_2O + H_2O \rightarrow H_3O^+ + OH^-$

C. $[Al(H_2O)_6]^{3+} \rightarrow [Al(H_2O)_5 OH]^{2+} + H^+$

D. $Cu^{2+} + 4NH_3 \rightarrow [Cu(NH_3)_4]^{2+}$

15. What is the concentration of HNO_3 acid with pH of 1?

A. 0.3M B. 0.2M C. 0.01M D. 0.1M

16. Which of the following is the most acidic?

A. $HClO_4$ B. H_3PO_4 C. H_2SO_4 D. $HBrO_4$

17. All of the following are weak bases. EXCEPT

A. $Sr(OH)_2$ B. $Cu(OH)_2$ C. $Mg(OH)_2$ D. $Fe(OH)_2$

18. The color of phenolphthalien in basic solution is

A. Colorless B. Pink C. yellow D. Blue

19. Which of following is the strongest Lewis base?

A. BH_3 B. H_2O C. CH_4 D. NH_3

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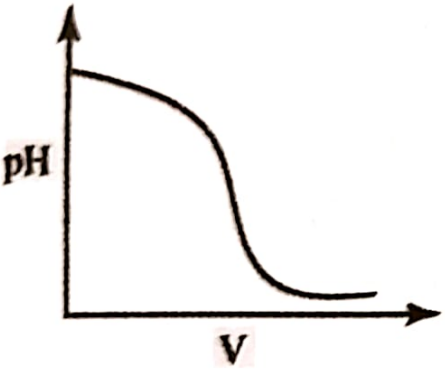
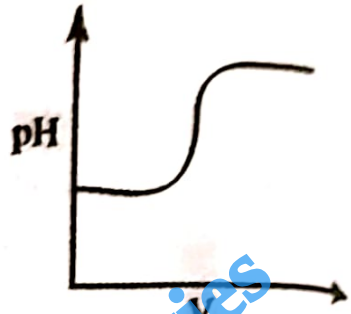
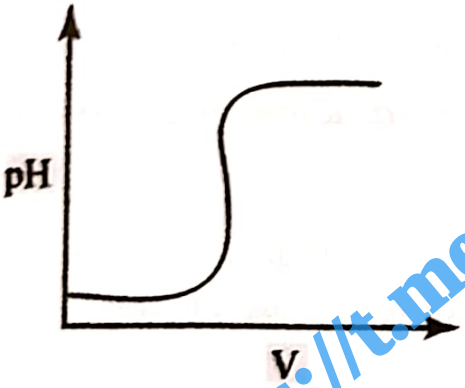
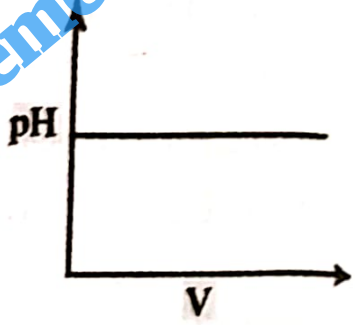
20. The graphs shown below are titration curues A weak base is initially in flask and a strong acid is added from the burette. Which graph below best represents the nature of the titration?

A. 

C. 

B. 

D. 

21. The pOH of 10^{-4} MHl solution is:

A. 4 B. 10 C. 14 D. 8

22. Which of the following equimolar solution can acts as acidic buffer?

A. NH_4Cl/NH_4OH C. HCl/NaCl

B. HCOOH/HCOOK D. $HNO_3/NaNO_3$

23. Which of the following aqueous solutions will have highest pH?

A. Sodium formate C. Ammonium nitrate

B. Calcium chloride D. Potassium chloride

24. The ion product of water changes when

A. An acid is added to it C. A salt is added to it

B. A base is added to it D. Temperature is raised

25. Which of the following will have nearly equal hydrogen ion concentration?

I. 100ml of 0.1 MHCl mixed with 50 ml water

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II. 50ml of 0.1M H_2SO_4 mixed with 50ml water

III. 50ml of 0.1M H_2SO_4 mixed with 100ml water

IV. 50ml of 0.1M HCl mixed with 50ml water

A. I and II B. I and III C. II and III D. III and IV

26. Which has highest degree of ionization?

A. 1M CH_3COOH C. 0.01M CH_3COOH

B. 0.1M CH_3COOH D. 0.001M CH_3COOH

27. A solution has a pH of 9. The solution is best described as:

A. Strongly alkaline C. Strongly acidic

B. Weakly alkaline D. Weakly acidic

28. The point at which a titration is complete is called the

A. End point C. Calibrated point

B. Equilibrium point D. Chemical point

29. A buffer solution

A. Changes pH rapidly with the addition of an acid.

B. Does not change pH at all

C. Resists change in pH

D. Chang pH only with the addition of a strong base

30. The common ionic reaction of an acid with a base involves ions of

A. Hydrogen and hydroxide C. Hydrogen and hydronium

B. Sodium and chloride D. Hydroxide and nitrate.

II. Short answer items

31. If 10ml of 1MHCl was required to titrate a 20ml NaOH solution of unknown concentration to its end point, what was the concentration of the NaOH?

32. The hydronium ion is represented by the formula _____

33. Determiner conjugate base for

a. H_2Se b. $N_2H^+_5$ c. $HC_2O^-_4$

34. Write the conjugate acid of

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a. PH_3 b. Br^- c. S^{2-}

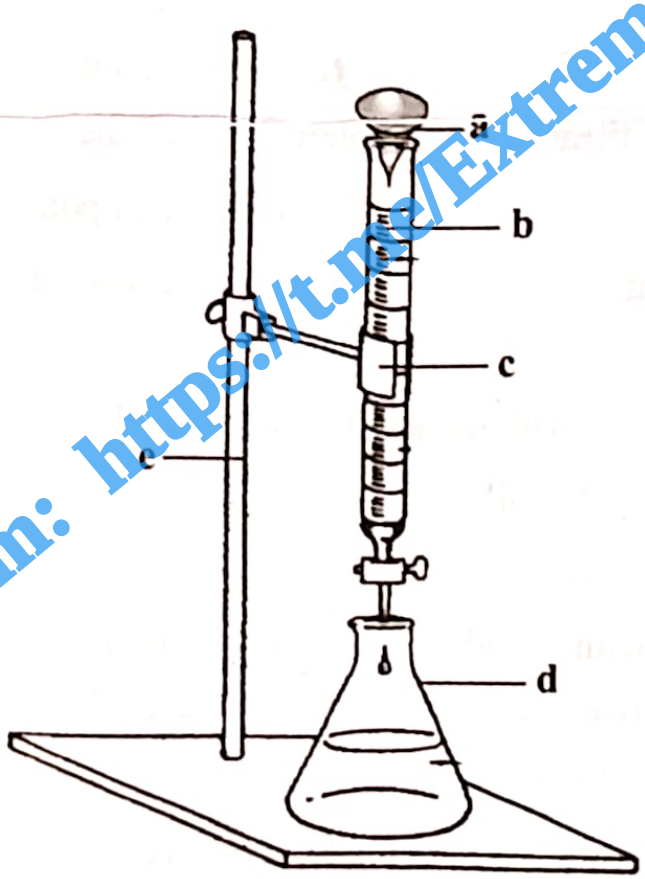
35. A more generalizer acid - base concept is that of _____

36. The pH of human arterial blood is 7.40 what is the pOH of the blood

37. What is the effect of adding sodiumnitriet, $NaNO_2$ to a 0.5M solution of nitrous acid, HNO_2? Explan the effect on:
    a. The dissociation of HNO_2
    b. The concentration of hydrogen ion
    c. The pH value of the solution

38. What are the two components of a buffer solution?

39. Write the name of the apparatus used in the acid - base titration satup below



40. Write the methods used to measure the strength of acids and the in aqueous solution

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Answer to Supplementary exercise

Part I

1. C	6. A	11. B	16. A	21. B	26. D
2. B	7. C	12. C	17. A	22. B	27. B
3. A	8. C	13. A	18. B	23. A	28. A
4. A	9. D	14. B	19. D	24. D	29. C
5. C	10. B	15. D	20. C	25. A	30. A

Part II

31. 0.5M

32. H_3O^+

33. a. HSe^- b. N_2H_5 c. $C_2O^{2-}_4$

34. a. PH^+_4 b. HBr c. HS^-

35. Lewis concept of acid and base

36. 6.60

37. a. Decreases b. Decrease c. Increases

38. Weak acid and its conjugate base or weak base and its conjugate acid.

39. a. Funnel b. Burette c. Clamp
    d. Conical flask e. Stand

40.
- H^+ and OH^- concentration
- pH and pOH
- Percent of ionization
- Dissociation or ionization constant

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UNIT 8

Electro chemistry

Introduction

The branch of chemistry that deals with the interconversion between electrical and chemical energy is known **electro chemistry.** The apparatus where the energy conversions take place is known as **electrochemical cell.** There are two types of electro chemical cells. They are **electrolytic and galvanic cells.**

In electrolytic cell, electrical energy is converted into chemical energy **but in galvanic cell,** chemical energy is converted into electrical energy.

The practical applications of electrochemistry are countless, ranging from batteries to the manufacture of key chemicals, refining of metals and preventing corrosion.

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8.1 Oxidation - Reduction Reaction (Redox Reaction)

Oxidation Number or Oxidation State

It is the charge the atom would have if electrons were transferred completely.

In may be negative, zero or positive integers and rarely fractions

Rule to Determine Oxidation Number

1. The oxidation number of free atoms or homo atomic molecules is zero.

   Example: Na, S, O_2, O_3, Cl_2: ox, oxidation number = 0

2. Oxidation number of ions is the charge of the ions.

   Example

   $Na^+ = +1$ $\quad Cl^- = -1$

   $Cu^{2+} = +2$ $\quad S^{2-} = -2$

   $Al^{3+} = +3$ $\quad SO_4^{2-} = -2$

   $NH_4^+ = +1$ $\quad PO_4^{3-} = -3$

3. Hydrogen in compounds is +1 except in metal hydrides (CaH_2, NaH, KH) which is -1
4. Oxygen in most compound is -2, in peroxides -1in super oxides -1/2 and in compound with fluorine (OF_2) is +2.
5. Alkali metals (Li, Na, K, Rb, Cs) in a compound is +1
6. Alkaline earth metals (Mg, Ca, Sr, Ba) in a compound is +2.
7. Fluorine all compounds is -1
8. Halogens in metal halides is -1
9. The algebraic sum of the oxidation number of elements in a neutral compound is zero.

   Example: Calculate the oxidation number of Mn in $KMnO_4$.

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$K^{+1}\overset{x}{M}nO_4^{-2}$

$1 + x + 4(-2) = 0$

$x = +7$

Thus, the oxidation number of Mn in $KMnO_4$ is $+7$

10. The sum of oxidation number of elements in polyatomic ion is equal to the charge on the ion

Example: Find the Oxidation number of Cr in $Cr_2O_7^{2-}$

$\overset{x}{C}r_2O_7^{2-}$

$2x + 7(-2) = -2$

$x = +6$, Thus the oxidation state of Cr in $Cr_2O_7^{2-}$ is $+6$

Illustrative Examples

1. The oxidation number of nitrogen in NH_3, HNO_3 and NO_2 are respectively.

A. -3, -5 + 4 B. +3, +5,+4 C. -3, +5, +4 D. -3, +4, -4

Solution

$\overset{x}{N}\overset{-1}{H_3}$: $x + 3(+1) = 0 \Rightarrow x = -3$

$H\overset{x}{N}O_3$; $1 + x + 3(-2) = 0 \Rightarrow x = +5$

$\overset{x}{N}O_2$; $x + 2(-2) = 0 \Rightarrow x = +4$

Answer: C

2. The oxidation number of Na, S^{2-} and NH_4^+ respectively are

A. 0, -2, +1 B. +1, -2, 0 C. -2, + 1, 0 D. +1, 0, -2

Answer: A

3. The oxidation number of chlorine in ClO_4^- is:

A. -1 B. +3 C. +5 D. +7

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Solution

$$ClO_4^- : x + 4(+2) = -1, \Rightarrow x = +7$$

Answer: D

4. Describe the word electro chemistry

Solution

Electro - chemistry contains two words electro - and chemistry electro - means electricity and chemistry - reaction Therefore, electrochemistry deals with how electricity causes chemical reaction and how chemical reactions generate electricity.

5. Calculate the oxidation number of

a. C in $H_2C_2O_4$
b. Cr in Na_2CrO_4
c. N in NH_4^+
d. Fe in $K_4(Fe(CN)_6)$
e. Fe in Fe_3O_4
f. P in $Ca_3(PO_4)_2$

a) +3 b) +6 c) -3 d) +2 e) $+\frac{8}{3}$ f) +5

6. What type of energy conversion takes place in electro chemical cell?

Solution

Chemical to electrical and electrical to chemical energy.

Oxidation and Reduction half - Reactions

1. Oxidation

Early Definition

Oxidation is the addition of oxygen to a substance or removal of hydrogen from a substance. **Example:** $C + O_2 \rightarrow CO_2$

Modern Definition

Oxidation is loss of electron by an atom ion or molecule leading to an increase in oxidation number.

Use "Oil" to remember oxidation easiry. O - oxidation, i - is and l - loss

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Example

$2Cl^- \longrightarrow Cl_2 + 2e$, $\quad Sn^{2+} \longrightarrow Sn^{4+} + 2e$

2. Reduction

Early Definition of Reduction

Reduction is the addition of hydrogen or the removal of oxygen from a substance.

Example $WO_3 + 3H_2 \longrightarrow 3H_2O + W$

Modern Definition of Reduction

Reduction is gain of electrons by an atom or an ion or a molecule leading decreases in oxidation number.

Examples: $Cu^{2+} + 2e \longrightarrow Cu$ $\quad Fe^{3+} + 3e \longrightarrow Fe$

Use "Rig" to remember reduction easily R - reduction, i - is, g - gain

Oxidizing agent is a species being reduced but causes oxidation.

- **It gains electrons**
- **It shows decrease in oxidation number**

Reducing agent: is a species being oxidized but cause reduction.

- **It gains electrons**
- **It shows increase in oxidation number**

Example: Identify the reducing and oxidizing agents in

$$Zn + Cu^{2+} \longrightarrow Zn^{2+} + Cu$$

Reducing agent: Zn because its oxidation number increases from 0 to + 2.

Oxidizing agent: Cu^{2+} because its oxidation number decreases from +2 to 0.

Oxidation - reduction (redox) reaction: is a reaction that involves oxidation and reduction.

Example

$$3H_2S^{-2} + 2HN^{+5}O_3 \longrightarrow 3S^{\circ} + 2N^{+2}O + 4H_2O$$

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✓ Sulphur is oxidized and nitrogen is reduced.

If the same element is oxidized and reduced at the same time, the reaction is known as disproportionation reaction.

Example: $Cl_2 + H_2O \longrightarrow HCl + HClO$

All disproportionation reactions are redox reaction but not the reverse.

Non - redox reactions are reactions that do not involve oxidation and reduction.

Example: Double displacement reactions

$$AgNO_3 + NaCl \longrightarrow AgCl + NaNO_3$$

Illustrative Examples

7. Which of the reactions is redox reaction?

A. $2Al + 6H_3O^+ \longrightarrow 2Al^{3+} + 3H_2 + 6H_2O$

B. $CH_3COOH + OH^- \longrightarrow CH_3COO^- + H_3O^+$

C. $3Fe^{2+} + 2PO_4^{3-} \longrightarrow Fe_3(PO_4)_2$

D. $Mg(OH)_2 + H_3O^+ \longrightarrow Mg^{2+} + 4H_2O$

Solution

➥ Acid - base reactions (B and D) are non - redox reaction.

➥ C is also non - redox reaction.

Answer: A

8. Identify the disproportionation reaction among the following reaction

A. $3OCl^- \longrightarrow ClO_3^- + 2Cl^-$

B. $SO_2 + H_2O \longrightarrow H^+ + HSO_3^-$

C. $NaI + H_3PO_4 \longrightarrow HI + NaH_2PO_4$

D. $2NO_2 \longrightarrow N_2 + 2O_2$

Solution

➥ If the same substance is oxidized and reduced in the given reaction,

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is disproportionation reaction

➥ The oxidation number of Cl in ClO^- is +1, in ClO^-_3 +5 and in Cl^- is -1

➥ Chlorine is oxidized and reduced.

Answer: A

9. For which conversion is an oxidizing agent required?

A. $2H^+ \longrightarrow H_2$

B. $2Br^- \longrightarrow Br_2$

C. $SO_3 \longrightarrow SO^{2-}_4$

D. $MnO_2 \longrightarrow Mn^{2+}$

Solution

Oxidizing agent is required for oxidation of substance.

Answer: B

10. In the following reaction:

$3Cl^- + 4CrO^{2-}_4 + 23H^+ \longrightarrow 3HClO_2 + 4Cr^{3+} + 10H_2O$

What is the behavior of Cl^- ?

A. An acid

B. A base

C. An oxidizing agent

D. A reducing agent

Solution

Cl^- is oxidized because its oxidation number increases from -1 to +3. As a result, it becomes reducing agent.

Answer: D

11. In the reaction: $H_2S + I_2 \longrightarrow S + 2H^+ + 2I^-$

The oxidizing agent is

A. H_2S B. I_2 C. S D. I^-

Answer: B

12. Identify the oxidizing and reducing agents in the following reactions.

a. $2H_2O + Al + MnO^-_4 \longrightarrow Al(OH)^-_4 + MnO_2$

b. $C + 2N_2O \longrightarrow CO_2 + 2N_2$

c. $I_2O_5 + 5CO \longrightarrow I_2 + 5CO_2$

d. $Mg + Fe^{2+} \longrightarrow Mg^{2+} + Fe$

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e. $2KMnO_4 + 10KCl + 6H_2SO_4 \longrightarrow 5K_2SO_4 + 2MnSO_4 + 5Cl_2 + 6H_2O$

Solution

Reducing agent (Oxidized substance)	Oxidizing agent (Reduced substance)
a. Al	MnO^-_4
b. C	N_2O
c. CO	I_2O_5
d. Mg	Fe^{2+}
e. KCl	$KMnO_4$

Balancing Redox Reactions

- ✓ There are two importance methods used to balance redox reactions. These are oxidation number method and Ion - electron method.
- ✓ In a balanced chemical reaction, **atoms, mass** and **charge** are conserved.

1. Oxidation Number Method

To balance the redox reaction with this method, follow the steps below.

1. Write the unbalance chemical reaction if it is given in word equation.
2. Identify the oxidized and reduced elements
3. Separate into oxidation and reduction half reactions
4. Balance the electrons reduced and oxidized
5. Balance the charges of the half - reactions
6. Balance the electrons transferred
7. Write the coefficients of oxidized and reduced substances in their formula
8. Balance the remaining substance by inspection method

Example

Balance the following reaction by oxidation number method.

$H_2S + HNO_3 \longrightarrow S + NO + H_2O$

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Solution

$$\overset{-2}{H_2S} + H\overset{+5}{N}O_3 \longrightarrow \overset{0}{S} + \overset{+2}{N}O + H_2O$$

S is oxidized and N is reduced

$$\overset{2-}{S} \longrightarrow \overset{0}{S} + 2e^-$$

$$\overset{5+}{N} + 3e^- \longrightarrow \overset{2+}{N}$$

Balance the electrons lost and gained by finding their LCM

LCM of 3 and 2 is $6.6 \div 2 = 3$ and $6 \div 3 = 2$, Thus, multiply the first with 3 and the second with 2.

$$(S^{2-} \rightarrow S^0 + 2e) \times 3 \Rightarrow 3S^{2-} \rightarrow 3S^0 + 6e^-$$

$$(N^{5+} + 3e^- \rightarrow N^{2+}) \times 2 = 2N^{5+} + 6e^- \rightarrow 2N^{2+}$$

Coefficient of H_2S and S is 3, HNO_3 and NO is 2. Balance H_2O

With inspection method.

$$3H_2S + 2HNO_3 \longrightarrow 3S + 2NO + 4H_2O \text{ (Balanced)}$$

2. The ion - Electron Method

This method is used to balance redox ionic equation.

Follow the Steps Below

1. Separate into oxidation and reduction half - reactions.
2. Balance the atoms oxidized and reduced if necessary
3. Write the electrons lost and gained
4. Balance charge by adding hydrogen ion (H^+) in acidic medium, hydroxide ion (OH^-) in basic medium if necessary.
5. Balance hydrogen by adding water to the defficient side
6. Balance the electrons transferred
7. Add the half - reactions and check

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Example

1. Balance the following in acidic medium

$$H_2C_2O_4 + MnO_4^- \longrightarrow CO_2 + Mn^{2+}$$

Solution

Step1: $H_2C_2O_4 \longrightarrow CO_2$

$MnO_4^- \longrightarrow Mn^{2+}$

Step2: $H_2C_2O_4 \longrightarrow 2CO_2$

$MnO_4^- \longrightarrow Mn^{2+}$ (Mn is already balanced)

Step3: $H_2\overset{+3}{C_2}O_4 \longrightarrow 2\overset{+4}{C}O_2 + 2e^-$ $(2\times4 - 2\times3 = 2$ (lost))

$\overset{+7}{M}nO_4^- + 5e^- \longrightarrow \overset{2+}{M}n$ $(2 - 7 = -5$(gained))

Step 4: $H_2C_2O_4 \longrightarrow 2CO_2 + 2e^- + 2H^+$ $(0 = -2 + 2)$

$8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^{2+}$ $(8 + (-1 + -5) = +2)$

Step 5: $H_2C_2O_4 \longrightarrow 2CO_2 + 2H^+ + 2e^-$ (H is already balanced)

$8H^+ + MnO_4^- + 5e^- \longrightarrow Mn^{2+} + 4H_2O$ $(8H = (4 \times 2)H)$

Step 6: LCM of 2 and 5 is 10 Multiply the first by 5 and the second by 2

$5H_2C_2O_4 \longrightarrow 10CO_2 + 10H^+ + 10e^-$

$16H^+ + 2MnO_4^- + 10e^- \longrightarrow 2Mn^{2+} + 8H_2O$

Step 7: $5H_2C_2O_4 + (16H^+ - 10H^+) + 2MnO_4^- \longrightarrow 10CO_2 + 8H_2O + 2Mn^{2+}$

$5H_2C_2O_4 + 6H^+ + 2MnO_4^- \longrightarrow 10CO_2 + 8H_2O + 2Mn^{2+}$ (balanced)

2. Balance the equation below in basic medium $Cr_2O_7^{2-} + I^- \longrightarrow Cr^{3+} + I_2$

Solution

Step 1: $Cr_2O_7^{2-} \longrightarrow Cr^{3+}$

$I^- \longrightarrow I_2$

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Step2: $Cr_2O_7^{2-} \longrightarrow 2Cr^{3+}$

$2I^- \longrightarrow I_2$

Step3: $Cr^{+6}{}_2O^{2-}{}_7 + 6e^- \longrightarrow 2Cr^{3+}$ $(2 \times (+3) - 2 \times (+6) = -6)$

$2I^- \longrightarrow I^0{}_2 + 2e^-$ $(0 - (2 \times (-1)) = 2)$

Step4: $Cr_2O^{2-}{}_7 + 6e^- \longrightarrow 2Cr^{3+}$

$2I^- \longrightarrow I_2 + 2e^-$ (charge is already balanced)

Step5: $Cr_2O^{2-}{}_7 + 7H_2O + 6e^- \longrightarrow 2Cr^{3+} + 14H_2O^-$

$2I^- \longrightarrow I_2 + 2e^-$

Step 6: LCM of 6 and 2 is 6. Thus, multiply the first with 1 and the second with 3.

$Cr_2O^{2-}{}_7 + 7H_2O + 6e^- \longrightarrow 2Cr^{3+} + 14OH^-$

$6I^- \longrightarrow 3I_2 + 6e^-$

Step7: $7H^+O + Cr_2O^{2-}{}_7 + 6I^- \longrightarrow 2Cr^{3+} + 14OH^- + 3I_2$ (balanced)

Finding LCM of electron lost and gained is important to get simplest coefficient ratio of reactants and products.

Illustrative Examples

13. Balanced the following equation by providing the missing coefficients (left to right) $NH_3 + O_3 \longrightarrow NO + H_2O$

A. 5, 4, 5, 6 B. 6, 5, 4, 4 C. 4, 5, 4, 6 D. 5, 5, 4, 6

Solution

$\overset{-3}{N}H_3 + \overset{0}{O}_2 \longrightarrow \overset{+2}{N}\overset{-2}{O} + H_2O$

$\left(\overset{-3}{N} \longrightarrow N^{+2} + 5e^-\right) \times 4 = 4N^{-3} \longrightarrow 4N^{+2} + 20e^-$

$\left(O_2 + 4e^- \longrightarrow 2\overset{2+}{O}\right) \times 5 \quad = 5O_2 + 20e^- \longrightarrow 10\overset{2-}{O}$

$\Rightarrow 4NH_3 + 5O_2 \longrightarrow 4NO + 6H_2O$

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Answer: C

14. Acetylene, C_2H_2 reacts with oxygen to produce carbon dioxide and water. What is the coefficient ratio of O_2 to C_2H_2?

A) $\frac{3}{2}$ B) $\frac{2}{1}$ C) $\frac{5}{2}$ D) $\frac{3}{2}$

Solution

$$2C_2H_2 + 5O_2 \longrightarrow 4CO_2 + 2H_2O$$

Answer: C

15. The sum of the coefficients of reactants and products after balancing the reaction below is $KClO_3 + HBr \longrightarrow KCl + Br_2 + H_2O$

A. 5 B. 16 C. 15 D. 8

Solution

$$KClO_3 + 6HBr \longrightarrow 2KCl + 3Br_2 + 3H_2O$$

$1 + 6 + 2 + 3 + 3 = 15$

Answer: C

16. The coefficient of HCl after balancing the redox reaction below is

$$Na_2Cr_2O_7 + FeCl_2 + HCl \longrightarrow CrCl_3 + NaCl + FeCl_3 + H_2O$$

A. 2 B. 14 C. 6 D. 7

Solution

$$Na_2Cr_2O_7 + 6FeCl_2 + 14HCl \longrightarrow 2CrCl_3 + 2NaCl + 6FeCl_3 + 7H_2O$$

Answer: B

17. How many electrons will appear when the following half - reaction is balanced? $S_4O^{2-}_6 \longrightarrow S_2O^{2-}_3$

A. 2 B. 3 C. 4 D. 4

Solution

Balanced the atoms by inspection method and then balance the charge.

$$S_4O_6^{2-} + 2e^- \longrightarrow 2S_2O_3^{2-}$$

Answer: A

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18. How many moles of MnO_2 are produced in a basic medium when MnO_4^- and CN^- react to form MnO_2 and CNO^-?

$$MnO_4^- + CN^- \longrightarrow MnO_2 + CNO^-$$

A. 5 B. 1 C. 3 D. 5

Solution

$$\left(2H_2O + MnO_4^- + 3e^- \longrightarrow MnO_2 + 4OH^-\right)\times 2$$

$$\left(2OH^- + CN^- \longrightarrow H_2O + CNO^- + 2e^-\right)\times 3$$

$$4H_2O + 2MnO_4^- + 6e^- \longrightarrow 2MnO^2 + 8OH^-$$

$$6OH^- + 3CN^- \longrightarrow 3CNO^- + 6e^- + 3H2O$$

$$H_2O + 2MnO_4^- + 3CN^- \longrightarrow 2MnO_2 + 2OH^- + 3CNO^-$$

Answer: D

19. Balance the following by oxidation number method.

a. $Sn + HNO_3 \longrightarrow Sn(NO_3)_2 + NO_2 + H_2O$

b. $K_2CrO_4 + H_2O + S \longrightarrow SO_2 + KOH + Cr_2O_3$

Solution

a. $\left(Sn \longrightarrow Sn^{2+} + 2e^-\right)\times 1$

$\left(N^{+5} + e \longrightarrow N^{+4}\right)\times 2$

$Sn \longrightarrow Sn^{2+} + 2e^-$

$2N^{+5} + 2e \longrightarrow 2N^{+4}$

$Sn + 2HNO_3 \longrightarrow Sn(NO_3)_2 + 2NO_2 + H_2O$

N is not balanced, So change the coefficient

$Sn + 4HNO_3 \longrightarrow Sn(NO_3)_2 + 2NO_2 + 2H_2O$

b. $\left(2Cr^{+6} + 6e^- \longrightarrow Cr_2^{+3}\right)\times 2$

$\left(S \longrightarrow \overset{+4}{S} + 4e^-\right)\times 3$

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$4Cr^{+6} + 12e^- \longrightarrow 2Cr_2^{+3}$

$3S \longrightarrow 3S^{+4} + 12e^-$

$4K_2CrO_4 + 4H_2O + 3S \longrightarrow 3SO_2 + 8KOH + 2Cr_2O_3$

20. Why we balance a chemical reaction?

Answer: To obey the lay of conservation of mass

21. Balance the following by ion - electron method

a. $NO_2^- + Cr_2O_7^{2-} \longrightarrow Cr^{3+} + NO_3^-$ (acidic solution)

b. $I^- + NO_2^- \longrightarrow I_2 + NO$ (acidic solution)

c. $Sn^{2+} + Bi^{3+} \longrightarrow [Sn(OH)_6]^{2-} + Bi$ (basic solution)

d. $MnO_4^- + SO_3^{2-} \longrightarrow Mn^{2+} + SO_4^{2-}$ (basic solution)

Solution

a. $8H^+ + 3NO_2^- \longrightarrow Cr_2O_7^{2-} + 2Cr^{3+} + 4H_2O$

b. $4H^+ + 2NO_2^- + 2I^- \longrightarrow I_2 + 2NO + 2H_2O$

c. $18OH^- + 3Sn^{2+} + 2Bi^{3+} \longrightarrow 3[Sn(OH)_6]^{2-} + Bi$

d. $3H_2O + 2MnO_4^- + 5SO_3^{2-} \longrightarrow 2Mn^{2+} + 6OH^- + 5SO_4^{2-}$

22. In the following equation

$ClO_3^- + 6H^+ + x \longrightarrow Cl^- + 3H_2O$, What is x?

Solution

In a given chemical equation, atoms, mass and charge must be conserved

$x + 6(+1) + 1x(-1) = -1 \Rightarrow x = 6e^-$

$ClO_3^- + 6H^+ + 6e^- \longrightarrow Cl^- + 3H_2O$

23. In the equation: $Cr_2O_7^{2-} + Fe^{2+} + H^+ \longrightarrow Cr^{3+} + Fe^{3+} + H_2O$

What is the sum of the coefficients of reactants?

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Solution

$$Cr_2O_7^{2-} + 6e^- \longrightarrow 2Cr^{3+}$$

$$6Fe^{2+} \longrightarrow 6Fe^{3+} + 6e^-$$

$$Cr_2O_7^{2-} + 6Fe^{2+} + 14H^+ \longrightarrow 2Cr^{3+} + 6Fe^{3+} + 7H_2O$$

➥ Sum of coefficient of reactants 1 + 6 + 14 = 21

8.2 The Conduction Electricity

✓ Substance that transmit electricity are called **electrical conductors.**

✓ There are two types of conductors. These are metallic and electrolytic conductors.

Table 8.1 The Difference Between Metallic and Electrolytic Conductors

Metallic Conduction (Electronic Conduction)	Electrolytic conduction (Ionic Conduction)
➥ Transmit electricity without undergoing chemical change ➥ Charge carries are free electrons ➥ Transmit electricity in solid and other states **Example** Metals	➥ Transmit electricity by under going chemical changes. ➥ Charges carriers are mobile ions. ➥ Transmit electricity in molten or aqueous state **Example** Acids, bases and salt

➥ Solutions that contain large amount of ions are called **strong electrolytes.**

Example

➥ Molten salts, strong acids, strong bases and soluble salts are strong electrolytes.

➥ Solutions that have smaller amount of ions are known as **weak electrolytes.**

Example

➥ Solution of weak acids, weak bases and partially soluble salts are weak electrolytes

➥ Solutions that do not transmit electrolytes are known as **non -**

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electrolytes.

Example

- Sugar solution, ethanol solution etc.

Conductivity apparatus: is a device used to test conductivity of substances. Its components are

- Connecting wire
- Electrodes (anode and cathode)
- Direct current source (battery)
- Bulb and switch

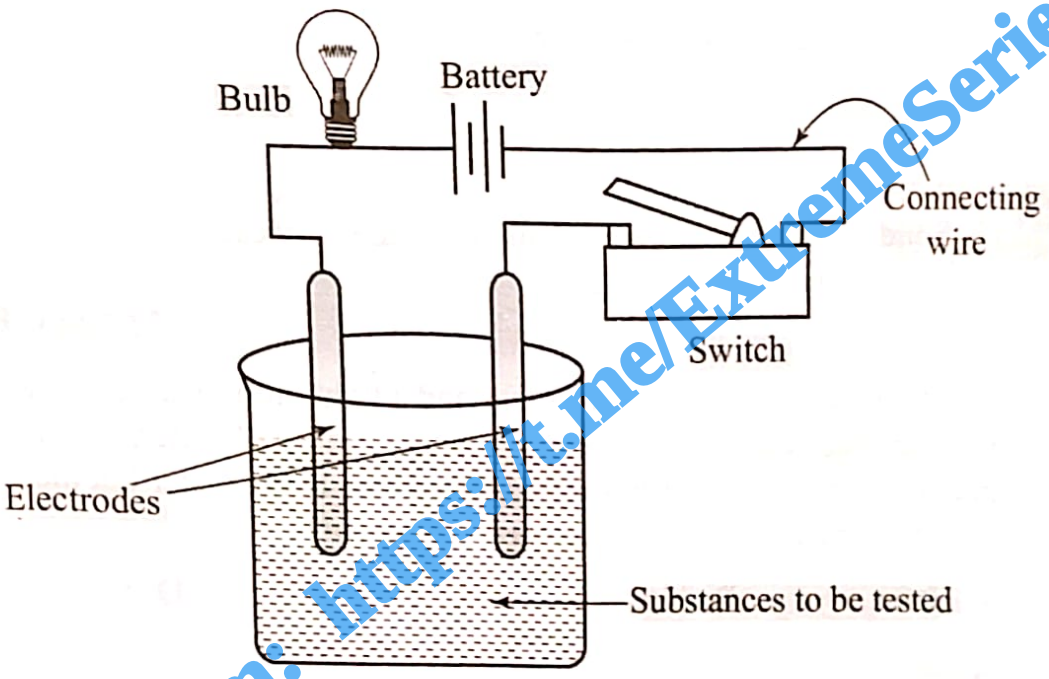

Fig 8.1 Conductivity Apparatus

Illustrative Examples

24. One of the basic differences between metallic conduction and electrolytic conduction is: (ESSLCE)

A. The conduction of electricity is due to the movement of ions in metallic conduction while the flow of mobile electrons in electrolytic conduction

B. The charge carriers in metals are localized electrons while strongly held ions are in electrolytic conduction

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C. The charge carriers in metals are cations and anions while mobile electrons are in electrolytic conduction

D. The conduction of electricity is due to the movement of ions in electrolytic conduction while the flow of mobile electrons in metallic conduction

Solution

The charge carriers in metallic and electrolytic conductors are respectively mobile electrons and ions (cations and anions)

Answer: D

25. Which of the following substance is a non - conductor of electricity?

A. Graphite B. $MgCl_2(s)$ C. Silver (s) D. $H_2SO_4(aq)$

Solution

Solid ionic compounds are non - conductors because the ions are not mobile to carry charge

Answer: B

26. Two wires are connected to a battery and a light bulb in series. When the free ends of the wires are dipped in to a beaker of water, the bulb remains unlit. A substance was then dissolved in the water but the bulb remains unlit. Which substance was added?

A. HNO_3 B. NaOH C. KCl D. C_2H_5OH

Solution

The bulb does not give light, if the solution in non - electrolyte.

Answer: D

27. Which of the following transmit electricity without undergoing chemical change?

A. Copper B. Acid C. Salt D. Sugar

Solution

Metallic conductors transmit electricity without undergoing chemical change.

Answer: A

28. Why electrolytic conductors are also called ionic conductors?

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Solution

Because the charge carriers are mobile ions.

29. List the components of conductivity apparatus.

Solution

Electrodes, connecting wire, light bulb, battery and switch.

8.3 Electrolytic Cell

An electrochemical cell that converts electrical energy into chemical energy is termed as electrolytic Cell

In electrolytic cell, electrical energy causes non - spontaneous chemical reaction to takes place.

The components of electrolytic cell are:

- Connecting wire
- Electrodes
- Direct current source
- Electrolyte

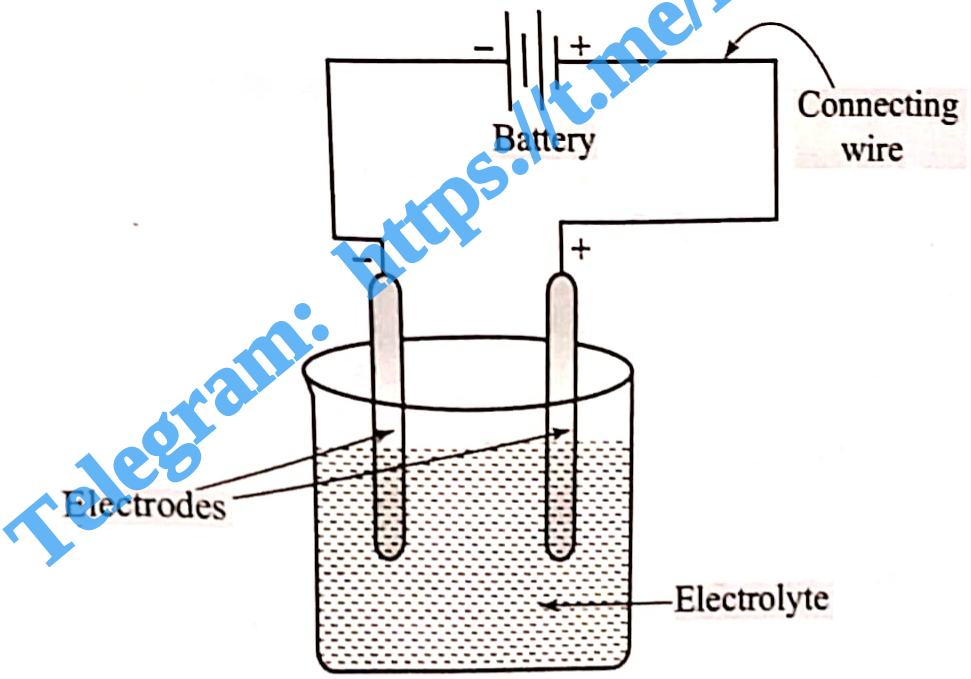

Fig 8.2 Electrolytic Cell

Electrodes are strip of metals or graphite that allow the entrance and existence of electrons into and out of the electrolyte.

Anode is electrode where **oxidation** half - reaction occurs.

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To remember it relate with 'Anox' (Anode is oxidation)

Cathode is electrode where **reduction** half - reaction takes place.

To remember it easily relate with 'car' (cathode is reduction)

In electrolytic cell, anode is cannected to the positive terminal of the battery and cathode is connected to the negative terminal of the battery.

Electrolysis of Molten Electrolyte

Electrolysis is the process of converting electrical energy into chemical energy.

During Electrolysis:

- **Anions** migrate toward anode and discharged by losing electrons
- **Cations** migrate toward cathode and discharged by gaining electrons.

In electrolysis of molten electrolyte, there is one cation to be discharged at cathode and one anion to be discharged at anode.

Example

Write the anode, cathode, and cell reactions for electrolysis of molten or fused sodium chloride

Solution

$NaCl \longrightarrow Na^{-} + Cl^{-}$

- Na^{-} migrates toward cathode
- Cl^{-} migrates toward anode

Cathode reaction: $Na^{-} + e^{-} \longrightarrow Na$, reduction

Anode reaction: $2Cl^{-} \longrightarrow Cl_2 + 2e^{-}$, oxidation

To write over all or cell reaction, first balance the electrons transferred.

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$$2Na^{+} + 2e^{-} \longrightarrow 2Na$$

$$2Cl^{-} \longrightarrow 2Cl_2 + 2e$$

Cell reaction : $2Na^{+} + 2Cl^{-} \xrightarrow{\text{electrolysis}} 2Na + Cl_2$
(over all reation)

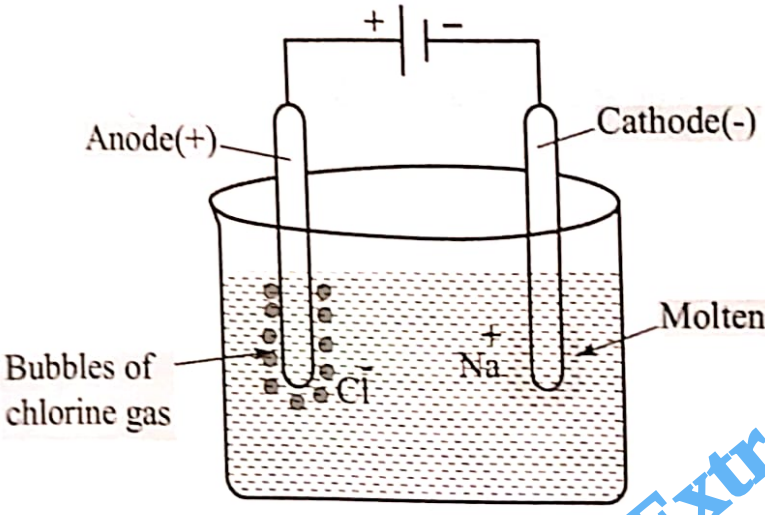

Fig 8.3 Electrolysis of Molten Sodium Cholride

Illustrative Examples

30. The process in which a compound absorbs electrical energy and decomposes into its element is (ESSLCE)

A. Electrolysis
B. Solvolysis
C. Electroplating
D. Hydrolysis

Solution

$$MX \xrightarrow{\text{Electrolysis}} M + X$$

Answer: A

31. All of the following are the components of electrolytic cell. **EXCEPT**

A. Electrodes
B. Connecting wire
C. Battery
D. Light bulb

Answer: D

32. Which of the is not true during electrolysis?

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A. Anions are discharged at anode by losing electrons

B. Cations are discharged at cathod by gaining electrons

C. Electrolysis is the process of causing non - spontaneous reaction to takes place by means of electrical energy.

D. During electrolysis electrical energy is generated.

Answer: D

33. Write anode, cathode and cell reactions for electrolysis of molten lead (II) bromide, $PbBr_2$.

Solution

$$PbBr_2 \xrightarrow{\text{Melting}} Pb^{2+} + 2Br^-$$

Cathode reaction: $Pb^{2+} + 2e^- \longrightarrow Br_2 + 2e$

Anode reaction: $2Br^- \longrightarrow Br_2 + 2e$

Cell reaction: $Pb^{2+} + 2Br^- \longrightarrow Pb + Br_2$

34. What are the two types of half - reactions that take place in electrolytic cell?

Solution

The oxidation - half reaction at anode and the reduction half - reaction at cathode.

35. Define electrode.

Solution

Electrode is part of an electro chemical cell that conducts the electricity between the cell and the surrounding.

36. What is the driving force that pushes electrons from anode to cathode in the external circuit?

Solution

The direct current source or battery.

37. The electrode connected to the negative terminal of the battery in electrolytic cell is _____

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Answer: Cathode

Electrolysis of Aqueous Solution

In aqueous solution, the sources of ions are both the solute and the solvent. As a result, the electrodes have a chance to select and discharge one type of ion based on different factors. The selection of one type of ion by electrodes to discharge is known as **Preferential discharge.**

Factors Affecting Preferential Discharge

1. Nature of Electrodes

Based on their nature, electrodes are classified in to two;

i. **Active electrodes** affect product of electrolysis

Example

Copper, Zinc, magnesium, aluminum etc.

ii. **Inert electrodes** do not affect product of electrolysis.

Example

Graphite, platinum etc.

Consider electrolysis of copper (II) sulphate solution to understand how nature of electrode affects preferential discharge

a. Electrolysis of Copper (II) Sulphate Solution Using Inert Electrode such as Graphite

Four ions are present in the solution from water: H^+ and OH^- from copper (11) sulphate: Cu^{2+} and SO_4^{2-}

Cu^{2+} and H^+ move toward cathode

OH^- and SO^{2-}_4 move toward anode

From the experiment, the following took place:

Cathode reaction: $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

Anode reaction: $4OH^-(aq) \longrightarrow 2H_2O(1) + O_2(g) + 4e^-$

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Note:

- O_2 is liberated at anode
- The blue color due to cupper (II) ion disappears since it is reduced.
- Copper is deposited at cathode
- The resuting solution is acidic

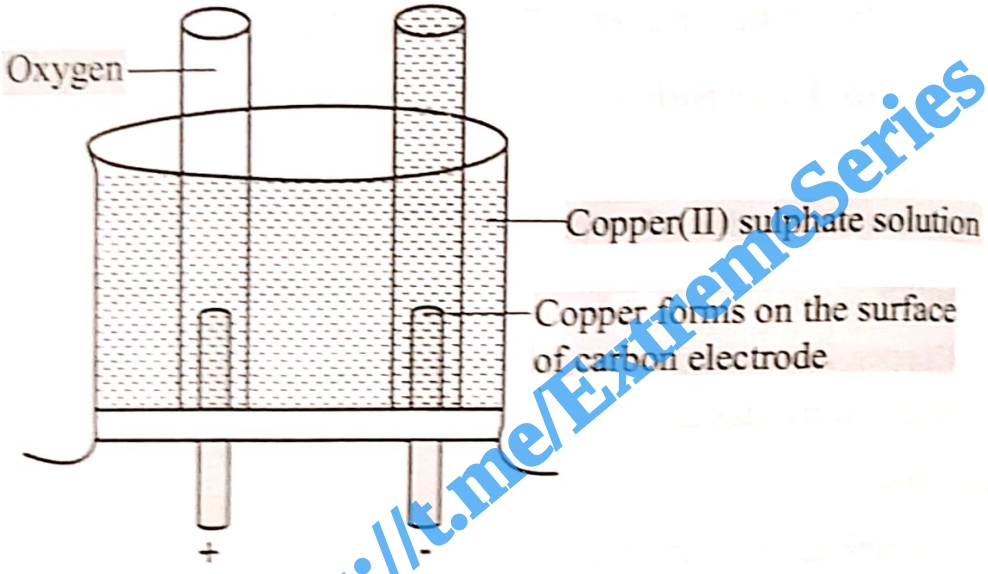

Fig 8.4 The Electrolysis of copper (II) Sulphate Solution Using Inert Electrodes

b. Electrolysis of Copper (II) sulphate Solution Using Copper Electrode

The same species are present as (a) above

Experimentally, the following happens at electrodes

Cathode reaction: $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

Anode reaction: Itself oxidized and loses its mass.

$$Cu(s) \longrightarrow Cu^{2+}(aq) + 2e^-$$

Note

- Copper is deposited at cathode
- Copper is oxidized at anode
- Mass of cathode increases

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- Mass of anode decreases
- The blue color of the solution remains blue.

2. Nature of Ions (Position of ions in the Electro Chemical series)

Ions (both cation and anion) at the bottom of the electrochemical series are easily discharged at the respective electrode.

Cations		Anions	
Li^+	The case of discharge increase downward	F^-	The case of discharge increase downward
Na^+		SO_4^{2-}	
Mg^{2+}		NO_3^-	
Al^{3+}		CO_3^{2-}	
Mn^{2+}		Cl^-	
Cr^{3+}		Br^-	
Fe^{2+}		I^-	
Cd^{2+}		OH^-	
Fe^+		S^{2-}	
H^+			
Cu^{2+}			
Ag^+			
Au^{3+}			

Fig 8.5 Reactivity series of metals

Example

- If the same concentration of Li^+ and Au^{3+} are found in the given solution, cathode selects Au^{3+}.
- Au^{3+} is the strongest oxidizing agent while Li^+ is the weakest oxidizing agent
- S^{2-} is the strongest reducing agent but F^- is the weakest reducing agent
- **Reactivity series** is a series of metals in decreasing order of reactivity. More reactive metal displace less reactive metal in chemical reaction.

Example

- $Zn + CuSO_4 \longrightarrow ZnSO_4 + Cu$, Zinc is more reactive than copper.

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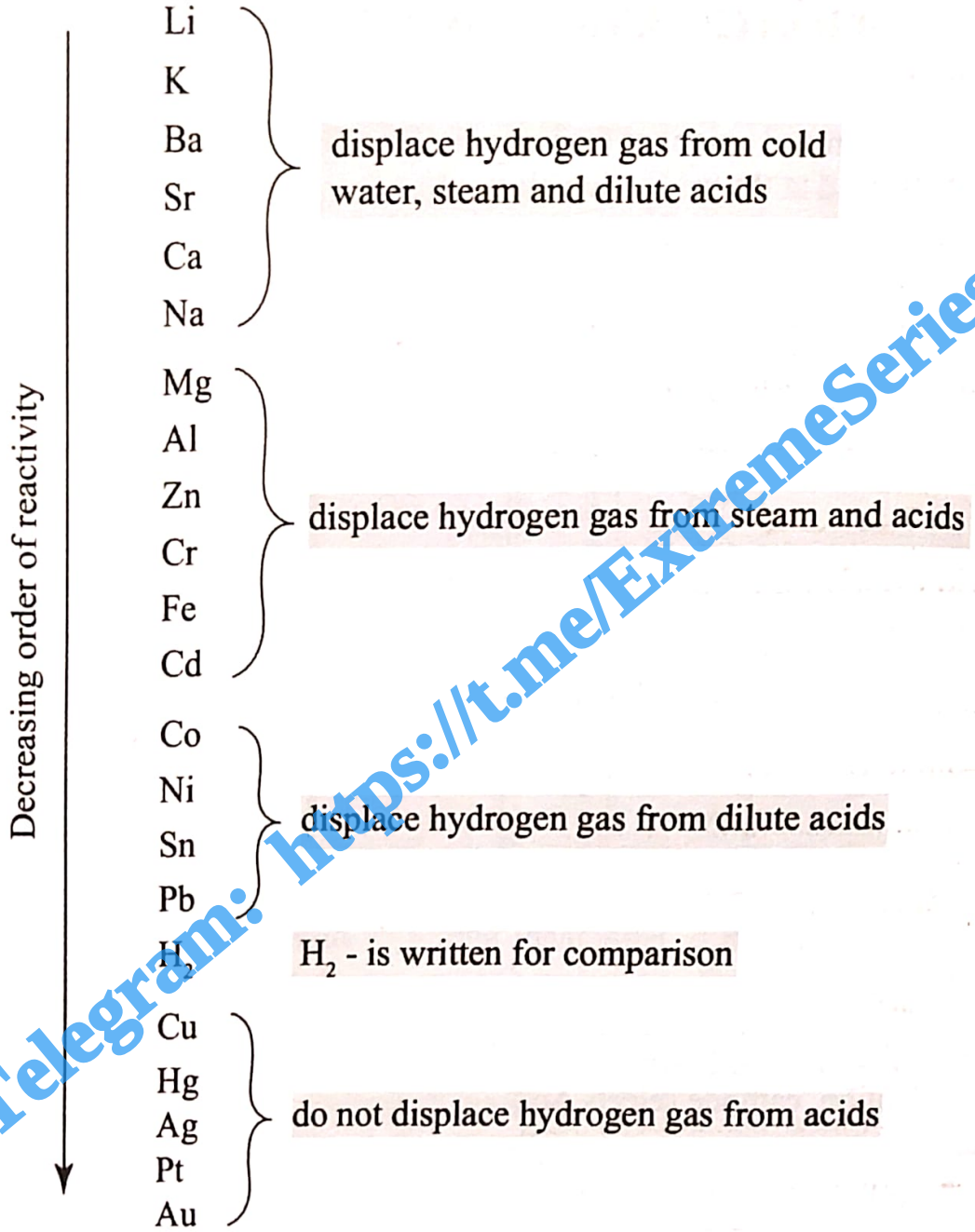
➥ $Cu + ZnSO_4 \longrightarrow$ no reacton, because

Cu is less reactive than Zinc



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Note: In neutral form:

- ➥ Li is the powerful reducing agent
- ➥ F_2 is the powerful oxidizing agent

3. Concentrations of Ions

In concentrated solution, usually ions of solute are discharged but in dilute solutions most of the time ions of solvent are discharged.

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Example

a. Electrolysis of Dilute Sodium Chloride

➥ **Ions present:** Na^+ and Cl^- from solute (NaCl)

H^+ and OH^- from solvent (H_2O)

➥ Experimentally, the following occur at electrodes

Cathode reaction: $2H^+(aq) + 2e^- \longrightarrow H_2(g)$

Anode reaction: $4OH^-(aq) \longrightarrow 2H_2O(1) + O_2(g) + 4e^-$

Cell reaction: $2H_2O(1) \longrightarrow 2H_2(g) + O_2(g)$

Note: O_2 is given off at anode

H_2 is liberated at cathode.

The resulting solution is neutral.

b. Electrolysis of Concentrated Sodium Chloride Solution (brine)

Concentrated sodium chloride solution is known as **brine**

Anode reaction: $2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$

Cathode reaction: $2H^+(aq) + 2e^- \longrightarrow H_2(g)$

Cell reaction: $2H^+(aq) + 2Cl^-(aq) \longrightarrow H_2(g) + Cl_2(g)$

Note

➥ Cl_2 is given off at anode

➥ H_2 is formed at cathode

➥ NaOH is formed as residue

➥ The resulting solution is alkaline or basic.

Illustrative Examples

38. What product (s) is lare formed during electrolysis of a concentrated aqueous solution of sodium chloride?

A. $Cl_2(g)$

B. $Cl_2(g)$ and $H_2O(g)$

C. $NaOH(ag)$ and $Cl_2(g)$

D. $Cl_2(g)$, $H_2(g)$ and $NaOH(aq)$

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Solution

$Cl_2(g)$ at anode, $H_2(g)$ at Ca thode and NaOH(aq) are formed during electrolysis of brine

Answer: D

39. During the electrolysis of an aqueous solution of copper (11) sulphate solution using platinum electrodes, the reaction takes place at anode is

A. $Cu^{2+} + 2e \longrightarrow Cu$

B. $Cu \longrightarrow Cu^{2+} + 2e^-$

C. $4OH^- \longrightarrow 2H_2O + O_2 + 4e^-$

D. $4H^- + 4e^- \longrightarrow 2H_2$

Answer: C

40. Identify the best oxidizing agent among the following

A. Cu^{2+} B. Ag^+ C. Au^{3+} D. Zn^{2+}

Solution

Cation at the bottom of the electrochemical series is the best oxidizing agent.

Answer: C

41. Which of the following is incorrect about electrolysis of copper (11) solution using copper electrode?

A. Mass of anode decreases

B. Mass of cathode increases

C. Copper is oxidized at anode

D. Hydrogen gas is given off at cathode

Answer: D

42. Write anode, cathode and cell reaction for electrolysis of concentrated copper (II) chloride solution

Solution

Anode reaction: $2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$

Cathode reaction: $Cu^{2+}(aq) + 2e^- \longrightarrow Cu(s)$

Cell reaction: $Cu^{2+}(aq) + 2Cl^-(aq) \longrightarrow Cu(s) + Cl_2(g)$

43. What happens to the concentration of dilute sulphuric acid after

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electrolysis?

Solution

The concentration of sulphuric acid increases because water is hydrolyzed. $2H_2O(1) \rightarrow 2H_2(g) + O_2(g)$

44. Write the anode, cathode and cell reactions of electrolysis of dilute sodium hydroxide solution?

Solution

Cathode reaction: $2H^+(aq) + 2e^- \longrightarrow H_2(q)$

Anode reaction: $4OH^-(aq) \longrightarrow 2H_2O(1) + O_2(g) + 4e^-$

Cell reaction: $2H_2O(1) \longrightarrow 2H_2(g) + O_2(g)$

8.4 Quantitative Aspects of Electrolysis

The quantity of electricity flowing through an electrolysis cell is measured in coulombs (C)

Charge = Current × Time

$$\boxed{Q = I \times t}$$

$$1C = 1A.S$$

The quantitative aspects of electrolysis was developed by **Michael Faraday.**

Faraday's first law of electrolysis states that "the mass of a substance formed at each electrode during electrolysis is directly proportional to the quantity of electricity passed through the electrolyte"

Mathematically

$m \propto Q \Rightarrow m = KQ$, K = constant

$$K = \frac{M}{ZF}$$

$$\Rightarrow \boxed{m = \frac{MQ}{ZF}}$$

m - mass of substances formed at electrodes

M - molar mass of substances formed at electrodes

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$\frac{m}{M} = \frac{Q}{ZF}$ Z - electrons lost or gained

F - Faraday's constant

$n = \frac{Q}{ZF}$, because $n = \frac{m}{M}$

$m = \frac{EQ}{F}$, because $E = \frac{M}{Z}$

$\frac{m}{E} = \frac{Q}{F}$

$V = \frac{QRT}{ZFP}$

$V = \frac{It\,RT}{ZFP}$

For gaseous substance formed at electrodes to find the volume

$PV = nRT$

$V = \frac{nRT}{P}$

1F = charge of one mole of electrons

1mole of electrons = 6.02×10^{23} electrons

charge of one electron = $1.602 \times 10^{-19}C$

$\Rightarrow 1F = 1.602 \times 10^{-19} \frac{C}{\text{electron}} \times 6.02 \times 10^{23} \frac{\text{electrons}}{\text{mol}}$

$1F = 96485\ C/_{mol} \approx 96500 C/mol$

Example:

How many grams of copper will be deposited from a solution of copper (II) sulphates by a current of 2.5A flowing for 1hour. (M of Cu = 64 g/mol)

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$Cu^{2+} + 2e^- \longrightarrow Cu$

$Z = 2$

$t = 1 hour = 3600s$

$$m = \frac{MIt}{ZF} = \frac{64\frac{g}{mol} \times 2.5\,A \times 3600s}{2 \times 96500\,A.s/mol} = 2.98g$$

Faraday's Second Law of Electrolysis

"The masses of different substance oxidized or reduced by the same quantity of electricity are directly proportional to the equivalent masses of the substances"

Mathematically:

$m \alpha E$

From Faraday's first law, we know that

$$\frac{m}{E} = \frac{Q}{F}, \qquad Q = \frac{mF}{E}$$

$$Q_1 = \frac{m_1F}{F_1}, \; Q_2 = \frac{m_2F}{E_2}, \; Q_3 = \frac{m_3F}{E_3} \text{ ----}$$

But Q is constant

$$\Rightarrow Q_1 = Q_2 = Q_3 = Q_4 = \text{----}$$

$$\frac{m_1F}{E_1} = \frac{m_2F}{E_2} = \frac{m_3F}{E_3} = \frac{m_4F}{E_4} \text{ --------}$$

$$\boxed{\frac{m_1}{E_1} = \frac{m_2}{E_2} = \frac{m_3}{E_3} = \frac{m_4}{E_4}}$$

Example

Two cells are connected in series, one contains $AlCl_3$ and the other contains $CuSO_4$ as electrolytes. If 54g of Al is deposited in the cell containing $AlCl_3$, how much copper is deposited in the other cell?

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Solution:

$$Al^{3+} + 3e^- \longrightarrow Al$$

$$Z = 3,\ E_{Al} = \frac{M}{Z} = \frac{27}{3} = 9\ g/eq$$

$$Cu^{2+} + 2e^- \longrightarrow Cu$$

$$Z = 2,\ E_{cu} = \frac{M}{Z} = \frac{64}{2} = 32 g/eq$$

$$\frac{m_{cu}}{E_{cu}} = \frac{m_{Al}}{E_{Al}}$$

$$m_{cu} = \frac{m_{Al} \times E_{cu}}{EAl} = \frac{54g \times 32\,g/eq}{9\ g/eq}$$

$$m_{cu} = 192g$$

Illustrative Examples

45. What mass of magnesium is produced in the electrolysis of molten $MgCl_2$, if a current of 7.50×10^2A is passed through molten salt for 1hour?

A. 336g B. 133g C. 712g D. 560g

Solution

$$Mg^{+2} + 2e^- \longrightarrow Mg$$

$$Z = 2,\ t = 1hour = 3600s$$

$$m = \frac{MIt}{ZF} = \frac{24\ \frac{g}{mol} \times 7.50 \times 10^2\ A \times 3600s}{2 \times 96500\ \frac{A.s}{mol}} = 335.8g$$

Answer: A

46. Electrolysis of dilute aqueous NaCl solution was carried out by passing 10mA current. What is the time required to liberate 0.01mol of H_2 gas at the cathode?

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Solution

$$n = \frac{It}{ZF}, \; I = 10mA = 10 \times 10^{-3} \; A = 0.010A$$

$$t = \frac{n \times ZF}{I} = \frac{0.01 \text{mol} \times 2 \times 96500 \; A.s/\text{mol}}{0.01 \; A}$$

$$t = 1.93 \times 10^{5} \text{ seconds}$$

47. When electricity is passed through aqueous solution of $AlCl_3$, 13.5g of Al produced. What must be the number of faraday of electricity?

Solution

$$Al^{3+} + 3e \longrightarrow A$$

$$3F = 1\text{mol Al} = 27\text{g Al}$$

$$? \qquad 13.5\text{g Al}$$

$$\Rightarrow F = \frac{3F \times 13.5\text{g Al}}{27\text{g Al}} = 1.5F$$

48. How long, in hours, would be required for the electroplating of 78g of platinum from a solution of $[Pt\,Cl_6]^{2-}$ using an average current of 10A at 80% efficiency?

Solution

The oxidation state of Pt in $[Pt\,Cl6]^{2-}$ is + 4

$$Pt^{4+} + 4e^{-} \longrightarrow Pt$$

$$Z = 4$$

$$E = \frac{M}{Z} = \frac{195g}{4} = 48.75g$$

Electrode efficiency = 80%

$$E = 80\% \times 48.75g = 39g$$

$$t = \frac{mF}{EI} = \frac{78g \times 96500 \; A.S}{39g \times 10A} = 19300 \text{ sec} = 5.36\text{hrs}$$

49. Calculate the volume of H_2 and O_2 gases at 37^0C and 1140torr that will

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be collected at the cathode and anode respectively, when dilute sodium hydroxide is electrolyzed for 2hours with 5A current.

Solution

Cathode reaction: $2H^+(aq) + 2e \longrightarrow H_2(g)$

$Z = 2,\ t = 2hr = 7200s$

$$nH_2 = \frac{It}{ZF} = \frac{5A \times 7200S}{2 \times 96500 \frac{A.S}{mol}} = 0.187mol$$

From ideal gas law: $PV = nRT$, $P = 1140$ torr $= 1.5$ atm

$$VH_2 = \frac{nRT}{P} = \frac{0.187\ mol \times 0.082atm/mol \times 310K}{1.5\ atm}$$

$$VH_2 = 3.17L$$

Anode reaction: $4OH^-(aq) \longrightarrow 2H_2O(l) + O_2(g) + 4e^-$

$Z = 4,$

$VO_2 = \frac{nRT}{P} = 1.58L$ or you may use the formula

$$V = \frac{IT\ RT}{ZFP}$$

50. How many moles of electrons are required to discharge 1mol of the following ions?

a. Mg^{2+} b. K^+ c. Al^{3+} d. O^{2-}

Solution

a. $Mg^{2+} + 2e^- \longrightarrow Mg \Rightarrow$ 2moles of electrons

b. $K^+ + 1e^- \longrightarrow K \Rightarrow$ 1mole of electrons

c. $Al^{3+} + 3e^- \longrightarrow Al \Rightarrow$ 3moles of electrons

d. $2O^{2-} \longrightarrow O_2 + 4e^- \Rightarrow$ 4moles of electrons

51. Calculate the number of moles of electrons required to deposite 6.35g of copper a metal surface. (At mass of Cu = 63.5)

Solution

$Cu^{2+} + 2e^- \longrightarrow Cu$

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$2mol.\ e^- = 63.5g$

$X ? = 6.35g$

$$\Rightarrow \text{Number of mole of electrons} = \frac{2mol.e^- \times 6.35g}{63.5g}$$

$$= 0.2\,\text{mole of electrons}$$

52. A certain amount of electricity passes through a cell in which 27.9g of Fe is deposited from $FeSO_4(aq)$. How much Au would be deposited if the same amount of electricity passed through K $[AUCl_4]$ (aq)?

Solution

$$Fe^{2+} + 2e^- \longrightarrow Fe,\ E_{Fe} = \frac{56}{2} = 28$$

$$Au^{3+} + 3e^- \longrightarrow Au,\ EAu = \frac{197}{3} = 65.67$$

$$\frac{m_{Au}}{E_{Au}} = \frac{m_{Fe}}{E_{Fe}}$$

$$m_{Au} = \frac{m_{Fe}}{E_{Fe}} \times E_{Au} = \frac{27.9g}{28} \times 65.67 = 65.7g$$

53. Two cells are connected in series one contains $AgNO_3$ and the other MF_3 as electrolytes. If 1.44g of Ag was deposited in the cell containing $AgNO_3$ where as 0.12g of the unknown metal M was deposited in the cell containing MF_3, calculate the molar mass of M.

Solution

$$Ag^+ + e^- \longrightarrow Ag,\ E_{Fe} = \frac{108}{1} = 108$$

$$E_M = \frac{m_M \times EAg}{EAg} = \frac{0.12g \times 108}{1.44g} = 9g/eq$$

$$M^{3+} + 3e^- \longrightarrow M, Z=3\ \frac{eq}{mol}$$

$$M = E \times Z = 9g/eq \times \frac{3eq}{mol} = 27\,g/mol$$

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54. 0.5 mole of electrons are passed though three electrolytic cells in series that contain Ag^+, Cu^{2+} and Cr^{3+} ions. How many grams of each metal will be deposited at cathode?

Solution

$Ag^+ + e^- \longrightarrow Ag$,

$1 mole^- = 108g$

$0.5 mole^- = ?$

$$m_{Ag} = \frac{0.5 mole^- \times 108g}{1 mole^-}$$

$m_{Ag} = 54g$

$Cu^{2+} + 2e^- \longrightarrow Cu$

$2 mole^- = 63.5g$

$0.5 mole^- = ?$

$$m_{cu} = \frac{0.5 mole^- \times 63.5g}{2 mole^-}$$

$m_{cu} = 15.87g$

$Cr^{3+} + 3e^- \longrightarrow Cr$

$3 mole^- = 52g$

$0.5 mole^- = ?$

$$m_{cr} = \frac{05 mol.e^- \times 52g}{3 mole^-} = 8.7g$$

55. 0.5F of electricity was passed to deposit all the copper present in 500ml of $CuSO_4$ solution. What is the molarity of the solution?

Solution

$Cu^{2+} + 2e^- \longrightarrow Cu$

$2F = 1 mol\ Cu$

$0.5F = ?$

$$n_{Cu^{2+}} = \frac{0.5F \times 1mol}{2F} = 0.25mol$$

$$M = \frac{n}{V} = \frac{0.25mol}{0.50L} = 0.50M$$

56. 2.5mA of current pass through molten $CaCl_2$ for 1minute. Calculate the number of calcium atoms deposited on cathode.

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Solution

$t = 1min = 60s$, $I = 2.5mA = 2.5 \times 10^{-3}A$

$Ca^{2+} + 2e^- \longrightarrow Ca$, $Z = 2$

$$\text{Number of electrons} = n \times 6.02 \times 10^{23} \frac{atoms}{mol}$$

$$= \frac{It}{ZF} \times 6.02 \times 10^{23} \frac{atoms}{mol}$$

$$= \frac{2.5 \times 10^{-3} A \times 60s}{2 \times 96500 \frac{A.S}{mol}} \times 6.02 \times 10^{23} \frac{atom}{mol}$$

$$= 4.68 \times 10^{17} \text{ atoms}$$

8.5 Industrial Applications of Electrolysis

Electrolysis has the following industrial applications.

1. Production of Active Metals Such as Na, Ca and Al

- Metals are generally produced by reduction process
- Down's cell used for the electrolysis of molten NaCl.

$Na^+(aq) + e^- \longrightarrow Na(\ell)$

- The Hall - Herout cell is used in industry to extract aluminum by electrolysis.

$Al^{3+}(l) + 3e^- \longrightarrow Al(\ell)$

2. Production of non - metals such as fluorine, chlorine and oxygen

- Non - metals are generally produced by **oxidation process**.
- Electrolysis of brine

$2Cl^-(aq) \longrightarrow Cl_2(g) + 2e^-$

- $2KHF_2 \longrightarrow 2KF + HF$

$2KF \longrightarrow 2K^+ + 2F^-$

$2F^- \longrightarrow F_2 + 2e^-$

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3. Production of Compounds such as NaOH and KOH

- NaOH - from electrolysis of concentrated NaCl solution.
- KOH - from electrolysis of concentrated KCl solution

4. Electroplating

It is the process involving electrolysis to plate or coat, one metal with another or a plastic with a metal.

During Electroplating

- Anode: plating metal
- Cathode: metal or material to be plated
- Electrolyte: salt of plating metal

Example

- Electroplating of a spoon made of iron with silver.

  Anode: silver

  Cathode: spoon

  Electrolyte: silver nitrate ($AgNO_3$)

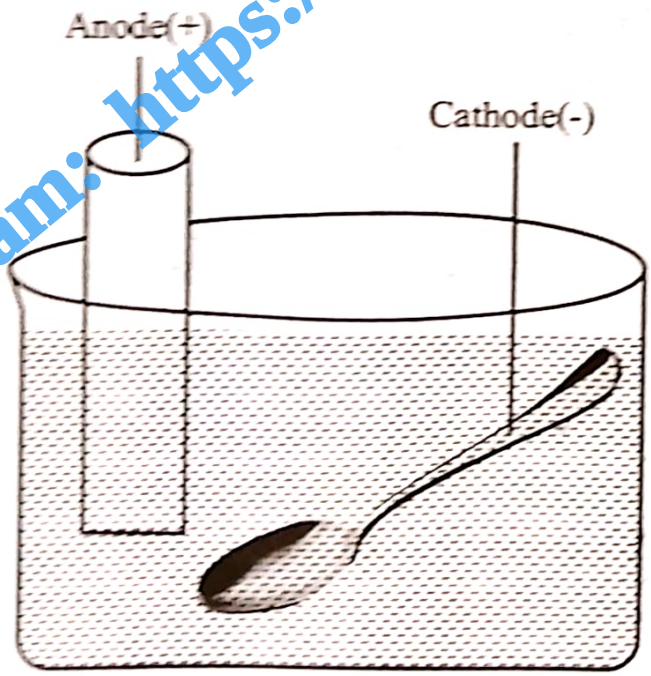

Fig 8.6 Electroplating of Spoon Made of Iron with Silver

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Anode reaction: $Ag \longrightarrow Ag^+ + e^-$

Cathode reaction: $Ag^+ + e^- \longrightarrow Ag$

5. Electro Refining of Metals

- ✓ It is purification of metals using electrolysis.
- ✓ During electrorefining of metals.

Anode: impure metal

Cathode: pure metal

Electrolyte: salt of the metal

Example

Purification of copper

Anode: impure copper

Cathode: Pure copper

Electrolyte: copper (11) sulphate solution

Anode reaction: $Cu \longrightarrow Cu^{2+} + 2e^-$

Cathode reaction: $Cu^{2+} + 2e^- \longrightarrow Cu$

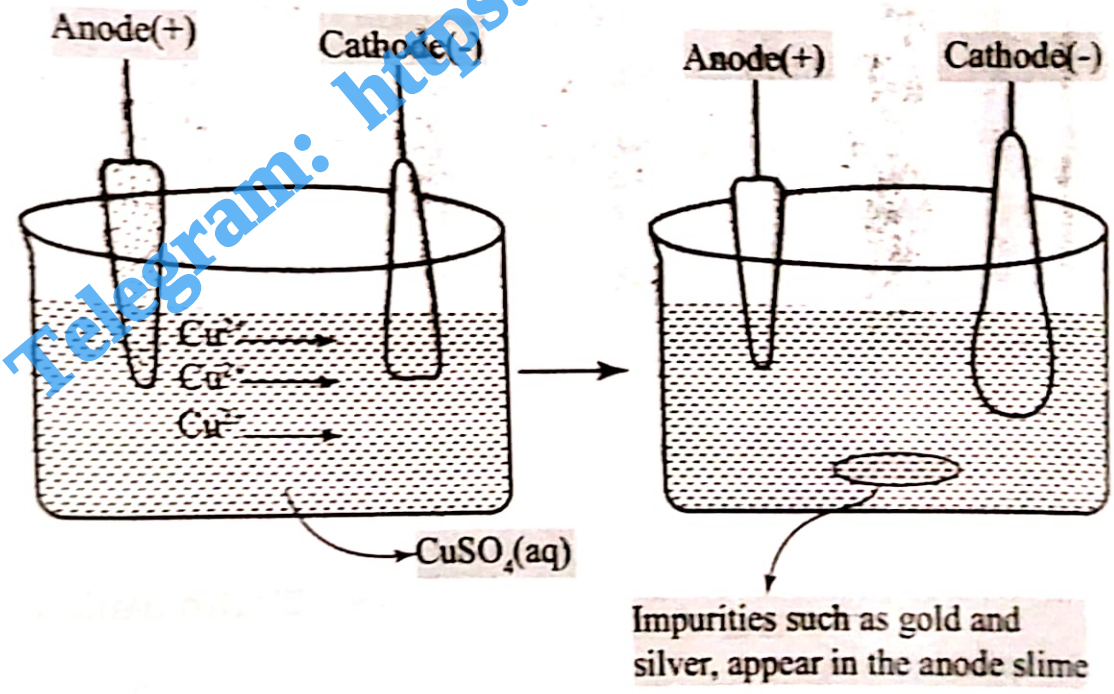

Fig 8.7 Copper Purification Process

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Illustrative Examples

57. Which one of the following explains how electroplating takes place?

A. Metal is deposited on the anode by reduction of ions in the electrolyte solution

B. The cations in the electrolyte are supplied either by the added salt or from oxidation of the anode

C. The object to electroplated either metal or graphite - coated plastic constitutes the anode

D. The electrolyte is an aqueous solution of a salt of the plated metal

Solution

- Anode: plating metal - is oxidized
- Cathode: material to be plated - metal is deposited on it
- Electrolyte: salt of plating metal
- The cation reduced at cathode is the one obtained from the electrolyte or oxidation of plating metal (anode)

Answer: B

58. How can silver be plated on nickel?

A. Electricity to a nickel anode in a solution of silver ions

B. Electricity to a sliver anode in a solution of nickel ions

C. A solution of nickel ions to react with a piece of silver

D. Electricity to a nickel cathode in a solution of silver ions.

Solution

Anode : Ag, cathode: Ni, electrolyte: solution containing silver ions

Answer: D

59. The process of depositing metals from solution in the from of a layer on other surface such as metal or plastic is called _____

Answer: Electro plating

60. Define the following

a. Tin plating b. Galvanizing c. Chrom plating

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Solution

a. Coating with tin
b. Coating with zinc
c. Coating with chromium

61. Write the uses of electroplating

- Protect metals or materials from corrosion
- Give materials pleasant appearance

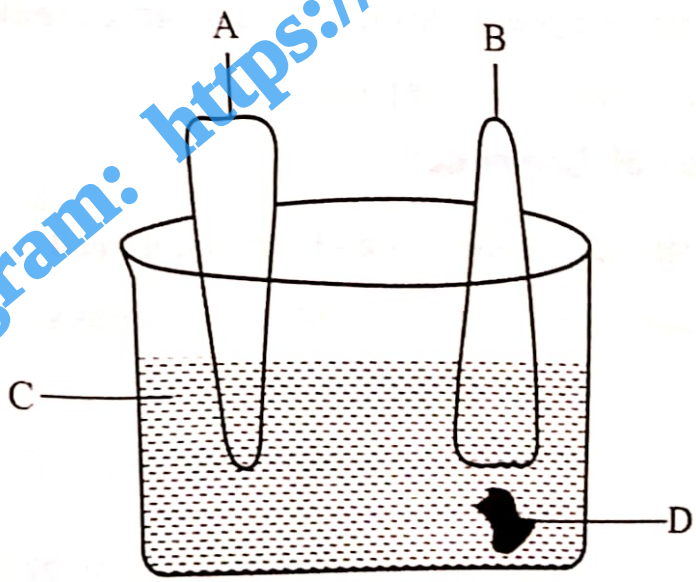
62. Write the formula of compounds manufactured industrially by electrolysis.

Solution

NaOH and KOH

63. Copper is purified by electrolysis, as in the example shown below

a. Name the materials used for the electrodes A and B
b. Name the electrolyte C and substance D
c. Why is substance D of economic importance in respect of this process.



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Solution

a. A is cathode and B is anode
b. C is $CuSO_4$ and D is impurities such as gold and silver
c. The impurities, gold and silver are precious metals used to make coins.

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64. The electrolytic cell in which sodium is extracted from molten sodium chloride is known as _____ .

Answer: Down's cell

65. An electrolytic cell used for the production of NaOH, H_2 and Cl_2 from electrolysis of brine is known as _____

Answer: Membrane cell

66. When a strip of metal X is placed in a solution containing ions of Y, metal Y plates out on the strip of X.

a. Which metal is being oxidized?

b. Which metal is being displaced?

Solution

a. Metal X b. Metal Y

8.6 Galvanic or Voltaic Cells

They are electro chemical cells which convert chemical energy to electrical energy.

In galvanic cells, spontaneus chemical reactions generate electrical energy.

Anode is negative and cathode is positive.

Construction of Daniel cell

The Daniel cell is also known as Zn - Cu galvanice cell.

It is wet galvanic cell because the electrelyte is aqueous solution

The components of Daniel cell are:

- Anode Zn in 1M $ZnSO_4$
- Cathode Cu in 1M $CuSO_4$
- Salt bridge - inverted U - tube filled with neutral salt solution

Anode reaction: $Zn(s) \longrightarrow Zn^{2+}(aq) + 2e^-$

Cathode reaction: $Cu^{2+}(eq) + 2e^- \longrightarrow Cu(s)$

Cell reaction: $Zn(s) + Cu^{2+}(eq) \longrightarrow Cu(s) + Zn^{2+}(eq)$

Cell potential (Voltage) = 1.10V

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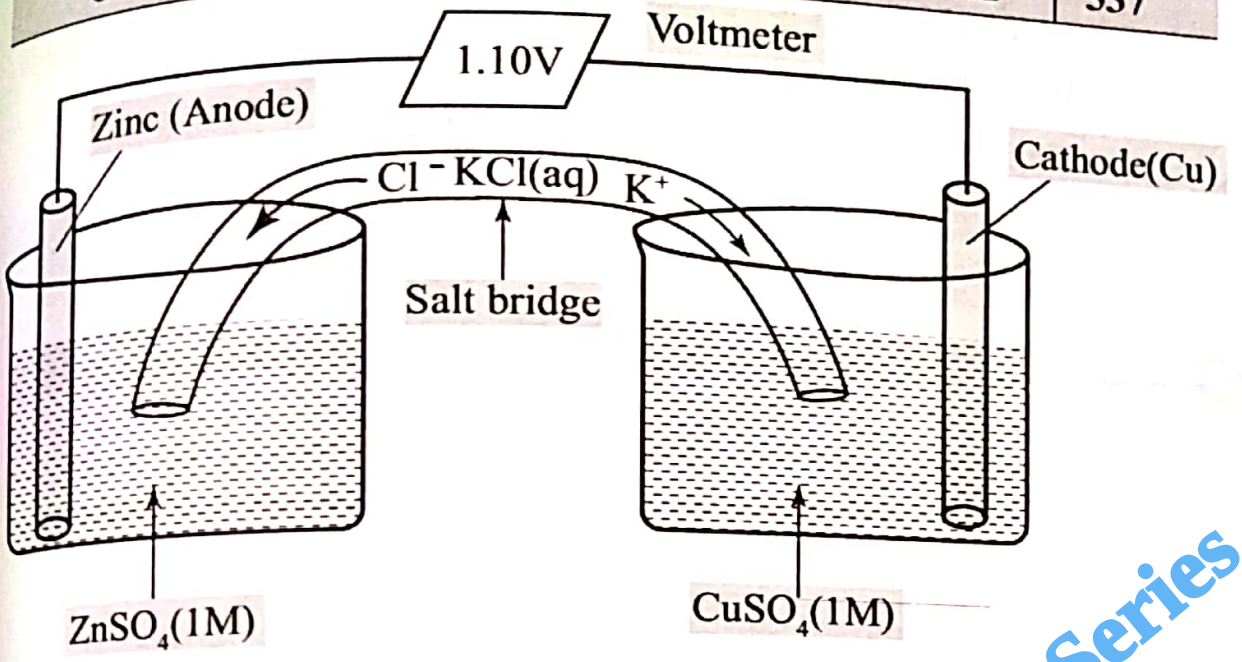

Fig 8.8 The Daniel Cell

Note:-The concentration of Zn^{2+} increases in the anode compartment and the salt bridge releases Cl^- to maintain electrical neutrality.

- The concentration of Cu^{2+} decreases in the cathode compartment and the salt bridge releases K^+ to maintain electrical neutrality.
- The mass of anode decreases and that of cathode increases.

Uses of Salt Bridge

1. Maintains electrical neutrality
2. Allows electrical contact between electrode solutions
3. Prevents mixing of the electrode solutions

Cell Notation or Cell Diagram

It is short hand representation of a Galvanic cell without drawing a diagram.

The formula of cell notation is:

Anode /Anode electrolyte// cathode electrolyte/ cathode

/: boundary between electrodes and their electrolytes

//: salt bridge

Example: Write cell notation for Daniel cell

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Solution

Anode: Zn(s), anode electrolyte, $ZnSO_4$(1M)

Cathode: Cu(s), Cathode electrolyte, $CuSO_4$(lM)

Cell notation: Zn(s) /$ZnSO_4$(1M)// $CuSO_4$(1M)/ Cu(s)

Illustrative Example

67. In galvanic cells, oxidation takes place at:

A. Anode B. Cathode C. Salt bridge D. Voltmeter

Solution

Whether it is galvanic or electrolytic cell, oxidation occurs at anode

Answer: A

68. For a voltaic cell using Ag, Ag^+(1.0M) and Zn, Zn^{2+}(1.0M) half - cells, which of the following statement is incorrect?

A. The zinc electrode is the anode

B. Electrons will flow through the external circuit from the zinc electrode to silver electrode

C. The mass of zinc electrode will decrease as the cell operates.

D. Reduction occurs at zinc electrode as the cell operates

Solution

➥ In galvanic cells, more active metal is used as anode to be oxidized easily and the less active metal is used as cathode. Thus Zinc is anode and Ag is cathode.

➥ In electro chemical cell, oxidation occurs at anode and reduction at cathode.

Answer: D

69. For the galvanic cell shown below, which one of the following is correct ?

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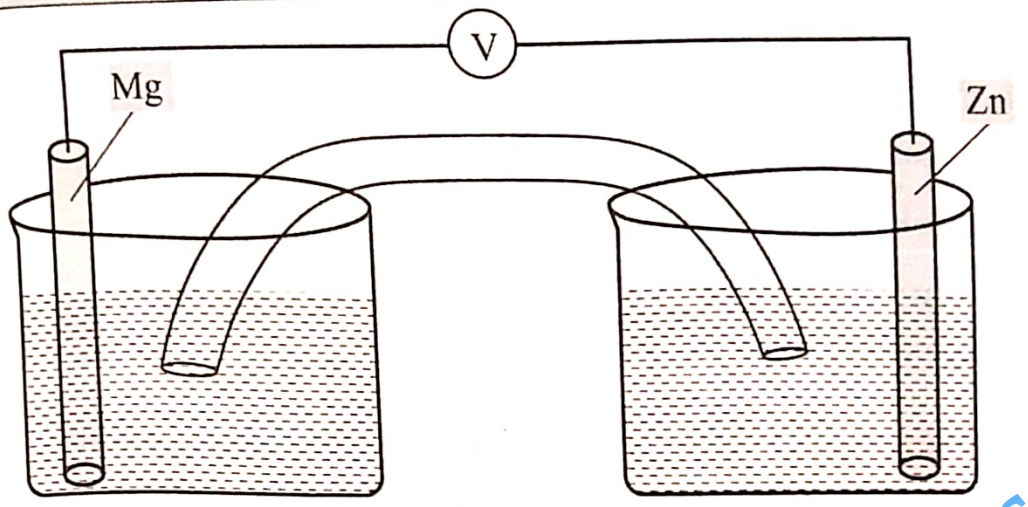

A. At the zinc electrode, zinc ions are formed

B. The electrode potential is measured by the voltmeter

C. The following reaction takes place at the magnesium electrode
$Mg^{2+} + 2e^- \rightarrow Mg$

D. Electrons flow from zinc electrode to magnesium electrode

Solution

- Mg is anode and Zn is cathode because of reactivity of the metals
- Anode reaction: $Mg \rightarrow Mg^{2+} + 2e^-$
- Cathode reaction: $Zn^{2+} + 2e \rightarrow Zn$
- Electron flow from Mg to Zn electrodes.

Answer: B

70. What type of energy conversion takes place in galvanic cell?

Answer: Chemical energy to electrical energy

71. What type of salt solution is used to fill salt bridge?

Solution

Neutral salt ($NaNO_3$, KCl) solution

72. Write cell notation for the galvanic cell with cell reaction

$2Al(s) + 3Zn^{2+}(0.1M) \longrightarrow 2Al^{3+}(0.1M) + 3Zn(s)$

Solution

$Al(s) \mid Al^{3+}(0.1M) \mid\mid Zn^{2+}(0.10M) \mid Zn(s)$

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73. Write anode, cathode and cell reaction: for the galvanic cells below

a. $Mg(s) \mid Mg^{2+}(1M) \mid \mid Zn^{2+}(aq) \mid Zn(s)$

b. $Pb(s) \mid Pb^{2+}aq \mid \mid Cl_2(s) \mid Cl^-(aq) \mid C(s)$

Solution

a. $Mg(s) \longrightarrow Mg^{2+}(1M) + 2e^-$: Anode reaction

$Zn^{2+}(1M) + 2e^- \longrightarrow Zn(s)$: Cathode reaction

$Mg(s) + Zn^{2+}(1M) \longrightarrow Mg^{2+}(1M) + Zn(s)$: cell reaction

b. $Pb(s) \longrightarrow Pb^{2+}(aq) + 2e^-$: Anode reaction

$Cl_2(g) + 2e^- \longrightarrow 2Cl^-(aq)$: Cathode reaction

$Pb(s) + Cl_2(g) \longrightarrow Pb^{2+}(aq) + 2Cl^-(aq)$: cell reaction

74. Consider the following cell notation

$Al(s) \mid Al^{3+}(aq) \mid \mid Zn^{2+}(aq) \mid Zn(s)$

a. What substances are the component of anode compartment

b. What substances are the component of the cathode compartment

c. Write the net ionic equation for the cell reaction.

Solution

a. Al electrode and Al^{3+} ions in the electrolyte solution

b. Zn electrode in the Zn^{2+} ions

c. $2Al(s) + 3Zn^{2+}(aq) \rightarrow 3Zn(s) + 2Al^{3+}(eq)$

8.7 Standard Reduction Potential (E°_{red})

Reduction potential is the measure of the ability of a species to gain electrons in reduction half - reaction.

Oxidation potential is the measure of the tendency of species to lose electrons in the oxidation half - reaction.

If the reagents are under standard conditions (temperature = 25°C (298K), Pressure = 1atm (760 torr), concentration = 1M) the half - cell potentials are known as **standard reduction potential (E^0_{red}), and standard oxidation potential (E^0_{ox})**

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Standard cell potential (E^0_{cell}) is the sun of the standard oxidation and reduction potentials since always the standard reduction potentials are given, it is also defined as the difference between reduction potential of cathode and anode

$$E^0_{cell} = E^0_{ox} + E^0_{red} = E^0_{red}(cathode) - E^0_{red}(anode)$$

Standard hydrogen electrode(SHE) is used as a reference electrode to determine the reduction potentials of other substances.

$$2H^+(aq) + 2e^- \rightleftharpoons H_2(g),\ E^0_{red}(SHE) = 0.00V$$

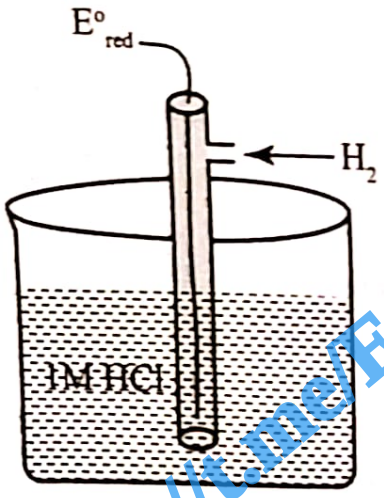

Fig 8.9 Standard Hydrogen Electrode

The standard reduction potential is measured by coupling each half - cells with the SHE. SHE is used as cathode for substance above it in the electrochemical series but as anode for substance below it.

Example

Determine the standard reduction potential of aluminum if the standard cell potential is 1.66V.

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Solution

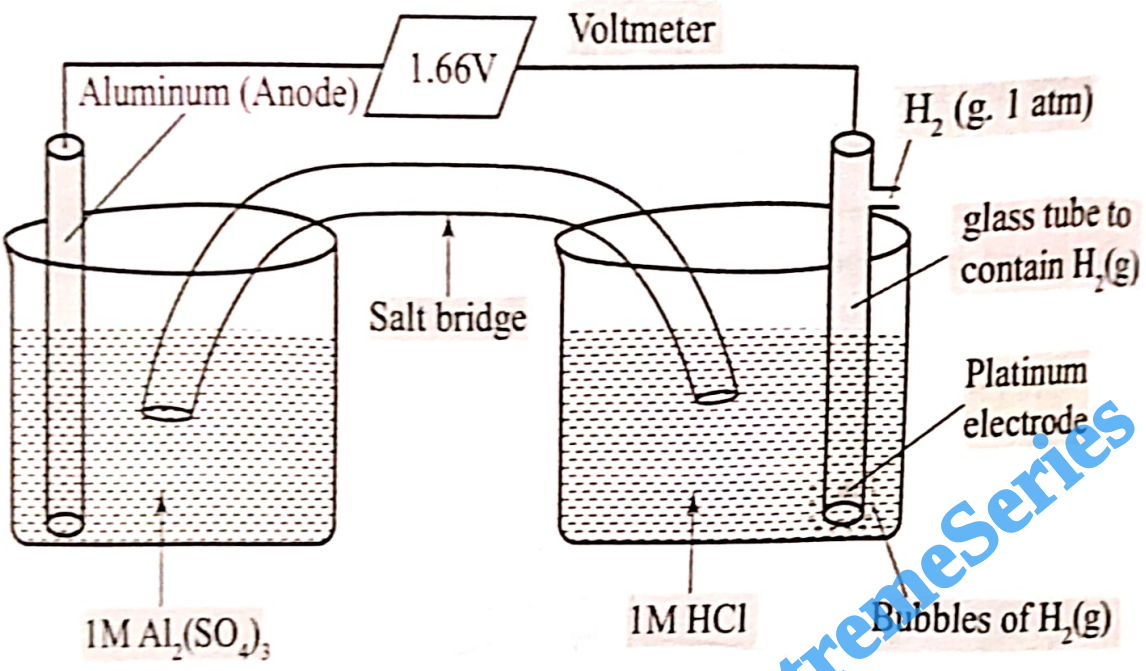

Fig 8.10 A cell Consisting of an Aluminium and Standard Hydrogen Electrode

$Al \rightarrow Al^{3+} + 3e^-$ anode reaction

$2H^+(ag) + 2e^- \rightarrow H_2(g)$ cathode reaction

Cell Notation: $Al(s) \mid Al^{3+}(1M) \mid\mid H^+(1M) \mid H_2(1atm) \mid Pt(s)$

$E^0_{cell} = E^0_{red}$ (SHE) - E^0_{red} (Al)

E^0_{red} (Al) = E^0_{red} (SHE) - E^0_{cell}

= 0.00V - 1.66V = -166V

Thus, the standard reduction potential of aluminium is - 1.66V.

Note:

1. The more positive the standardard reduction potential, the easier to be reduced (it is a better oxidizing agent)

Example

$Ag^+ + e^- \longrightarrow Ag$, $E^0_{red} = 0.80V$

$Zn^{2+} + 2e^- \longrightarrow Zn$, $E^0_{red} = -0.76V$

$Cu^{2+} + 2e^- \longrightarrow Cu$, $E^0_{red} = 0.34V$

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The strongest oxidizing agent is _____

Solution

Ag^+ because it has the highest positive E^0_{red}

Ag^+ is also the easiest to be reduced at cathode

2. Standard reduction potential is intensive property (independent on the amount of matter)

$$Cu^{2+} + 2e^- \longrightarrow Cu,\ E^0_{red} = 0.34V$$

$$2Cu^{2+} + 4e^- \longrightarrow 2Cu,\ E^0_{red} = 0.34V$$

3. When the reduction - half reaction is reversed, the magnitude remains same but the sign changes.

$$Cu^{2+} + 2e \longrightarrow Cu,\ E^0_{red} = 0.34V$$

$$Cu \longrightarrow Cu^{2+} + 2e^-,\ E^0_{ox} = -0.34V$$

$$\boxed{E^0_{ox} = -E^0_{red}}$$

4. The more positive the standard oxidation potential (E^0_{ox}), the easier is to be oxidized (better reducing agent.)

Example

$$Zn \longrightarrow Zn^{2+} + 2e,\ E^0_{ox} = 0.76V$$

$$Cu \longrightarrow Cu^{2+} + 2e^-,\ E^0_{ox} = -0.34V$$

Here, Zn is better reducing agent than Cu

5. $E^0_{cell} > 0$, the cell reaction is spontaneous

$E^0_{cell} < 0$, the cell reaction is non - spontaneous

$E^0_{cell} = 0$, the cell reaction is at equilibrium

6. By convention, the half - reaction are written as reduction

Table: Standard reduction potential (E^0_{red}) half - reactions

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Half - Reaction	E^0_{red} in (V)
$Li^+ (eq) + e \rightarrow Li(s)$	-3.05
$K^+ (eq) + e^- \rightarrow K(s)$	-2.93
$Ba^{2+}(eq) + 2e \rightarrow Ba(s)$	-2.90
$Sr^{2+}(eq) + 2e \rightarrow Sr(s)$	-2.89
$Ca^{2+}(eq) + 2e\ Ca(s) \rightarrow Ca(s)$	-2.87
$Na^+(aq) + e \rightarrow Na(s)$	-2.71
$Mg^{2+}(aq) + 2e^- \rightarrow Mg(s)$	-2.37
$Be^{2+} (aq) + 2e \rightarrow Be(s)$	-1.85
$Al^{3+} (aq) + 3e^- \rightarrow Al(s)$	-1.66
$Mn^{2+}(aq) + 2e \rightarrow Mn (s)$	-1.18
$2H_2O(1) + 2e^- \rightarrow H_2(g) + 2OH^-(aq)$	-0.83
$Zn^{2+}(aq) + 2e\text{-} \rightarrow Zn(s)$	-0.76
$Cr^{3+} (aq) + 3e^- \rightarrow Cr(s)$	-0.74
$Fe^{2+}(aq) + 2e^- \rightarrow Fe(s)$	0.44
$Cd^{2+}(aq) + 2e^- \rightarrow Cd(s)$	-0.40
$PbSO_4(s) + 2e\text{-} \rightarrow Pd(s) + SO_4^{2-}(aq)$	-0.31
$CO^{2+} (aq) + 2e^- \rightarrow Co(s)$	-0.28
$Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$	-0.25
$Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$	-0.14
$Pb^{2+}(aq) + 2e^- \rightarrow Pb (s)$	-0.13
$2H^+(aq) + 2e \rightarrow H_2(g)$	**0.00**
$Sn^{4+}(aq) + 2e \rightarrow Sn^{2+}(aq)$	+ 0.13
$SO_4^{2+}(aq) + 4H^+(aq) + 2e^- \rightarrow SO_2(g) + 2H_2O(1)$	+ 0.15

Increasing strength of oxidizing agent

Increasing strength of reducing agent

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Increasing strength of oxidizing agent

Increasing strength of reducing agent

$Cu^{2+}(aq) + e^- \rightarrow Cu^+(aq)$	+ 0.20
$AgCl(s) + e^- \rightarrow Ag(s) + Cl^-(aq)$	+ 0.22
$Cu^{2+}(aq) + 2e \rightarrow Cu(s)$	+ 0.34
$O_2(g) + 2H_2O(1) + 4e^- \rightarrow 4OH^-(aq)$	+ 0.40
$I_2(s) + 2e^- \rightarrow 2I^-(aq)$	+ 0.53
$Mno^-_4(aq) + 2H_2O(1) + 3^- \rightarrow Mno_2(s)$	+ 0.59
$O_2(g) + 2H^+(aq) + 2e^- \rightarrow H_2O_2(aq)$	+ 0.68
$Fe^{3+}(aq) + e^- \rightarrow Fe^{2+}(aq)$	+ 0.77
$Ag^+(aq) + e^- \rightarrow Ag(s)$	+ 0.80
$Hg_2^{2+}(aq) + 2e^- \rightarrow 2Hg(1)$	+ 0.85
$2Hg^{2+}(aq) + 2e^- \rightarrow Hg_2^{2+}(aq)$	+ 0.92
$NO^-_3(aq) + 4H^+(aq) + 3e \rightarrow NO(aq) + 2H_2O(1)$	+ 0.96
$Br_2(1) + 2e^- \rightarrow 2Br(aq)$	+ 1.07
$O_2(g) + 4H^+(aq) + 4e^- \rightarrow 2H_2O(1)$	+ 1. 23
$MnO_2(s) + 4H^+(aq) + 2e^- \rightarrow Mn^{2+}(aq) + H_2O(1)$	+ 1.23
$Cr_2O_7^{2-}(aq) + 14H^+(aq) + 2e^- \rightarrow Mn^{2+}(aq) + 7H_2O(1)$	+ 1.33
$2Cl^-(aq) + 2e^- \rightarrow Cl_2(g)$	+ 1.36
$Au^{3+}(aq) + 3e^- \rightarrow Au(s)$	+ 1.50
$MnO^-_4(aq) + 8H^+(eq) + 5e^- \rightarrow Mn^{2+}(aq) + 4H_2O(1)$	+ 1.51
$Ce^{3+}(aq) + 3e\text{-} \rightarrow Ce(s)$	+ 1.61
$PbO_2(s) + 4H^+(aq) + SO^{2-}_4(aq) + 2e^- \rightarrow PbSO_4(s) + 2H_2O(1)$	+ 1.70
$H_2O_2(aq) + 2H^+(aq) + 2e^- \rightarrow 2H_2O(1)$	+ 1.77
$Co^{3+}(aq) + e\text{-} \rightarrow Co^{2+}(aq)$	+ 1.82

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$O_3(g) + 2H^+(aq) + 2e^- \rightarrow O_2(g) + H_2O(1)$	+ 2.07
$F_2(g) + 2e^- \rightarrow 2F^-(aq)$	+ 2.87

From the standard Reduction Potential Table

- F_2 is the strongest oxidizing agent ($E^0_{red} = + 2.87V$)
- Li is the strongest reducing agent ($E^0_{ox} = 3.05V$)

Illustrative Examples

75. Which one of the following reactions would occur spontaneously in an aqueous solution at 25°C? Assume that the initial concentrations of the dissolved species are all 1.0M. (ESSLCE)

Half reaction	$E^0(V)$
$Sn^{2+}(aq) + 2e^- \rightarrow Sn(s)$	- 0.14
$Br_2(1) + 2e^- \rightarrow 2Br(aq)$	+ 1.07
$Aq^+(aq) + e^- \rightarrow Ag(s)$	+ 0.80
$Ni^{2+}(aq) + 2e^- \rightarrow Ni(s)$	- 0.25
$Ca^{2+}(aq) + 2e^- \rightarrow Ca(s)$	- 2.87
$Cd^{2+}(aq) + 2e^- \rightarrow Cd(s)$	- 0.40
$Cu^{2+}(aq) + K^- \rightarrow Cu^+(s)$	0.15
$Fe^{3+}(aq) + e^- \rightarrow F^{2+}(aq)$	0.77

A. $Ca(s) + Cd^{2+}(aq) \rightarrow Ca^{2+}(aq) + Cd(s)$

B. $Cu^{2+}(aq) + Fe^{2+}(aq) \rightarrow Cu^+(aq) + Fe^{3+}(aq)$

C. $2Br^-(aq) + Sn^{2+}(aq) \rightarrow Br_2(1) + Sn(s)$

D. $2Ag(s) + Ni^{2+}(aq) \rightarrow 2Ag^+(aq) + Ni(s)$

Solution

A. $E^0_{cell} = E^0_{red}$ (cathode) - E^0_{red}(anode) = 0.40 - (2.87) = + 2.47V

It is spontaneous cell reaction because $E^0_{cell} > 0$

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B) $E^0_{cell} = 0.15 - 0.77 = -0.64V$
C) $E^0_{cell} = -014 - 1.07 = -1.21V$
D) $E^0_{cell} = -0.25 - 0.50 = 1.05V$
} Non - spontaneous cell reactions

Answer: A

76. For the electro chemical cell represented by :

$Zn \mid ZnSO_4(aq) \mid\mid CuSO_4(aq) \mid Cu(s)$, the standard reduction potentials for $Cu^{2+} \mid Cu$ and $Zn^{2+} \mid Zn$ are + 0.34V and -0.76V, respectively. What will be the cell reaction that occurs spontaneously and its cell voltage?

A. $Cu + Zn^{2+} \rightleftharpoons Cu^{2+} + Zn$, -1.10V

B. $Zn + Cu^{2+} \rightleftharpoons Zn^{2+} + Cu$, + 0.42V

C. $Cu^{2+} + Zn^{2+} \rightleftharpoons Zn + Cu$, -0.42V

D. $Cu^{2+} + Zn \rightleftharpoons Cu + Zn^{2+}$, + 1.10V

Solution

For spontaneous cell reaction, the half - reaction with more positive E^0_{red} is used as cathode and the other as anode.

$E^0_{cell} = E^0_{red}$ (cathode) - E^0_{red} (anode)

$= 0.34V - (-0.76V) = 1.10V$

Answer: D

77. Which of the following of the following statements is true?

A. The more positive the value of E^0_{red}, the greater the driving force for reduction

B. The more exothermic the value of E^0_{red}, the greater the driving force for reduction

C. The more endothermic the value of E^0_{red}, the greater the driving force for reduction.

D. The more negative the value of E^0_{red}, the greater the driving force reduction.

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Solution

➥ The more positive the E^0_{red}, the greater the driving force for reduction.

➥ The more negative the E^0_{red}, the greater the driving force for oxidation

Answer: A

78. A solution in an electrolytic cell contains Cu^{2+} (E^0_{red} = 0.34V), Ag^+ (E^0_{red} = 0.80V) and Zn^{2+} (E^0_{red} = -0.76V). If the voltage is initially low and slowly increased, in which order will the metals be plated out on to the cathode?

A. $Zn^{2+} > Cu^{2+} > Ag^+$ C. $Ag^+ > Zn^{2+} > Cu^{2l}$

B. $Cu^{2+} > Zn^{2+} > Ag^+$ D. $Ag^+ > Cu^{2+} > Zn^{2+}$

Solution: The more positive the reduction potential, the easier to be plated out on to cathode.

Answer: D

79. Given the E^0_{red}:

$Cr^{3+} \mid Cr : E^0_{red} = -0.74V$

$Sn^{4+} \mid Sn^{2+} : E^0_{red} = +0.15V$

Calculate the standard cell potential by coupling the two half - cells

Solution

More positive E^0_{red} is used as cathode and the other as anode.

$E^0_{cell} = E^0_{red}$ (cathode) - E^0_{red} (anode)

= 0.15V - (-0.74V) = 0.85V

80. An electrochemical cell constructed for the reaction: $Cu^{2+}(aq) + M(s) \longrightarrow Cu(s) + M^{2+}(aq)$ has a E^0_{cell} of 0.75V. What is the standard reduction potential of $M^{2+}(aq)$?

Solution

From the cell reaction, M is used as anode and Cu as cathode.

$E^0_{red}\ Cu^{2+} \mid Cu = 0.34V$, $E^0_{cell} = 0.75V$

$E^0_{cell} = E^0_{red}$ (cathode) - E^0_{red} (anode)

E^0_{red} (anode) = E^0_{red} (cathode) - E^0_{cell}

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$= 0.34V - 0.75V$

$= -0.41V$

$\therefore E^0_{red}\ M^{2+} \mid M = -0.41V$

81. Consider the following half - reactions:

$Fe(OH)_2(s) + 2e^- \longrightarrow Fe(s) + 2OH^-(aq),\ E^0 = -0.88V,$

$NiO_2(s) + 2H_2O(1)\ 2e^- \longrightarrow Ni(OH)_2(s) + 2OH^-(aq),\ E^0 = 0.49V$

What is the standard cell potential for the given voltaic cell?

Solution

$E^0_{cell} = E^0_{red}(cathode) - E^0_{red}(anode) = (0.49V - (-0.88V) = +1.37V$

82. Given the following reduction potentials

$Ag^+(aq) + e^- \longrightarrow Ag(s),\ E^0 = 0.80V$

$Pb^{2+}(aq) + 2e^- \longrightarrow Pd(s),\ E^0 = 0.13V$

$V^{2+}(aq) + 2e^- \longrightarrow V(s),\ E^0 = -1.18V$

Write two cell reactions that will occur spontaneously and two cell reactions that will occur non - spontaneously.

Solution

Spontaneous cell reaction ($E^0_{cell} > 0$)

$Pb(s) + 2Ag^+(aq) \longrightarrow Pd^{2+}(aq) + 2Ag(s)$

$V(s) + 2Ag^+(aq) \longrightarrow V^{2+}(s) + 2Ag(s)$

Non - spontaneous cell reaction ($E^0_{cell} < 0$)

$V^{2+}(aq) + 2Ag(s) \longrightarrow V(s) + 2Ag^+(aq)$

$V^{2+}(aq) + Pb(s) \longrightarrow V(s) + Pb^{2+}(aq)$

83. Given the standard reduction potential.

$E^0\ Zn^{2+} \mid Zn = -0.76V$

$E^0\ Cu^{2+} \mid Cu = 0.34V$

$E^0\ Sn^{2+} \mid Sn = -0.14V$

$E^0\ Ni^{2+} \mid Ni = -0.26V$

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Identify the

a. Strongest oxidizing agent
b. Strongest reducing agent
c. Most difficult to reduce
d. Most difficult to oxidize

Solution

a. Cu^{2+} b. Zn c. Zn^{2+} d. Cu

84. Given the cell reaction:

$2Al^{3+}(aq) + 3Cu(s) \longrightarrow 2Al(s) + 3Cu^{2+}(aq)$

What is the standard cell potential for this cell?

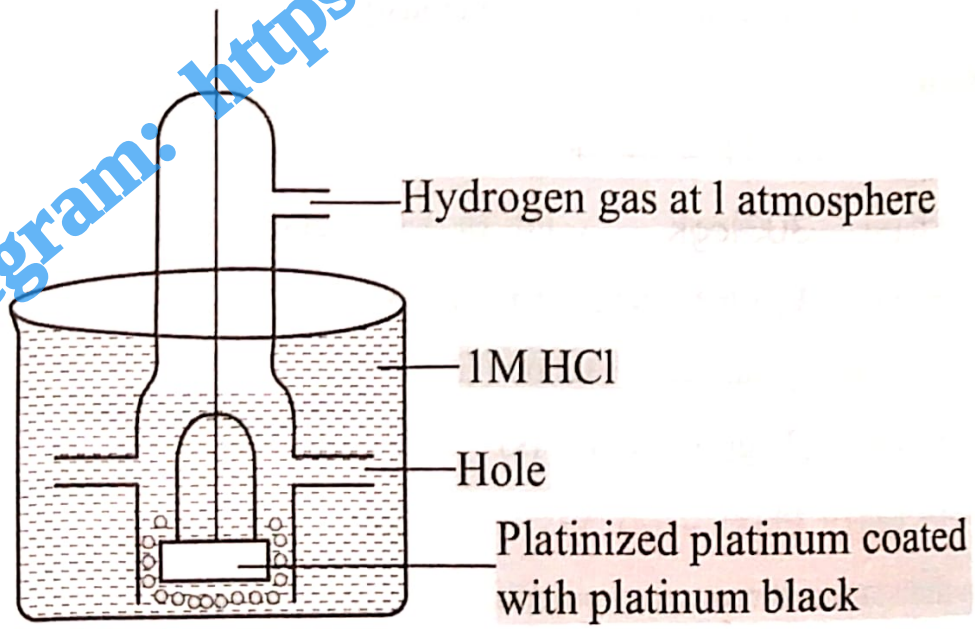
Solution

Anode: Cu, cathode: Al from the cell rxn

$E^0_{cell} = -1.66V - 0.34V = -2.00V$

85. Draw the standard hydrogen electrode and describe its component.

Solution



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Electromotive Force /EMF/

It is the maximum potential difference between two electrodes of the galvanic cell

It is the net voltage between the oxidation and reduction half - reactions

At standard condition EMF is called standard EMF or E^0_{cell}

$E^0_{cell} = E^0cathode - E^0anode = E^0_{ox} + E^0_{red}$

$\Delta G^0 = -nFE^0_{cell}$ ----- (1)

$\Delta G^0 = - RTlnK$ ------ (2)

Relate (1) and (2)

$- nFE^0_{cell} = -RTlnK$

$$E^0_{cell} = \frac{8.134\frac{J}{mol.K} \times 298K}{n \times 96500\frac{J}{V.mol}} lnK$$

$$\boxed{E^0_{cell} = \frac{0.0257V}{n} lnK}$$

$$\boxed{E^0_{cell} = \frac{0.0592V}{n} logK}$$

$1nk = 2.303logK$

n = total electrons gained or lost in the cell reaction

K- equilibrium constant

ΔG^0 - Gibb's free energy change at standard condition

Note

i. For spontaneous process

$E^0_{cell} > 0, \Delta G^0 < 0$ and $K > 1$

ii. For non - spontaneous process

$E^0_{cell} < 0, \Delta G^0 > 0$ and $K < 1$

iii. For equilibrium process

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$E^0_{cell} = 0, \Delta G^0 = 0$ and $K = 1$

Example

Calculate the standard free - energy change for the following reaction at 25°c

$$2Au(s) + 3Ca^{2+}(1M) \rightleftharpoons 2Au^{3+}(1M) + 3Ca(s)$$

$E^0Ca^{2+} \mid Ca = -2.87V$, $E^0Au^{3+} \mid Au = 1.50V$

Solution

$E^0_{cell} = E^0$cathode $- E^0$andode $= -2.87 - 1.50 = -4.37V$

$n = 2 \times 3 = 3 \times 2 = 6$

$$\Delta G^0 = -nFE^0_{cell} = -6 \times 96500 \frac{J}{V.mol} \times (-4.37V)$$

$$\Delta G^0 = 2.53 \times 10^6 J/mol$$

$$E^0_{cell} = \frac{0.0592V}{n} \log K$$

$$K = 10^{\frac{nE^0_{cell}}{0.0592V}} = 10^{\frac{6 \times (-4.37V)}{0.0592V}} = 1.23 \times 10^{-443}$$

Effect of Concentration on Cell Potential

Cell potential increases as concentration of reactants increases or concentration of products decreases and vice versa. This was explained by Nernst equation.

The Nernst Equation

Quantitatively, we use the nernst equation to explain the effect of concentration on cell potential.

$\Delta G = -nFE_{cell}$ ----- (1), at non - standard condition

$\Delta G^0 = -nFE^0_{cell}$ ----- (2), at standard codition

$\Delta G = \Delta G^0 + RT\ln Q$ ---- (3)

Substitute (1) and (2) in (3)

$-nFE_{cell} = -nFE^0_{cell} + RT\ln Q$

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$$E_{cell} = E^0_{cell} = \frac{RT}{nF} \ln Q \quad T = 25^0C = 298K$$

$$E_{cell} = E^0_{cell} - \frac{00257V}{n} \ln Q$$

$$E_{cell} = E^0_{cell} - \frac{0.0592V}{n} \log Q$$

Q - reaction quotient

Note:

$E_{cell} > 0$, spontaneous process

$E_{cell} < 0$, Non - spontaneous process

$E_{cell} = 0$, Equilibrium process

Example:

Given the cell reaction:

$$Co(s) + Fe^{2+}(aq) \rightleftharpoons Co^{2+}(aq) + Fe(s)$$

$[Co^{2+}] = 0.15M$, $[Fe^{2+}] = 0.30M$

$E^0Fe^{2+} \mid Fe = -0.44V$

$E^0Co^{2+} \mid Co = -0.28V$

Is the cell reaction occur spontaneously?

Solution

$E^0_{cell} = E^0$cathode $- E^0$ anode $= -0.44V - (-0.28V) = -0.16V$, $n = 2$

$$E_{cell} = E^0cell - \frac{0.0592V}{n} \log Q$$

$$E_{cell} = -0.16V - \frac{0.0592V}{2} \log \frac{[Co^{2+}]}{[Fe^{2+}]}$$

$$E_{cell} = -0.16V - \frac{0.0592V}{2} \log \frac{0.30V}{0.15V}$$

$$E_{cell} = -0.17V$$

Thus, the cell reaction is non - spontaneous

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Illustrative Examples

86. The standard cell potential (E^0cell) for the reaction below is 1.10V. What is the cell potential for this reaction when $[Cu^{2+}] = 1 \times 10^{-5}M$ and $[Zn^{2+}] = 1M$?

A. -0.95V B. 0.95V C. 0.80V D. -0.80V

Solution

$$E^0_{cell} = E^0_{cell} - \frac{0.0592v}{n} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$= 1.10V - \frac{0.052V}{V} \log \frac{1M}{1 \times 10^{-5}M} = 0.95V$$

Answer: B

87. The two standard electrodes potentials involved in the nickel - cadium rechargeable cell are given below. Calculate ΔG^0 in kJ of the cell.

$NiO_2(s) + 2H_2O(1) + 2e^- \longrightarrow Ni(OH)_2(s) + 2OH^-(aq)$ $E^0_{red} = 0.49V$

$Cd(OH)_2(s) + 2e^- \longrightarrow Cd(s) + 2OH^-(aq)$, $E^0_{red} = -0.76V$

A. -184 B. -153 C. -241 D. -206

Solution

$E^0_{cell} = E^0$cathod - E^0anode = 0.49 - (-0.76) = 1.25V

$$\Delta G^0 = -nFE^0_{cell} = -2 \times 96500 \frac{J}{mol} \times 1.25V$$

$$\Delta G^0 = -241250 \frac{J}{mol} = -241kJ/mol$$

Answer: C

88. The standard EMF for the reaction

$Zn(s) + Cu^{2+}(aq) \longrightarrow Cu^{2+}(aq) + Cu(s)$ is 1.10V

What would be the for that cell reaction when 0.1M Cu^{2+} and 0.1M Zn^{2+} solution are used at 298K?

A. 1.10V B. 0.10V C. -1.10V D. -1.11V

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Solution

$$E_{cell} = E^0_{cell} - \frac{0.0592V}{n} \log \frac{[Zn^{2+}]}{[Cu^{2+}]}$$

$$= \frac{1.10V - 0.0592V}{2} 10g \frac{0.1}{0.1}$$

$$= 1.10V$$

Answer: A

89. If a pieces of zinc metal is placed in 1M $Cu(NO_3)_2$ solution

$Zn^{2+} + 2e^- \longrightarrow Zn, E^0 = -0.76$

$Cu^{2+} + 2e \longrightarrow Cu, E^0 = 0.34V$

A. Zinc dissolves completely

B. Reaction occurs and when equilibrium is reached, $[Zn^{2+}] > [Cu^{2+}]$

C. Reaction occurs and when equilibrium is reached, $[Zn^{2+}] << [Cu^{2+}]$

D. Reaction occurs and until equilibrium is reached

Solution

$E^0_{cell} = 0.34V - (-0.76V) = 1.10V$

$E^0_{cell} > 0$, the net reaction will occur spontaneously as written that is zinc will dissolve in 1M $Cu(NO_3)_2$.

Answer: A

90. From where comes the electromotive force in a galvanic cell?

Solution

The EMF in a galvanic cell comes from the spontaneous redox reactions that push electrons from anode to cathode through the external circuit

91. Calculate the equilibrium constant for the following reaction at 298K.

$$Sn(s) + 2Ag^+(aq) \rightleftharpoons Sn^{2+}(aq) + 2Ag(s)$$

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Solution

$$K = 5.71 \times 10^{31}$$

Concentration Cells

Galvanic cells whose electrodes are made up of the same material are **concentration cells.** These cells have zero standard cell potential. The anode concentration should be dilute and the cathode concentration should be concentrated.

$$E_{cell} = \frac{-0.0592V}{n} \log \frac{[anode]_{diute}}{[cathode]_{conc}}$$

Determination of PH

PH can determined by using Nernst equation and solution

$$pH = \frac{E^{\circ}_{ox} - E_{cell}}{0.0592V}, \text{ if } n = 2$$

8 . 8 Batteries

Battery is an electro chemical device that a current or flow of electrons at constant voltage. There are two types of batteries.

1. Primary cells
2. Secondary cells

I. Primary Cells

They are galvanic cells that cannot be recharged because the reaction is **irreversible**

They are use and throw batteries and their power gradually diminishes.

Example

i. Zinc - carbon or Leclanche Dry Cell

It does not use any liquid electrolyte that is why it is called dry cell.

Anode : Zinc

Cathode : Graphite

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Electrolytes: NH_4Cl, MnO_2, $ZnCl_2$

Anode reaction: $Zn(s) \longrightarrow Zn^{2+}(aq)$

Cathode reaction: $2MnO_2(s) + H_2O(1) + 2e^- \longrightarrow Mn_2O_3(s) + 2OH^-(aq)$

An acid - base reaction occurs between OH^- and NH_4^+ derived from NH_4Cl

$NH^+_4(aq) + OH^-(aq) \longrightarrow NH_3(g) + H_2O(1)$

A build up of ammonia gas cannot be permitted to occur around the cathode because it disrupts the current. This can be prevented by the reaction of Zn^{2+} and ammonia.

$Zn^{2+}(aq) + 2NH_3(g) + 2Cl^-(aq) \longrightarrow Zn(NH_3)_2\ Cl_2(s)$

Cell reaction: $Zn(s) + 2MnO_2(s) + 2NH_4Cl(aq) \longrightarrow Zn(NH_3)_2\ Cl_2(s)$

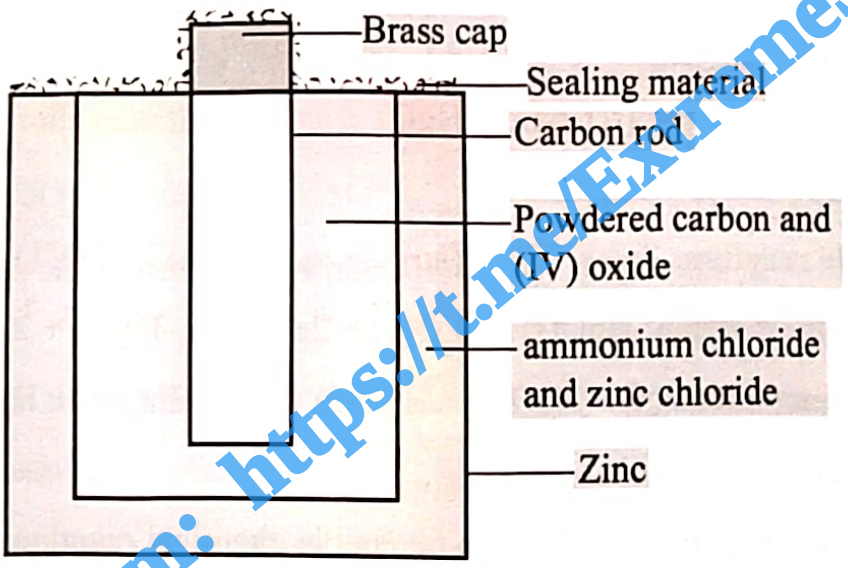

Fig 8.11 A Leclanche dry cell

ii. Alkaline Battery ($Zn - MnO_2$ Cell)

It is more expensive and durable than Laclanche dry cell.

Anode: Zinc (Zn)

Cathode: Manganese (IV) oxide (MnO_2)

Electrolyte: Sodium or potassium hydroxide (NaOH or KOH)

Anode reaction: $Zn\ (s) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(1) + 2e^-$

Cathode reaction: $2MnO_2(s) + H_2O(1) + 2e^- \longrightarrow Mn_2O_3(s) + 2OH^-(aq)$

Cell reaction: $Zn(s) + 2MnO_2(s) \longrightarrow ZnO(s) + Mn_2O_3(s)$

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Voltage = 1.50V

iii. Silver Oxide Battery($Zn - Ag_2O$ cell)

Anod: Zinc

Cathode: silver oxide

Electrolyte: KOH or NaOH

Anode reaction: $Zn(s) + 2OH^-(aq) \longrightarrow ZnO(s) + 2H_2O(1) + 2e^-$

Cathode reaction: $Ag_2O(s) + H_2O(1) + 2e^- \longrightarrow 2Ag(s) + 2OH^-(aq)$

Cell reaction: $Zn(s) + Ag_2O(s) \longrightarrow ZnO(s) + 2Ag(s)$

iv. Mercuric Cells (Zn - HgO Cells)

Anode: Zinc (Zn)

Cathode: mercury (II) oxide (HgO)

Electrolyte: NaOH or KOH

Anode reaction: $Zn(s) + 2OH^-(aq) \longrightarrow ZnO(s) + H_2O(1)$

Cathode reaction: $HgO(s) + H_2O(1) + 2e^- \longrightarrow Hg(s) + 2OH^-(aq)$

Cell reaction: $Zn(s) + HgO(s) + HgO(s) \longrightarrow ZnO(s) + Hg(s)$

2. Secondary Cells

- They are galvanic cells with reversible chemical reaction.
- The are rechargeable batteries
- They are called accumulators

i. Nickel - Cadmium (Nicad) Battery

It is used in battery - operated tools and portable computers.

Anode: Cadmium(Cd)

Cathode: Nickel (IV) oxide (NiO_2)

Electrolyte: NaOH or KOH

...action: $Cd(s) + 2OH^-(aq) \rightleftharpoons Cd(OH)_2(s) + 2e^-$

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Cathode reaction:

$$NiO_2(s) + 2H_2O_2(1) + 2e^- \rightleftharpoons Ni(OH)_2(s) + 2OH^-(aq)$$

Cell reaction:

$$Cd(s) + NiO_2(s) + 2H_2O(1) \rightleftharpoons Cd(OH)_2(s) + Ni(OH)_2(s)$$

ii. The Lead Storage Battery (car battery)

- The positive electrode is mad of lead IV oxide
- The negative terminal is a lead plate. These terminals are made of grids of lead - antimony alloy.
- The electrolyte is dilute sulphuric acid
- When the lead acid battery is in use, the following reactions occur.

Anode reaction: $Pb(s) + SO_4^{2-}(aq) \longrightarrow PbSO_4(s) + 2e^-$

Cathode reaction:

$$PbO_2(s) + 4H^+(aq) + SO^{2-}{}_4(aq) \longrightarrow PdSO_4(s) + 2H_2O(1)$$

Cell reaction: $Pb(s) + PbO_2(s) + 2SO_4^{2-}(aq) + 4H^+(aq) \longrightarrow 2PbSO_4(s) + 2H_2O(1)$

- When the car battery is charged, the following reactions occur.

Anode reaction:

$$PbSO_4(s) + 2H_2O(1) \longrightarrow PbO_2(s) + 4H^+(aq) + SO^2{}_4(aq) + 2e^-$$

Cathode reaction: $PbSO_4(s) + 2e^- \longrightarrow Pd(s) + SO^{2-}{}_4(aq)$

Cell reaction:

$$2PbSO_4(s) + 2H_2O(1) \longrightarrow Pd(s) + PbO_2(s) + 4H^+(aq) + 2SO_4^{2-}(aq)$$

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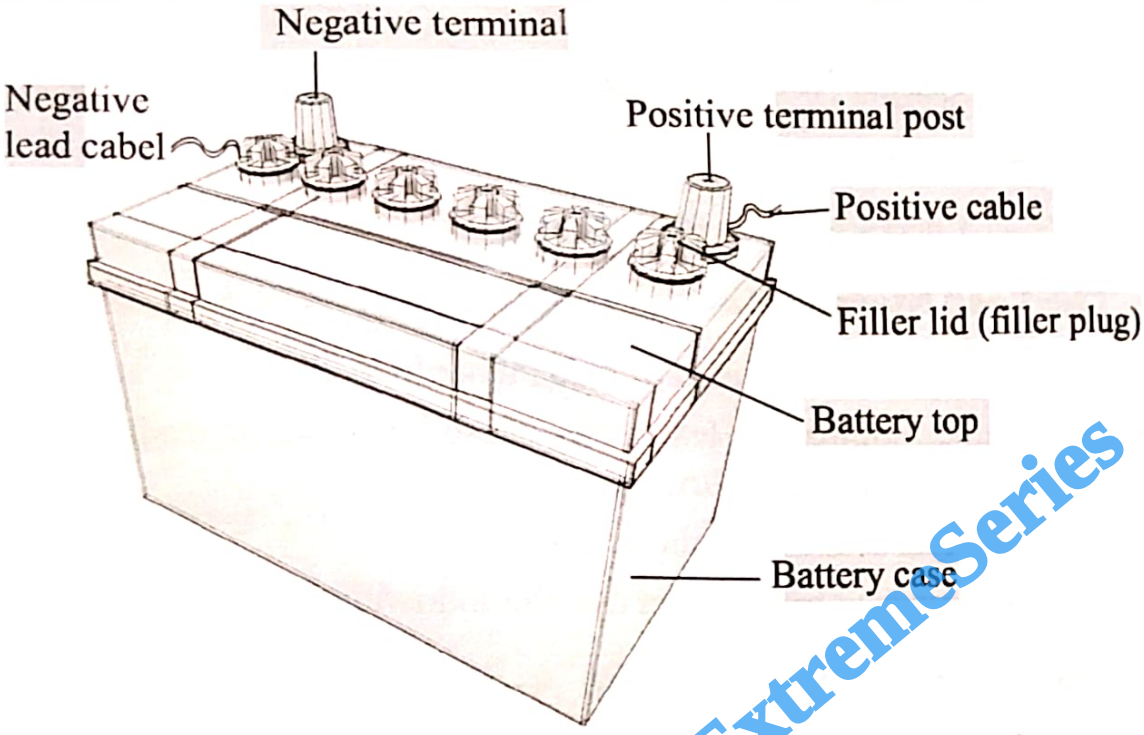

Fig 8.12 A cary Battery

Under normal operating condition each cell produces 2V: a total of 12V from a six cells is used to power the ignition circuit of the automobile and its electrical systems.

Fuel Cells

These are galvanic cell that require a continuous supply of reactants to keep functioning

- They do not store chemical energy like batteries
- The common example of fuel cell is the hydrogen oxygen fuel cell.
- Anode and cathode: inert electrodes

Anode reaction: $2H_2(g) + 4OH^-(aq) \longrightarrow 4H_2O(1) + 4e^-, E^0_{ox} = 0.83V$

Cathode reaction: $O_2(g) + 2H_2O(1) + 4e^- \longrightarrow 4OH^-(aq)$ $_{red} = 0.40V$

Cell reaction: $2H_2(g) + O_2(g) \longrightarrow 2H_2O(1), E^0_{cell} = E^0_{ox} + E^0_{red} = +1.23V$

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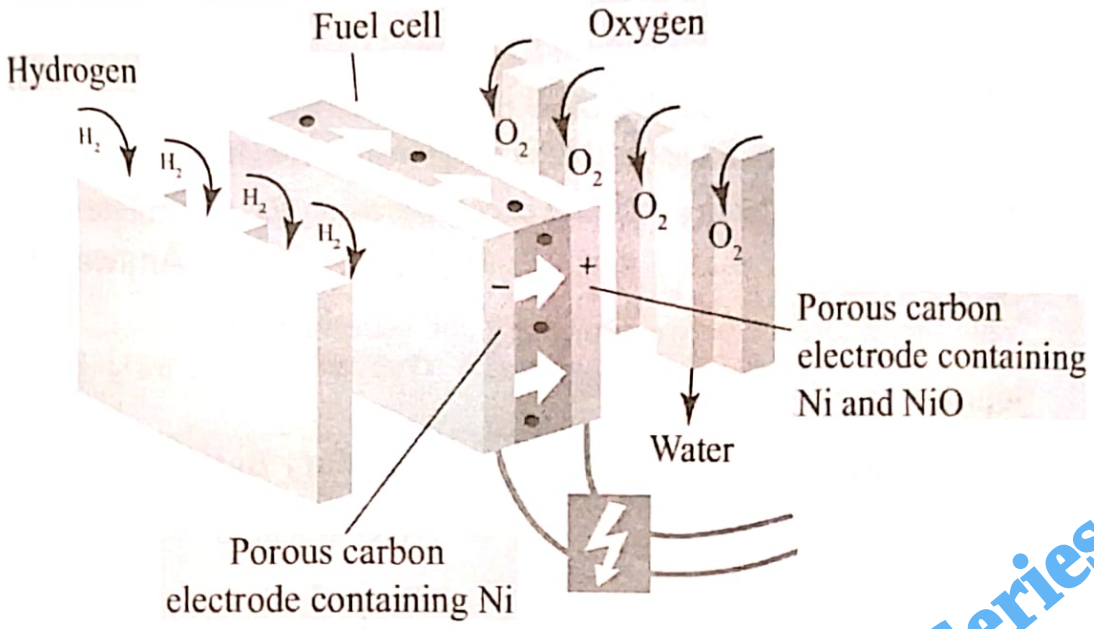

Fig 8.13 Hydrogen Oxygen Fuel Cell

92. Which one of the following galvanic cell is an example of secondary cell? (ESSLCE)

A. Zinc - carbon dry cell
B. Leclanche dry cell
C. lead storage cell
D. Daniel cell

Answer: C

93. Which of the following is true about galvanic cell used in commercial lead storage batteries? In lead storage batteries, (ESSLCE)

A. Galvanic cells are connected in parallel
B. Galvanic cells are connected in series
C. The galvanic cells are made up of PbO_2 of anodes
D. Half of the galvanic cells are connected in series.

Answer: B

94. Which of the following statement is true?

A. A battery is rechargeable
B. A battery produces electricity spontaneously
C. A battery has generally no liquid components
D. A battery produces the same amount of electricity regardless of

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composition

Solution

➥ A battery can be rechargeable or non - reachargeable

➥ A battery may be dry cell or wet cell based on the electrolyte in it.

Answer: B

95. Write the difference between primary and secondary batteries.

Solution

Primary Battery	Secondary Battery
* Not recharged	* Rechargeable
* Not reused	* Reused
* Irreversible reaction	* Reversible reaction

96. Expain the difference between battery and fuel cell

Solution

Batteries	Fuel Cells
* Store chemical energy	* Do not store chemical energy
* Pollute environment	* No impact on environment

8 . 9 Corrosion

The deterioration of metals by an oxidation - reduction reaction is known as **corrosion.**

Rusting of iron is an example of corrosion.

Moisture and air (Oxygen) are cause for rusting

Iron rusts spontaneously.

Anode reaction: $Fe(s) \longrightarrow Fe^{2+}(aq) + 2e^-, E^0_{ox} = 0.44V$

Cathode reaction: $O_2(g) + 4H^+(aq) \longrightarrow 2H_2O(1), E^0_{red} = 1.23V$

Cell reaction: $2Fe(s) + O_2(g) + 4H^+(aq) \longrightarrow 2Fe^{2+}(aq) + 2H_2O(l)$

$$E^0_{cell} = E^0_{ox} + E^0_{red} = 0.44 + 1.23 = 1.67V$$

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Note: This reaction occurs in acidic medium, the H^+ ions are supplied in part to form carbonic acid (H_2CO_3).

The Fe^{2+} ions formed at the anode are further oxidized by oxygen

$$4Fe^{2+}(aq) + O_2(g) + (4 + 2x)\, H_2O(l) \longrightarrow 2Fe_2O_3 \cdot x\, H_2O(s) + 8H^+(aq)$$

Ways Used to Prevent Rusting of Iron

1. **Electro - plating:** such as galvanizing, tin - planting and chrom plating are used to coat iron containers
2. **Oiling**
3. **Painting**
4. **Cathodic protection** is a process in which the metal that is to be protected from corrosion is made the cathode.
   - ➥ The metal that is oxidized while protecting the cathode is called **sacrificial anode**
   - ➥ Cathodic protection is used to protect under ground pipe.

Illustrative Examples

97. How does the tendency of iron to rust depend on the pH of solution?

Solution

The tendency of iron to rust increases as the value of pH decreases.

98. What is the general formula of rust?

Solution

$Fe_2O_3 \cdot xH_2O$

99. List some metals that protect iron from rust by cathodic protection

Solution

In general, metals that are more reactive than iron or with more positive oxidation potential are used as sacrificial anode.

100. Given the standard reduction potentials

$E^0Mg^{2+} \mid Mg = -2.37V$

$E^0Fe^{2+} \mid Fe = -0.44V$

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$E^0Cu^{2+} \mid Cu = 0.34V$

$E^0Cd^{2+} \mid Cd = -0.40V$

Which metal is used as sacrificial anode for iron?

Solution

Magnesium (Mg) because it has higher oxidation potential than iron

Solved Problems

101. Balance the following by oxidation number method

a. $I_2 + O_3 + H_2O \longrightarrow HIO_3 + O_2$

b. $KClO_3 \longrightarrow KClO_4 + KCl$

c. $C + H_2SO_4 \longrightarrow CO_2 + SO_2 + H_2O$

d. $As_2S_5 + HNO_3 \longrightarrow H_3AsO_4 + H_2SO_4 + H_2O + NO_2$

e. $CuS + HNO_3 \longrightarrow Cu(NO_3)_2 + S + H_2O + NO$

Solution

a. $I_2 + 5O_2 + H_2O \longrightarrow 2HIO_3 + 4SO_2$

b. $4KClO_3 \longrightarrow 3KClO_4 + KCl$

c. $C + 2H_2SO_4 \longrightarrow CO_2 + 2SO_2 + 2H_2O$

d. $As_2S_5 + 40HNO_3 \longrightarrow 2H_3AsO_4 + 5H_2SO_4 + 12H_2O + 40NO_2$

e. $3CuS + 8HNO_3 \longrightarrow 3Cu(NO_3)_2 + 3S + 2NO + 4H_2O$

102. Balance the following in acidic medium by ion - electron method.

a. $P \longrightarrow PH_3 + H_2PO_2^-$ b. $Cl_2 + IO_3^- \longrightarrow IO_4^-$

Solution

a) $4P + 6H_2O \longrightarrow PH_3 + 3H_2PO_2^- + 3H^+$

b. $H_2O + IO_3^- \longrightarrow IO_4^- + 2e^- + 2H^+$

$$\frac{Cl_2 + 2e \longrightarrow 2Cl^-}{H_2O + Cl_2 + IO_3^- \longrightarrow IO_4^- + 2H^+ + 2Cl^-}$$

103. Balance the following in basic medium by ion - electron method.

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a. $MnO_2 + HO_2^- \longrightarrow MnO_4^-$

b. $Cl_2O_7 + H_2O_2 \longrightarrow ClO_2^- + O_2$

Solution

a. $(MnO_2 + 4OH^- \longrightarrow MnO_4^- + 3e^- + 2H_2O) \times 2$

$(H_2O + HO_2^- + 2e^- \longrightarrow 3OH^-) \times 3$

$2MnO_2 + 8OH^- \longrightarrow 2MnO_4^- + 6\not{e} + 4H_2O$

$3H_2O + 3HO_2^- + 6\not{e} \longrightarrow 9OH^-$

$2MnO_2 + 3HO_2^- \longrightarrow 2MnO_4^- + H_2O + OH^-$

b. $Cl_2O_7 + H_2O_2 \longrightarrow ClO_2^- + O_2$

$(3H_2O + Cl_2O_7 + 8e^- \longrightarrow 2ClO_2^- + 6HO^-) \times 1$

$(2OH^- + H_2O_2 \longrightarrow O_2 + 2e^- + 2H_2O) \times 4$

$3H_2O + Cl_2O_7 + 8\not{e}^- \longrightarrow 2ClO_2^- + 6OH^-$

$8OH^- + 4H_2O_2 \longrightarrow 4O_2 + 8\not{e}^- + 8H_2O$

$Cl_2O_7 + 2OH^- + 4H_2O_2 \longrightarrow 5H_2O + O_2 + 2ClO_2^-$

104. How long will it take to deposit the copper from 500ml of a 0.2M $CuSO_4$ solution using a current of 10A?

Solution

$$n = M \times V = 0.2\frac{mol}{L} \times 0.5L = 0.1mol$$

$Cu^{2+} + 2e^- \longrightarrow Cu,\ Z = 2$

$$n = \frac{It}{ZF} \Rightarrow t = \frac{nZF}{I}$$

$$t = \frac{0.1mol \times 2 \times 96500\ \frac{A.s}{mol}}{10A} = 1930s$$

105. What current is required to plate out 0.05mol of gold from $AuCl_3$ solution in 2hours.

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Solution

$$Au^{3+} + 3e^- \longrightarrow Au,\ Z = 3$$

$$t = 2h = 2 \times 3600s = 7200s$$

$$n = \frac{It}{ZF} \Rightarrow I = \frac{nZF}{t}$$

$$I = \frac{0.05mol \times 3 \times 96500 \frac{A.s}{mol}}{7200s} = 2A$$

106. How much copper can be deposited by 2.5F of electricity?

Solution

$$Cu^{2+} + 2e^- \longrightarrow Cu$$

$$2F = 1mol$$

$$2.5F = ?$$

$$n = \frac{2.5F \times 1mol}{2F} = 1.25mol$$

107. The charge required for the reduction of 0.4mol of $K_2Cr_2O_7$ to Cr^{3+} ions is _____

Solution

$$K_2Cr_2O_7 + 6e^- \longrightarrow 2Cr^{3+},\ Z = 6$$

$$Q = nZF = 0.4mol \times 6 \times 96500 \frac{C}{mol} = 2.3 \times 10^5 C$$

108. Balance the following redox reaction with oxidation number method

a. $Zn + HNO_3 \longrightarrow Zn(NO_3)_2 + NH_4NO_3 + H_2O$

b. $KMnO_4 + HCl \longrightarrow KCl + MnCl_2 + H_2O + Cl_2$

c. $HCl + MnO_2 \longrightarrow H_2O + MnCl_2 + Cl_2$

Solution

The coefficient respectively from left to right are

a. 4, 10, 4, 1, 3

b. 2, 16, 2, 2, 8, 5

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c. 4, 1, 2, 1, 1

109. How many moles of electrons are required to reduce 3.0 grams of nickel ions from melted $NiCl_2$?

Solution

$$n = \frac{m}{M} = \frac{3g}{59\,g/mol} = 0.05mol$$

$$Ni^{2+} + 2e^- \longrightarrow Ni$$

$$2mol\ e^- = 1mol$$

$$? = 0.05mol$$

$$? = \frac{2mol\ e^- \times 0.05mol}{1mol} = 0.1mol \text{ of electrons}$$

110. Given the reaction, $Sn + 4HNO_3 \longrightarrow Sn(NO_3)_2 + 2NO_2 + 2H_2O$

Identify

a. Reduced substance
b. Oxidized substance
c. Oxidizing agent
d. Reducing agent

Solution

a. N b. Sn c. HNO_3 d. Sn

111. A current of 5A flowing for 30min deposits 3g of Zinc at the cathode. Calculate the equivalent mass of zinc.

Solution

$$\frac{m}{E} = \frac{It}{F}$$

$$E = \frac{mF}{It} = \frac{3g \times 96500A.s}{5A \times 1800s} = 32.2g/eq$$

112. For electrolysis of fused sodium bromide write the products formed at anode and cathode

Solution

Anode reaction: $2Br^- \longrightarrow Br_2 + 2e^-$

Cathode reaction: $Na^+ + e \longrightarrow Na$

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113. Write anode, cathode and cell reactions for electrolysis of

a. Dilute hydrochlorid acid solution

b. Concentrated hydro chloric acid solution

Solution

Anode reaction: $2OH^-(q) \longrightarrow 2H_2O(1) + O_2(g) + 4e^-$

Cathode reaction: $2H^+(q) + 2e^- \longrightarrow H_2O(g)$

Cell reaction: $2H_2O(1) \longrightarrow 2H_2(g) + O_2(g)$

b. Anode reaction: $2Cl^-(eq) \longrightarrow Cl_2(g) + 2e^-$

Cathode reaction: $2H^+(aq) + 2e^- \longrightarrow H_2(g)$

Cell reaction: $2H^+(aq) + 2Cl^-(aq) \longrightarrow H_2(g) + Cl_2(g)$

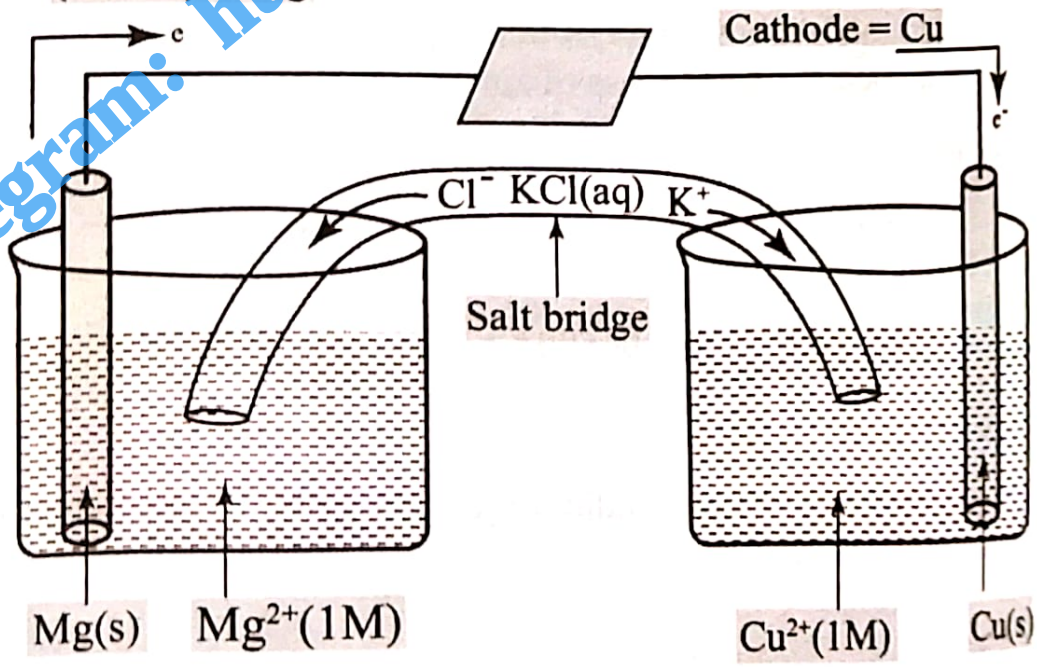
114. Sketch the galvanic cell based on the following overall reaction. show the direction of electron flow, the direction of ion migration through the salt bride by identying the anode and cathode.

$Mg(s) + Cu^{2+}(1M) \longrightarrow Mg^{2+}(1M) + Cu(s)$

Solution

Anode = Mg, Cathode = Cu

(Anode) = Mg



115. In the figure below magnesium is oxidized and zinc ions are reduced

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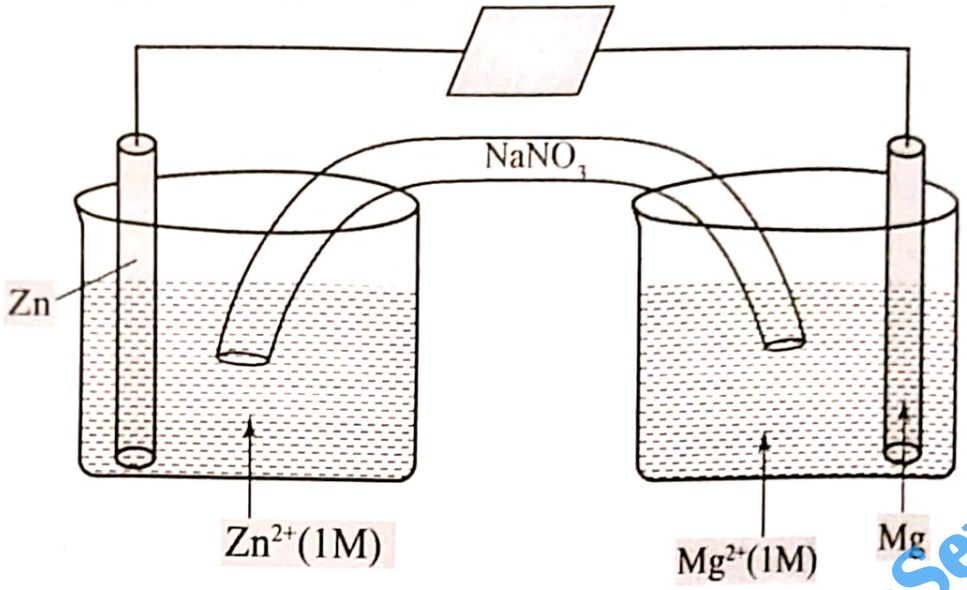

a. Label each part of the cell
b. Write the cell notation
c. Write the half - reaction and net reaction

Solution

a. Anode = Mg, cathode = Zn
b. $Mg(s) \mid Mg^{2+}(1M) \mid\mid Zn^{2+}(1M) \mid Zn(s)$
c. Anode reaction: $Mg(s) \longrightarrow Mg^{2+}(aq) + 2e^-$

Cathode reaction: $Zn^{2+}(aq) + 2e^- \longrightarrow Zn(s)$

Cell reaction: $Mg(s) + Zn^{2+}(aq) \longrightarrow Mg^{2+}(aq) + Zn(s)$

16. What is the standard potential of the cell containing $Fe^{2+} \mid Fe$ and $Br_2 \mid Br^-$ couples?

$E^0 Fe^{2+} \mid Fe = -0.44V$ $E^0 Br_2 \mid Br^- = 1.08V$

Solution

More positive E^0red is used as cathode and the other as anode.

$$E^0_{cell} = E^0_{red}(cathode) - E^0_{red}(anode)$$
$$= 1.08V - (-0.44V) = +1.52V$$

17. $Pb + Sn^{2+} \longrightarrow Pb^{2+} + Sn$, calculate the ratio of $[Pd^{2+}]$ to $[Sn^{2+}]$ for which $E_{cell} = 0$.

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Solution

$$E_{cell} = E^0_{cell} - \frac{0.0592V}{n} \log \frac{[Pb^{2+}]}{[Sn^{2+}]}$$

$$E^0_{cell} = -0.14V - (-0.13V) = -0.01V$$

$$\Rightarrow E^0_{cell} = \frac{0.0592}{2} \log \frac{[Pd^{2+}]}{[Sn^{2+}]}$$

$$\frac{-0.01 \times 2}{0.0592} = \log \frac{[Pd^{2+}]}{[Sn^{2+}]}$$

$$-0.34 = \log \frac{[Pd^{2+}]}{[Sn^{2+}]}$$

$$\frac{[Pd^{2+}]}{[Sn^{2+}]} = 10^{-0.34} = 1:2$$

118. Calculate E_{cell} for the reaction

$Zn(s) + Cu^{2+}(aq) \longrightarrow Zn^{2+}(aq) + Cu(s)$ if it runs under conditions such that $\Delta G = -532kJ$

Solution

$$\Delta G = -532KJ = -5.32 \times 10^5 J$$

$$n = 2, \Delta G = -nFE_{cell}$$

$$E_{cell} = \frac{-\Delta G}{nF} = \frac{-(-5.3L \times 10^5 J)}{2 \times 96500 \frac{J}{V}} = +2.76V$$

119. A current of 2A passing for 5 hours though the molten tin salt deposits 22.2g of tin. Calculate the oxidation state of tin.

Solution

$$I = 2A, t = 5h = 5 \times 3600s = 18000s$$

$$m = \frac{MIt}{ZF} \qquad \Rightarrow Z = \frac{MIt}{mF}$$

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$$Z = \frac{188.7\ g/mol \times 2A \times 18000s}{22.2g \times 96500 \frac{A.s}{mol}} = 2$$

120. The number of electrons to be added to the reactant side in the following half - reaction is $NO^-_3 + 6H_2O \longrightarrow NH_3 + 9OH^-$

Solution

8 electrons

121. Given the concentration cell Zn^{2+} (1.0M) $\rightarrow$ Zn^{2+} (0.1M) . Calculate the cell potential

Solution

$$E_{cell} = \frac{-0.0592V}{n} \log \frac{[Zn^{2+}]dilute}{[Zn^{2+}]con} = \frac{0.0592V}{2} \log \frac{0.1}{1} = 0.0296V$$

Supplementary Exercise

I. Multiple Choice item

1. Ionic compounds will conduct an electric current when they are
   A. Solidified C. Frozen
   B. Melted D. Dehydrated
2. The anode in an electrochemical cell is the electrode where
   A. Always positive C. Reduction occurs
   B. Always negative D. Oxidation occurs
3. Of the following, the element that does NOT react with hydrochloric acid to produce hydrogen gas is
   A. Zn B. Fe C. Hg D. Ca
4. In the electrochemical cell shown below which of the following half - reaction occurs at the anode?

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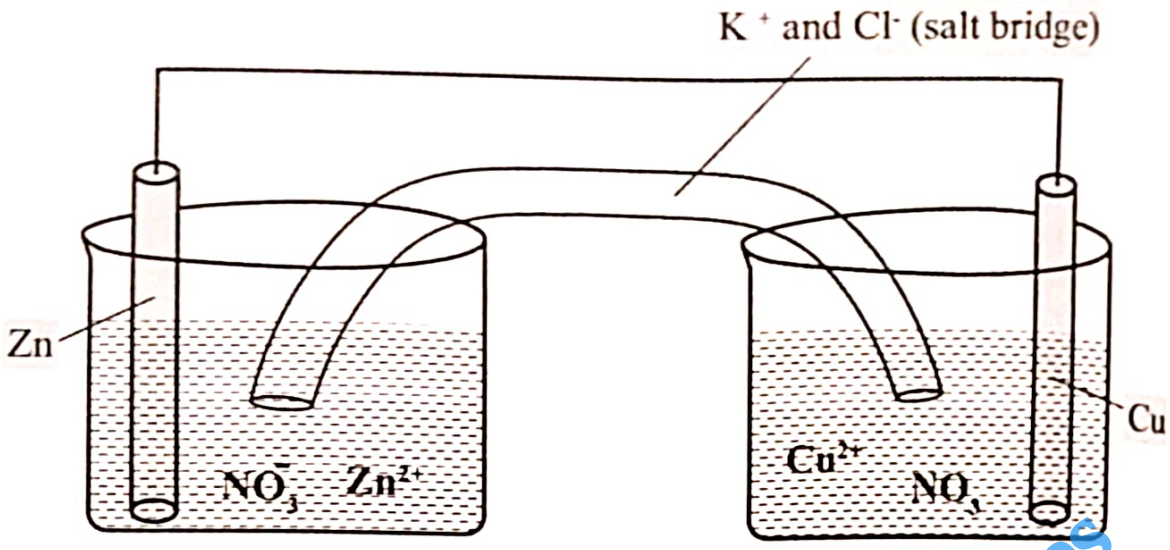

A. $Cu^{2+} + 2e^- \longrightarrow Cu$

B. $Zn \longrightarrow Zn^{2+} + 2e^-$

C. $Cu \longrightarrow Cu^{2+} + 2e$

D. $Zn^{2+} + 2e^- \longrightarrow Zn$

5. Which metal has the highest oxidation state?

A. Ti B. Cr C. Mn D. Cu

6. What is oxidizing agent the reaction below?

$$Cr_2O_7^{2-} + 6S_2O_3^{2-} + 14H^+ \longrightarrow 2Cr^{3+} + 3S_4O_6^{2-} + H_2O$$

A. $Cr_2O_7^{2-}$ B. $S_2O_3^{2-}$ C. H^+ D. Cr^{3+}

7. Which of the following is a redox reaction?

A. $3Se + 4HNO_3 \longrightarrow 3SeO_2 + 4NO + 2H_2O$

B. $CaCl_2 + 2AgNO_3 \longrightarrow Ca(NO_3)_2 + CaCl_2$

C. $BaCO_3 \longrightarrow BaO + CO_2$

D. $3Mg(OH)_2 + 2NH_3 \longrightarrow Mg_3N_2 + 6H_2O$

8. For spontaneous process

A. $E_{cell} > 0$ B. $\Delta G > 0$ C. $E_{cell} = 0$ D. $\Delta G = 0$

9. $MnO_4^- + ye^- \longrightarrow MnO_2$, the value of y is

A. 1 B. 2 C. 3 D. 4

10. If the reaction, $Cu + HNO_3 \longrightarrow Cu(NO_3)_2 + NO + H_2O$ is balanced, the sum of the coefficient will be

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A. 12 B. 16 C. 10 D. 20

11. Which of the following is the strongest oxidizing agent?

A. Au B. Ag C. Cl_2 D. F_2

12. All of the following are active electrodes **EXCEPT**

A. Copper B. Zinc C. Magnesium D. Graphite

13. Which reactions shows increase in oxidation number?

A. $Cl_2 \longrightarrow Cl^-$

B. $SO_4^{2-} \longrightarrow SO_3^{2-}$

C. $MnO_2 \longrightarrow MnO^-_4$

D. $Cr_2O_7^{2-} \longrightarrow CrO_4^{2-}$

14. Which of the following process always involve the decrease in oxidation number?

A. Hydrolysis

B. Oxidation

C. Reduction

D. Decomposition

15. In which of the following compound, Oxidation number of chlorine is +3?

A. IC B. ClO^-_3 C. IF_3 D. $HClO_4$

16. Which of the following process represents disproportiontion?

A. $Cu + 4HNO_3 \longrightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$

B. $3I_2 + 6OH^- \longrightarrow IO^-_3 + 5I^- + 3H_2O$

C. $Cl_2 + I_2 \longrightarrow 2ICl$

D. $Zn + 2HCl \longrightarrow ZnCl_2 + H_2$

17. The value of $E^0\ Zn/Zn^{2+} = 0.76V$ and that of $E^0Fe^{2+}/Fe = -0.41V$. The E^0_{cell} of the cell with net cell reaction:

$Zn + Fe^{2+} \longrightarrow Zn^{2+} + Fe$ is

A. -0.35V B. -1.17V C. +1.17V D. +0.35V

18. The conductivity of strong electrolyte

A. Increases on dilution slightly

B. Does not change on dilution

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C. Decrease on dilution

D. Depends on density of electrolyte itself

19. $XI_2 + ySO_2 + ZH_2O \longrightarrow aSO^{2-}_4 + bI^- + cH^+$. The values of z,b and c are respectively

A. 1, 1, 2 B. 1, 2, 3 C. 2, 2, 4 D. 2, 4,2

20. Given four half cell reactions. the highest cell potential is resulted by combining

i. $A + e^- \longrightarrow A^-$, $E^0 = -0.24V$

ii. $C^- + 2e^- \longrightarrow C^{3-}$, $E^0 = -1.25V$

iii. $B^- + e^- \longrightarrow B^{2-}$, $E^0 = 1.25V$

iv. $D + 2e \longrightarrow D^{2-}$, $E^0 = 0.68V$

A. i and ii

B. ii and iii

C. iii and iv

D. i and iv

II. Short Answer items

21. The oxidation state of iron in the brown ring complex $[Fe(H_2O)_5NO]\ SO_4$ is _____

22. Determine the oxidation state of the underlined element.

a. $K_4\underline{P_2}O_7$ b. $Ba_2\underline{Xe}O_6$ c. $\underline{Mn}O^{2-}_4$ d. $\underline{U}O^{2-}_2$

23. Given the reaction:

$$2KMnO_4 + 10KCl + 8H_2SO_4 \longrightarrow 2MnSO_4 + 6K_2SO_4 + 5Cl_2 + 8H_2O$$

Identify:

a. Reducing agent b. Oxidizing agent

24. Consider the following half - reactions

$Ce^{4+} + e^- \longrightarrow Ce^{3+}$, $E^0 = 1.61V$

$Ag^+ + 2e^- \longrightarrow Ag$, $E^0 = 0.80V$

$Mg^{2+} + 2e^- \longrightarrow Mg$, $E^0 = -2.37V$

$Sn^{2+} + 2e^- \longrightarrow Sn$, $E^0 = -0.14V$

a. Which one is the weakest oxidizing agent?

b. Which one is the strongest oxidizing agent?

c. Which is the weakest reducing agent?

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d. Which one is the strongest reducing agent?

25. Classify the following as wet cell or dry cell

a. Car battery
b. Mobile battery
c. Leclanche cell
d. Daniel cell

Answer to Supplementary exercise on Unit 8

Part I

1. B	6. A	11. D	16. B
2. D	7. A	12. D	17. D
3. C	8. A	13. C	18. C
4. B	9. C	14. C	19. C
5. C	10. D	15. C	20. B

Part II

21. +2

22. a. +5 b. +8 c. +7 d. +2

23. a. KCl b. $KMnO_4$

24. a. Mg^{2+} b. Ce^{4+} c. Ce^{3+} Mg

25. a and d wet cells

b and c dry cells

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UNIT 9

Industrial Chemistry

9.1 Introduction

A chemical industry is an institution that produces new chemical substance through chemical processes .

Generally a Chemical Industry is Characterized by the Following:

- Uses naturally available raw materials to produce the desired products.
- Involves chemical reaction to change reactants into finished and semi-finished products
- Uses safe operation methods
- Uses large quantities of energy during the manufacturing processes
- Tests its products during and after manufacture in its quality control laboratory.

Industrial Chemistry is the application of chemistry to the manufacturing and processing of chemicals for industrial purposes.

It involves the analysis of raw materials, the use of thermodynamic principles and computer modelling to simulate chemical plant operations and the synthesis of chemicals.

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It is responsible for the discovery and development of new and improved synthetic fibers, paints, adhesives, drugs, cosmetics, electronic components, lubricants, soaps, fertilizers and so on.

It has also improved process for oil refining and petrochemical processes that saves energy and reduces pollution.

Illustrative Examples

1. Which of the following is not true about chemical industries? chemical industry:

   A. Converts reactants in to finished and semi - finished products.
   B. Uses naturally existing raw maw materials to produce desired products
   C. Uses either natural resources or other substances as raw material
   D. Focuses always on chemical reactions that release heat energy.

   Answer: D

2. Which of the following is not a characteristic of chemical industry?

   A. It uses naturally - available raw materials
   B. It transforms raw materials into finished and semi - finished products.
   C. It consumes relatively small quantities of energy and water.
   D. It uses safe operation methods.

   Solution

   Chemical industry uses large quantities of energy during the manufacturing process.

   Answer: C

3. _____ involves the taking of raw material from the environment and turning it into a useable product by chemical means.

   Solution

   Chemical industry

4. What are the main raw materials of the industry?

   Solution

   Fossil fuels (coal, petroleum, and natural gas), air, water, salt, limestone, sulphur, chlorine and some special raw materials such

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Illustrative Examples

6. Of the following substances, which one is NOT a natural resource?

A. Metal ore C. Glass

B. Soil D. Stone

Solution

Glass is made by man in industry. Industry products are artificial resources.

Answer: C

7. Which of the following natural resources is non - renewable?

A. Petroleum B. Water C. Plants D. Animals

Answer: A

8. All of the following are non - renewable natural resources. EXCEPT

A. Coal B. Minerals C. Natural gas D. Plants

Answer: D

9. List the components of the environment that are used as source of raw materials.

Solution

From the atmosphere: gases

From the hydrosphere: water and dissolved salts

From the lithosphere: Mineral ores, coal, natural gas, petroleum, carbon and hydrocarbon

From the biosphere: plants and animals

10. Natural resource can be classified as _____ and _____ resources.

Solution

Renewable and non - renewable

Industry

- The highly industrialized countries in particular, many classify chemical industries in to five groups, Each of which may be further subdivided.
- Many of the chemical discoveries were made quite recently.

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as phosphates.

5. What are the roles of industrial chemistry for the society?

Solution

- To manufacture different substance
- To protect the environment.

9.2 Natural Resources

Natural resources are resources that exist in nature.

Table : 9.1 Natural Resources Obtained From Different Components of Environment

Environment	Natural Resource Obtained
Atmosphere	Gases such as N_2, O_2, CO_2, Ne, Ar, Kr, Xe
Hydrosphere	Water, dissolved material (NaCl, Mg and Br_2)
Lithosphere	Elements, mineral ores, carbon and hydrocarons, coal, natural gas and crude petroleum
Biosphere	Vegetation and animals contribute raw materials to agro - based industries oils, fats, waxes, resins, sugar, natural fibers and leather.

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Classification of Natural Resources

Natural resources are classified into two. These are

1. Renewable natural resource
2. Non - renewable natural resource

1. Renewable Natural Resource

- They are replaced by natural process

Example:- Animals, plants, soil, firewood and water

2. Non - renewable Natural Resources

- They cannot be replaced by natural processes.

Example:- Petroleum, minerals, , natural gas, coal

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Illustrative Examples

6. Of the following substances, which one is NOT a natural resource?

A. Metal ore
B. Soil
C. Glass
D. Stone

Solution

Glass is made by man in industry. Industry products are artificial resources.

Answer: C

7. Which of the following natural resources is non - renewable?

A. Petroleum B. Water C. Plants D. Animals

Answer: A

8. All of the following are non - renewable natural resources. EXCEPT

A. Coal B. Minerals C. Natural gas D. Plants

Answer: D

9. List the components of the environment that are used as source of raw materials.

Solution

From the atmosphere: gases

From the hydrosphere: water and dissolved salts

From the lithosphere: Mineral ores, coal, natural gas, petroleum, carbon and hydrocarbon

From the biosphere: plants and animals

10. Natural resource can be classified as _____ and _____ resources.

Solution

Renewable and non - renewable

Industry

- The highly industrialized countries in particular, many classify chemical industries in to five groups, Each of which may be further subdivided.
- Many of the chemical discoveries were made quite recently.

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➥ There is no clear - cut line between some of the classifications.

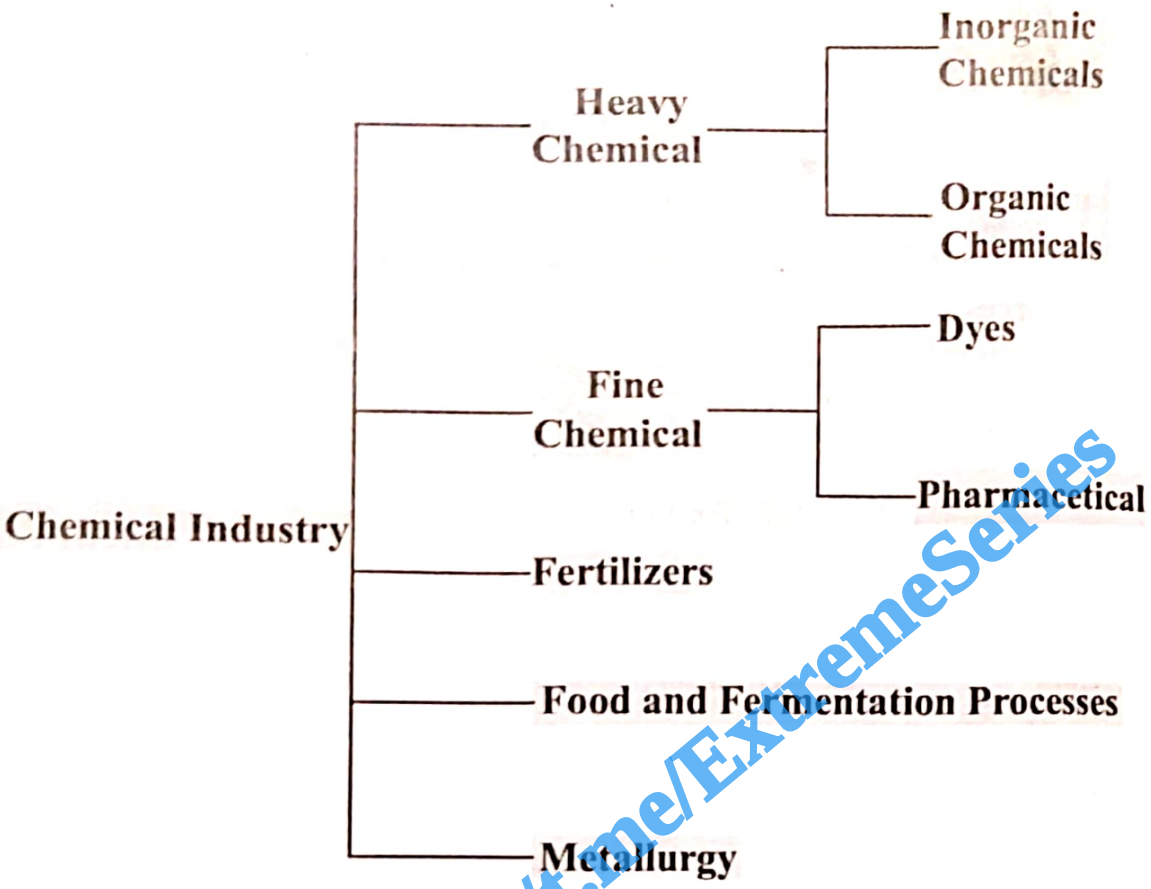

Fig 9. 1 Classification of Chemical industries

Based on the products obtained, manufacturing industries can be classified as follows.

- ➥ Food, beverage and tobacco industries
- ➥ Textiles, wearing apparel, leather goods
- ➥ Paper products, printing and publishing
- ➥ Chemical, petroleum, rubber and plastic products
- ➥ Non - metallic mineral products other than petroleum products
- ➥ Basic metal products, machines and equipment

Based on the raw materials used, chemical industry may be classified in to two

1. Chemical industries that use natural resource directly

   Example: Sugar industry, paper and pulp factory, leather factory etc.

2. Chemical industries that use products of other chemical industrials as raw materials.

   Example: Soap and detergent factories.

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Illustrative Examples

11. Which of the following chemical industries uses the products of other chemical industries as raw materials?

A. Sugar factory
B. Leather factory
C. Paper and pulp factory
D. Soap and detergent factory

Answer: D

12. Industrialized countries may categorize chemical industries

A. Four B. Five C. Two D. Three

Answer: B

13. How products are obtained from chemical industries?

Solution

The products are obtained from:

- Chemical reactions of organic substance, inorganic substance or both
- Extraction, separation or purification of natural products without the help of chemical reactions.
- The preparation of specifically valuable materials.

14. The chemicals obtained from chemical industry are called _____.

Answer: Industrial Chemicals

15. Define manufacturing industry

Solution

It is the sector of the economy that transforms raw materials into finished products, through the use of tools, human labor, machinery and chemical processing.

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9.3 Manufacturing of Valuable Products/ Chemicals

The production of the valuable produces may involve physical and chemical steps. **Reactants → Process → products**

The most important chemicals discussed here are ammonia (NH_3), sulphuric acid, nitric acid, sodium carbonate, and diammoniun monohydr phosphate.

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I. Ammonia (NH_3)

Properties of Ammonia

- It is a colorless gas with a characterstic shocking smell.
- It is lighter than air hence collected by upward delivery
- It is completey miscible with water.
- It is an alkaline gas.
- It does not support combustion and extinguishes a burning splint.

Preparation of Ammonia

1. Laboratory Preparation

In laboratory, it is manufactured by heating ammonium chloride with calcium or sodium or potassium hydroxide

$$2Ca(OH)_2 + 2NH_4Cl \rightarrow NaCl_2 + 2H_2O + 2NH_3(g)$$

$$NH_4Cl + NaOH \rightarrow NaCl + H_2O + NH_3$$

2. Industrial Preparation

- Industrially ammonia is manufactured by **Haber process.**

The Stages in the Haber Process are:

1. A mixture of nitrogen (1 Volume) and hydrogen (3 volumes) is compressed.
2. The compressed gases pass into a large tank called a converter. This contains trays of iron catalyst. The temperature in the converter is 450°C and the pressure is 200 atmospheres.

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$$

3. The mixture is passed to a cooling chamber and the ammonia condenser here and is removed.
4. The unreacted nitrogen and hydrogen are returned to the converter

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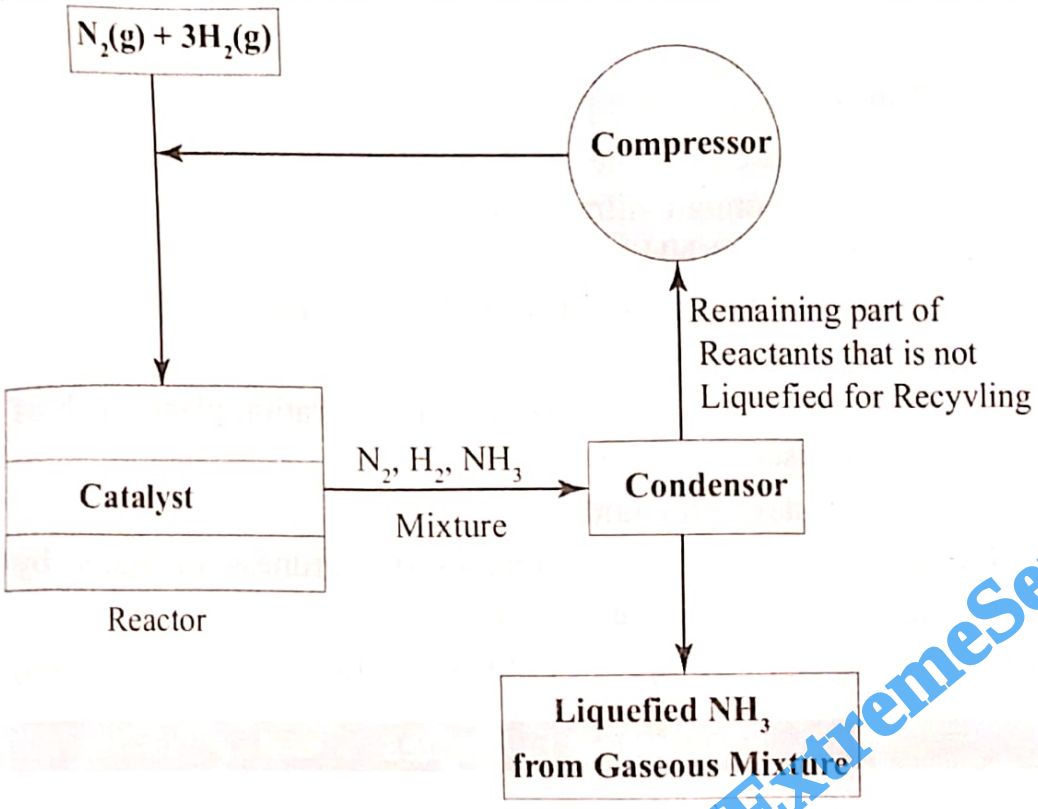

Fig 9. 2 Ammonia Synthesis Reaction

Note: Formation of ammonia is favored at low temperature and high pressure

- Ammonia gas is highly soluble in water. This can be be tested using fountain experiment.
- It is the most soluble of all gases.

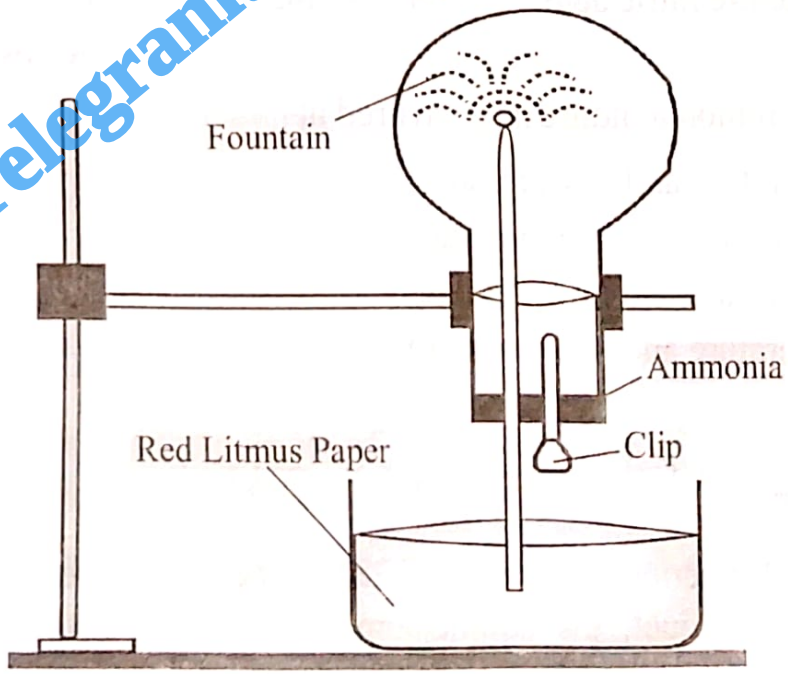

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Fig 9. 3 Fountain Experiment

Uses of Ammonia

1. Ammonia gas is used in the manufacture of artificial fertilizers which include ammonium nitrate, NH_4NO_3, ammonium phosphate, $(NH_4)_3 PO_4$ and urea $CO(NH_2)_2$.
2. It is used in the manufacture of nitric acid, HNO_3 and sodium carbonate or soda ash, Na_2CO_3
3. It is used as a liquid refrigerant in large refrigeration plants such as ships and ware houses.
4. To remove greasy stains in laundry
5. Its solution is used to remove temporary hardness of water by precipitating hydrogen carbonate as calcium carbonate.

$$Ca(HCO_3)_2(aq) + NH_4OH(aq) \rightarrow CaCO_3(s) + 2H_2O(1) + (NH_4)_2 CO_3(aq)$$

Illustrative Examples

16. Which of the following chemicals is industrially manufactured by Haber process?

A. N_2 B. NH_3 C. HNO_3 D. H_2SO_4

Answer: B

17. All of the following are the uses of ammnia **EXCEPT**

A. To make fertilizers
B. To manufacture nitric acid,
C. To make sodium carbonate
D. To make sulphuric acid,

Answer: D

18. Formation of ammonia industrially favored at

A. Low temperature and low pressure
B. Low temperature and high pressure
C. High temperature and low pressure
D. High temperature and high pressure

Solution

$$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g) + 92\,kJ, \text{ since}$$

The forward reaction is exothermic and there is fewer number of moles of product, the equilibrium shifts to the right at low

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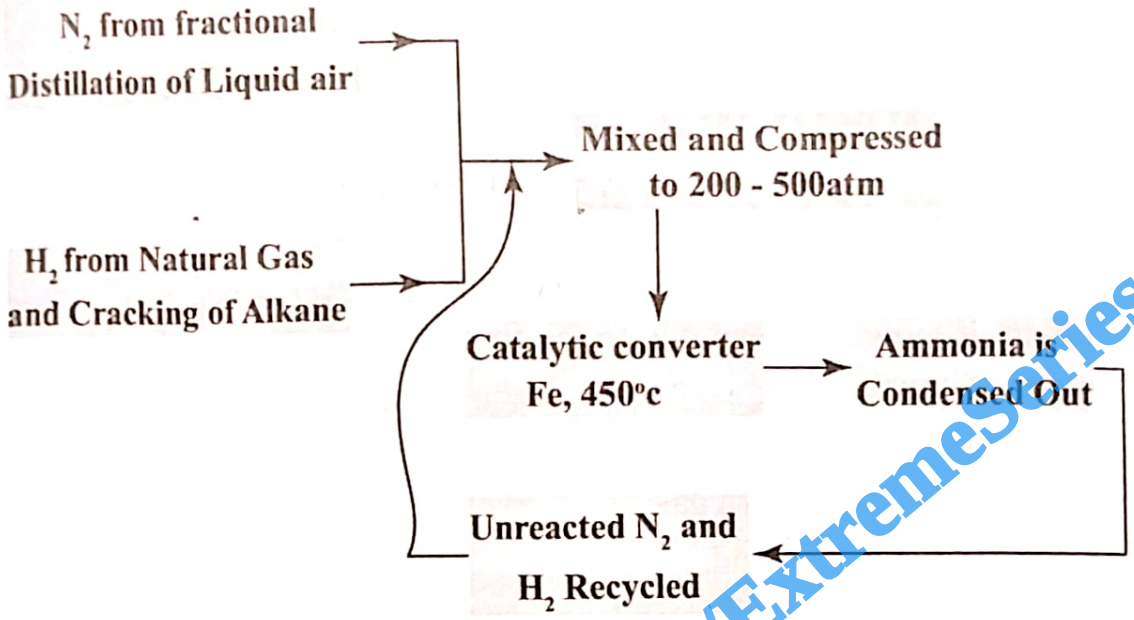
temperature high pressure.

Answer: B

19. Draw a flow diagram of the Haber process

Solution



20. Complete and balance the following

a. $Ca(OH)_2 + (NH_4)_2 SO_4 \rightarrow$

b. $NH_3 + H_2O \rightarrow$

c. $NH_3 + HCl \rightarrow$

d. $NH_3 + HNO_3 \rightarrow$

Solution

a. $Ca(OH)_2 + (NH_4)_2 SO_4 \rightarrow CaSO_4 + 2H_2O + 2NH_3$

b. $NH_3 + H_2O \rightarrow NH_4^+ + OH$

c. $NH_3 + HCl \rightarrow NH_4Cl$

d. $NH_3 + HNO_3 \rightarrow NH_4NO_3$

21. In the laboratory preparation of ammonia from calcium hydroxide and ammonium chloride

a. Why a dried boiling tube required?

b. What is the purpose of putting the lump of CaO into the boiling tube containing $Ca(OH)_2$ and NH_4Cl?

c. What is the white fume produced when the NH_3 gas is collected in to the HCl?

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Explanation

a. Because ammonia has a strong smell and poisonous in quantity

b. To prevent the water produced from coming out of the delivery tube as water vapor.

c. Ammonium chloride (NH_4Cl)

$NH_3(g) + HCl\ (g) \rightarrow NH_4Cl(s)$

2. Nitric Acid (HNO_3)

Properties of Nitric Acid

It is colorless liquid with a pungent odor, however, when it is exposed to light, it turns to brown due to the formation of nitrogen dioxide upon decomposition.

$$4HNO_3(aq) \rightarrow 4NO_2(g) + O_2(g) + 2H_2O(1)$$

Brown gas

Nitric acid is a strong acid.

$$HNO_3(aq) + H_2O(1) \rightarrow NO^-_3(aq) + H_3O^+(aq)$$

It reacts with bases and basic oxides to given salts, called **nitrates** and water

Example

$$HNO_3(aq) + NaOH(aq) \rightarrow NaNO_3(aq) + H_2O(1)$$

$$CaO(aq) + 2HNO_3(aq) \rightarrow Ca(NO_3)_2(aq) + H_2O(1)$$

It reacts with carbonates and hydrogen carbonates to given salt, carbondioxide and water.

Example

$$Na_2CO_3(s) + 2HNO_3(aq) \rightarrow 2NaNO_3(aq) + CO_2\ (g) + H_2O(1)$$

It reacts with active metals to give hydrogen gas and salt.

Example

$$Mg(s) + 2HNO_3(aq) \rightarrow Mg(NO_3)_2(aq) + H_2(g)$$

It reacts with less active metal, such copper

$$3Cu + 8HNO_3(diute) \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$$

$$Cu + 4HNO_3\ (conc) \rightarrow Cu(NO_3)_2 + 2NO_2 + 2H_2O$$

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It oxides carbon to carbondioxide.

$$C(s) + 4HNO_3(l) \rightarrow CO_2(g) + 4NO_2(g) + 2H_2O(l)$$

Preparation of Nitric Acid

1. Laboratory Preparation of Nitric Acid

Potassium nitrate reacts with condensed sulphuric acid giving off nitric acid fumes $KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$

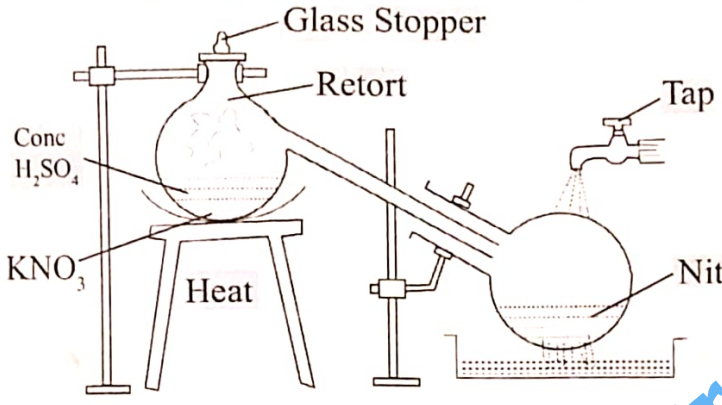

Fig 9. 4 Laboratory preparation of Nitric Acid

2. Industrial Preparation of Nitric Acid

Industrially, nitric acid is prepared by **Ostwald process.**

Steps

1. Catalytic oxidation of ammonia into nitric oxide

$$4NH_3(g) + 5O_2(g) \xrightarrow{Pt} 4NO(g) + 6H_2O(g)$$

2. Oxidation of nitric oxide in to nitrogen dioxide

$$2NO(g) + O_2(g) \rightarrow 2NO_2(g)$$

3. Dissolution of nitrogendioxide in water.

$$3NO_2(g) + H_2O(l) \rightarrow 2HNO_3(aq) + NO(g)$$

The NO formed here is recycled in the second stage

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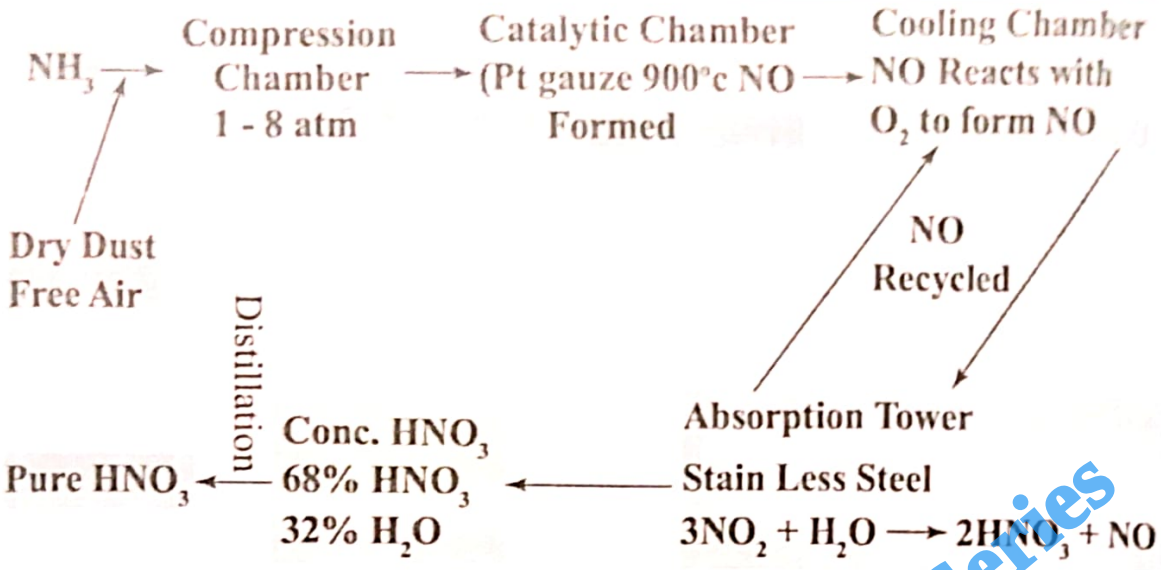

Fig 9. 5 Flow Chart for Production of Nitric Acid

Uses of Nitric Acid

1. To manufacture nitrogenous fertilizers

   Example

   - Ammonium nitrate (NH_4NO_3)
   - Calcium nitrate ($Ca(NO_3)_2$)
   - Sodiumnitrate ($NaNO_3$)

2. It is used in the manufacture of explosives such as trinitrotoluene (TNT) and nitroglycerine.
3. It is important in plastic production and aquaregia ($3HCl + HNO_3$)
4. It is used in production fungicides and pharmaceuticals.

Illustrative Examples

22. Which of the following mixture is given the name aquaregia?

A. $H_3PO_4 + H_2SO_4$
B. $3HCl + HNO_3$
C. $HCl + 3HNO_3$
D. $HCl + CH_3COOH$

Solution

The mixture of HCl and HNO_3 in 3 to 1 mole ratio is known as a quaregia.

Answer: B

23. The catalyst used in to the Ostwald process for the manufacture of

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HNO_3 is

A. Platinum black
B. Vanadium pentoxide
C. Finely divided nickel
D. Platinum gauze

Solution

The catalyst used to speed up formation of nitric oxide from oxidation of ammonia is platinum gauze.

Answer: D

24. The name aquafortis is given to

A. HCl B. H_2SO_4 C. HNO_3 D. HNO_2

Answer: C

25. Salts of nitric acid are called

A. Nitrates B. Nitrites C. Nitrides D. None

Answer: A

26. A mixture of HNO_3, N_2O_4 and H_2O_2 is knowns as _____.

Answer: Red fuming nitric acid

27. List some uses of nitric acid

Solution

- Cleaning and etching of metal surfaces
- Refining of precious metals
- Manufacturing of household cleaning products

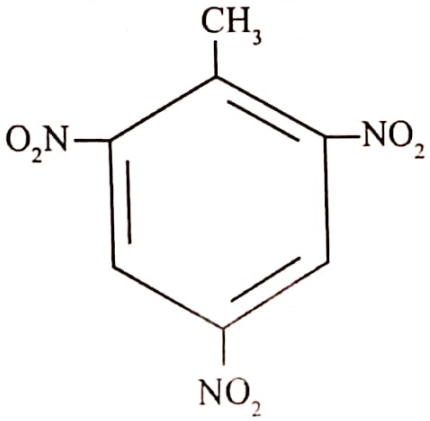
28. Write the other name and molecular formula of trinitrotoluene.

Solution

Trinitrotoluene is also known as 2,4, 6 - trinitrotoluene

Molecular formula: $C_6H_2(NO_2)_3CH_3$

Structural formula:



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29. Why nitric acid is displaced by sulphuric acid from nitrates?

Solution

Reason: Nitric acid is more volatile than sulphuric acid.

$$KNO_3 + H_2SO_4 \rightarrow KHSO_4 + HNO_3$$

30. Balance and complete the following equation:

a. $Cu + HNO_3(dilute) \rightarrow$

b. $Zn + HNO_3 \rightarrow$

c. $KOH + HNO_3 \rightarrow$

Solution

a. $3Cu + 8HNO_3 \rightarrow 3Cu(NO_3)_2 + 2NO + 4H_2O$

b. $Zn + 2HNO_3 \rightarrow Zn(NO_3)_2 + H_2$

c. $KOH + HNO_3 \rightarrow KNO_3 + H_2O$

3. Nitrogen - Based Fertilizers

The production of ammonia and nitric acid are important in the production of many **artificial fertizers**

The use of artificial fertilizers is essential if farmer are to produce sufficient crops to feed the ever increasing population. Crops remove nutrients from the soil as they grow these include nitrogen, phosphorus and potassium. Artificial fertilizers are added to the soil to replace these nutrients and others.

Table : 9.2 Some Nitrogenous Fertilizers

Fertiliser	Formula
Ammonium nitrate	NH_4NO_3
Diammonium monohydrogen phosphate (DAP)	$(NH_4)_2HPO_4$
Ammonium sulphate	$(NH_4)_2SO_4$
Urea	$CO(NH_2)_2$

The fertilizer which add the three main nutrient (N,P and K) are called **NPK fertilizers**

NPK fertilizers = $NH_4NO_3 + (NH_4)_2HPO_4 + KCl$

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A. Production of Ammonium Nitrate

$NH_3 + HNO_3 \longrightarrow NH_4NO_3$

B. Production of DAP Ammonium nitrate (Nitram)

$3NH_3 + 2H_3PO_4 \longrightarrow NH_4H_2PO_4 + (NH_4)HPO_4$

$NH_4H_2PO_4 + NH_3 \longrightarrow (NH_4)_2HPO_4$

DAP

C. Production of Ammonium Sulphate

$2NH_3 + H_2SO_4 \longrightarrow (NH_4)_2SO_4$

D. Production of Urea

$2NH_3 + CO_2 \longrightarrow NH_2COONH_4$ (ammonium carbamate)

$NH_2COONH_4 \xrightarrow[\text{high temperature}]{\text{high pressure}} (NH_2)_2CO + H_2O$

E. Urea - ammonium Nitrate (UAN) Solutions

It is important nitrogen fertilizer

Urea - ammonium nitrate (UAN) solution

⇓

Urea + ammonium nitrate + water

Illustrative Examples

31. Which of the following nitrogen containing fertilizer is known as nitram?

A. NH_4NO_3 B. $(NH_4)_2SO_4$ C. $(NH_4)_2HPO_4$ D. $NaNO_4$

Answer: A

32. NPK fertilizers contain all of the following EXCEPT

A. NH_4NO_3 B. $(NH_4)_2HPO_4$ C. NaCl D. KCl

Answer: C

33. Given the diagram below.

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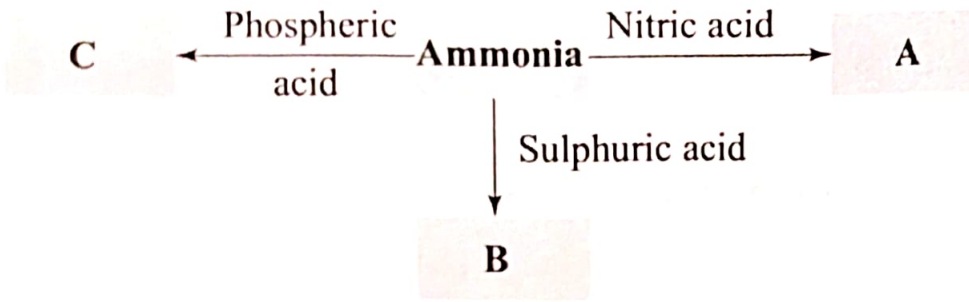

Solution

A. NH_4NO_3 B. $(NH_4)_2SO_4$ C. $(NH_4)_2HPO_4$

34. Calculate the percentage composition by mass of nitrogen and phosphorus in DAP.

Solution

$(NH_4)_2HPO_4$ N : $2 \times 14 = 28g$

H : $9 \times 1 = 9g$

P : $1 \times 31 = 31g$

O : $4 \times 16 = 64g$

$132g$

$$\%N = \frac{\text{mass N}}{\text{mass DAP}} \times 100 = \frac{28g}{132g} \times 100 = 21.2\%$$

$$\%P = \frac{\text{mass P}}{\text{mass DAP}} \times 100 = \frac{31}{132} \times 100 = 23.5\%$$

4. Sulphuric Acid (H_2SO_4)

Properties of Sulphuric Acids

- Sulphuric is colorless, odorless and oily liquid
- It is heavier than water and soluble in all proportions in water.
- It is a diprotic strong acid
- It reacts with bases and basic oxides to yield salt and water.

Example

$$H_2SO_4 + 2NaOH \longrightarrow Na_2SO_4 + 2H_2O$$

$$H_2SO_4 + CaO \longrightarrow CaSO_4 + H_2O$$

- Sulphuric acid reacts with active metals

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$$Zn + H_2SO_4 \longrightarrow Zn\,SO_4 + H_2$$

- Sulphuric acid reacts with carbonates and hydrogen carbonates

$$H_2SO_4 + Na_2CO_3 \longrightarrow Na_2\,SO_4 + CO_4 + H_2O$$

- Sulphuric acid oxidizes metals and non - metals

$$Cu + 2H_2SO_4 \longrightarrow Cu\,SO_4 + SO_2 + H_2O$$

$$C + 2H_2\,SO_4 \longrightarrow 2SO_2 + CO_2 + 2H_2O$$

- Concentrated sulphuric acid is a powerful **dehydrating agent.**

Example

$$\text{Sucrose (sugar)} \xrightarrow{\text{conc}\,H_2SO_4} \text{carbon + water}$$

$$C_{12}\,H_{22}\,O_{11}(s) \xrightarrow{\text{conc}\,H_2SO_4} 12C(s) + 11\,H_2O\,(1)$$

Preparation of Sulphuric Acid

☑ Industrially, sulphuric acid is manufactured by **contact process.**

☑ The process has the following stages

Step 1: sulphur dioxide is first produced, primarily by the reaction of Sulphur with air

$$\text{Sulphur + oxygen} \longrightarrow \text{sulphurdioxde}$$

$$S\,(s) + O_2(g) \longrightarrow SO_2(g)$$

Step 2: conversion of sulphurdioxide in to sulphurtrioxide

$$2SO_2(g) + O_2(g) \underset{}{\overset{V_2O_5}{\rightleftharpoons}} (s)\ 2SO_3(g) + 197\ kJ/mol$$

Formation of SO_3 is favored at low temperature and high pressure

Step 3: If SO_3 is added directly to water, sulphuric acid is produced. This reaction, however, is very violent and a thick mist is produced.

$$SO_3(g) + H_2O(1) \longrightarrow H_2SO_4(1)$$

This acid mist is very difficult to deal with and so a different route to sulphuric acid is employed. Instead, the SO_3 is dissolved in concentrated sulphuric acid (98%) to give a substance called **oleum.** ($H_2S_2O_7$)

$$H_2SO_4(aq) + SO_3(g) \longrightarrow H_2S_2O_7(1)$$

Step 4: The oleum formed is then added to the correct amount of water to

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- For pickling that is, cleaning of metal surfaces before electroplating.
- In chemical industries for the preparation of hydrochloric acid, nitric acid, sulphates, ether, dyes and paints.
- It is used as catalyst in organic chemical reaction such as esterification.

Illustrative Example

35. In contact process, which of the following favours the better yield of sulphuric acid?

A. Low temperature, high pressure
B. Low temperature, low pressure
C. High temperature, low pressure
D. High temperature, high pressure

Solution

The second step in contact process is reversible. Formation of SO_3 is favoured at low temperature and high pressure.

Answer: A

36. Which of the following describes part of the contact process.

A. SO_2 is absorbed in distilled water, then reacts with water to give H_2SO_3
B. SO_3 is absorbed in distilled water, then reacts with water to give H_2SO_4
C. H_2S is burned in air to produce SO_2
D. The oxidation of SO_2 to SO_3 is performed in the presence of catalyst.

Solution

One of the reaction in the contact process is oxidation of SO_2 to SO_3 in the presence of a catalyst.

$$2SO_2(g) + O_2(g) \rightleftharpoons^{V_2O_5(s)} 2SO_3(g)$$

Answer: D

37. Which chemical determines the development of a country?

A. H_2SO_4 B. $(NH_4)_2HPO_4$ C. NH_3 D. HNO_3

Answer: A

38. The chemical reaction taking place in contact chamber of contact process is

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Solution

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$$2SO_2(g) + O_2(g) \underset{}{\overset{V_2O_5(s)}{\rightleftharpoons}} 2SO_3(g)$$

Answer: D

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produce sulphuric acid

$$H_2S_2O_7(l) + H_2O(l) \longrightarrow 2H_2SO_4(l)$$

Note: The raw materials for the contact process are sulfur, air and water

The Source of Sulfur are

- Sulfur from beneath the ground obtained by **Frasch process.**
- Sulfide ores.
- Hydrogen sulfide

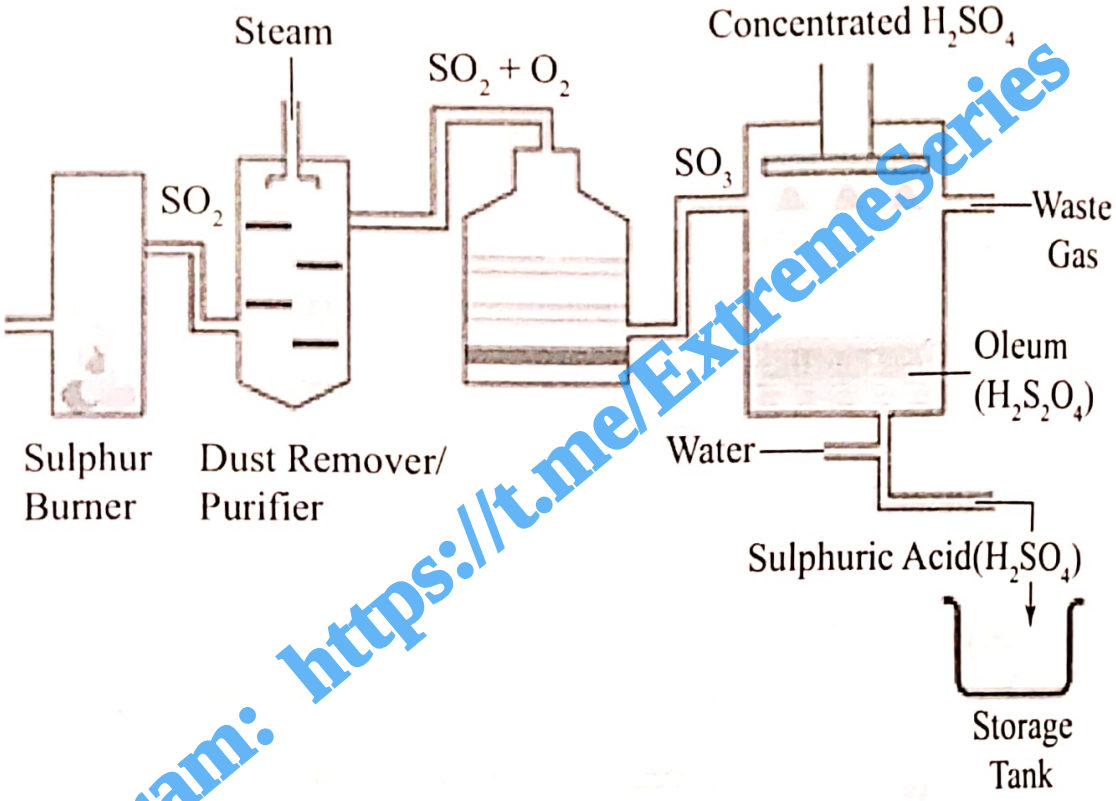

Fig 9.6 The Manufacture of Sulfuric acid by the Contact Process

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Uses of Sulfuric Acid

- **Sulfuric acid** is one of the most important industrial chemical that determine the development of a country.
- Sulphuric acid is used to make fertilizers, paper, synthetic fibers, and textiles, insecticides, detergents, food additives, drugs, anti freezing agent, synthetic rubber, printing inks, photographic films, explosives and automobile batteries.
- In metallurgy, for electrolytic refining of metals.

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$$2SO_2(g) + O_2(g) \rightleftharpoons^{V_2O_5(s)} 2SO_3(g)$$

Answer: D

37. Which chemical determines the development of a country?

A. H_2SO_4 B. $(NH_4)_2HPO_4$ C. NH_3 D. HNO_3

Answer: A

38. The chemical reaction taking place in contact chamber of contact process is

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A. $2S + 3O_2 \longrightarrow 2SO_3$

B. $S + O_2 \longrightarrow SO_2$

C. $SO_3 + H_2SO_4 \longrightarrow H_2S_2O_7$

D. $2SO_2 + O_2 \longrightarrow 2SO_3$

Solution

The process is called contact process because SO_2 and O_2 make contact on the surface of the solid catalyst V_2O_5.

Answer: D

39. In the manufacture of sulphuric acid by contact process.

I. V_2O_5 for catalytic oxidation of SO_2 to SO_3

II. SO_2 is produced in the sulphur burner

III. SO_3 is directly absorbed in water

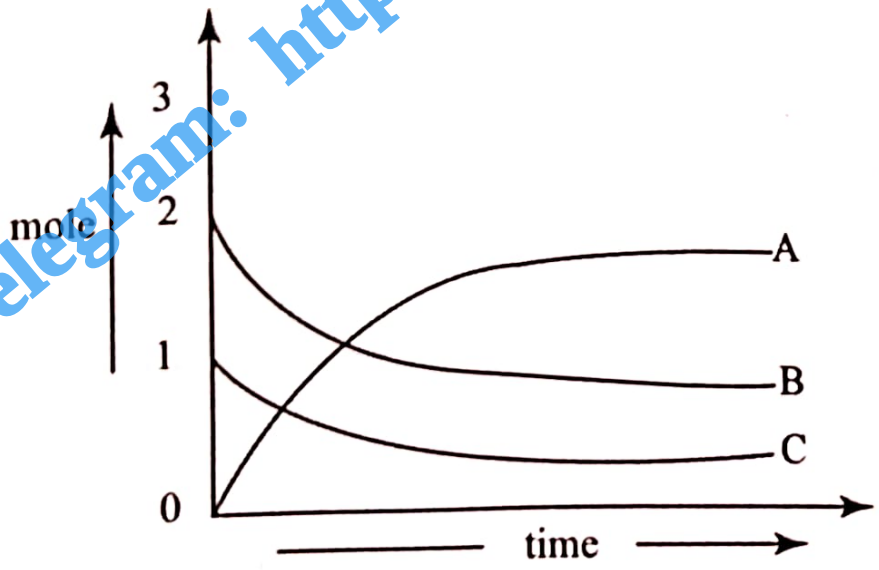
IV. $H_2S_2O_7$ is dissolved in water

Which statements is/are incorrect?

A. I and III B. III and IV C. II only D. III only

Answer: D

40. For the reaction involving the conversion sulphurdioxide to sulphur trioxide, the changes in number of moles Vs time is shown by different curves in the figure below. What are the substances represented by the lefters A, B and C?



Solution

$$2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$$

A = SO_3 because it is a product that increases with time

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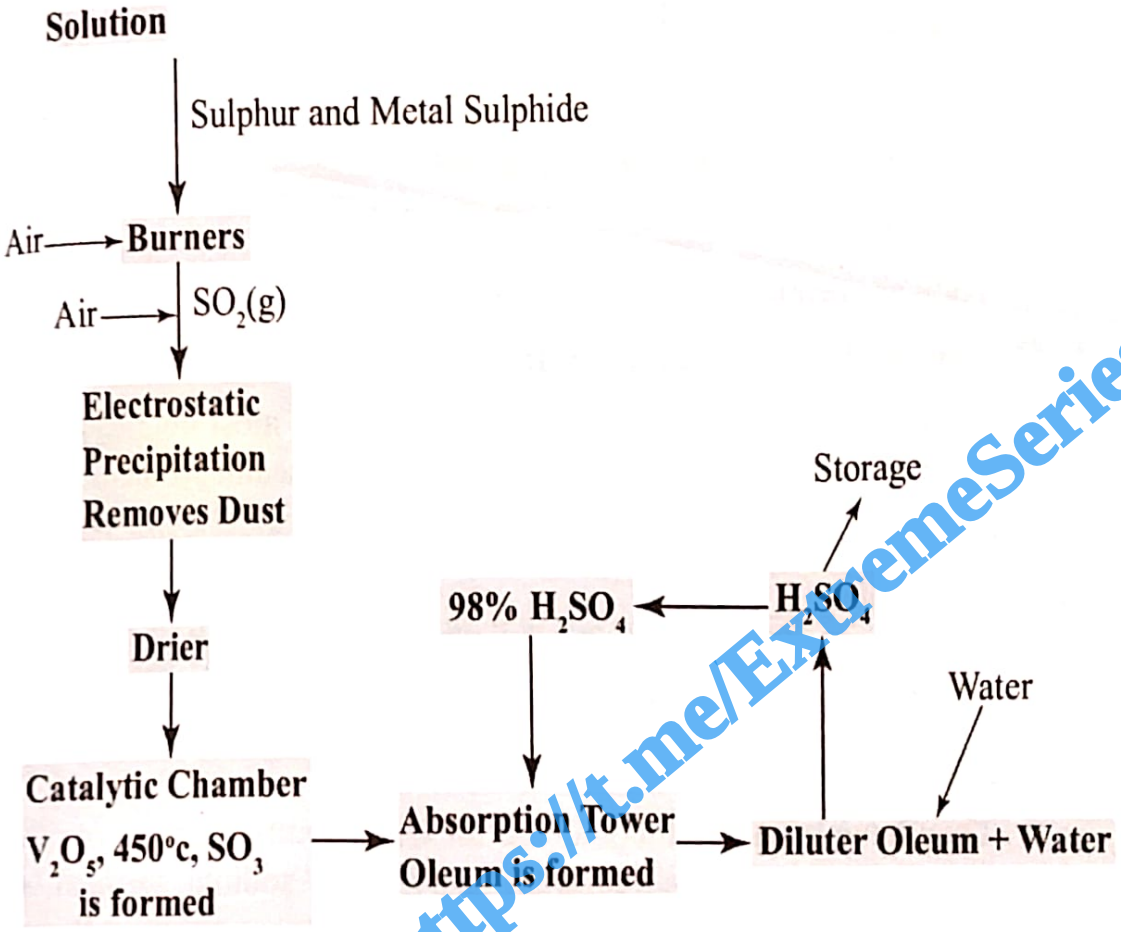
B = SO_2 because its coefficient is 2 and decreases rapidly.

C = O_2 because its coefficient is 1.

41. Draw the flow chart that shows production of sulphuric acid by contact process



5. Some Common Pesticides and Herbicides

Pesticides

- ☑ They are used to prevent pests, diseases and other plant phatogens.
- ☑ They reduce product losses and maintain product quality
- ☑ The are grouped according to the types of pests which they kill
  - ➥ **Insecticides** - insects
  - ➥ **Herbicides** - weeds
  - ➥ **Rodenticides** - rodents (rats and mice)
  - ➥ **Fungicides** - fungi
  - ➥ **Bactericides** - bacteria
  - ➥ **Larvicides** - larvae

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☑ Based on how biodegradable they are

- **Biodegradable** - break down by microbe in to harmless compounds
- **Persistent (non - biodegradable)** - not break down in to harmless compounds

Chemically - related Pesticides

1. Organophosphate

- Most of them are insecticides and affect the nervous system

2. Carbonate

- They are pesticides and affect the nervous system.

3. Organochlorine Insecticides

- They were commonly used earlier but now, most countries have been removed from their market due to their health problem.

  Example : DDT, chlordane, toxaphene

4. Phrethroid

- These are a synthetic version of pyrethrin, a naturally occuring pesticide, found in chrysanthemums (flower).

5. Sulflonylurea Herbicides

- They have been commercialized for weed control such as pyrithiobac - sodium, cyclosulfamuron, bispyribac - sodium, terbacil, sulfome - turon methyl sulfosulfuron, rimsulfuron, pyrazosulfuron - ethyl, imazosulfuron etc.

6. Biopesticides

- The are derived from animals, plants, bacteria and certian minerals

Neam Leaf

- It can be used as an organic pesticide.
- Neam leaves and neem oil have also been used traditionally to protect stored legumes from insect pests.

Herbicides

- Herbicides remove unwanted plants or weeds
- The common herbicides used by the Ethiopia farmer are

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- Chloro phenoxy acids - 2,4 - D and 2,4,5 - T
- Triazines - Atrazine, hexazinone and simazine
- Organic phosphorus chemicals - glyposate
- Thiocarbomates - Butylate
- Amides - Alachlor and metalchlor
- Dintro anilines - Trifuralin
- Chloroaliphaticas - Dalapon and trichloro acetate
- Inorganic chemicals - various arsenicals, cyanates and chlortes.

Illustrative Examples

42. All of the following are pesticides **EXCEPT**

A. Insecticides
B. Herbicides
C. Fungicides
D. Fertilizers

Solution

Fertilizers are plant nutrients that increase crop yield

Answer: D

43. Which of the following is not a group of herbicides?

A. Chlorophenoxy acids
B. Organic phosphorus chemicals
C. Amides
D. Bactericides

Solution

Bactericides kill bacteria They are pesticides

Answer: D

44. Arsenicals, cyanates and cholrates are

A. Inorganic chemicals
B. Organic phosphorus chemical
C. Chlorophenoxy acids
D. Triazines

Answer: A

45. Which of the following kill mice and rats

A. Insesticides
B. Rodenticides
C. Herbicides
D. Fungicides

Answer: B

46. What is the name of natural fertilizer that most of the Ethiopian farmers

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use to increase the fertility of soil?

Answer: compost

47. Any toxic substance that kill insects is known as _____

Answer: Insecticides

48. List the components of organochlorines.

Solution

Oranochlorines are composed of carbon, hydrogen, chlorine and oxygen atoms.

49. What are botanical pesticides?

Solution

They are pesticides extracted from various plant parts, stems, roots, leaves and flower heads.

50. List some common natural pesticides used in some areas of Ethiopia

Solution

- Neem leaf, salt spray, onion and Garlic spray.

7. Sodium Carbate (Na_2CO_3)

- It is commonly known as **washing soda**
- It is white crystalline solid powder.
- It occurs as hydrate form ($Na_2CO_3 . 10H_2O$)
- It has high melting and boiling points
- It has hygroscopic (absorbs water molecules from the air) properties in nature.
- It soluble in water

$$Na_2CO_3(s) + 2H_2O(1) \rightarrow H_2OCO_3(aq) + 2Na^+(aq) + 2OH^-(aq)$$

$$Na_2CO_3(aq) + H_2O(1) + CO_2(g) \rightarrow 2Na\,HCO_3(aq)$$

Manufacture of Sodium Carbonate

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- ☑ It is manufactured industrially by **the solvay process**
- ☑ It manufacture irygeniously using calcium carbonate ($CaCO_3$), sodium chloride (NaCl), carbondioxide and ammonia (NH_3) in a continuous process, with the **solvay tower** the centre of reactions
- ☑ Sodium carbonate is made as economically as possible because both the carbondioxide and the ammonia components are recycled continuously. The reactions are given below.

$$CaCO_3 \rightarrow CaO + CO_2$$

$$CaO + H_2O \rightarrow Ca(OH)_2$$

$$2NaCl + 2CO_2 + 2NH_3 + 2H_2O \rightarrow 2NaHCO_3 + 2NH_4Cl$$

$$2NaHCO_3 \rightarrow Na_2CO_3 + H_2O + CO_2$$

$$Ca(OH)_2 + 2NH_4Cl \rightarrow CaCl_2 + 2NH_3 + 2H_2O$$

Net reaction

$$CaCO_3 + 2NaCl \rightarrow Na_2CO_3 + CaCl_2$$

Uses of Sodium Carbonate

- ☑ It is used in the manufacture of soaps, detergent, dyes, drugs and other chemicals.
- ☑ It is also used to make glass, pulp and paper, textiles.
- ☑ It is used in brick industry.

8. Sodium hydroxide (NaOH)

Properties

- ☑ Sodium hydroxide is commonly known as **caustic soda or Lye.**
- ☑ It is white crystalline solid that dissolves in water
- ☑ It is strong base.

Manufacture of Sodium Hydroxide

- ☑ The electrolysis of concentrated sodium chloride solution **(brine)** is the basis of a major industry.
- ☑ The ions in this concentrated NaCl solutions are:

  Na^+ and Cl^- from solute (NaCl)

  H^+ and OH from solvent (H_2O)

- ☑ When the current flows, the chloride ion, Cl^- attracted to anode.

$$2Cl^-(aq) \rightarrow Cl_2(g) + 2e^-$$

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☑ This leaves a high concentration of sodium ion Na^+, around the anode.

☑ The hydrogen ions, H^+ are attracted to the cathode and hydrogen gas is produced

$$2H^+(aq) + 2e^- \rightarrow H_2(g)$$

☑ This leaves a high concentration of hydroxide ion around the cathod. The sodium and hydroxide ions combines to form sodiumhydroxide (NaOH)

Uses of Sodium Hydroxide

☑ It is used in many industrial process such as pulp and paper production, alumiuna extraction, testile industry, drinking water production, soap and detergent factory, waster gases scrubbing saponification.

☑ It is also used in manufacturing of petroleum products.

Illustrative Examples

51. Which of the following is used to make sodium carbonate industrially?

A. Haber process
B. Contact process
C. Ostwald process
D. Solvay process

Solution

Haber process - Ammonia
Contact process - Sulfuric acid
Ostwald process - Nitric acid

Answer: D

52. Washing soda or soda ash is the common name of

A. $NaHCO_3$ B. NaOH C. Na_2CO_3 D. NaCl

Solution

$NaHCO_3$ - baking soda
NaOH - Caustic soda
NaCl - table salt

Answer: C

53. All of the following are raw materials used to manufacture sodium carbonate. EXCEPT

A. Sodium chloride solution
B. Calcium chloride

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C. Ammonia gas D. Water

Answer: B

54. The main by product in the Solvay process is

A. $NaHCO_3$ B. NH_4Cl C. $CaCl_2$ D. NH_3

Answer: C

55. Sodium hydroxide is industrially manufactured from electrolysis of

A. Molten sodium chloride
B. Concentrated sodium chloride solution
C. Dilute sodium chloride solution
D. Alkaline water solution

Answer: B

56. Write the uses of:

a. NaOH b. Na_2CO_3

Solution

a. For neutralization, pulp and paper manufacturing oil refining, soap and detergent production

b. As metallurgical detergents, flat glass, glass fiber, glass container, water purification

9.4 Some Manufacturing Industries in Ethiopia

☑ Some of the manufacturing industries can be represented by diagram as follows

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Sulphuric acid	H_2SO_4 and $Al_2(SO_4)_3$	Awash Melkasa
Tabor ceramic factory	Ceramics	Hawassa
Alemeda textile factory	Textile	Adwa
Arbamich textile factory	Textile	Arbaminch
Caustics soda fanctory	Caustic soda	Zeway
Adamitulu Plasticide factory	Pesticides	Adamitulu
Mojo Tannery	Processed leather	Mojo
Dashen brewery	Beer	Gonder
Harar brewery	Beer	Harar
Meta brewery	Beer	Sebeta
Bedelle brewery	Beer	Bedelle
Wonji sugar	Sugar	Wonji
Metehara sugar	Sugar	Mete hara
Fincha sugar	Sugar	Fincha
Cament factories	Cement	Tigray, Muger, Dired Dawa, Derba

1. Glass Manufacturing

Glass is amorphous solid, consists of two or more silicates.

Types of Glass

1. **Soda - lime glass** - is a mixture of sodium silicate (Na_2SiO_3) and calciumsilicate ($CaSiO_3$)
   - ➥ Used to make bottles, dishes etc.
2. **Boro silicate glass (pyrex glass)** - Consists of boron oxide(B_2O_3) and resists chemical and temperature changes.
   - ➥ Used to make laboratory equipments such as test tubes, flasks and beakers.
3. **Quart glass** - is formed from pure silica and is highly transparent.
   - ➥ **Colored glasses** - are obtained by adding different chemicals.

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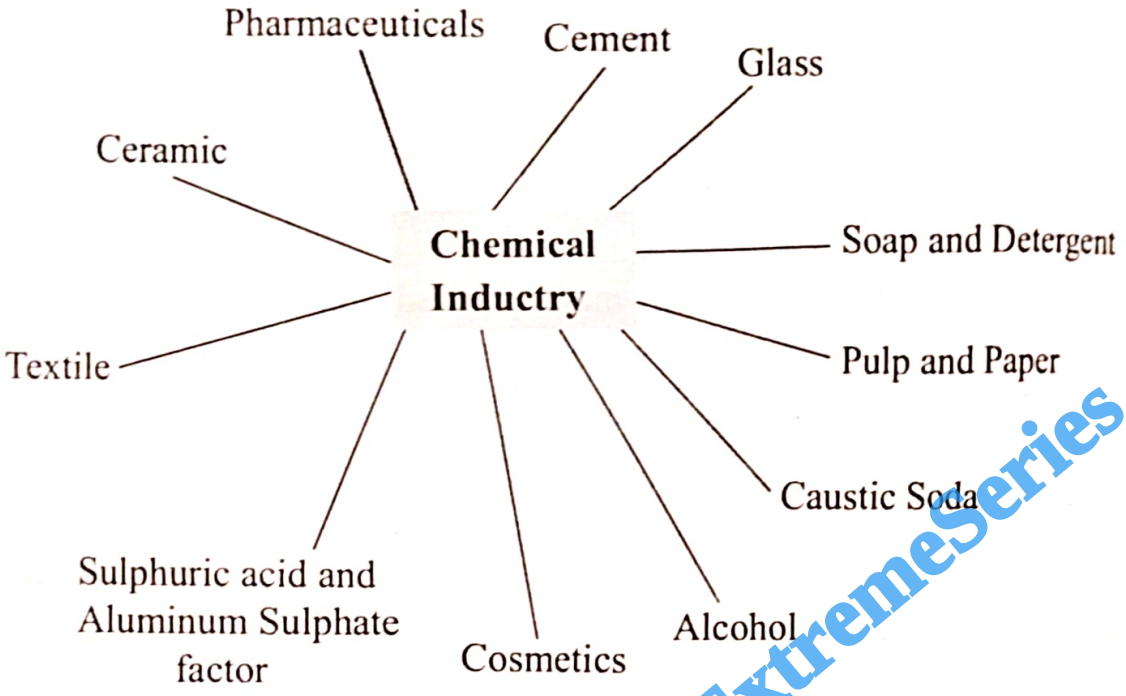

Table : 9.3 Some Industries in Ethiopia

Industry	Product	Location
Nifas silk paint factory	Paint	Addis Ababa
Dilpaint factory	Paint	Addis Ababa
Tseday paint factory	Paint	Addis Ababa
Repi soap and detergent factory	Soap and detergent	Addis Ababa
Gulele soap factory	Soap	Addis Ababa
Matador addis goma	Tyres	Addis Ababa
Saint george brewry	Beer	Addis Ababa
Awash tannery	Processed leather	Addis Ababa
Addis Ababa foam and plastic	Plastics	Addis Ababa
Ethiopia plastic	Plastics	Addis Ababa
Ethio gas and arts	Plastic, art and CO_2	Addis Ababa
Choragas and chemical	O_2,C_2,H_2 shoepolish	Addis Ababa
Addis glass factory	Glass	Addis Ababa
Nas foods factory	Bicut	Addis Ababa

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Sulphuric acid	H_2SO_4 and $Al_2(SO_4)_3$	Awash Melkasa
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Arbamich textile factory	Textile	Arbaminch
Caustics soda fanctory	Caustic soda	Zeway
Adamitulu Plasticide factory	Pesticides	Adamitulu
Mojo Tannery	Processed leather	Mojo
Dashen brewery	Beer	Gonder
Harar brewery	Beer	Harar
Meta brewery	Beer	Sebeta
Bedelle brewery	Beer	Bedelle
Wonji sugar	Sugar	Wonji
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C. Cobaltous oxide D. Manganese oxide

Solution

Cuprie oxide - green glass

Chromic oxide - green glass

Mangenes oxide - reddish - purple glass

Answer: C

59. At which stage of glass production, internal stresses are removed by heating the glass?

A. Glass forming C. Glass melting

B. Annealing D. Inspecting

Answer: B

60. Sulphuric acid and aluminum sulphate factory is found in _____.

Solution

Oromiya region, Awash melkasa.

61. A type of glass that resists temperature change and used to make laboratory glass wares is _____.

Answer: Boro silicate glass or pyrex glass

2. Manufacturing Ceramics

- ☑ Clay, talc and feldespar are used as traditional ceramics.
- ☑ The raw materials for industrial ceramics are: silicon carbide, alumina, bariumtitanate and titanium carbide.
- ☑ Ceramics are mainly used to make dishes, kichen tilies and bathroom structures. To lesser extent, used as insulators and artificial bones
- ☑ In the future, ceramics may be used to remove impurities from drinking water and to substitute diseased heart valves.

Steps in Ceramics Production

Step 1- Moulding - Adding some wax to bind ceramic powder.

➥ Powder can be shaped into different objects.

Step 2 - Densification - Using intense heat to condense ceramic object in to strong product.

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Steps in Sugar Production

A. Collecting the harvest

- Cutting the sugar cane by removing leaves and transporting to the sugar factory.

B. Cleaning and Grinding

- Cleaning the sugar cane and chopping into pieces.

C. Juicing

- Extracting juice and clarifying it.

D. Clarifying

- Using carbondioxid and milk of lime for purifying

E. Evaporation

- The clarified juice is evaporated into thick brown syrup.

F. Crystallization

- Forming cystal of sugar by removing extra water.

G. Refinery

- Raw sugar is transported in to a cane sugar, refinery for the removal of mollasses, minerals and other non - sugars that contiminate it.
- The brown syrup is bleached by **sulfurdioxide (SO_2)**

H. Separating and Packaging

- Large and small crystal of sugar, packed and labeled as white pure sugar.

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Illustrative Examples

62. Compounds that occur widely in glass, cement and ceramics are

A. Silicates B. Sulphaters C. Carbonates D. Phosphates

Answer: A

63. Cement chiefly consists of which of the following mixtures?

A. $Ca(OH)_2$ and $CaSiO_3$
B. Na_3AlF_6 and $Al_2O_3\,2H_2O$
C. Na_2SiO_3 and $CaSiO_3$
D. $CaSiO_3$ and $Ca(AlO_2)_3$

Solution

Cement mainly contains calcium silicate, calcium aluminate, iron (III) oxide and magnesium oxide

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Table : 9.4 Chemical Added and Colored Glasses

Chemical Added	Color of Glass
Chromic oxide	Green
Manganese oxide	Reddish - purple
Copper(11) oxide	Green
Cadmium sulphide	Yellow
Cobaltous oxide	Blue
Gold and cuprous	Rubby
Oxides	Fluorescent
Uraniumoxide	Yellowish green

Steps in Glass Production

1. **Batch preparation** - preparing materials for the production of glass.
2. **Glass melting** - raw materials and recycled glasses are fed into a glass melting furnace.
3. **Glass forming** - the melted glass is changed into required shape.
4. **Annealing** - removing internal stress by reheating the glass.
5. **Inspection** - testing of the glass product
6. **Packing and dispatching** - final step in glass production.

Illustrative Examples

57. The chief composition of glass are

A. Sodium chloride and calcium silicate
B. Sodium silicate and calcium silicate
C. Calcium silicate and calcium aluminate
D. Calcium and silicon oxides

Solution

The main components of glass are sodium and calcium silicates.

Answer: B

58. Which of the following chemical results blue color glass?

A. Cupric oxide
B. Chromic oxide

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C. Cobaltous oxide D. Manganese oxide

Solution

Cupric oxide - green glass

Chromic oxide - green glass

Manganes oxide - reddish - purple glass

Answer: C

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➥ Powder can be shaped into different objects.

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3. Cement Manufacturing

Cement is made by heating limestone, alumina and silica bearing material such as clay in a **Kiln** at very high temperature.

Port land cement - is important building material

☑ It is a mixture calcium and aluminm silicates.

☑ Cement mainly contains.

- Calcium silicate - $CaSiO_3$
- Calcium aluminate - $Ca\,Al_2O_4$
- Iron (III) oxide - Fe_2O_3
- Magnesium oxide - MgO

Manufacture Process

1. **In the upper part of the kiln**
   - raw materials are heated to remove moisture.
2. **In the middle part of the kiln**
   - lime stone is decomposed by heat.
3. **In the lower part of the kiln**
   - at high temperature the following reactions occur.

$2CaO + 2SiO_2 \rightarrow 2CaO.SiO_2$

$3CaO + 3SiO_2 \rightarrow 3CaO.SiO_2$

$4CaO + 4Al_2O_3 + 4e_2O_3 \rightarrow 4CaO.Fe_2O_3$

$MgO + SiO_2 \rightarrow MgSiO_3$

   - The end mixture of all silicates is called **cement clinker.**
   - After cooling, the cement clinker is mixed with **gypsum** and powdered.
   - Cement is used manly for construction

Cement is composed of CaO (50 - 60%), SiO_2 (20 - 25%), Al_2O_3 (5 - 10%), HgO (2 - 3%), Fe_2O_3 (1 - 2%) and SO_3 (1 - 2%)

4. Sugar Manufacturing

Raw materials: sugar cane and sugar beet

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Steps in Sugar Production

A. Collecting the harvest

- Cutting the sugar cane by removing leaves and transporting to the sugar factory.

B. Cleaning and Grinding

- Cleaning the sugar cane and chopping into pieces.

C. Juicing

- Extracting juice and clarifying it.

D. Clarifying

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Solution

Cement mainly contains calcium silicate, calcium aluminate, iron (III) oxide and magnesium oxide

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Answer: D

64. The molecular formula of sugar /sucrose/ is

A. $C_6H_{12}O_6$ C. $C_5H_{10}O_5$

B. $C_{12}H_{22}O_{11}$ D. $C_{10}H_{20}O_{10}$

Solution

Sucrose is a dimer of glucose and fructose with a molecular formula of $C_{12}H_{22}O_{11}$

Answer: B

65. The bleaching agent in sugar production is

A. SO_2 B. NaClO C. H_2O_2 D. O_3

Answer: A

66. Write the processes of manufacturing ceramics

Solution

Moulding and densification

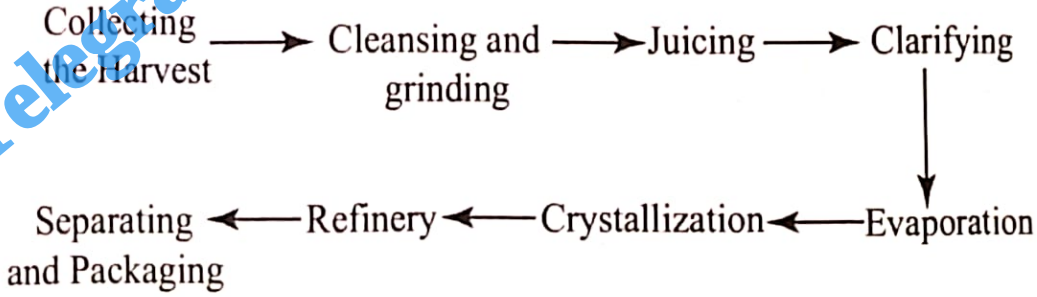
67. Write three traditional ceramics

Solution

Porcelain, tiles and pottery which are formed from minerals such as clay, talc and feldspare.

68. Write the steps in sugar production.

Solution



5. Pulp and Paper Production

Raw material - wood

Wood pulp is dry fibrous material used for making paper

Pulpwood is the timber resource used to make wood pulp.

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- **Spruce, pine, fir, larch and hemlock eucalyptus, aspen and birch** are used to make wood pulp. The first five are soft woods and the last three are hard woods.

Steps in Making Pulp and Paper

1. Harvesting

☑ Cutting trees and transporting into the pulp and paper factory.

2. Preparing for Pulping

☑ Bark of tree removed, the wood is chipped and screened.

3. Pulping

☑ Making wood pulp

This can be done by two processes

A. Mechanical Pulping

➥ Uses steam, pressure and temperature without chemical to tear fibers.

Disadvantage

☑ The quality of fiber decreases due to short, weak fibers are obtained.

B. Chemical pulping

➥ Uses chemicals to tear fibers.

Divided in to two

(i) The kraft process

➥ It is the main chemical pulping method.

➥ It is widely used method.

➥ It uses aqueous sodiumhydroxide and sodium sulphide as a digestion solution

(ii) The sulphate process

➥ It uses sodium or magnesium bisulphate digester.

4. Bleaching

➥ The bleaching agents in pulp and paper factory are **chlorine, chlorine oxide, ozone**

5. Making Paper from Pulp

➥ Producing paper

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6. Tannery

- Tanning is the process of changing animal skin into leather using tannin.
- Tannin is an acidic chemical that changes protein structure of skin permanently

Steps in Leather Production

1. Preparation Stages

- Preparation stages

a. Curing

- Salting or drying the skin using brine

b. Soaking

- Placing the hides in water for several hours.

c. Flesh Removal

- Removing flesh from hides using sharp machine or knives.

d. Hair Removal

- The hair is removed from hides using lime and water.

e. Scudding

- Removing any left hair with machines

f. Deliming

- Neutralizing the lime using vat of acid

2. Tanning

- Converting the protein of raw skin to stable material. This can be done using

(i) Natural Tanning or Vegetable Tanning

- This method uses the bark and leaves of many plants as tannin

(ii) Mineral Tanning

- This method uses chemicals such as chromium sulphate and other salts of chromium as tannin.

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➥ It is faster than natural tanning

3. Crusting

➥ It is the last step and includes coloring and rolling the leather.

7. Food Processing and Preservation

➥ There are two methods used to preserve food

A. Traditional methods - include

(i) **Salting** - removes moisture.

(ii) **Pickling** - preserving food in an edible liquid that inhibits the growth of micro - organisms.

(iii) **Sugaring** - sugar is used to preserve fruits such as apples, pears and peackls.

(iv) **Smoking** - preserving food by exposing to smoke.

(v) **Drying** - used to preserve meat by removing water from it.

(vi) **Canning** - sealing food in sterile cans or jars and boiling the contents to kill or weaken any remaining bacteria in the form of sterilization.

B. Modern Methods of Food Preservation

(i) **Freezing**

➥ Used to preserve wide range foods

(ii) **Vacuum - packing**

➥ Stores food in a vacuum environment most of the times in an air - tight bag bottle.

(iii) **Freeze - drying** - to preserve coffee, tea, juice extracts, vegetables

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Illustrative Examples

69. Which of the following steps are correct in the preparatory stages of tanning?

A. Deliming → Curing → hair removal → scudding → soaking → faesh removal.

B. Curing → soaking → flesh removal → hair removal → scudding → delimining

C. Soaking → deliming → curing → flesh removal → scudding →

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delining

D. Hair removal → flesh removal → curing → soaking → scudding → delining

Solution

The correct order in the preparatory stages of tanning is:

Curing → soaking → flesh removal → hair removal → scudding → deliming.

Answer: B

70. How does protein substance in animal hide or skin be prevented from bacterial infection during tanning?

A. By curing brine
B. By scudding using platic tool
C. By soaking in water
D. By deliming using an acid.

Solution

Brine curing is the simple and fastest method to prevent protein substance, collagen from bacterial infection.

Answer: A

71. In the tanning process, the purpose of putting the skin into slacked line solution is to

A. Facilitate the removal of unwanted flesh
B. Dehydrate the skin
C. Facilitate the removal of hair
D. Soften the skin

Answer: C

72. The purpose of adding salt to the skin in the process of tanning is to

A. Make the skin soft
B. Dehydrate the skin
C. Remove the unwanted flesh from the skin
D. Loosen the outer layer of the skin

Solution

Adding salt to the animal skin kill bacteria and dehydrate it.

Answer: B

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73. Food can be preserved by one of the following methods. Which one is it?

A. Applying an oxidizing agent
B. Providing heat and oxygen
C. Keeping wet and moist
D. Storing in an air tight bag or bottle.

Answer: D

74. The purpose of freezing in food preservation is to

A. Suppress the growth of bacteria and micro - organism
B. Keep it from the access of house hold pests
C. Increase air circulation in food substance
D. Improve the taste and odour of the food item

Answer: A

75. List the traditional methods used to preserve food.

Solution

Salting, pickling, sugaring, smoking, drying and canning.

76. What is food preservation?

Solution

Food preservation is the process of treating and handling food to stop or greatly reduce spoilage, loss of quality, edibility or nutritive value caused or accelerated by micro - organisms.

77. What are the bleaching agents in pulp and paper factory?

Solution

- Hydrogen peroxide (H_2O_2)
- Chlorine oxide (Cl_2O)
- Chlorie (Cl_2)
- Ozone (O_3)

78. What are the steps in pulp and paper factory?

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Solution

Harvesting → Preparing for Pulping → Pulping → Bleaching → Making paper from pulp

8. Manufacture of Ethanol

Ethanol is the second monohydric alcohol.

- It is miscible with water.
- It is a neutral, colorless, volatile liquid which does not conduct electricity.
- Ethanol burns quite readily with a clean hot flame.

$$CH_3CH_2OH(l) + 3O_2(g) \rightarrow 2CO_2(g) + 3H_2O(l) + \text{heat}$$

- Ethanol is oxidized in to ethanoic acid by strong oxidizing agent.

$$CH_3CH_2OH \xrightarrow{[O]} CH_3COOH$$

Ethanol — Ethanoic acid

Industrial Preparation of Ethanol

1. Fermentation of Carbonhydrates.

$$C_{12}H_{22}O_{22} \xrightarrow{\text{maltase}} 2C_6H_{12}O_6\ (aq)$$

maltose (sugar) — Glucose

$$C_{12}H_{22}O_{22}(aq) \xrightarrow{\text{zymase}} 2C_2H_5OH(aq) + 2CO_2(g)$$

$$C_{12}H_{22}O_{22} + H_2O \longrightarrow 2C_6H_{12}O_6$$

sucros — (glucose + frutose)

$$C_{12}H_{22}O_{22}(aq) \xrightarrow{\text{zymase}} 2CO_2(g) + 2C_2H_5OH(aq)$$

- The ethanol obtained is purified further by distillation to obtain more **pure ethanol.**

2. Catalytic Hydration of Ethene

$$CH_2{=}CH_2 + H_2O \xrightarrow{H_2SO_4} CH_3{-}CH_2{-}OH$$

Ethene — **Ethanol**

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Beer

- It is an alcoholic beverage produced by the process of brewing and fermentation of starch present in the grains such as malted barely
- The average alcohol content of beer is **2 - 6%** by volume.

Wine

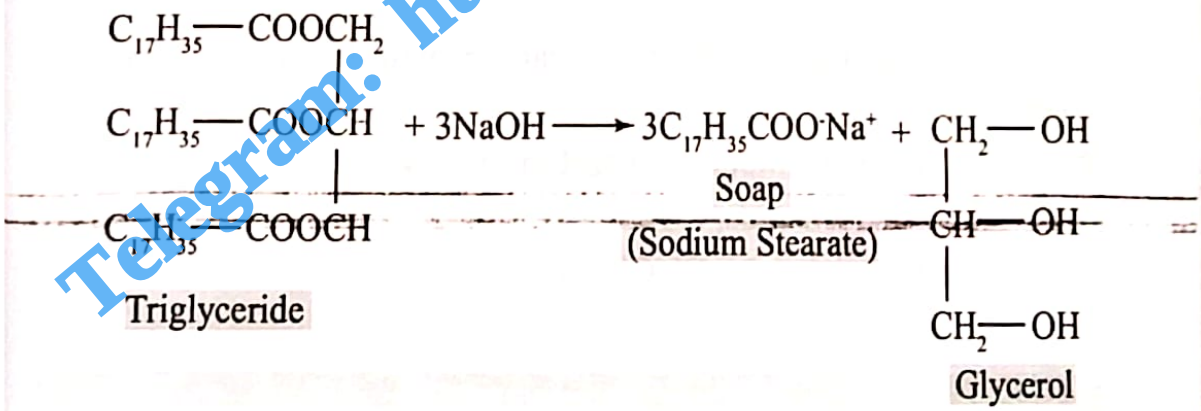
- It is alcoholic beverage.
- Wine making or vinification is the production of wine, starting with the selection of the fruit, its fermentation into alcohol and the bottling of the finished liquid.
- Most wine contain **10 - 66%** by volume.

Liquor

- It has higher alcohol than beer and wine between **30 - 45%** alcohol by volume .

9. Soaps and Detergents

- **Soap** is manufactured by heating natural fats and oils of either plants or animals with a strong alkali. These fats and oils called triglycerides, triesters or triacylglcerols
- The reaction between triglycerides and strong alkali is known as **saponification**.



The cleaning properties of the soap depend on its structure and bonding. Sodium stearate consists of a long hydrocarbon chain which is **hydrophobic** (water hating) attached to anionic head which is **hydrophilic** (water loving)

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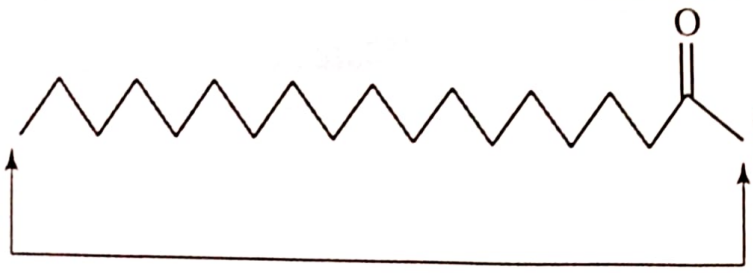
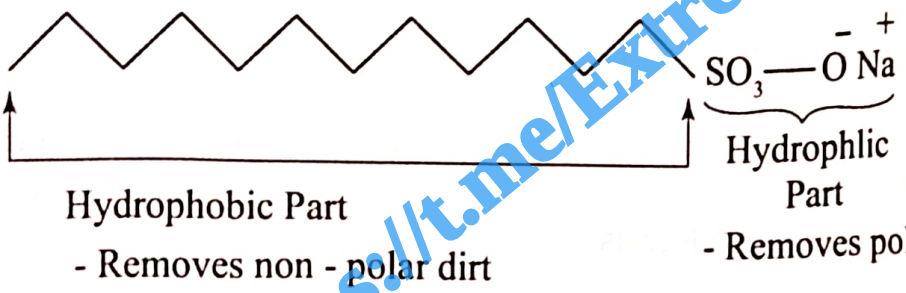


Detergents are neutral substance that make lather with hard water but not soaps.

$$C_{12}H_{25}—OH + H_2SO_4 \longrightarrow C_{12}H_{25}—O—SO_2—OH + H_2O$$

$$C_{12}H_{25}—O—SO_2—OH + NaOH \longrightarrow C_{12}H_{25}—SO_3—O^{-}Na^{+} + H_2O$$

Sodium Lauryl Sulphate



10. Dry Cleaning

It is removing of organic dirts using organic chemicals in the abscence of water.

The most common chemicals used in dry cleaning are

- Carbon tetrachloride, CCl_4
- Tetrachloro ethylene, $Cl_2C = CCl_2$
- Benzene and gasoline

Illustrative Examples

79. Which of the following alcohol is found in all alcoholic beverages?

A. Methanol B. Ethanol C. Glycerol D. Propanol

Answer: A

80. $C_6H_{12}O_6 \xrightarrow{\text{yeast}} X + CO_2$, X is

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A. Acetic acid B. Water C. Ethanol D. Ribase

Solution

Fermentation of carbohydrates yields ethanol

Answer: C

81. The reaction between triglycerides and alkali is known as

A. Esterification
B. Saponification
C. Hydrogenation
D. Halogenation

Answer: B

82. All of the following chemicals are used in dry cleaning. **EXCEPT**

A. CCl_4
B. C_6H_6
C. $Cl_2C = CCl_2$
D. C_2H_5OH

Answer: D

83. Write the four basic raw materials used to make soaps.

Solution

- Fats and oils (triglyceride)
- Alkalis (NaOH or KOH)
- Brine to recover glycerine
- Additives: sodium carbonate, sodium silicate, dyes, perfumes.

84. What are soaps?

Solution

Soaps are either sodium or potassium salts of long chain carbonxylic acids (fatfy acids). Sodium soaps are hard soap but that of potassium are soft soaps.

85. Detergent are better that soaps. How?

Solution

Detergent make lather with hard water but not soaps.

Solved problems

86. Study the following reaction scheme.

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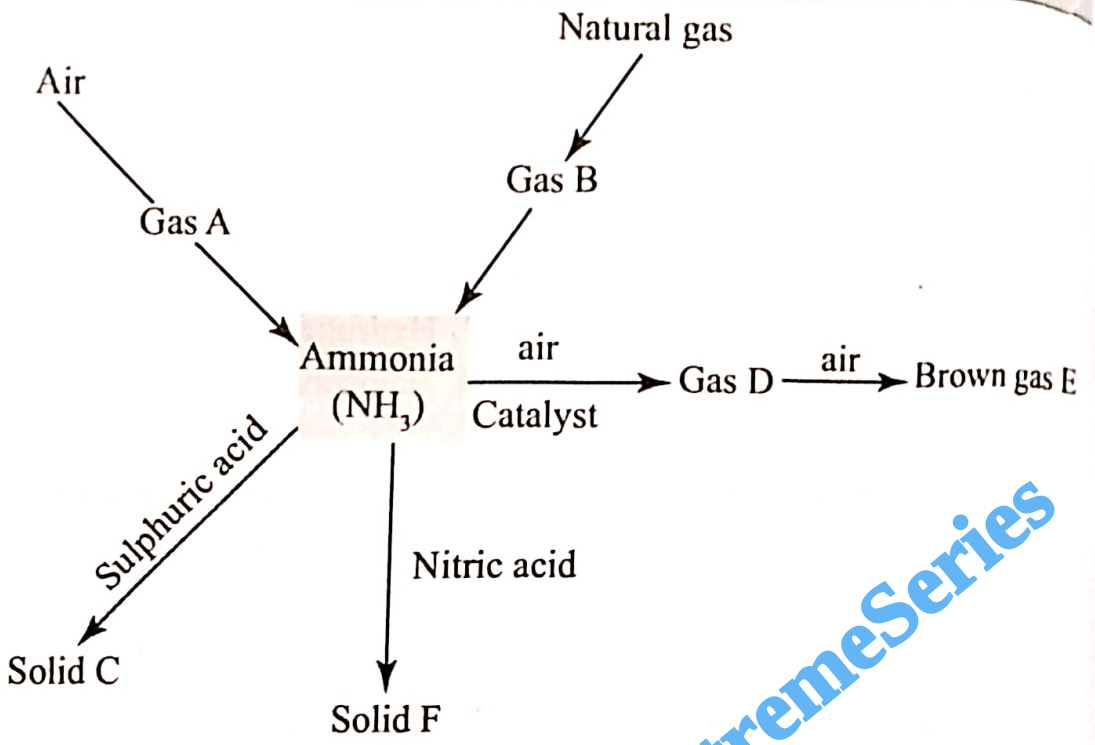

a. Identify the substance A to F

b. How is gas A obtained from the air?

c. Write the balanced chemical equation for the formation of C and F.

Solution

a. A = Nitrogen gas (N_2)
B = Hydrogen gas (H_2)
C = Ammonium sulphate ($(NH_4)_2SO_4$)
D = Nitric oxide (NO)
E = Nitrogen dioxide (NO_2)
F = Ammoniumnitrate (NH_4NO_3)

b. By fractional distillation of liquid air

c. $2NH_3 + H_2SO_4 \rightarrow (NH_4)_2\ SO_4$ (s) : C
$NH_3 + HNO_3 \rightarrow NH_4NO_3$: F

87. Calculate the percentage by mass of nitrogen in the following fertilizers.

a. NH_4NO_3 b. $(NH_4)_2\ SO_4$ c. $(NH_2)_2\ CO$ d. KNO_3

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Solution

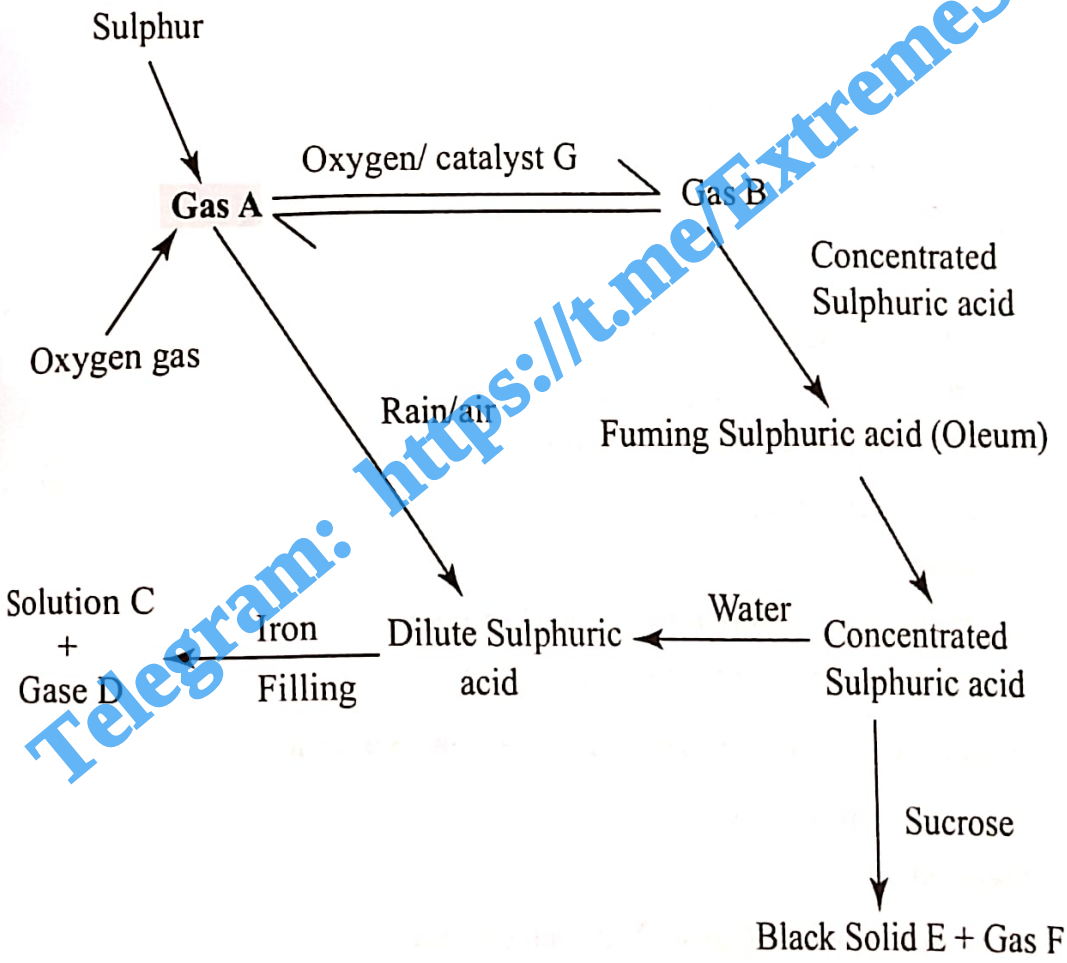
a. $\%N = \frac{\text{mass N}}{\text{mass } NH_4NO_3} \times 100 = \frac{28g}{80g} \times 100 = 35\%$

b. $\%N = \frac{\text{mass N}}{\text{mass } (NH_4)_2SO_4} \times 100 = \frac{28g}{132g} \times 100 = 21.2\%$

c. $\%N = \frac{\text{mass N}}{\text{mass } (NH_4)_2CO} \times 100 = \frac{28g}{60g} \times 100 = 46.7\%$

d. $\%N = \frac{\text{mass N}}{\text{mass } KNO_3} \times 100 = \frac{14g}{101g} \times 100 = 14\%$

88. Study the following reaction scheme



a. Identify the substances A to G by given their names and formula.

b. Write a balance chemical equation for the formation of gas B

c. Which path way shows the formation of acid rain?

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d. In which way is the concentrated sulphuric acid acting in its reaction with sucrose?

Solution

a. A : sulphurdioxide (SO_2)

B : Sulphuricioxide (SO_3)

C : Iron (II) sulphate ($FeSO_4$)

D : hydrogen gas (H_2O)

E : carbon (C (s))

F : water vapour ($H_2O(g)$)

G : Vanadium (V) oxide, V_2O_5 (s)

b. $2SO_2(g) + O_2(g) \rightleftharpoons 2SO_3(g)$

c. Gas A (SO_2) + rain/ air $\rightarrow$ H_2SO_4 (dilute) Acid rain

d. Dehydrating agent

89. Write the main uses of ammonia.

Solution

- To make fertilizers
- To manufacture nitric acid
- To remove greasy stains in laundry
- To remove temporary hardness.

90. State two uses of nitric acid

Solution

- To manufacture nitrogenous fertilizers
- To manufacture explosives and dyes.

91. **Liquid A $\longrightarrow$ Brown Gas B $\longrightarrow$ Nitric acid**

Write the name of liquid A and gas B.

Solution

Liquid A: water, Gas B: Nitrogen dioxide

92. What are the three advantages of Solvay process?

Solution

Its raw materials (NaCl and $CaCO_3$) are cheap

It does not involve large consumption of energy.

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Pollution problems have been considerably reduced.

93. List the factors that influence the location of chemical industry.

Solution

- Power supply
- Government policy
- Market
- Raw materials
- Space
- Labour supply
- Good transport system
- Concentration of industry
- Climate
- Pollution problems

94. Define ceramics

Solution

Ceramics are inorganic and non - metallic solids prepared by the action of heat followed by cooling.

95. What raw materials are used to make most industrial ceramics?

Solution

Silicon carbide, alumina, barium titanate and titanium carbide

96. List some local alcoholic beverages in Ethiopia

Solution

Areki or katicala, Tella, Tej, Bordie, Korefe

97. The traditional distilled alcoholic beverage in Ethiopia is _____

Answer: Areki

98. What is the purpose of yeast in the fermentation process to make ethanol?

Solution

Yeast catalyses the fermentation process

99. List some of the cement factors found in Ethiopia.

Solution

Muger, Dangote, Messobo, Derba cement factories

100. Write the uses of chlorine which is produced during the manufacture of sodium hydrogen

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- To treat drinking water and swimming pool water by killing bacteria.
- 20% of chlorine produced is used to make PVC
- To make chlorofrom (an anesthetic) and carbonterachloride (a dry - cleaning solvent)

Supplementary exercise

I. Multiple choice items

1. Which one of the following is weed killer?
   A. Insecticides
   B. Herbicides
   C. Fungicides
   D. Rodenticides
2. Which one of the following is not the major use of sulpuric acid? To manufacture
   A. Fertilizers
   B. Detergent
   C. Plastics
   D. Electrolytes of dry cells
3. Given the steps in the contact process
   I. $2SO_2 + O_2 \rightarrow 2SO_3$
   II. $H_2SO_4 + SO_3 \rightarrow H_2S_2O_7$
   III. $S + O_2 \rightarrow SO_2$
   IV. $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$
   The correct order is
   A. III, I, II and IV
   B. II, III, I and IV
   C. IV, III, I and IV
   D. I, III, IV and III
4. Urea is manufactured by reacting
   A. Ammonia and carbondioxide
   B. Nitric acid and carbondioxide
   C. Ammonium nitrate and carbondioxed
   D. Ammonium and carbonic acid
5. Given the reaction: $2NH_4Cl + Ca(OH)_2 \rightarrow X + CaCl_2 + 2H_2O$ X is:
   A. HNO_3 B. NH_3 C. NH_4NO_3 D. CO_2
6. Which of the following converts nitrates back to nitrogen gas?

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A. Haber process
B. Lightening
C. Nitrigying bacteria
D. Denitrying bacteria

7. The catalyst used in the Haber process is

A. V_2O_5 B. MnO_2 C. Fe D. Ni

8. The dense oily intermediate comound in the preparation of sulphuric acid has formula

A. SO_2 B. SO_3 C. H_2SO_4 D. $H_2S_2O_7$

9. Which of the following is not true about ammonia?

A. It is colorless gas
B. It has pungent smell
C. It is soluble in water
D. It is heavier than air

10. Aqueous solution of ammonia consists of

A. H^+_{ions} only
B. OH^-_{ions} only
C. NH^+_4 and OH^- ions
D. NH^+_4 and H^+ ions

11. Salts of sulphuris acid are

A. Sulphates B. Sulphites C. Sulphides D. Sulphurous

12. All of the following are alcoholic beverages **EXCEPT**

A. Gine B. Beer C. Liquor D. Pepsi

13. Which of the following physical techniques used to separate ethanol from other components?

A. Evaporation
B. Filtration
C. Distillation
D. Sublimation

14. Which of the following is not a modern method of food preservation?

A. Freezing
B. Drying
C. Freezer drying
D. Vacuum - packing

15. Which type of glass has the highest strength and transparency?

A. Quartz glass
B. Soda - lime glass
C. Boro silicate glass
D. None

II. Short answer items

16. _____ is a durable and flexible material created by the tanning of animal hides and skin.

17. _____ is the process of removing coloring matter from wood pulp

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and increasing its brightness.

18. Write the common forms of N - based fertilizers.

19. Write the natural environment that are used as source of natural resources

20. The most abundant gas in the atmosphere is _____

Answer key for Supplementary Exercise

Part I

1. B	6. D	11. A
2. D	7. C	12. D
3. A	8. D	13. C
4. A	9. D	14. B
5. B	10. C	15. A

Part II

16. Leather

17. Bleaching

18. NH_4NO_3, $(NH_4)_2HPO_4$, $(NH_2)_2CO$, $(NH_4)_2SO_4$

19. Atmosphere Lithosphere
Hydrosphere Biosphere

20. Nitrogen

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UNIT 10

Polymers

10.1 Introduction

A **Polymer** is a large molecule made by linking together repeating units of small molecules called **monomers**.

The process of forming a polymer by linking the monomers together is known as **polymerization**.

$$\underset{\text{Monomer}}{nM} \xrightarrow{\text{Polymerization}} \left[-M-M-M-M-M-M-\right]_{\text{Polymer}}$$

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10.2 Classification of Polymers

1. **Natural Polymers (biopolymers)** are synthesized by organisms.

   Example

   DNA, RNA, proteins, ploysaccharides, natural rubber

2. **Synthetic Polymers** are made by man.

   Example

Nylon	Teflon	Poystrene
dacron	PVC	Polyethylene
polyester	Bakelite	Synthetic rubber

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3. **Homo polymers** are polymers formed from one type of monomer.

Example

$$n\,CH_2 = CH_2 \xrightarrow{\text{Catalyst}} \left[CH_2 - CH_2\right]_n$$

Ethene — Polyethylene

4. **Copolymers** are formed from more than one type of monomers.

Example

$$n\,CH = CH_2 + m\,CH_2 = CH - CH_3 \downarrow$$

$$\left[CH_2 - CH_2\right]_n \left[CH_2 - \underset{\displaystyle CH_3}{\underset{|}{CH}}\right]_m$$

Copolymers are further classified into four:

i **Random copolymer** – monomers are arranged in random fashion.

ii **Alternating copolymer** – the two monomers alternate.

AB AB AB AB AB AB AB AB

iii **Block copolymer** – blocks of each type of monomer.

AAAAA BBBBB AAAAABBBBB AA

iv **Graft polymer** – contains branches derived one monomer grafted onto a back bone derived from another monomer

A A A A A A A A A A A A A A A A A A

B B B
B B B
B B B
B B B
B B B
B B B
B B B
B B B

Illustrative Examples

1. Which of the following is not natural polymer?

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A. Cellulose
B. DNA
C. Teflon
D. Protein

Answer: C

2. The type of polymer formed from one type of monomer is:
   A. Random polymer
   B. Regular polymer
   C. Natural polymer
   D. Homo polymer

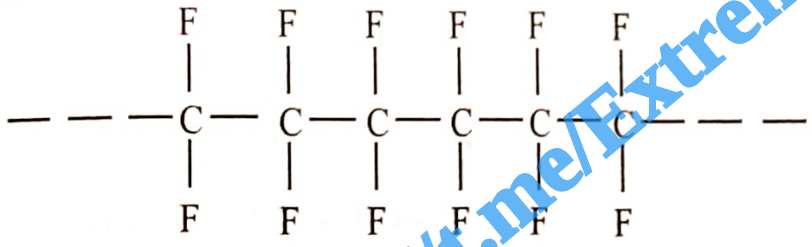
Answer: D

3. Synthetic polymers are considered to be pollutants but not natural polymers. Why?

Solution:

➥ Most synthetic polymers are non-biodegradable while natural polymers are biodegradable.

4. Teflon has the structure below

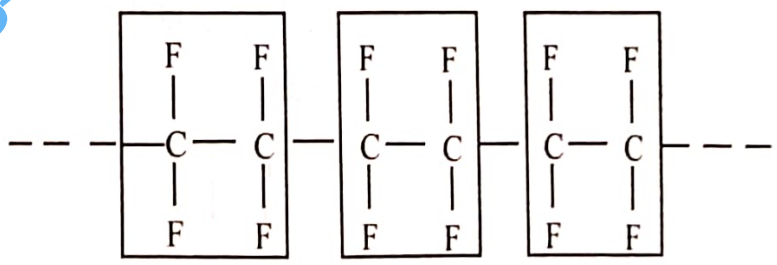


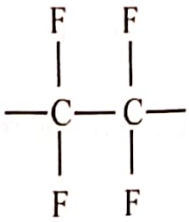
a. Draw the structure of the monomer from which Teflon is made.

b. How many monomer units are in Teflon polymer that has a molar mass of 2.82×10^4 g / mol?

Solution:

a. Identify the repeating unit.



Thus, the repeating unit is  which has carbon- carbon double bond

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⇒ The structure of the monomer is $CF_2 = CF_2$

b. n (monomer) = molar mass of polymer

$n(CF_2 = CF_2) = 2.8 \times 10^4$ g/mol

$100n$ g/mol $= 2.82 \times 10^4$ g/ mol

$n = 282$

Thus, 282 monomer units are linked together to form the polymer.

5. Classify the following polymers as natural or synthetic polymers.

a. Cotton c. Wool e. Bakelite

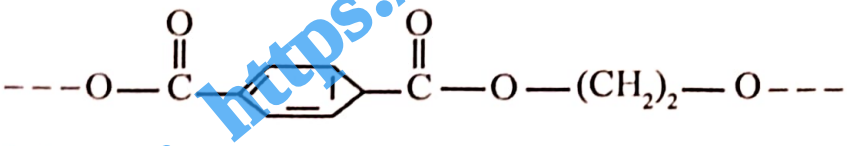
b. PVC d. Polyester f. Starch

Solution:

- a , c and f are natural polymers.
- b , d and e are synthetic polymers.

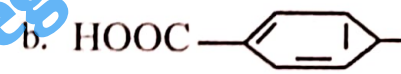
6. Identify the monomers of the following copolymers.

a.

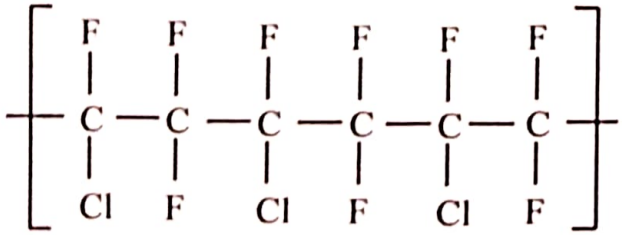
b. 

Solution:

a. HO CH_2 CH_2 OH and HOOC CH_2 CH_2 COOH

b.  and HO CH_2 CH_2 OH

7. A polymer has the following structure:



What is the name and structure of the monomer for this polymer?

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Solution:

$CFCl = CF_2$, 2 – chloro –1, 1,2 -tri fluoro ethene

10.3 Polymerization

Synthetic polymers can be classified into two major classes, depending on their method of preparation.

These are:

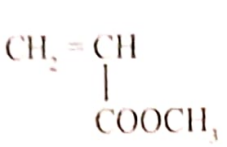
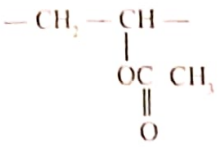
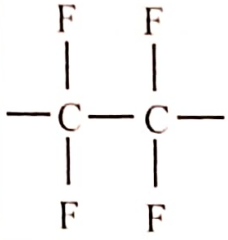
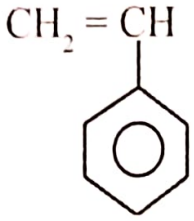
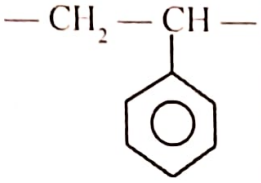
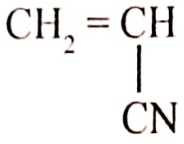
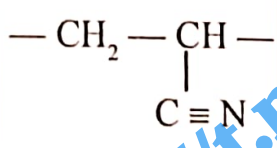
1. Addition polymers
2. Condensation polymers

1. Addition Polymers (chain growth polymers)

- They are made by chain reaction, the end of a growing chain
- The monomers usually contain multiple bonds
- There is no side product, only the polymer

Example

Monomer	Repeating unit	Polymer name	Use
$CH_2 = CH_2$	$-CH_2- CH_2-$	Polyethylene	bottles, plastic bags, electrical insulating film
$CH_3 -CH = CH_2$	$-CH_2-CH-$ $\quad\quad\quad

		Polyvinyl acetate	latex, paints adhesives
		Tetrafluoro ethylene (Teflon)	non sticking surfaces liners, electrical insulation
		Polystyrene	Packaging, toys, clear cups, CD and DVD cases, flower pots
		Polyacrylonitrile (Orlon, Acrilan)	Rugs, blankets, yarn, apparel, simulated fur

2. **Condensation Polymers (Step - growth Polymers)** are made by combining two molecules, while in most cases, removing a small molecule, generally water or alcohol.

   ➦ The monomers are usually bi functional, contains two functional groups. **Example** diols, dicarbobylic acids, amino acids etc.

Example

Condensation polymers:

a. Nylons

➦ **Nylon** is the generic name for polyamides.

➦ Nylon 66 - its monomers are hexamethylenediamine $(NH_2-(CH_2)_6-NH_2)$ and adipic acid $(HOOC(CH_2)_4-COOH)$.

➦ It is nylon 66 because both monomers contain 6 carbon atoms.

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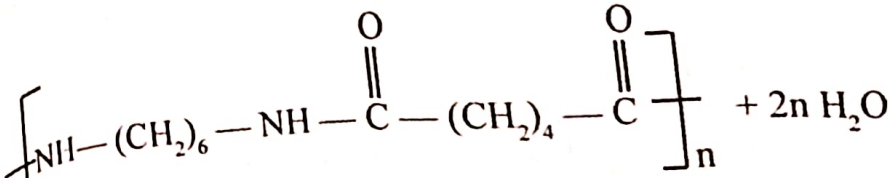
$$n(H_2N-(CH_2)_6-NH_2 + HOOC(CH_2)_4COOH)$$

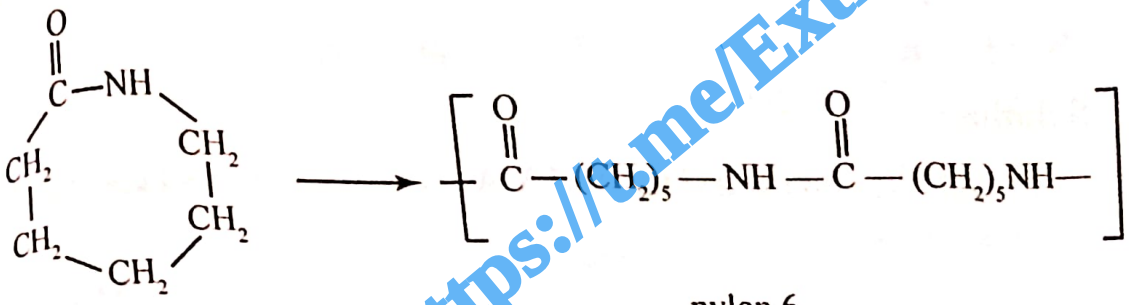


Nylon 66

- Nylon 66 is used in home furnishings, mostly carpets, clothes and tire cord.

Nylon 6 is made from **caprolactam**, which polymerizes on heating in water.

- It is called nylon 6 because of the presence of six carbon atoms in the monomer.



Caprolactam

nylon 6

- Nylon 6 can be machined and used for small, tough, plastic parts in addition to a textile fabric.

b. Polyesters

They are produced by condensation of ethylene glycol ($HO\,CH_2\,CH_2\,OH$) and dimethylterephthalate ($CH_3\,O_2\,C\,C_6H_4\,CO_2\,CH_3$) In the presence of acid or base.

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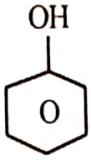
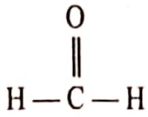
Answer: D

11. What are the monomers of bakelite? write the uses of bakelite.

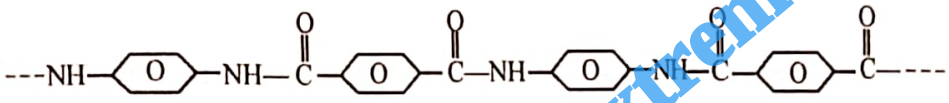
Solution:

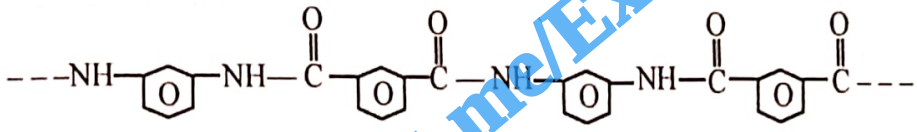
- The monomers of bakelite are phenol and formaldehyde

Phenol:  and formal dehyde: 

- Bakelite is used to make rotary - dial telephones, whistles, billard balls, dominos, chess, checkers,etc

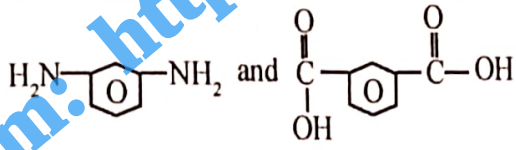
12. Draw the monomers the following condensation polymers.

a. 

b. 

Solution:

a.

b. 

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10.4 Thermoplastics and Thermosetting plastics

Thermoplastics	Thermosetting plastics
➦ soften on heating	➦ donot melt when heated
➦ held together by weak van der Waals forces at higher temperature these forces are not strong enough, the polymers soften on heating but solidfy on heating but solidify on cooling.	➦ cannot be reshaped ➦ unaffected by heat below the temperature at which they decompose

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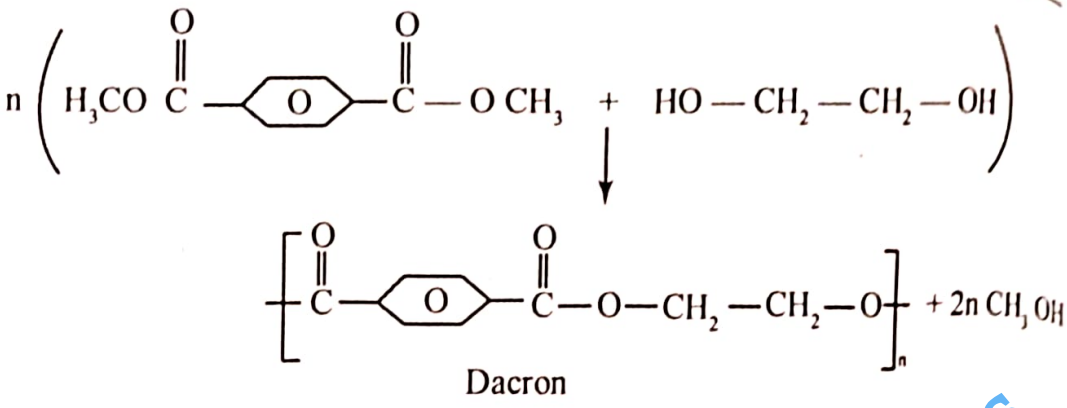

Dacron

➥ Most of the polyester produced goes into making clothes. The rest goes into making carpets or tire cord.

Illustrative Examples

8. What is the other name for polymethyl methacrylate (PM MA)?

A. Perspex
B. Teflon
C. Bakelite
D. Nylon -6

Solution:

Polymethyl methacrylate (PM MA) is also known as Perspex, plexi glass or Lucite.

Answer: A

9. What is the monomer of polyvinyl chloride (PVC)?

A. Chloro ethane
B. Ethylenedichloride
C. Chloro form
D. Chloro ethene

Solution

The monomer of PVC is chloro ethene or vinyl chloride

($CH_2 = CHCl$)

Answer: D

10. Which of the following statements about poly vinyl chloride is NOT correct?

A. PVC can be used in making water pipes
B. PVC is stiff
C. PVC is softened on heating
D. The monomer of PVC is CHCl = CHCl.

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Answer: D

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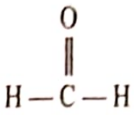
Phenol: and formal dehyde: 

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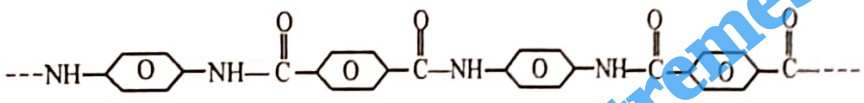
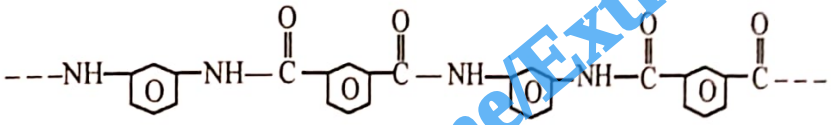
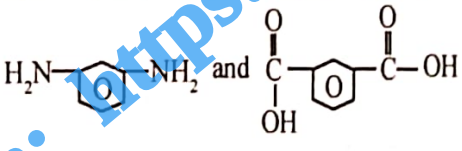
a. 

b. 

Solution:

a.

b. 

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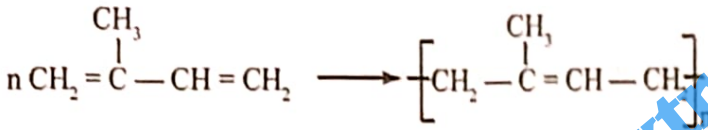
Solution:

- Cause cancer, birth defects, damage nervous system, blood and immune system.
- Pollute land, water and air.

10.5 Natural Polymers

1. Natural Rubber

- It is obtained from rubber tree
- It is addition homo polymer.
- Its monomer is isoprene (2 - methyl - 1.3 - butadiene)



Isoprene Natural rubber

- Natural rubber is hardened or vulcanized by sulfur.

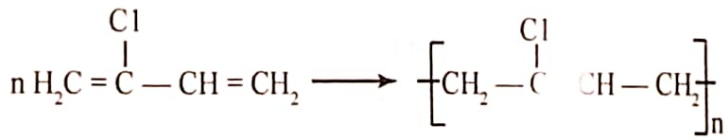
Properties of vulcanized rubber

- It is hard and non-sticky.
- It has high tensil strength.
- It is insoluble in solvents.
- It is not oxidized easily.

Synthetic rubber

i Neoprene

- It is the first synthetic rubber
- It is addition homo-polymer.
- Its monomer is chloroprene (2 - chloro - 1,3 - butadiene)



Chloroprene oprene

- It is used to make hoses for chemicals, petrols and oils.

ii Styrene butadiene rubber (SBR)

- It is addition copolymer.

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➥ have wide range of applications	➥ are often highly cross linked polymers
Example Polyethylen, polypropylene, polyvinyl chloride, polystyrene, Teflon, polymethylmetha crylate, nylon, dacron etc.	➥ are hard, strong and brittle **Example** Bakelite, melmac, rubber and various super glues.

Impacts of Synthetic Polymers on Environments

- ➥ Plastics such as PVC and polyethene give off highly toxic and potentially deadly gases when burned.
- ➥ Many of the poisonous chemicals released in plastic production can cause cancer, birth defects and damage the nervous system, blood, kidneys and immune systems
- ➥ Every year consumers throw millions of tons of plastics away. As municipal landfills reach capacity and additional land fill space diminishes, alternative methods for reducing and disposing plastics are explored. Some of these options include reducing consumption of plastics, using biodegradable plastics and incinerating or recycling plastics waste.
- ➥ Researchers are working to develop biodegradable plastics.

Illustrative Examples

13. Which of the following thermoset polymer?

A. PVC
B. PMMA
C. Bakelite
D. Teflon

Thermoset polymer is the same as thermosetting plastic

Answer: C

14. How thermoplastics and thermosetting plastics are recycled?

Solution:

- ➥ Thermoplastice can be remelted and made into products.
- ➥ Thermosetting plastics can be ground and then used as filler in modable thermoplastic materials.

15. State the impacts of synthetic polymers

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Solution:

- Cause cancer, birth defects, damage nervous system, blood and immune system.
- Pollute land, water and air.

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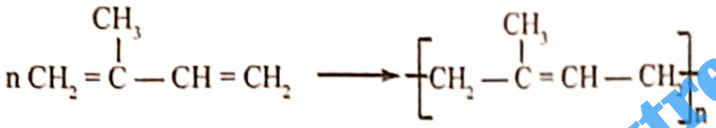

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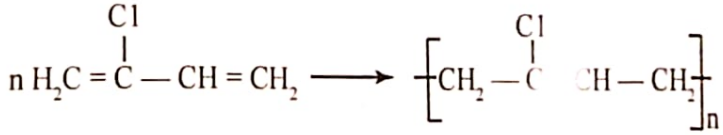

Chloroprene oprene

- It is used to make hoses for chemicals, petrols and oils.

ii Styrene butadiene rubber (SBR)

- It is addition copolymer.

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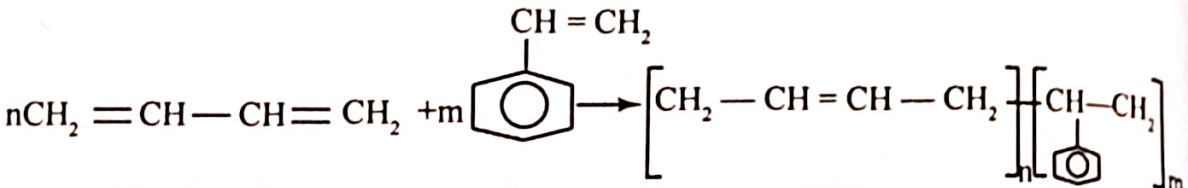
- It is vulcanized with Sulfur.
- Its monomers are styrene and 1, 3 - butadiene

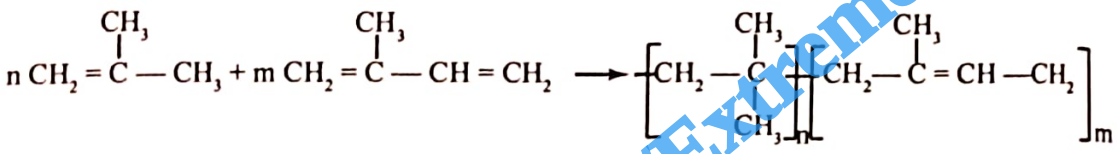


1.3 – butadiene Styrene SBR

- It is mainly used to make tires

iii Butyl rubber

- It is addition copolymer
- Its monomers are 2– methylpropene and isoprene



- It is used to make inner tubes.

10.6 Carbohydrates

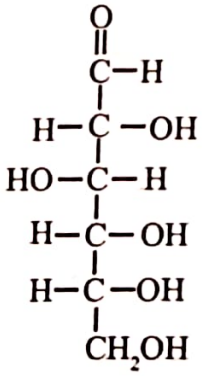
- They are organic compounds containing hydrogen, carbon and oxygen.
- They are polyhydroxy aldehydes or polyhydroxy ketones
- They are classified into three.

I. Mono Saccharides

- They are simple sugars containing three to six carbon atoms.
- They cannot be hydrolyzed into smaller molecules.
- The most common mono-saccharides are:

i Glucose ($C_6H_{12}O_6$)

- Most abundant monosaccharide.
- It is commonly known as blood sugar.
- It is aldhexose sugar.



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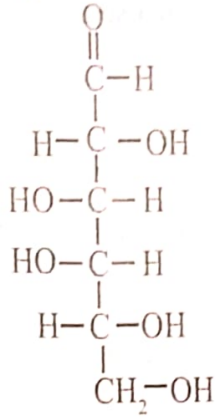
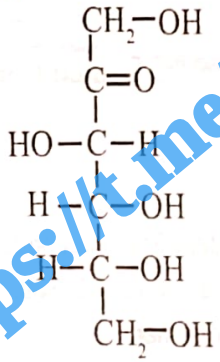
ii Galactose ($C_6H_{12}O_6$)

- It is commonly known as milk sugar.
- It is aldohexose sugar.



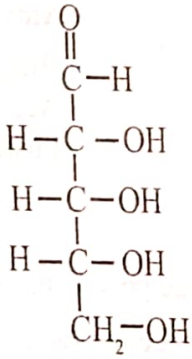
iii Fructose ($C_6H_{12}O_6$)

- It is commonly known as fruit sugar.
- It is ketohexose sugar.



iv Ribose ($C_5H_{10}O_5$)

- It is component of RNA and DNA.
- It is aldopentose.



2. Disaccharides

- They are formed from two monosaccharide units by glycosidic linkage or bond.

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- The most common disaccharides are.

i **Maltaose** ($C_{12}H_{22}O_{11}$)

- It is also called malt sugar
- It is formed from two glucose units by, $\alpha - 1, 4 -$ linkage.

ii **Sucrose** ($C_{12}H_{22}O_{11}$)

- It is also called table sugar
- It is the most abundant disaccharide
- It is formed from glucose and fructose by $\alpha, \beta - 1,2 -$ linkage.

iii **Lactose** ($C_{12}H_{22}O_{11}$)

- Formed from β - galactose and α or β - glucose by β - 1, 4 - linkage.
- It is also called milk sugar.

iv **Cellulobiose**

- Formed from two glucose units by β - 1, 4 - linkage

3. Polysaccharides

- They are natural polymers formed from monosaccharides

Example

i **Cellulose**

- It is the most abundant Poly saccharide
- It is structural component of cell wall.
- It is linear glucose polymer.

ii **Starch**

- It is a polyaccharide used as energy storage in plants.

Starch

Amylose	Amylopectin
Straight chain glucose Polymer	Branched glucose polymer
Accounts for 20 – 25% of starch.	Accounts for 75– 80% of starch
Connected by α –1.4 – glycosidie	Linked by α – 1.4 – linkege.

iii **Glycogen**

- It is a glucose storage polysacharide in man and animals
- Its units are joined together by $\alpha - 1, 4 -$ and $\alpha - 1, 6 -$ glycosidic bonds.
- Liver and muscle cells are the storage sites for glycogen in

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humans.

Illustrative Examples

16. Which of the following is the monomer of natural rubber?

A. Chloroprene
B. Isoprene
C. Styrene
D. Butadiene

Answer: B

17. Which of the following element is used to vulcanize rubber?

A. Phosphorps
B. Sulphur
C. Carbons
D. Nitrogen

Answer: B

18. Sucrose is _____

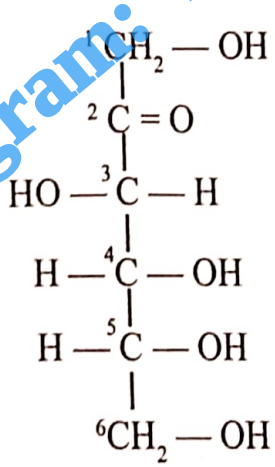
A. a disaccharide with one glucose and one fructose
B. a disaccharide with two units of glucose.
C. a disaccharide with two units of fructose.
D. a polysaccharide with a repeating α – D – glucose

Solution:

Sucrose is non - reducing sugar formed from one glucose and one fructose units.

Answer: A

19. How many chiral carbon atoms are there in the open chain form of fructose?



A. 2
B. 3
C. 4
D. 5

Explanation: The carbon atom that is surrounded by four different atoms or group of atoms is called asymmetric. Asymmetric atoms are known as

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chiral atoms and the molecules are chiral molecules.

There are three chiral atoms in open chain fructose.

Answer: B

20. What is the type of linkage that chemically connects the sub units of carbohydrates?

Solution: Glycosidic linkage.

21. What is the molecular formula of disaccharides?

Solution: $C_{12} H_{22} O_{11}$

22. The monosaccharide that circulates in the blood stream and supplies the body as a source of energy is _____

Answer: glucose

23. Write three properties of vulcanized rubber.

Solution:

- It is hard and non sticky
- It has higher elasticity
- It has high tensile strength
- It is insoluble in solvents

10.7 Proteins

- They contain carbon, hydrogen, oxygen and nitrogen.
- They are formed from the 20 naturally occurring α –amino acids.
- Amino – acids are bi-functional compounds containing amino ($-NH_2$) and carbocylic ($-COOH$) groups.

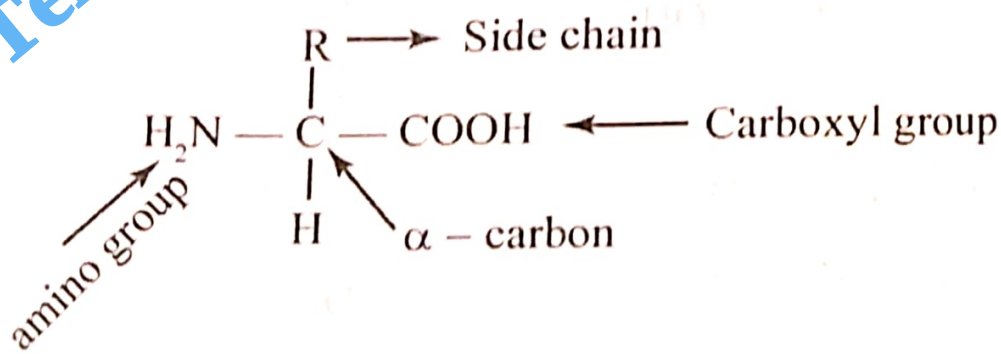

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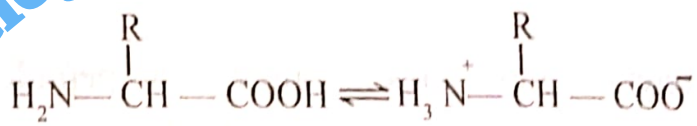
Table 12.1 Type of amino – acids

Non- polar amino acids	Polar neutral amino acids	Polar acidic amino acids	Polar basic amino acids
➦ have one amino and one carboxyl group ➦ have non-polar side chain.	➦ have one amino and one caboxyl group ➦ have polar neutral side chain	➦ have one amino and two carboxyl groups.	➦ have two amino and one carboxylic groups.
Example ➦ Glycine ➦ Alanine ➦ Valine ➦ Leucine ➦ Isoleucine ➦ Methionine ➦ Proline ➦ Phenylalanine	**Example** ➦ serine ➦ cryteine ➦ Threonine ➦ Asparagine ➦ Glutamine ➦ Tyrosine ➦ Trypotophan	**Example** ➦ Aspartic acid ➦ Glutamic acid	**Example** ➦ Histidine ➦ Lysinee ➦ Arginine

Physical properties of amino acids

- ➦ They are colourles crystalline substance
- ➦ They have high melting points.
- ➦ They are soluble in water
- ➦ They undergo internal protonation



Zwitter ion

- ➦ Zwitter ion has zero net charge

Peptide bond formation

- ➦ A peptide bond (amide) is a bond between the carbonyl of amino- acid and the amino group of another amino acid
- ➦ A compound containing two and three amino acids called **dipeptides and tri peptides** respectively

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- A **polypeptide** is a sequence of amino acids containing up to 50 amino acid units.
- Proteins are polypeptides.
- Enzymes and hormones are proteins

Illustrative Examples

24. What is the monomer unit in a natural protein polymer? (EUEE)

A. Amino acids
B. Essential amino-acids
C. α – amino acids.
D. Nucleic acids

Answer: C

25. The primary structure of proteins is

A. the way active sites are arranged in proteins
B. the way peptide chains pack together
C. the way α - helices ford.
D. the order in which amino acids are arranged in proteins.

Solution:

The primary structure of protein refers to the order or sequence of the amino acid units in the polymer

Answer: D

26. How many different tetra peptides can be made from 20 different amino acids? (EUEE)

A. 20 B. 20^2 C. 20^3 D. 20^4

Solution:

For a different amino - acids n^4 different tetrapeptide (peptides centainging four amino acids can be made.)

20 different amino acids form = 20^4 different tetrapeptides.

Answer: D

27. What type of polymer is formed from amino - acids?

Answer: Proteins

28. Write the functional groups in proteins.

Solution: Carboxyl and amino

Solved Problems

29. Identify the monomers of the following polymers.

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a. $\left[CH_2 - CH_2\right]_n$ b. $\left[CH_2 - CH = CH - CH_2\right]_n$

Solution:

a. $CH_2 = CH_2$ b. $CH_2 = CH - CH = CH_2$

30. What is the charge of an amino acid at pH 14?

Solution:

$P^H = 14$ is basic solution:

$$H_2N - \underset{\displaystyle R}{\underset{|}{CH}} - COOH + \bar{O}H \rightleftharpoons H_2N - \underset{\displaystyle R}{\underset{|}{CH}} - C\bar{O}O + H_2O$$

➦ The charge of the amino acid is −1.

31. What functional groups are involved in the addition polymerization processes?

Solution:

⇒ Monomers should contain multiple bonds, but in condensation polymerization monomers contain two functional groups for example diols, dicorboxylic acids and amino acids.

32. _____ are commonly known as polyamides

Answer: Nylons

Supplementary Exercise on Unit - 12

I. Multiple Choices Items

1. Which of the following is not a polymer?
   A. cellulose C. aminoacid
   B. PVC D. Dacron
2. Which of the following is not thermoplastic?
   A. Bakelite C. Teflon
   B. PVC D. Nylon
3. Which of the following cannot polymerize?
   A. Ethane C. Propene
   B. Ethene D. Isoprene

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4. Which of the following can not be hydrolyzed?

A. sucrose C. lactose

B. glucose D. Maltose

5. All of the following are synthetic polymer **EXCEPT**

A. nylon C. cotton

B. PVC D. Dacron

6. Which type of amino acids contain two amino groups and one carboxyl group?

A. Non-polar C. Polar acidic

B. polar neutral D. Polar basic

7. Which structure represents the sequence of amino acids a long a protein chain?

A. α- helix C. secondary

B. Primary D. Tertiary

8. Which of the following is used for making rotary – dial telephones?

A. Bakelite B. PVC C. Teflon D. Nylon

9. The monomer of PVC is

A. $CH_2 = CH_2$ C. $CH_2 = CH - CH_3$

B. $CH_2 = CHCl$ D. $CH_2 = CClCH_3$

10. The monomer of neoprene is

A. isoprene C. chloroprene

B. ethene D. styrene

II. Short answer Items

11. The process in which monomer units combine to form polymers is _____

12. What is the molar mass of a polypropylene sample, when n = 2400?

13. List the properties of non-vulcanzed rubber.

14. What are the monomers of Butyl rubber?

15. State the uses of

a. Nylon b. Dacron

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Answer To Supplementary Exercise

Part I

1. A
2. A
3. A
4. B
5. C
6. D
7. A
8. A
9. B
10. C

Part II

11. Polymerization
12. 1.008×10^5 g/mol
13. Non- sticky, not oxidized
14. Iso prene and 2- methyl propene
15. a. to make clothes
    b. to make clothes, carpets etc

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UNIT 11

Introduction to Environmental Chemistry

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11.1 Introduction

Chemistry is the study of the composition, structure and properties of materials and how they undergo chemical and physical changes.

Environmental chemistry is the study of those changes that have had an effect on both living organisms and non - living matter in the environment.

There are many ways in which human and other living organisms are exposed to chemicals such as:

- Detergents
- Paints
- Drugs
- Exhaust fumes
- Industrial effluents
- Pesticides

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- Natural toxin in plants and animal in their everyday existence.

These chemical may be Cause of Pollution due to

- Human error
- Lack of knowledge and understanding
- Inefficient technology

Many chemical are dangerous but many are beneficial. As a result, we have to use them properly and with great care.

Components of the Environment

The environment has four main components

1. Atmosphere or the air
2. Hydrosphere or the water
3. Lithosphere or the rocks and soil
4. Biosphere or the living component of the environment.

Air, water, soil and climate are the physical component of the environment.

The biological component of the environment includes all living things interacting ecosystem.

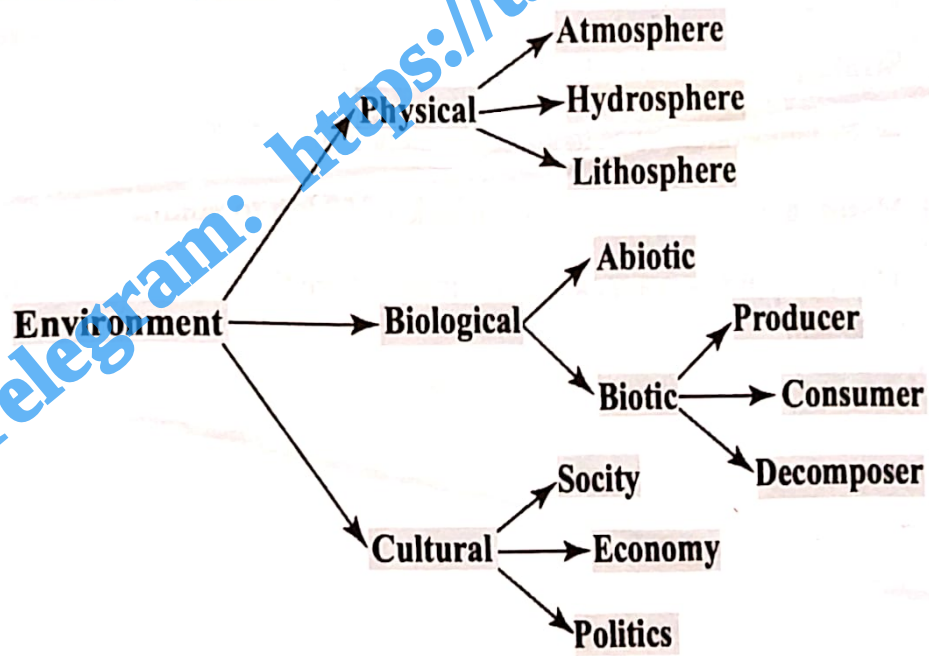

Fig 11 . 1 Environment and its Components

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1. The Atmosphere or the Air

It is a thin layer of gases that surrounds the earth and protects life from harmful radiation and extreme temperature.

Table 11.1 Components of the Earth's Atmosphere

Gas	Percentage by volume
Nitrogen	78
Oxygen	21
Argon	0.93
Carbon dioxide	0.03

The Atmosphere is Separated in to Four Main Regions.

i. **Troposphere** - is the lowest region of the atmosphere that contains about 90% of all the matter in the atmosphere.

   ➥ It extends to a height of between 8 and 15km depending on the altitude.

   ➥ Its temperature is between 200K and 220K (-73°C to -53°C)

ii. **Stratosphere** - extends to a height of about 50km.

   ➥ Its temperature rises to approximately 290K (17°C)

iii. **Mesosphere** - extends from an altitude of 50 km to 90km.

iv. **Thermosphere** - is a region above mesosphere that extends from 90km to 800km.

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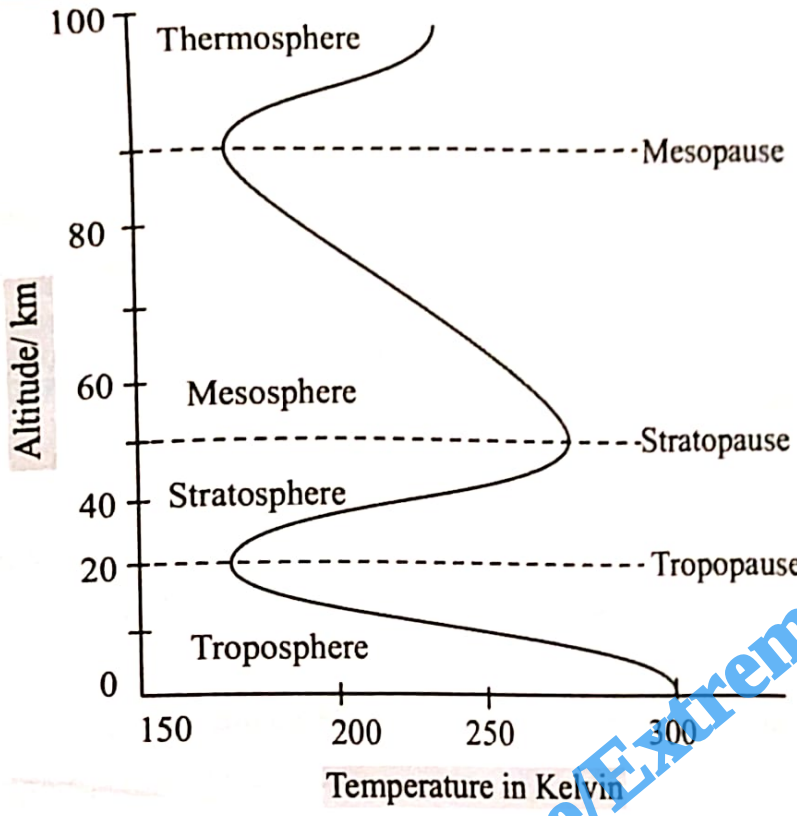

Fig 11 . 2 Vertical Structure of the Earth's Atmosphere

Illustrative Examples

1. A branch of chemistry that studies about production, transformation, reactions, effects and fates of chemical species in air, water, land and biosphere is

   A. Biochemistry
   B. Physical chemistry
   C. Environmental chemistry
   D. Industrial chemistry

   Answer: C

2. All of the following are components of the environments. **EXCEPT**

   A. Atmosphere
   B. Biosphere
   C. Lithosphere
   D. Troposphere

   Solution

   The four components of the environment are atmosphere, lithosphere, hydrosphere and biosphere.

   Answer: D

3. The most abundant gas in the atmosphere is

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A. Oxygen
C. Argon
B. Nitrogen
D. Carbondioxide

Solution

78% by volume of the atmosphere is nitrogen

Answer: B

4. Which one of the following is not the physical components of the environment?

A. Plants B. Air C. Soil D. Water

Solution

Air, soil, water and climate are physical components but living things are biological components of the environment.

Answer: A

5. Write the four regions of atmosphere from bottom to top.

Solution

Troposphere ⟶ Stratosphere ⟶ Mesosphere ⟶ Thermosphere

6. There are many ways in which human and other living things exposed to chemicals. List some of the chemicals.

Solution

Detergents, paints, drugs, insecticides, fertilizers, industrial effluents.

2. Hydrosphere or the Water

- It is the component of an environment that includes water bodies such as oceans, lakes, rivers, glaciers etc.
- These water bodies are found on, under and above the surface of the earth as well as in the atmosphere in the form of water vapour.
- But only around 1% of this water is fresh water. The ocean is the greatest source of water which is hard water. It is used as a source of many elements such as sodium. magnesium, chlorine, bromine etc.
- The major processes involved in hydrosphere are precipitation, evaporation, interception, transpiration, infiltration, percolation, retention, detention, overland flow, through flow and turn off.
- The nitrification process also occurs

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$$2NH_3 + 3O_2 \longrightarrow 2NO_3^- + 2H^+ + 2H_2O$$

3. The Lithosphere or the Rocks and Soils

It is the solid, outer part of the earth or any other rocky planet or natural satellite.

It is the rigid outer part of the earth consisting of the crust and upper layer of the mantle.

The chemical Reactions Occurring in Lithosphere are:

- Weathering of rocks to form soil.
- Reaction between carbondioxide and rain to from carbonic acid, which further reacts with carbonates in the rock.

$$CO_2 + H_2O \longrightarrow H_2CO_3$$

$$H_2CO_3 + CaCO_3 \longrightarrow Ca(HCO_3)_2$$

4. The Biosphere or the Living Component of the Environment

- It is part of the earth where life exists and grows
- It is global ecosystem composed of living organisms and the non - living factors from which they derive energy and nutrients.
- Photosynthesis and respiration are the common chemical reactions in the biosphere.

Photosynthesis

$$6CO_2(g) + 6H_2O(1) \xrightarrow{\text{sunlight}} \underset{\text{glucose}}{C_6H_{12}O_6(s)} + O_2(g)$$

Respiration

$$C_6H_{12}O_6(s) + 6O_2(g) \longrightarrow 6CO_2(g) + 6H_2O(1) + \text{heat}$$

Photosynthesis is endothermic reaction but respiration is an exothermic reaction

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Illustrative Examples

7. Which of the following is domain of living organisms?

A. Lithosphere
B. Hydrosphere
C. Atmosphere
D. Biosphere

Answer: D

8. Which of the following does not pertain to biosphere?

A. Hydrosphere
B. Thermosphere
C. Lithosphere
D. Atmosphere

Solution

Thermosphere is not part of a biosphere. It is the region of the atmosphere above the mesosphere.

Answer: B

9. Weathring of rocks into soil occurs in

A. Atmosphere
B. Biosphere
C. Lithosphere
D. Hydrosphere.

Answer: C

10. Write the two common reactions that occur in biosphere

Solution

Photosynthesis; $6CO_2(g) + 6H_2O(1) \longrightarrow C_6H_{12}O_6(s) + O_2(g)$

Respiration: $C_6H_{12}O_6(s) + O_2(g) \longrightarrow 6CO_2(g) + H_2O(1) + \text{heat}$

11. What is biosphere?

Solution

Biosphere refers to the realm of living organisms and their interaction with the environment.

12. What are the components of hydrosphere?

Solution

Water that is on the surface of the planet (earth), underground and in the air

13. Write two reactions that occur in the lithosphere

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Solution

1. Oxidation of Minerals

$$2FeCO_3 + \frac{1}{2}O_2 + 3H_2O \xrightarrow{\text{Ferrobacilha}} 2Fe(OH)_3 + 2CO_2$$

2. Chemical Weathering of Rocks

$Fe_2O_3 + 3H_2O \rightarrow Fe_2O_3\ 3H_2O$ (hydration of rocks)

11.2 Natural Cycles in the Environment

A natural process in which substances are continuously cycled in various forms between different compartments of the environment are known as **natural cycles.**

Some of the natural cycles are: the hydrologic cycle, oxygen cycle, nitrogen cycle, phosphorus cycle, sulphur cycle and carbon cycle.

1. The Hydrologic Cycle /water cycle/

- Water is the most important polar solvent.
- The driving force behind the water cycle is the **heat of the sun.**
- Heat from the sun causes evaporation from oceans, seas and lakes.
- Water vapour is also formed from **leaves (transpiration), through respiration and through combustion.**
- The water vapour rises and cools, and condenses to form tiny droplets of water. These droplets form **clouds.**
- The clouds are moved along by air currents. As they cool, the tiny droplets join to form **large droplets, which fall as rain.**
- The water that falls as rain runs into streams and rivers and then in to lakes, seas and oceans.

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